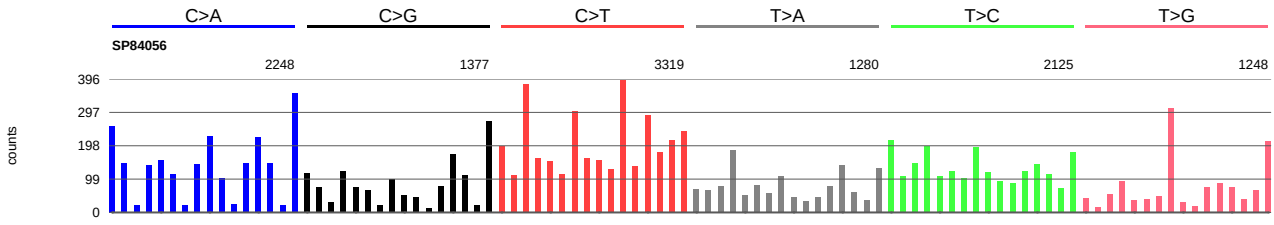
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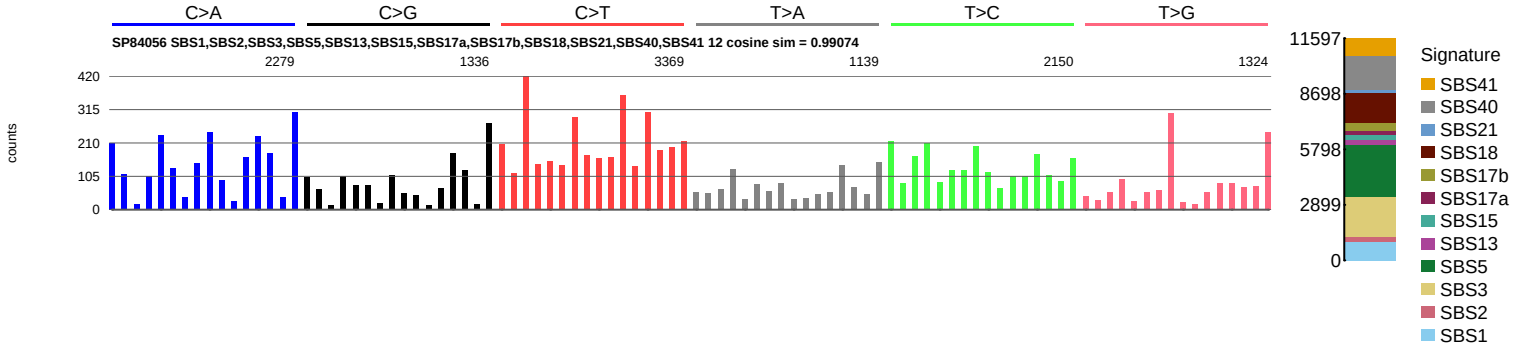
SP84056
2248
1377
3319
1280
2125
1248
counts
396
297
198
99
0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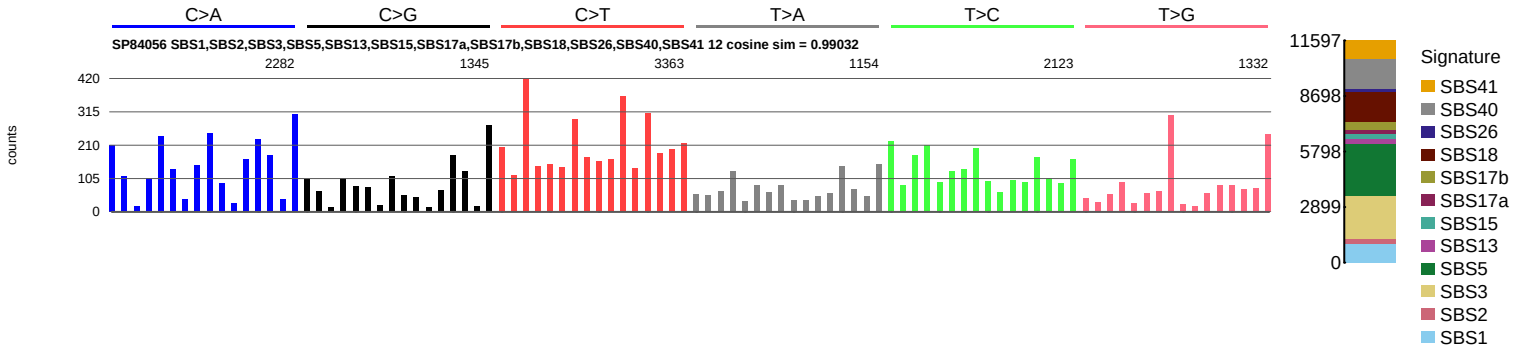
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 12 cosine sim = 0.99074
2279
1336
3369
1139
2150
1324
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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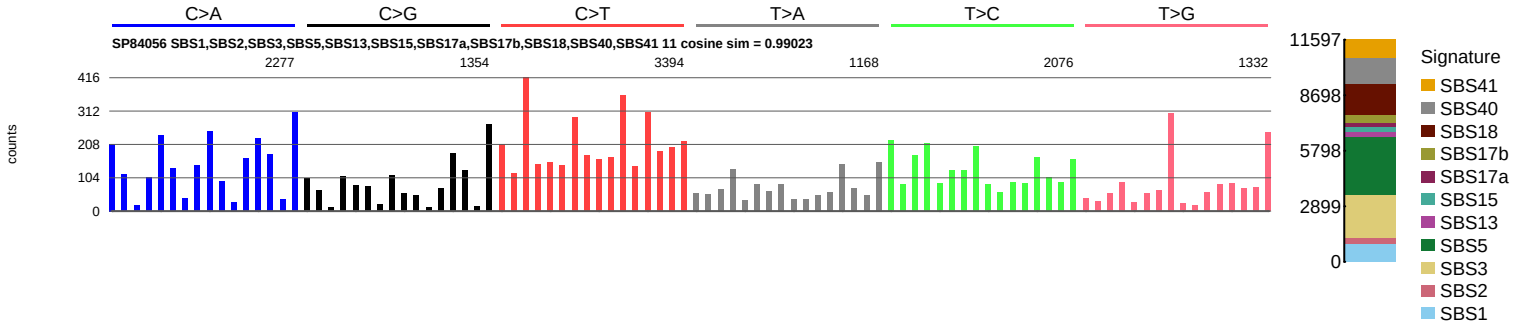
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 12 cosine sim = 0.99032
2282
1345
3363
1154
2123
1332
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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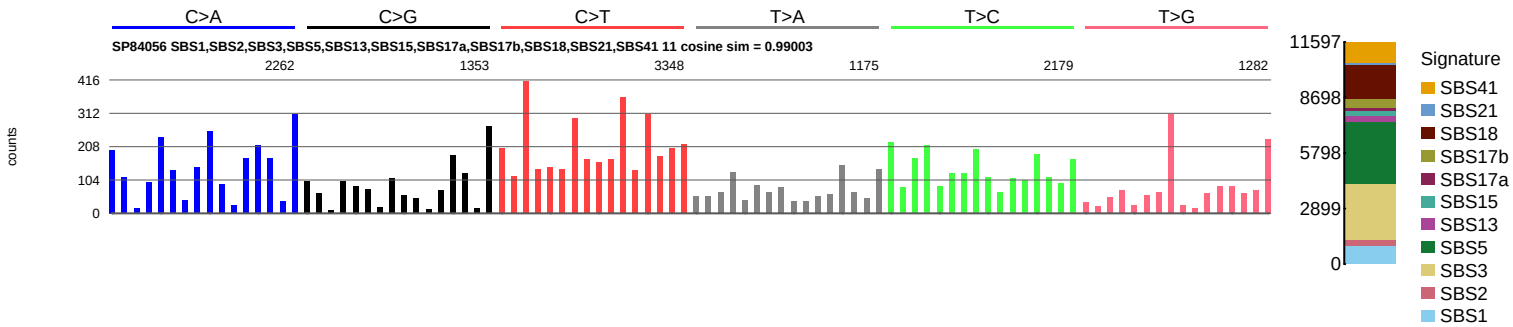
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.99023
2277
1354
3394
1168
2076
1332
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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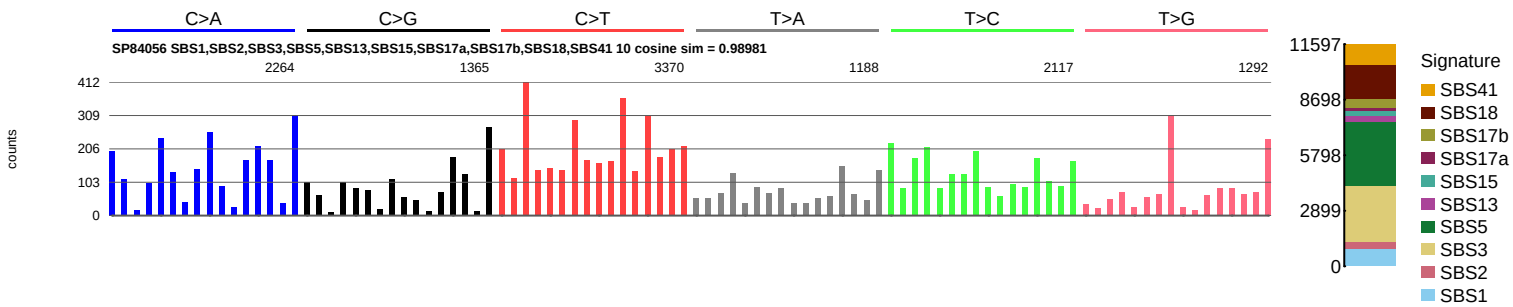
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99003
2262
1353
3348
1175
2179
1282
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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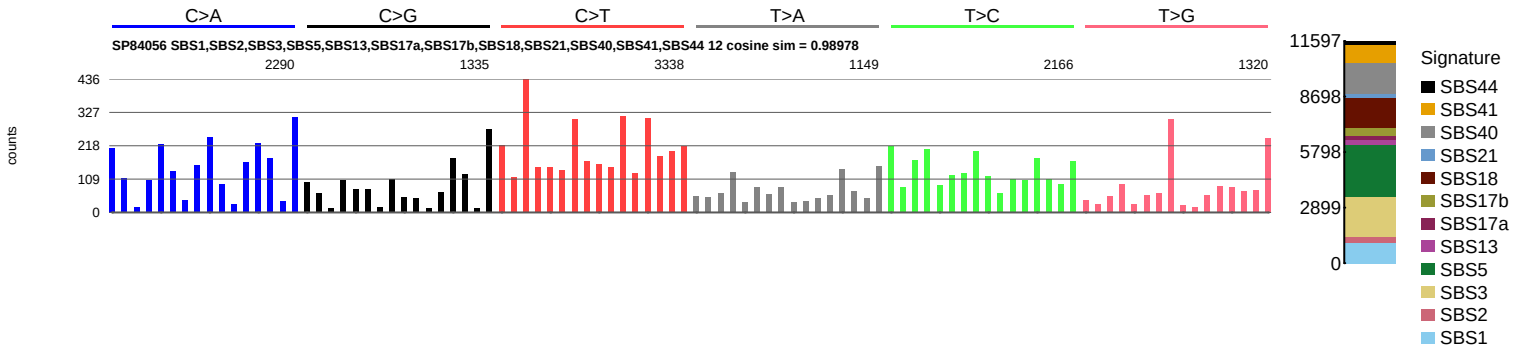
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98981
2264
1365
3370
1188
2117
1292
counts
412
309
206
103
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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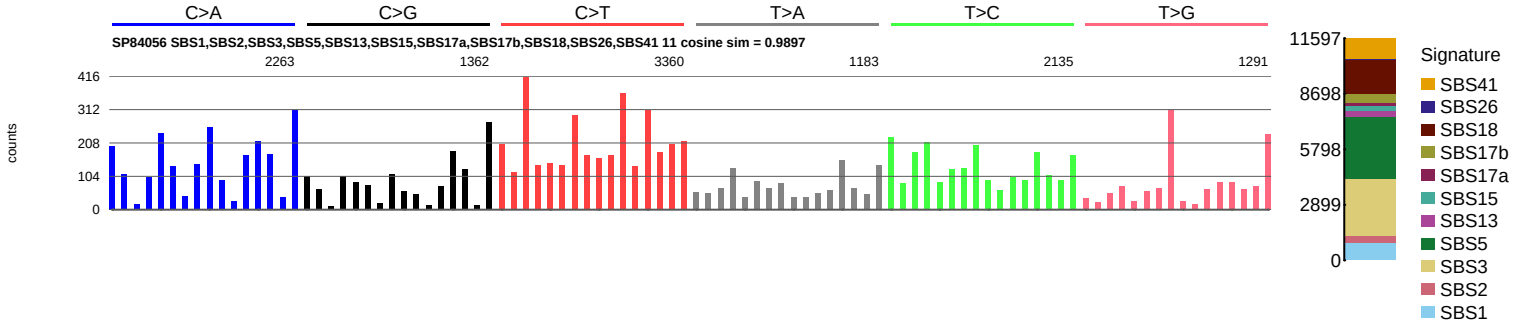
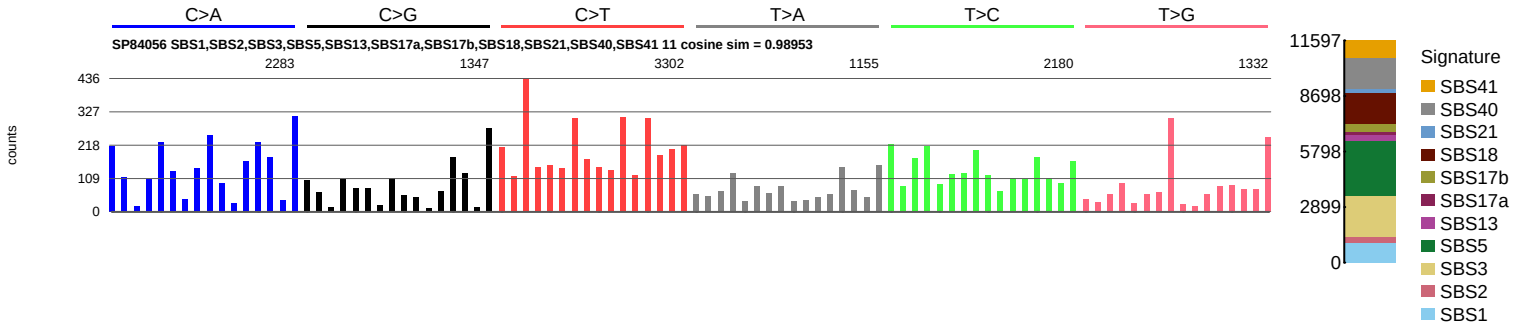
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98978
2290
1335
3338
1149
2166
1320
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 11 cosine sim = 0.9897
2263
1362
3360
1183
2135
1291
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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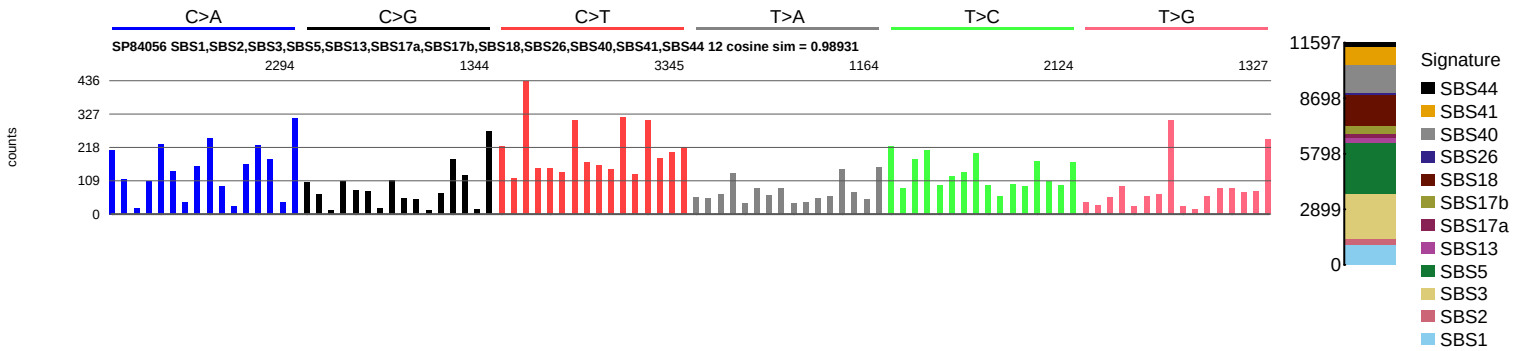
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98953
2283
1347
3302
1155
2180
1332
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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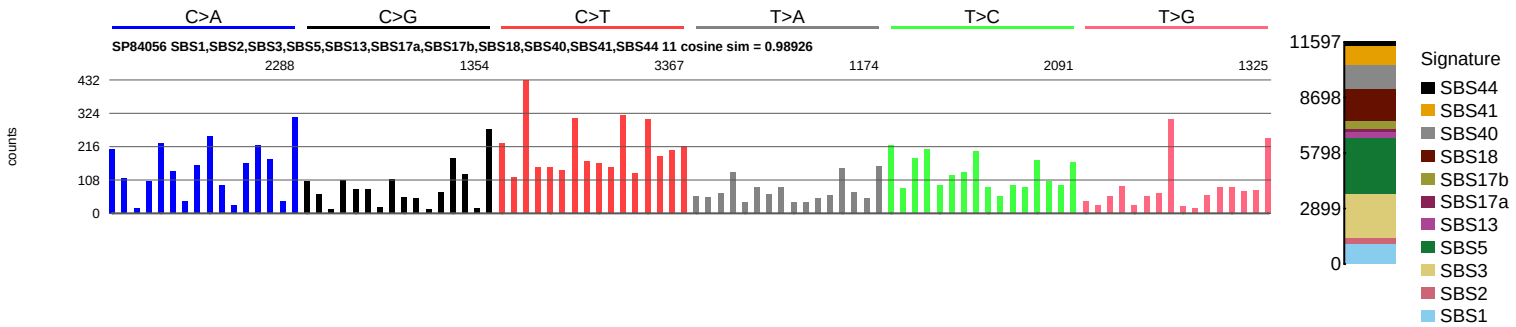
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98931
2294
1344
3345
1164
2124
1327
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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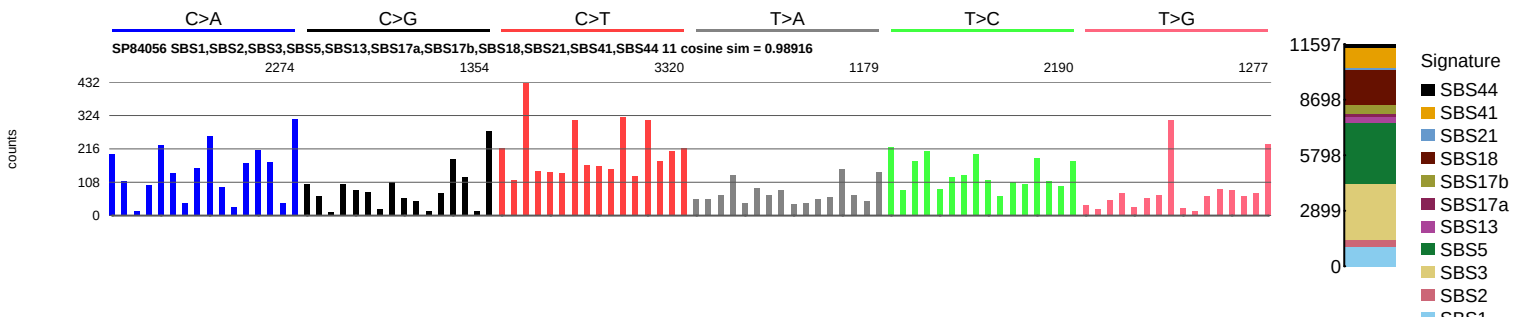
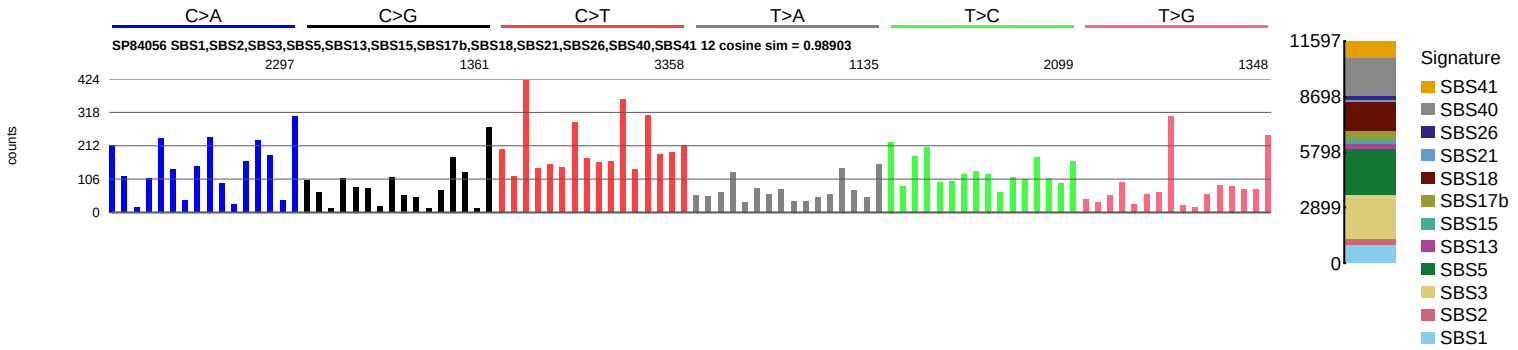
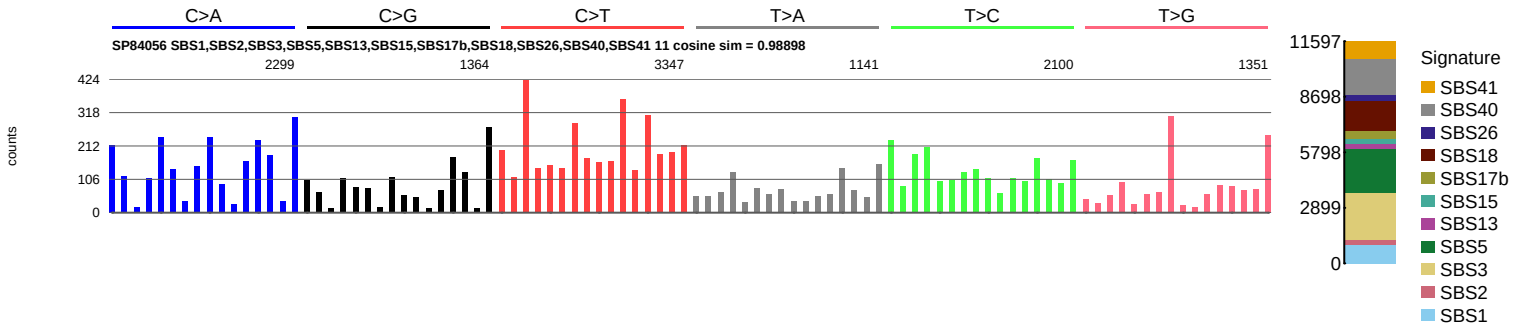
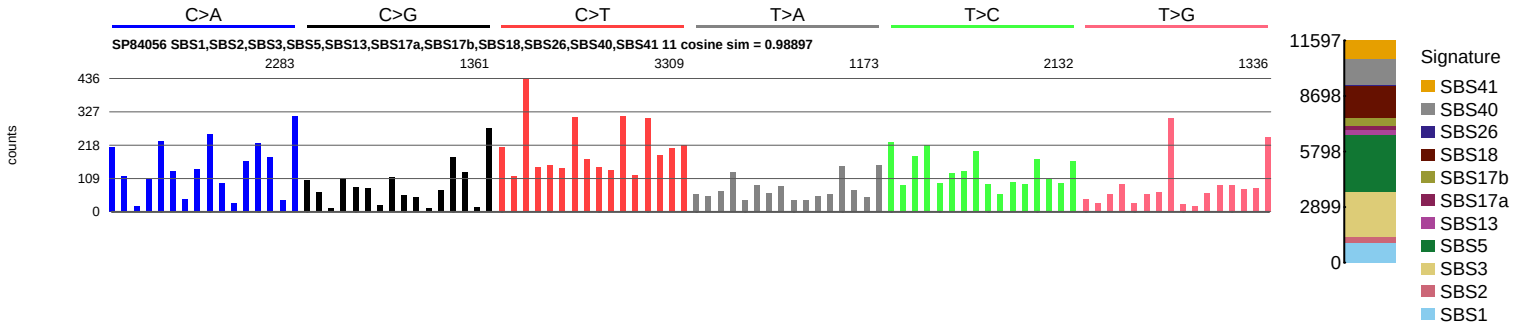
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98926
2288
1354
3367
1174
2091
1325
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98916
2274
1354
3320
1179
2190
1277
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.98903
2297
1361
3358
1135
2099
1348
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98898
2299
1364
3347
1141
2100
1351
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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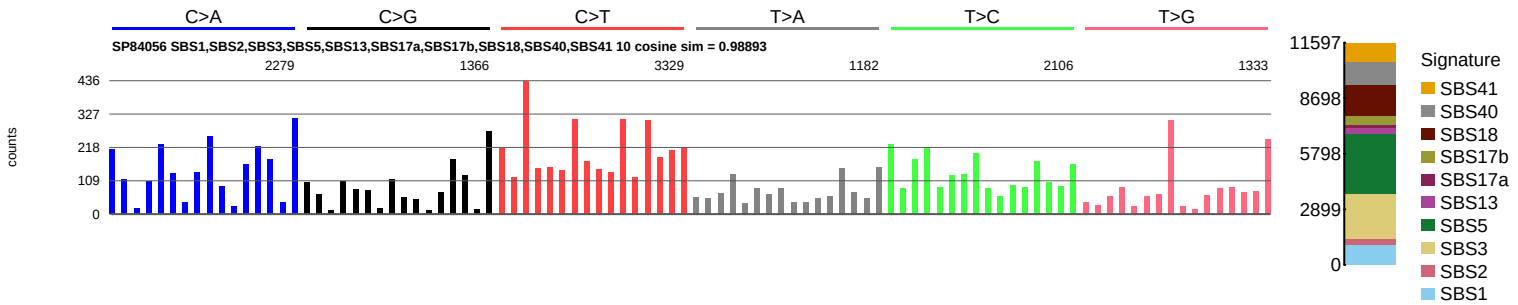
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98897
2283
1361
3309
1173
2132
1336
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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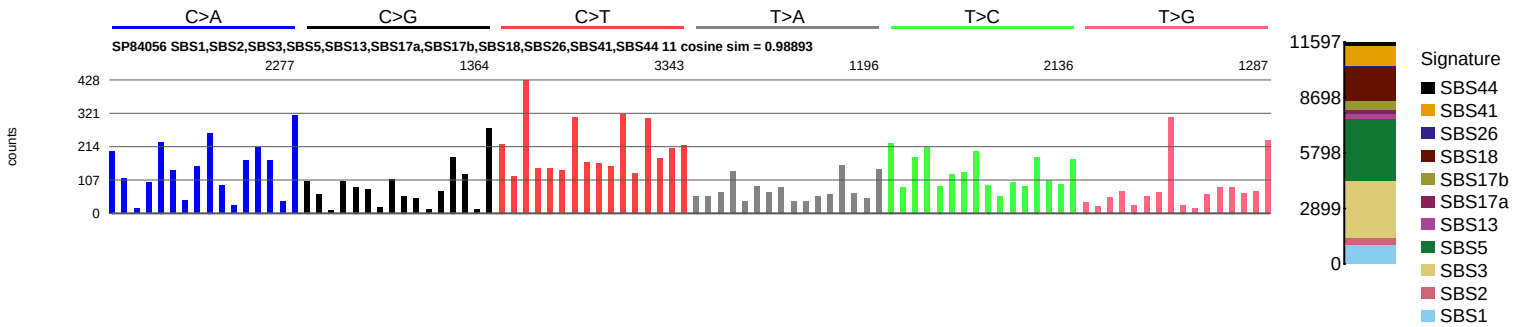
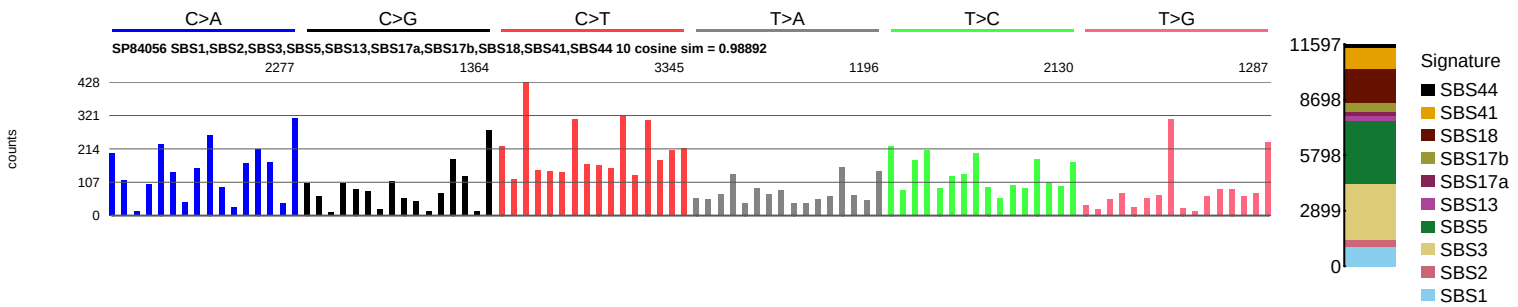
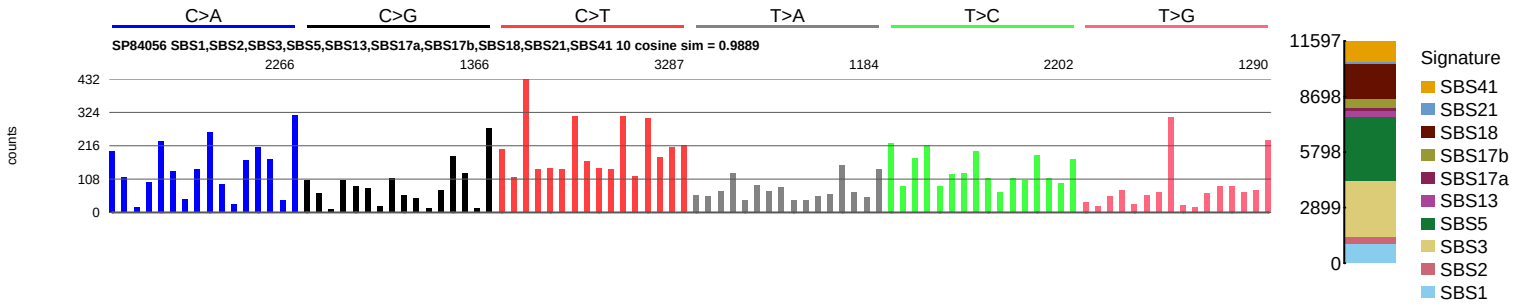
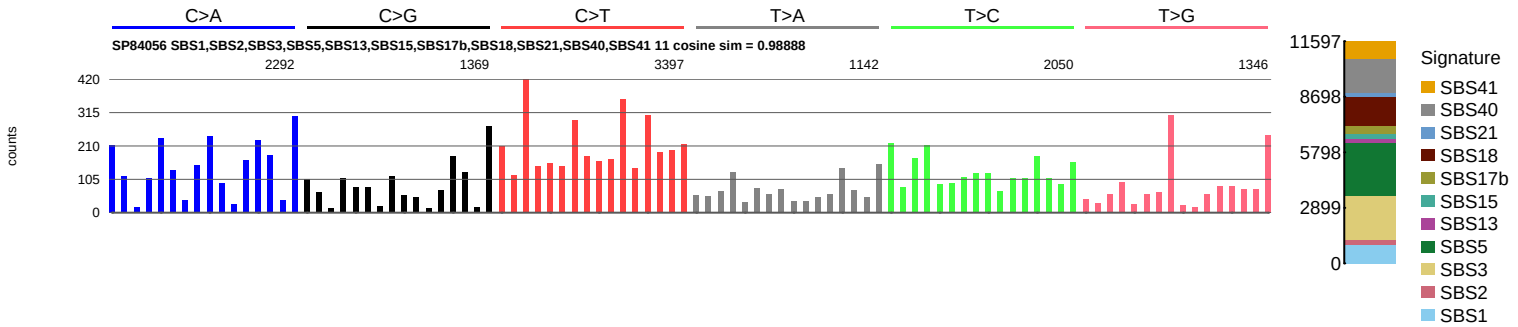
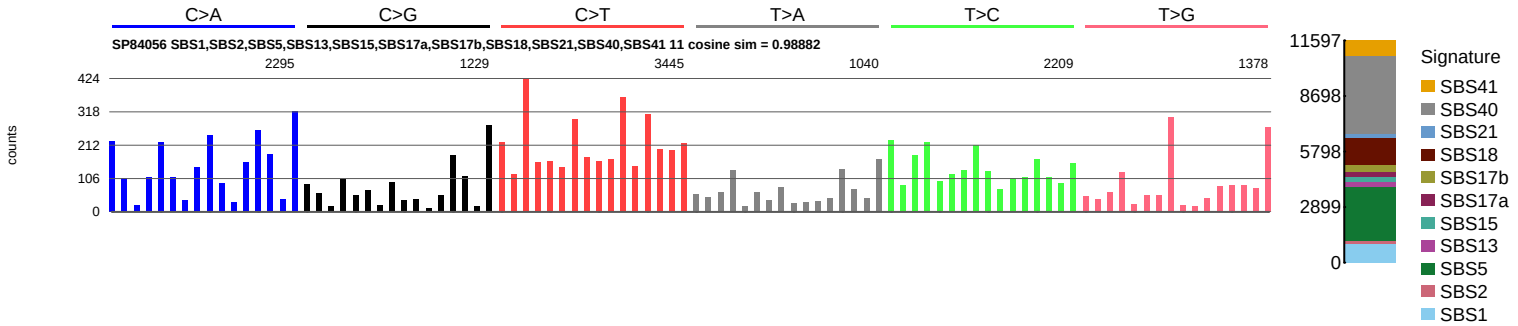
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98893
2279
1366
3329
1182
2106
1333
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

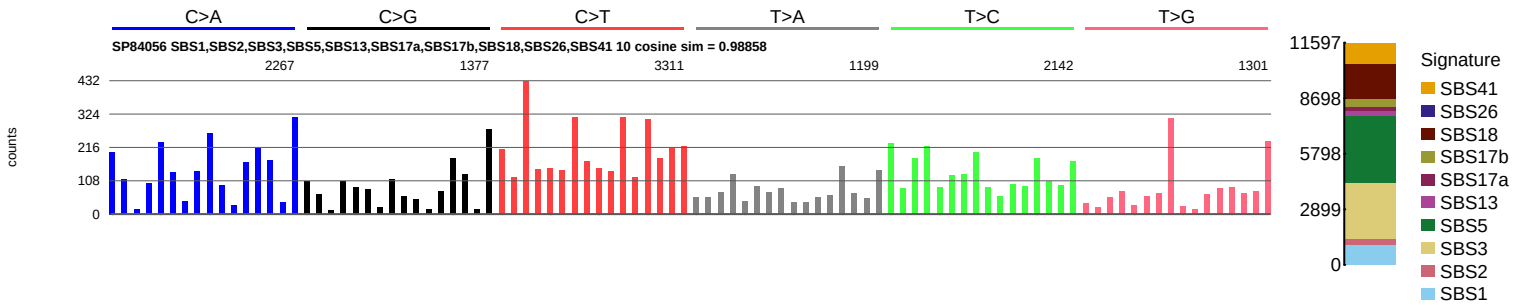

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98893
2277
1364
3343
1196
2136
1287
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

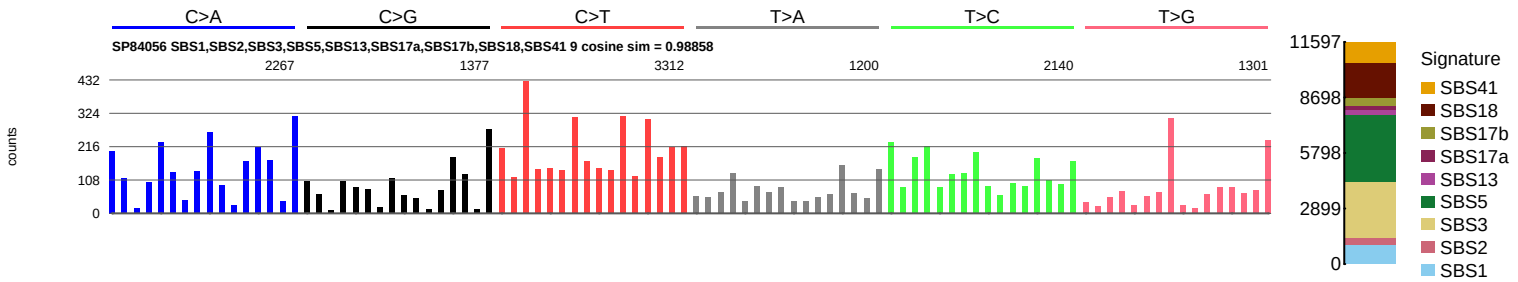

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98892
2277
1364
3345
1196
2130
1287
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

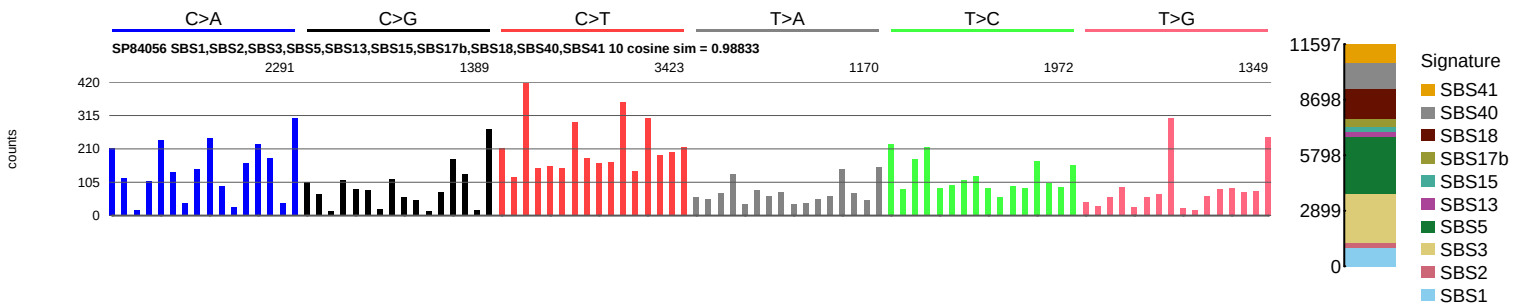

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9889
2266
1366
3287
1184
2202
1290
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

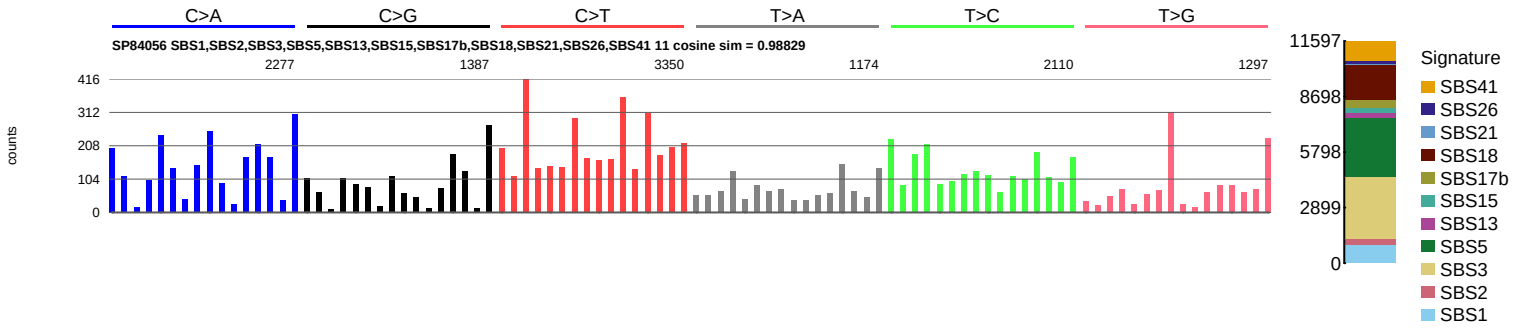

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98888
2292
1369
3397
1142
2050
1346
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98882
2295
1229
3445
1040
2209
1378
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98858
2267
1377
3311
1199
2142
1301
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98858
2267
1377
3312
1200
2140
1301
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98833
2291
1389
3423
1170
1972
1349
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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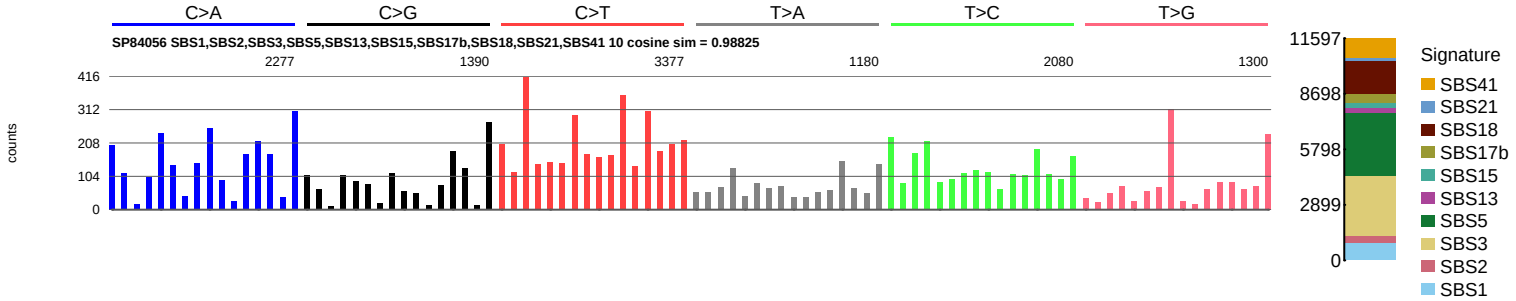
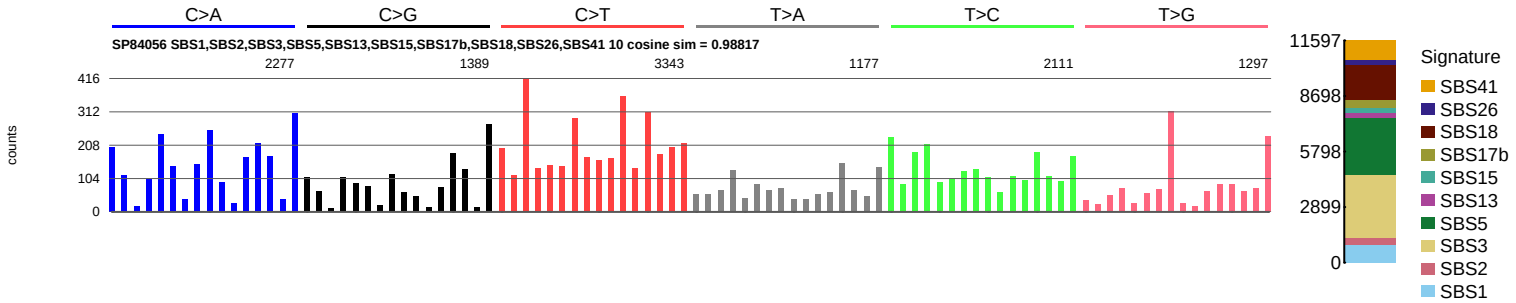
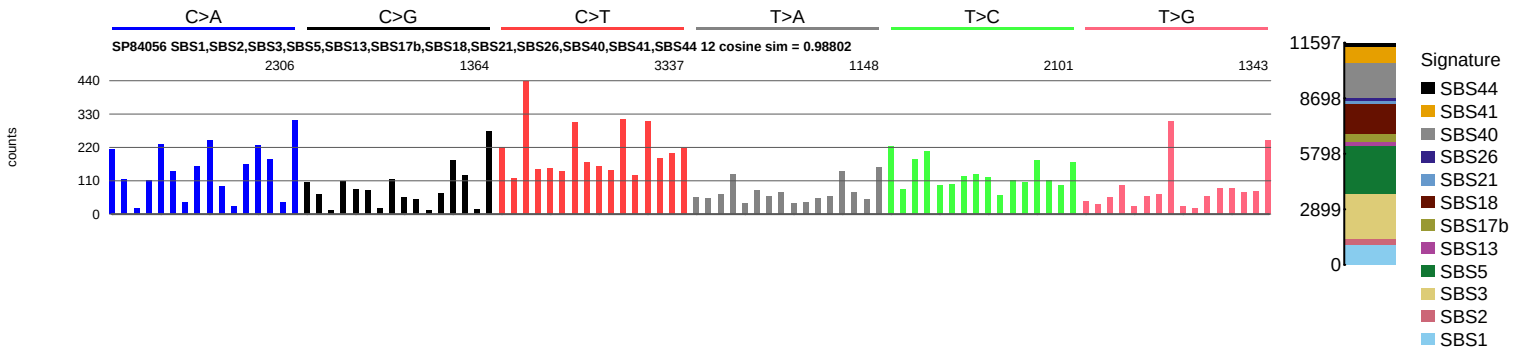
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.98829
2277
1387
3350
1174
2110
1297
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98825
2277
1390
3377
1180
2080
1300
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98817
2277
1389
3343
1177
2111
1297
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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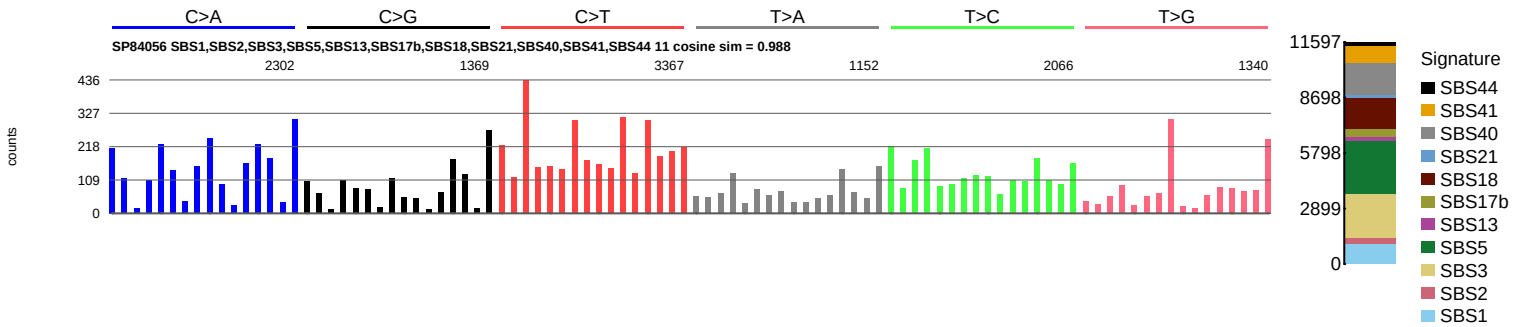
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98802
2306
1364
3337
1148
2101
1343
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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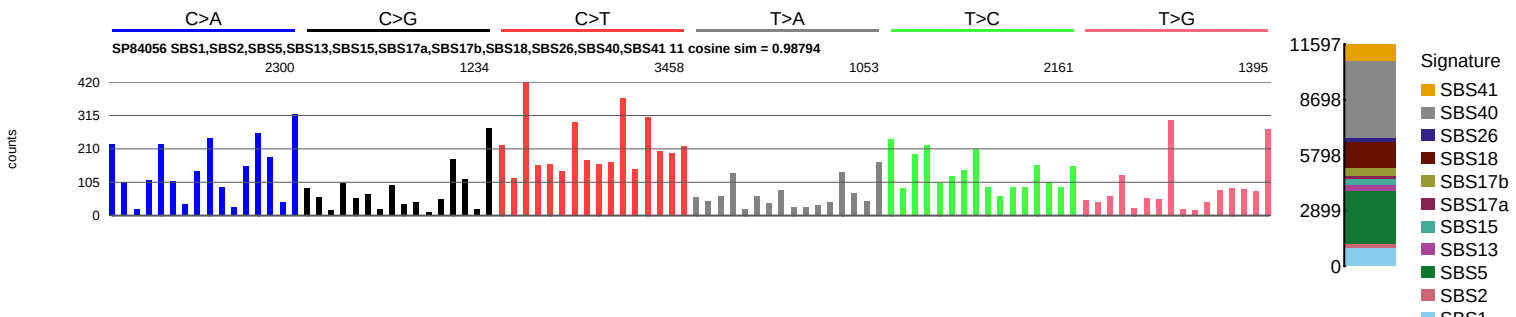
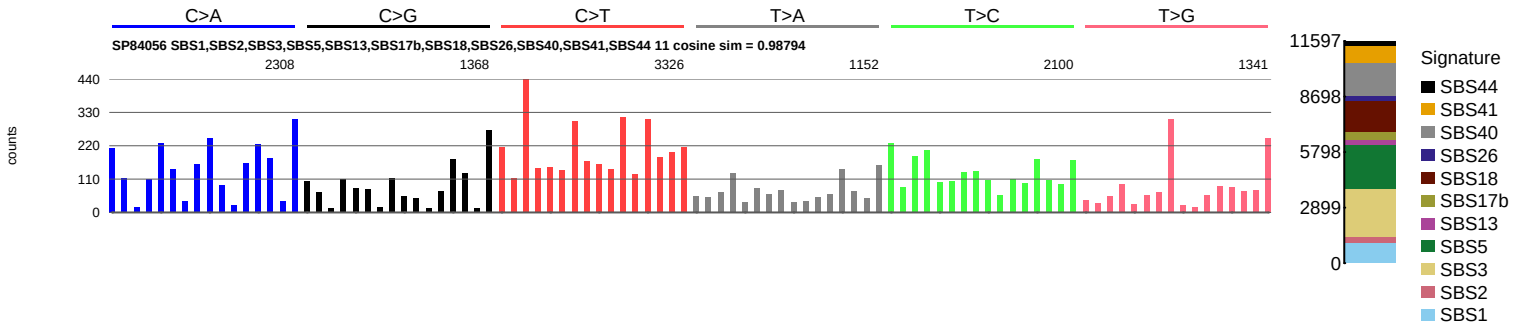
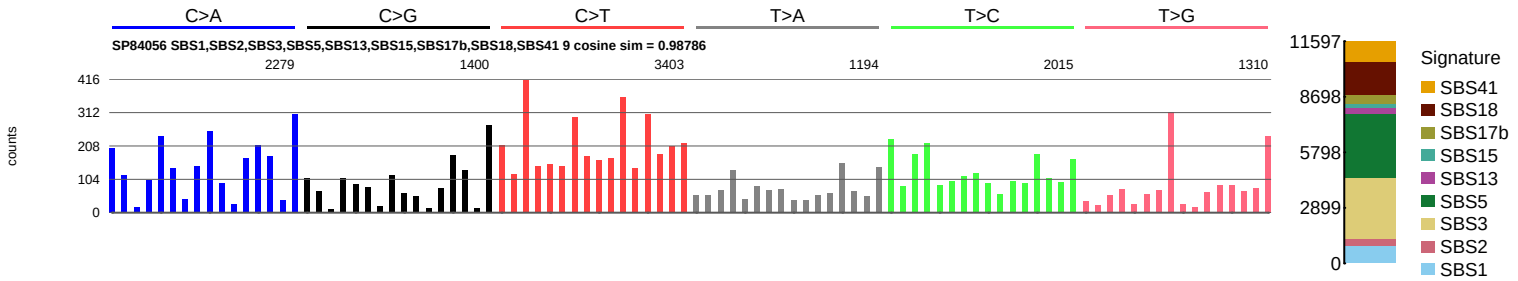
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.988
2302
1369
3367
1152
2066
1340
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98794
2300
1234
3458
1053
2161
1395
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98794
2308
1368
3326
1152
2100
1341
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98786
2279
1400
3403
1194
2015
1310
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
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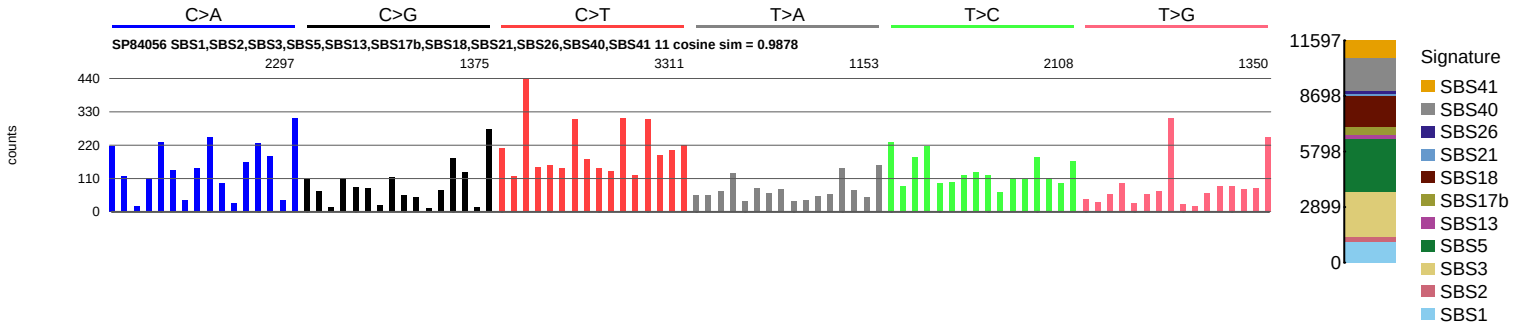
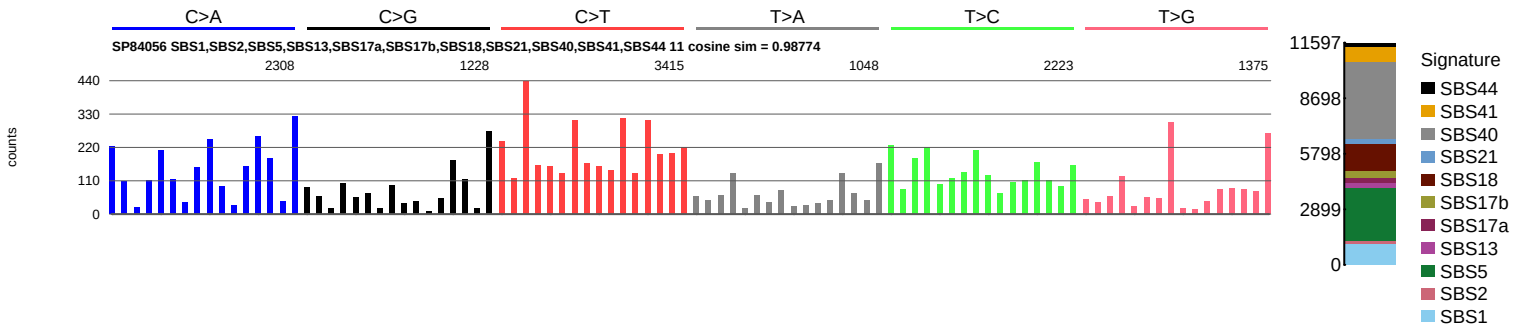
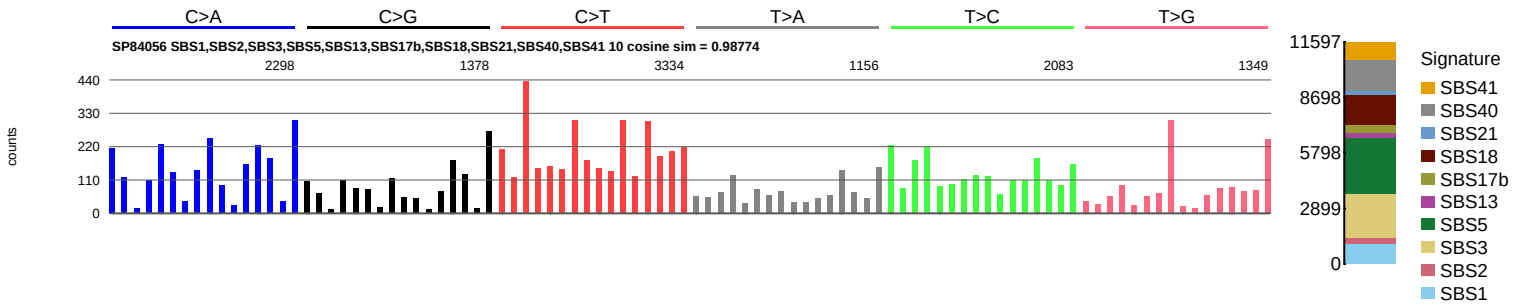
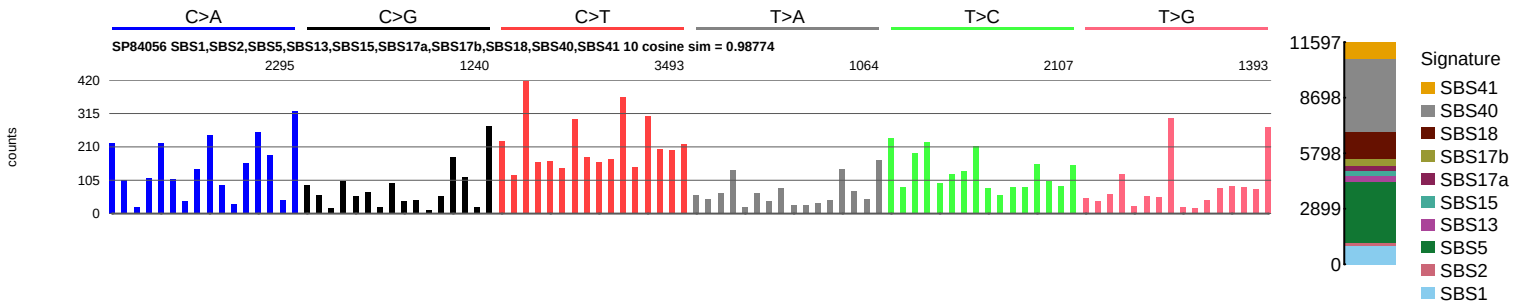
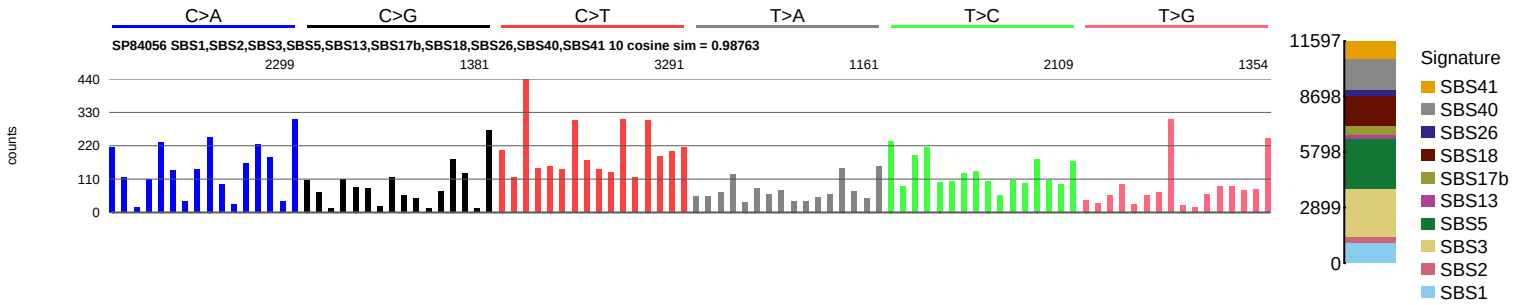
SBS3
SBS2
SBS1

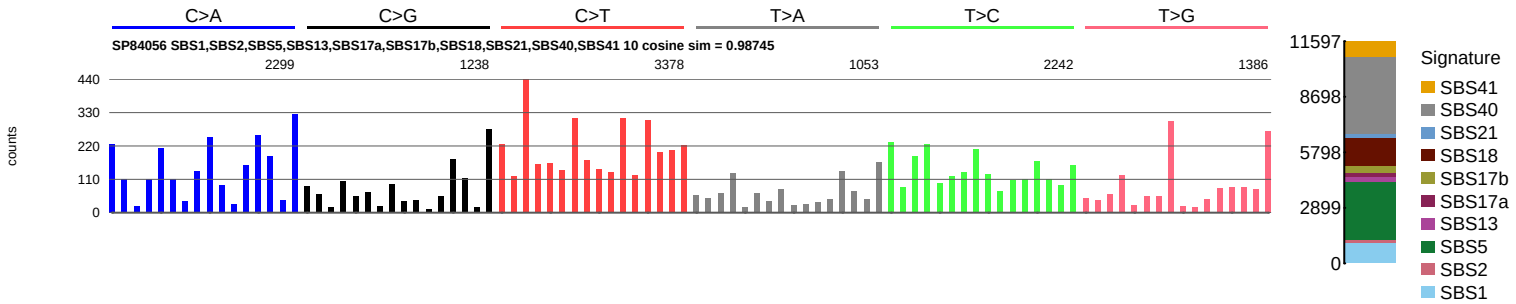

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9878
2297
1375
3311
1153
2108
1350
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98774
2308
1228
3415
1048
2223
1375
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98774
2298
1378
3334
1156
2083
1349
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98774
2295
1240
3493
1064
2107
1393
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98763
2299
1381
3291
1161
2109
1354
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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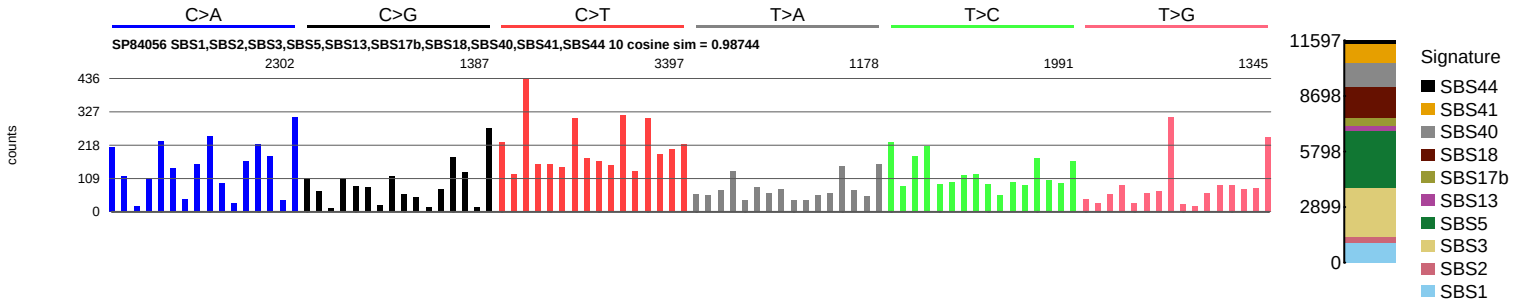
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98745
2299
1238
3378
1053
2242
1386
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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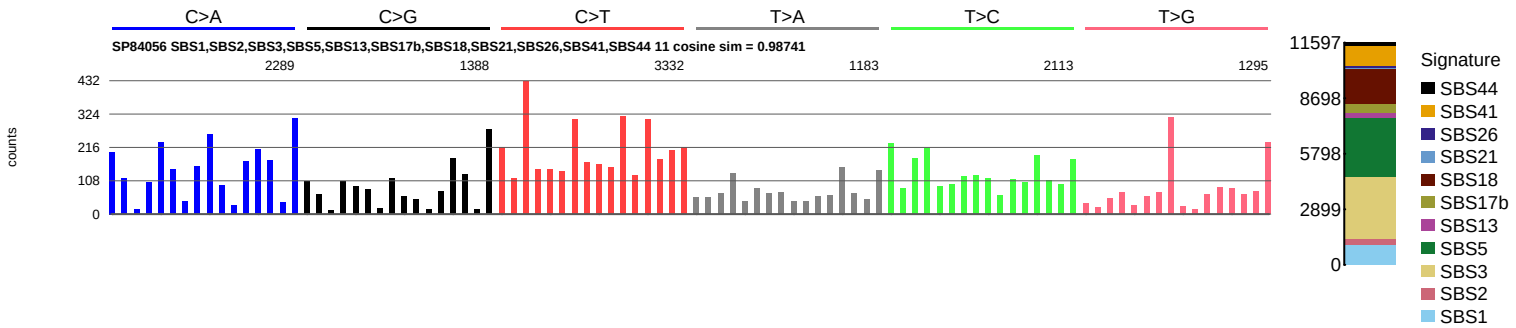
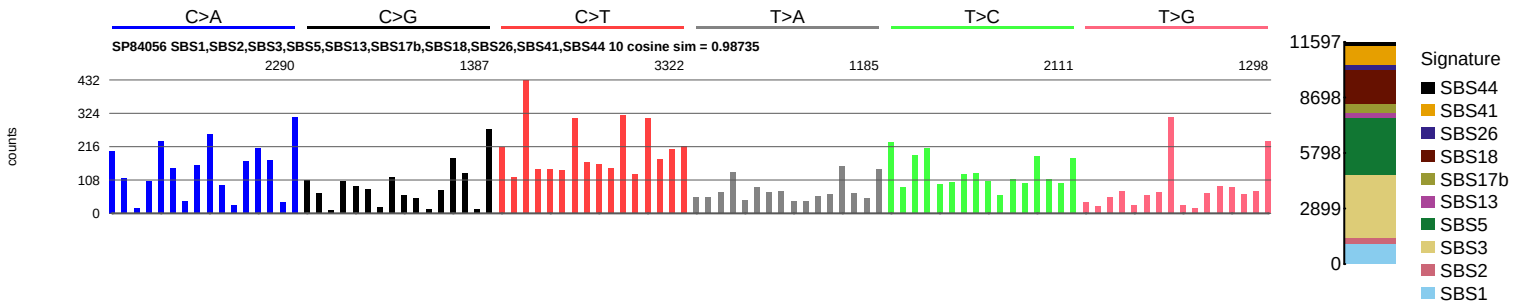
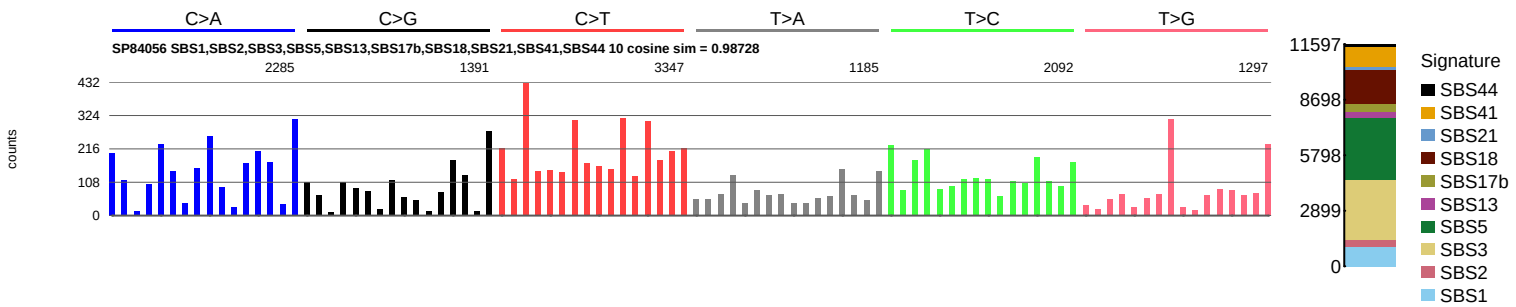
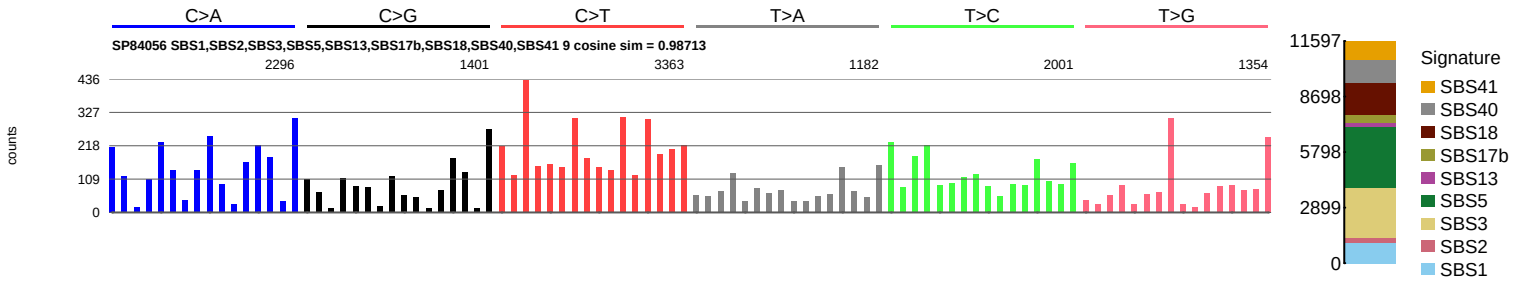
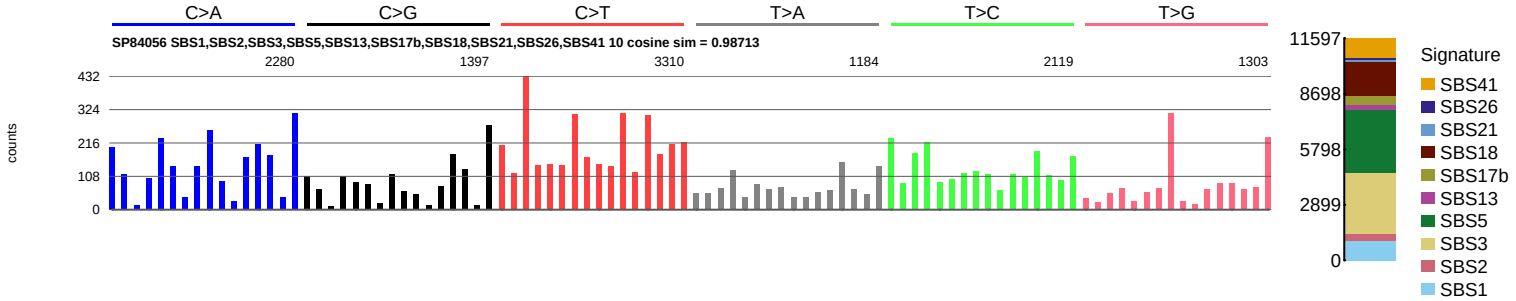
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98744
2302
1387
3397
1178
1991
1345
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

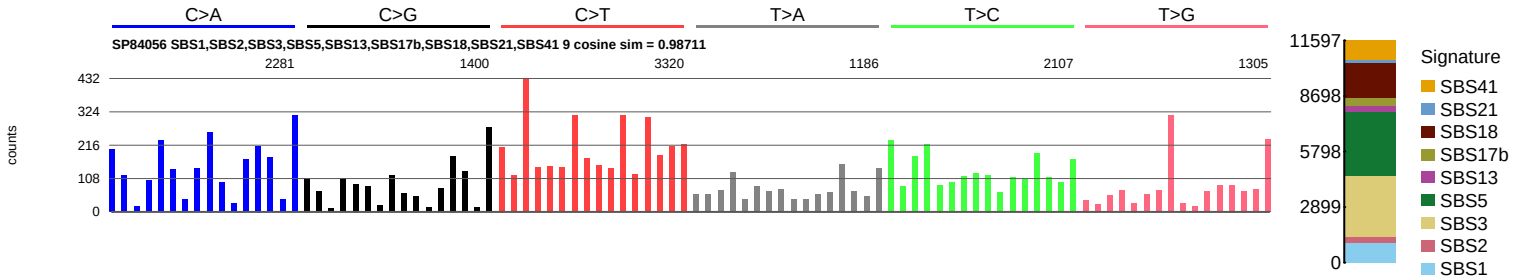

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98741
2289
1388
3332
1183
2113
1295
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

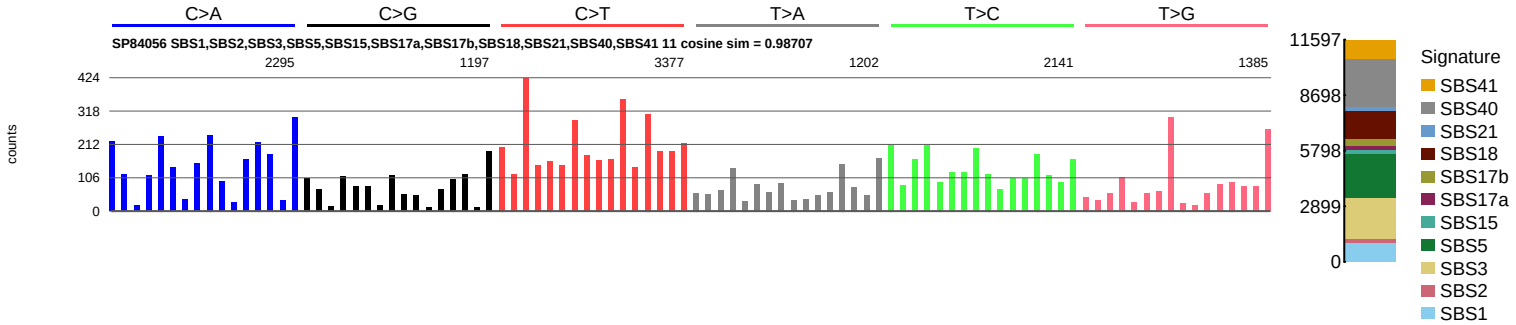

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98735
2290
1387
3322
1185
2111
1298
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

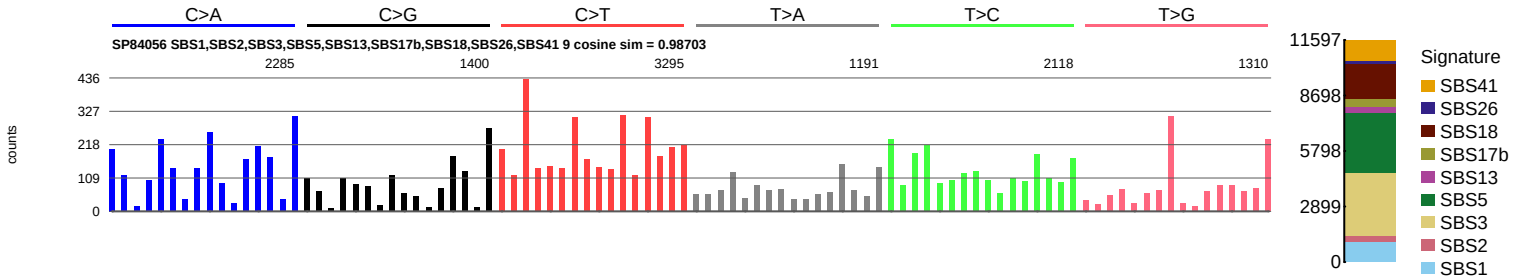

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98728
2285
1391
3347
1185
2092
1297
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

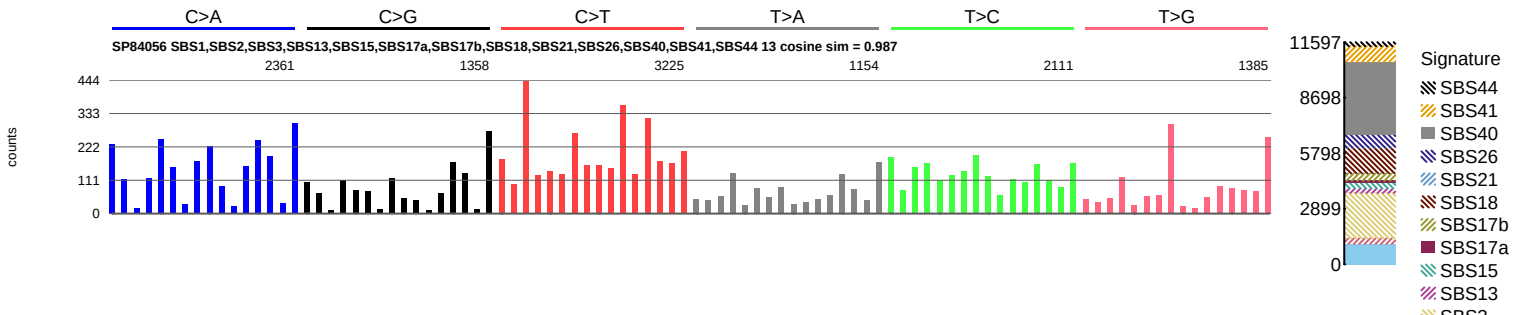

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98713
2296
1401
3363
1182
2001
1354
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98713
2280
1397
3310
1184
2119
1303
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98711
2281
1400
3320
1186
2107
1305
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98707
2295
1197
3377
1202
2141
1385
424
318
212
106
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98703
2285
1400
3295
1191
2118
1310
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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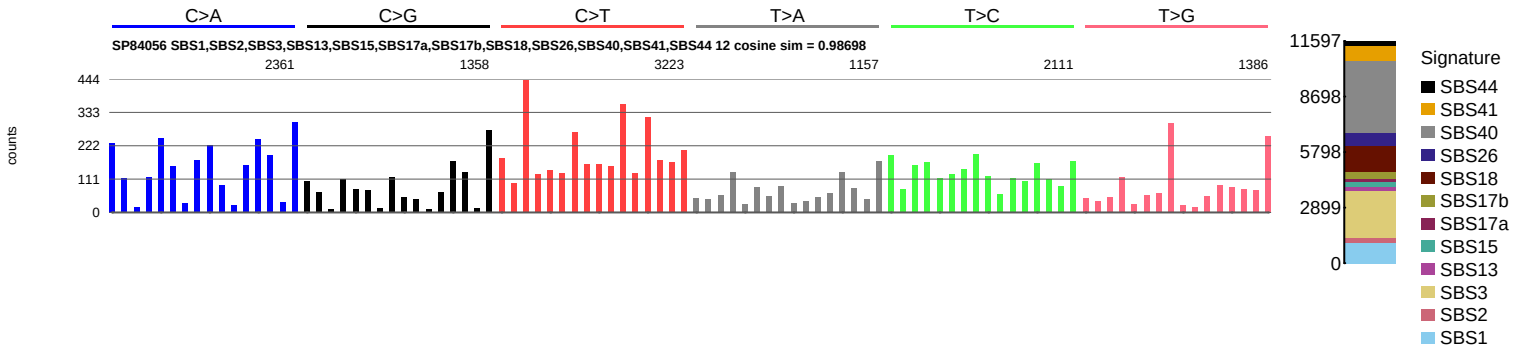
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.987
2361
1358
3225
1154
2111
1385
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13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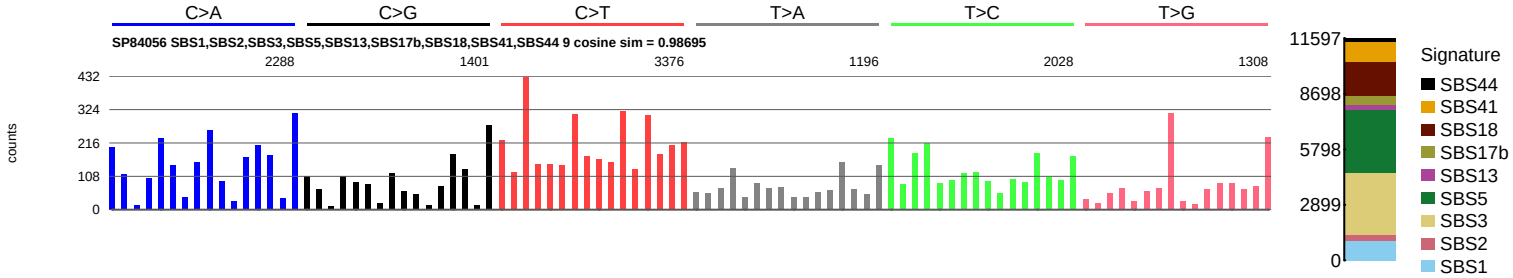
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98698
2361
1358
3223
1157
2111
1386
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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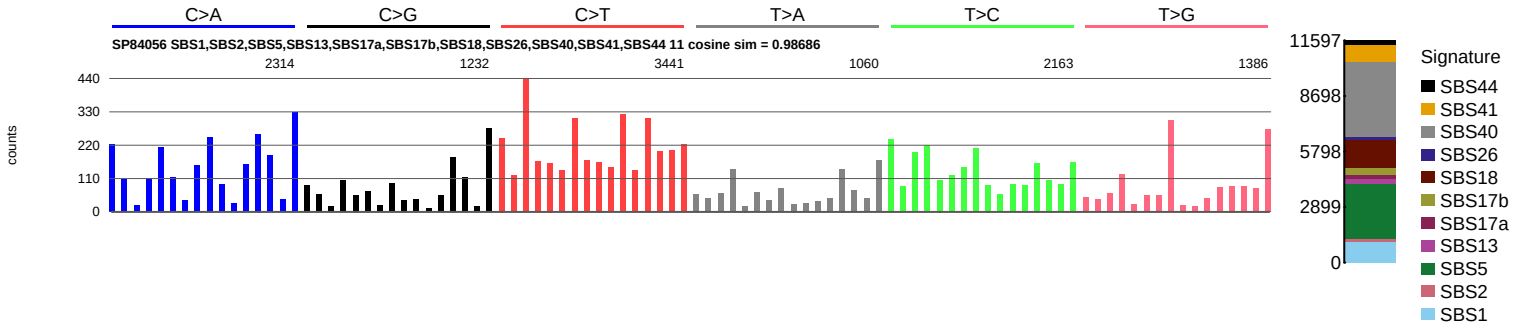
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98695
2288
1401
3376
1196
2028
1308
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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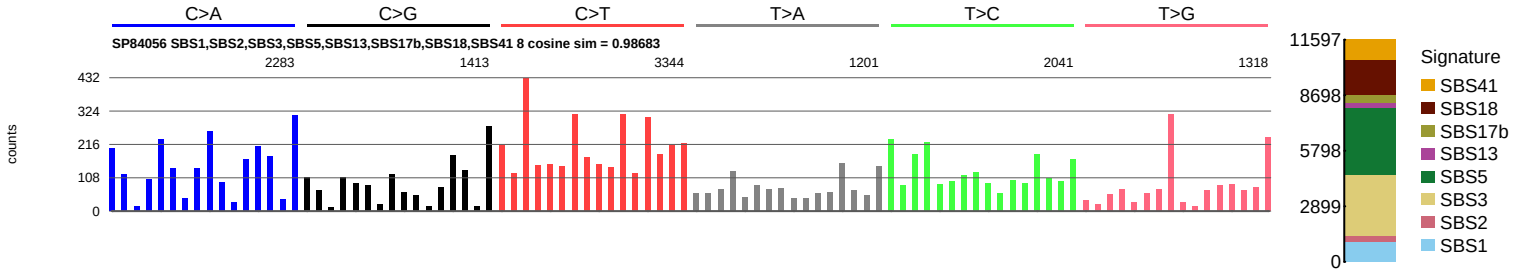
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98686
2314
1232
3441
1060
2163
1386
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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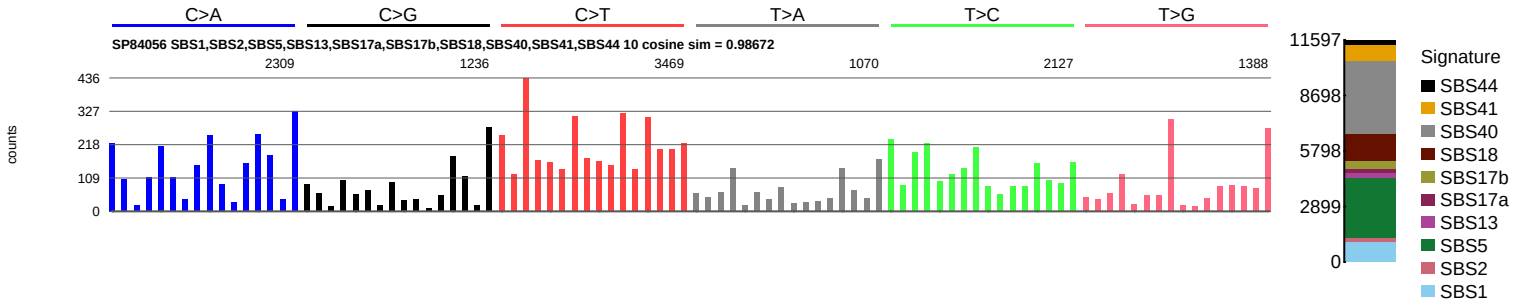
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98683
2283
1413
3344
1201
2041
1318
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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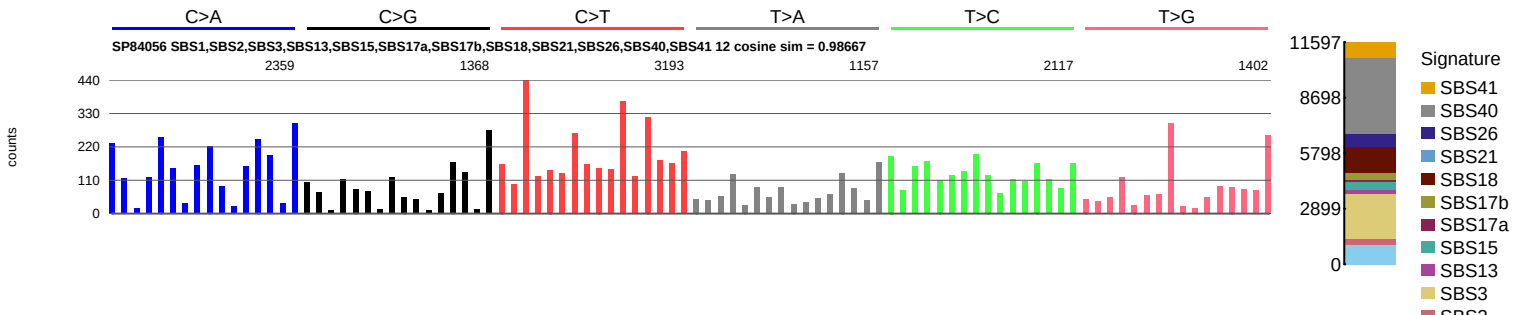
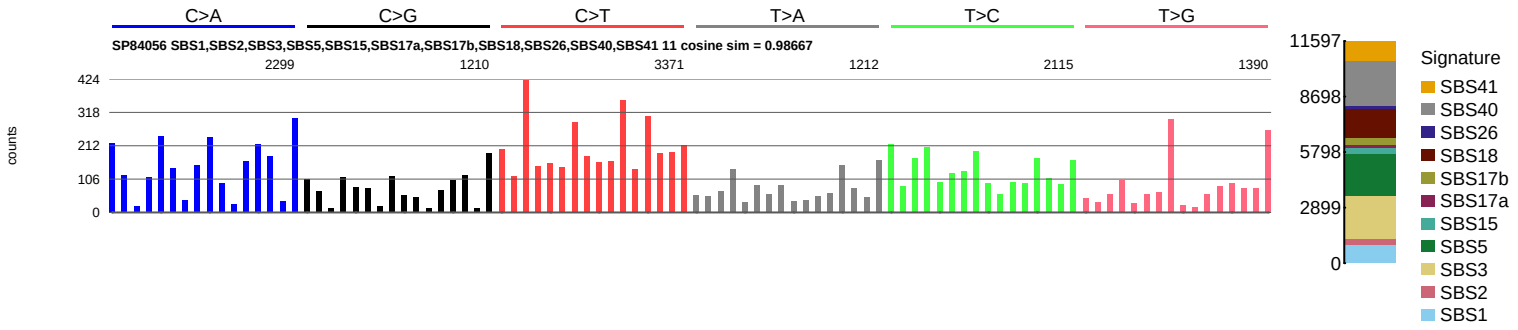
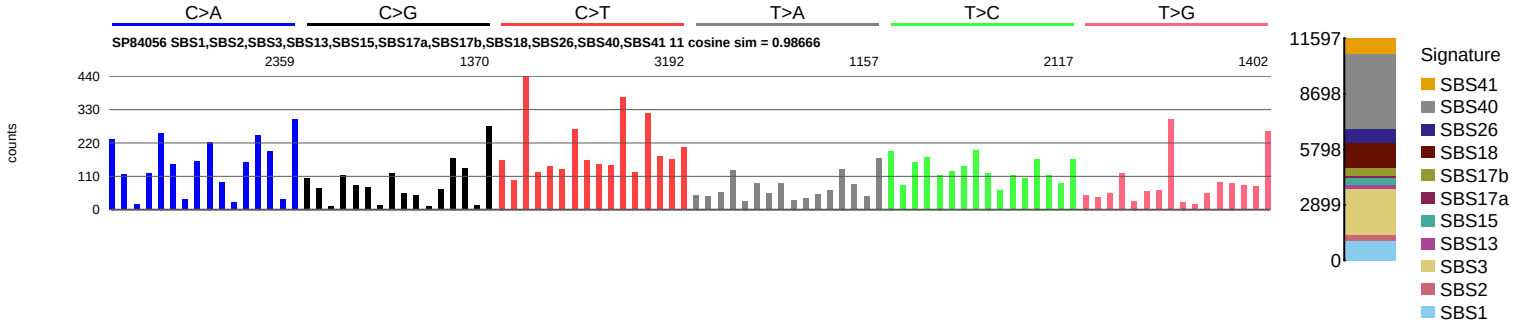
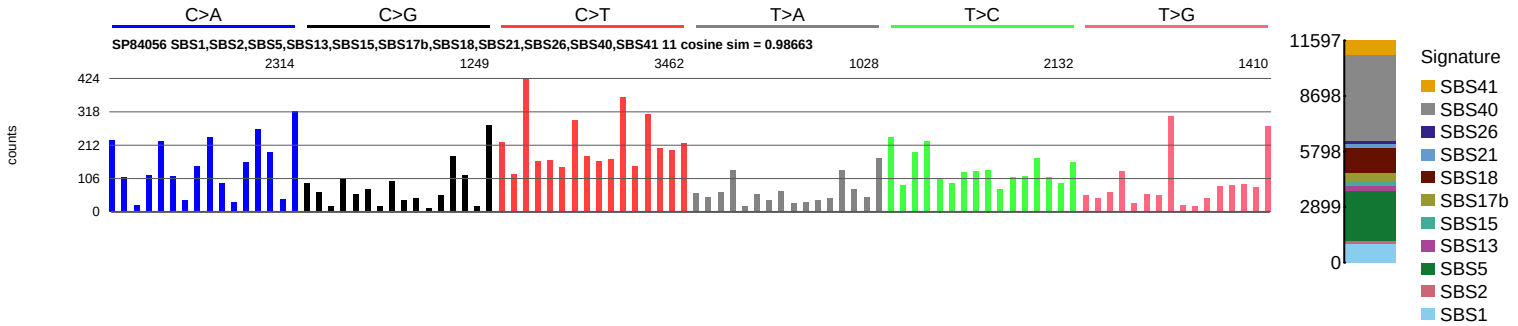
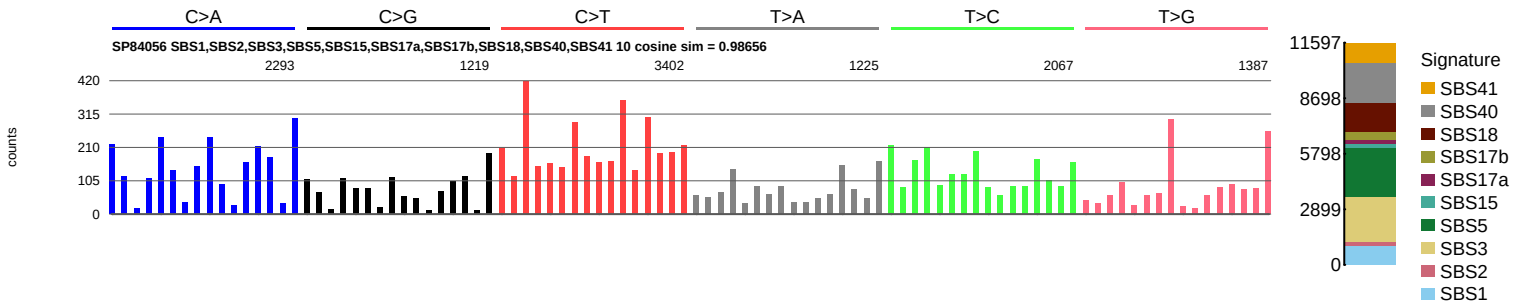
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98672
2309
1236
3469
1070
2127
1388
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

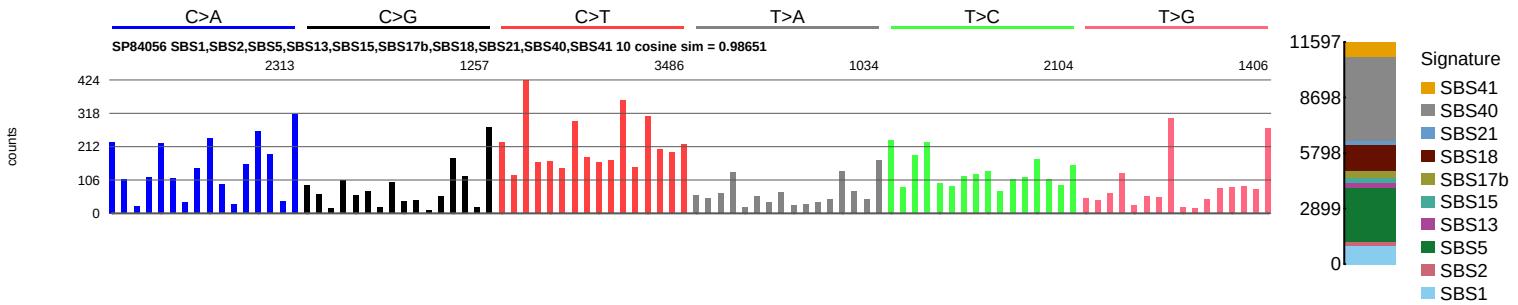

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.98667
2359
1368
3193
1157
2117
1402
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

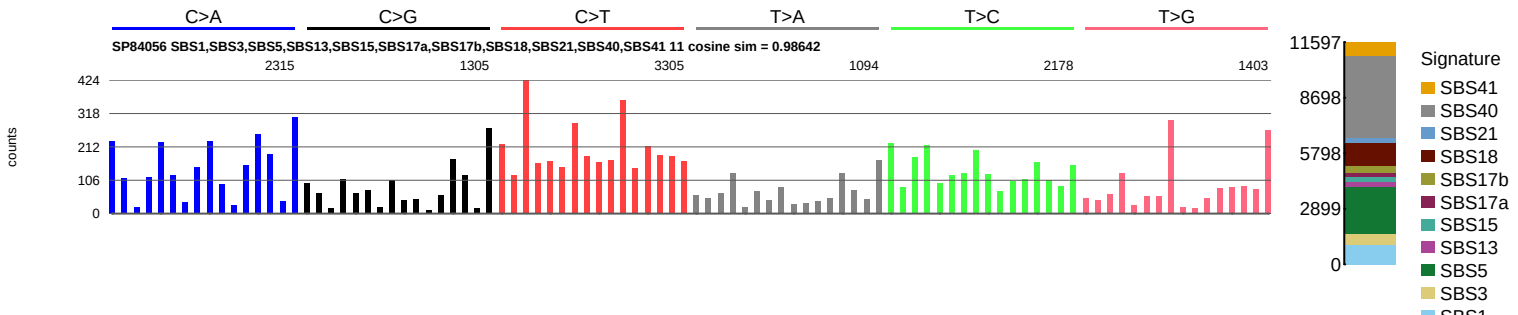

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98667
2299
1210
3371
1212
2115
1390
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

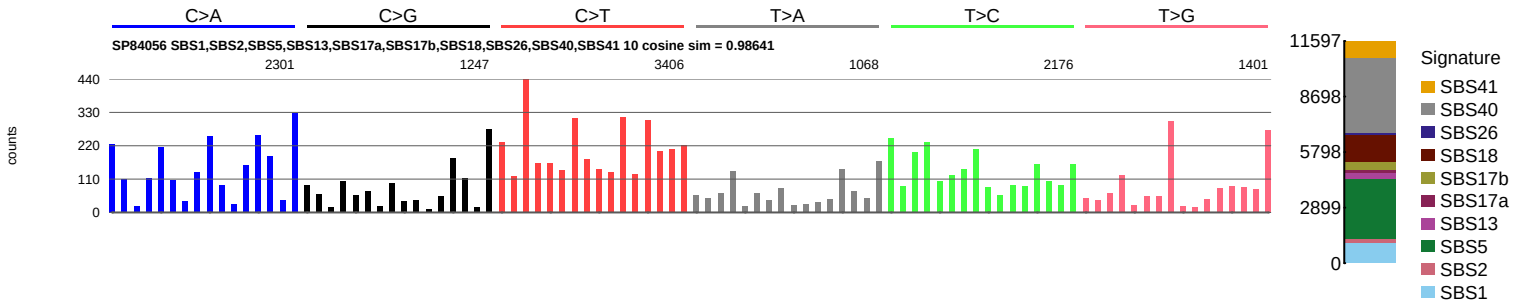

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98666
2359
1370
3192
1157
2117
1402
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

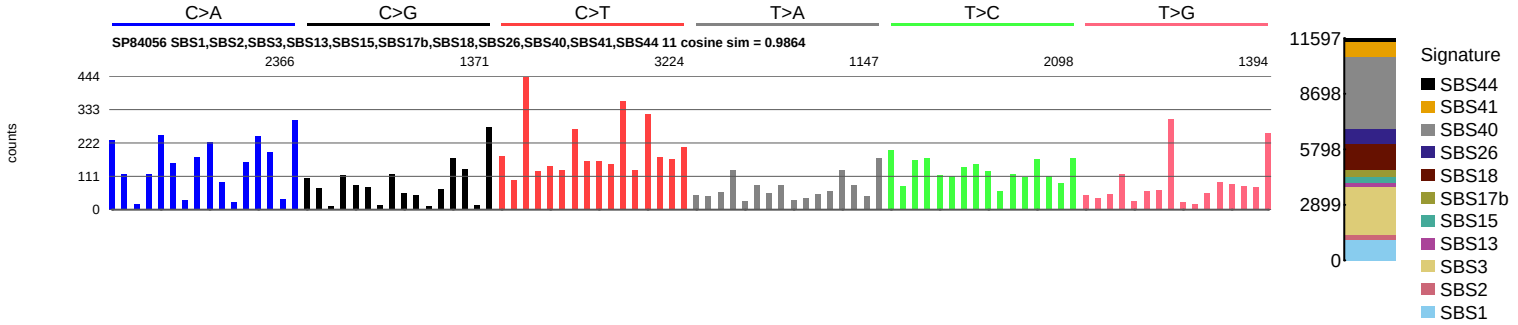

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98663
2314
1249
3462
1028
2132
1410
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

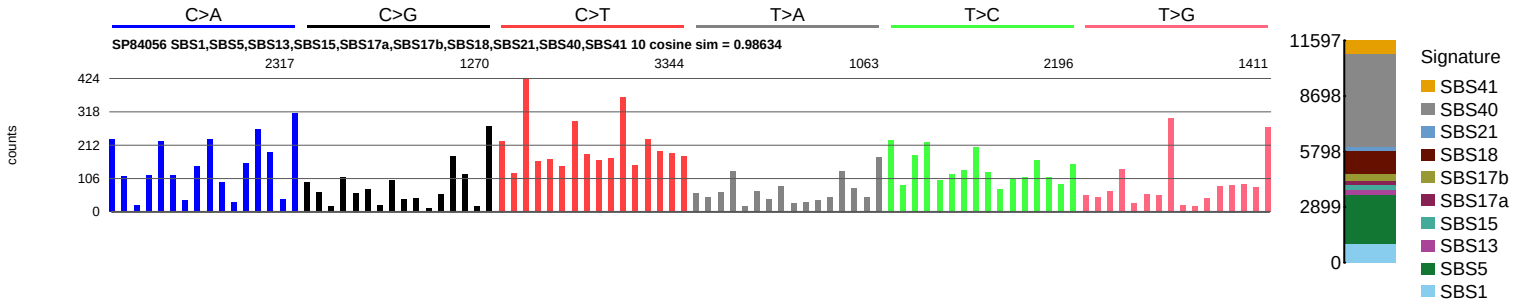

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98656
2293
1219
3402
1225
2067
1387
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98651
2313
1257
3486
1034
2104
1406
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98642
2315
1305
3305
1094
2178
1403
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98641
2301
1247
3406
1068
2176
1401
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9864
2366
1371
3224
1147
2098
1394
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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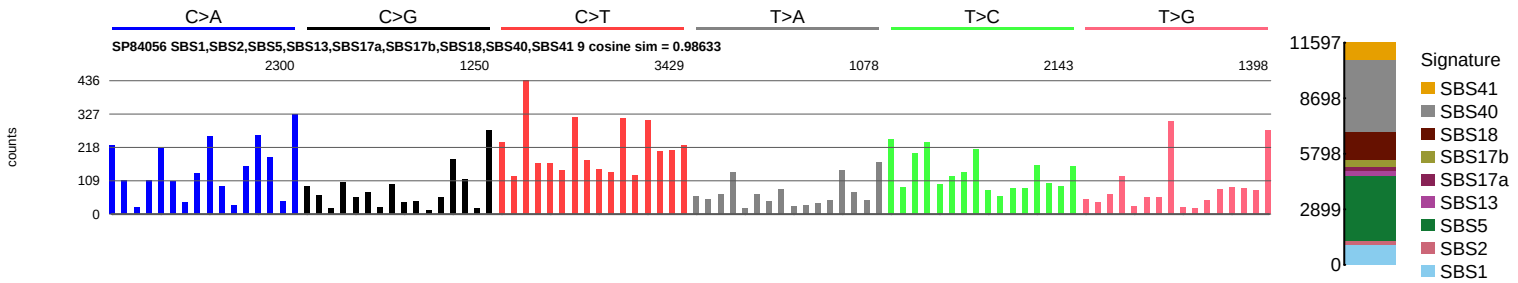
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98634
2317
1270
3344
1063
2196
1411
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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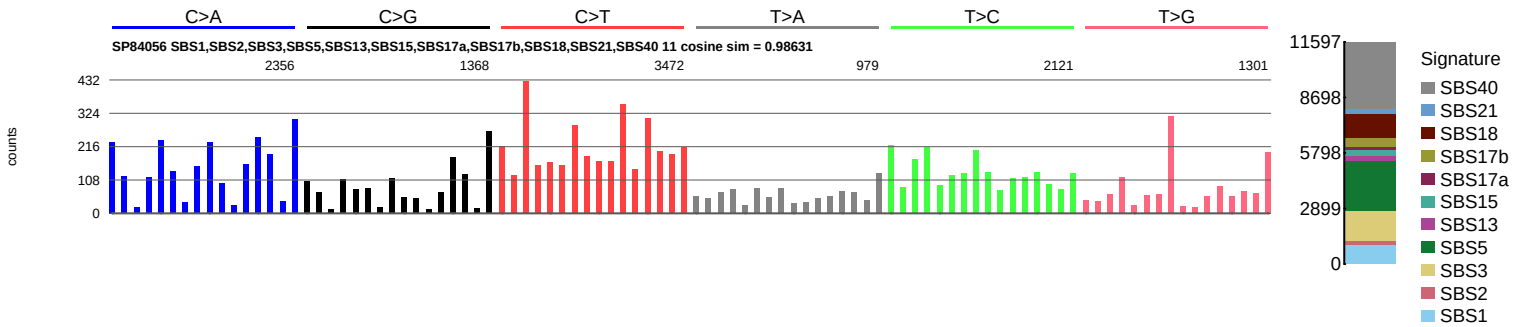
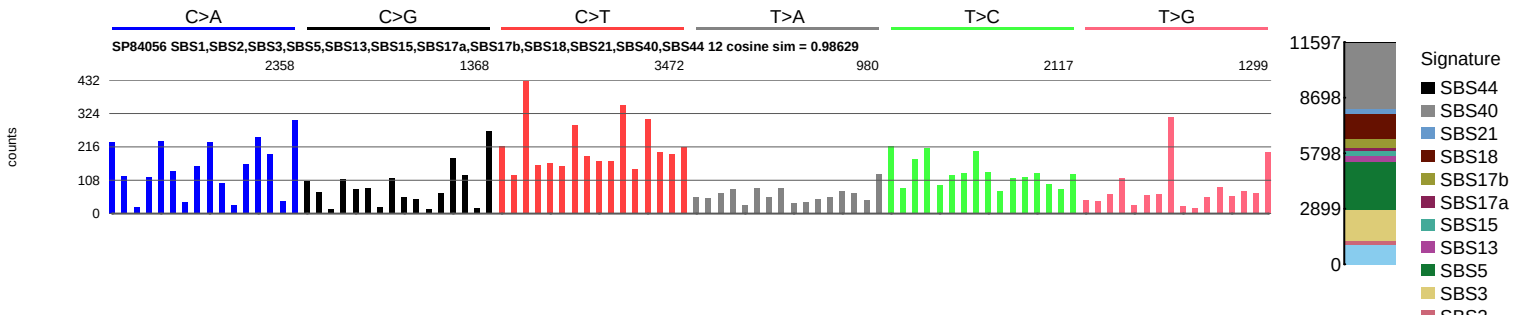
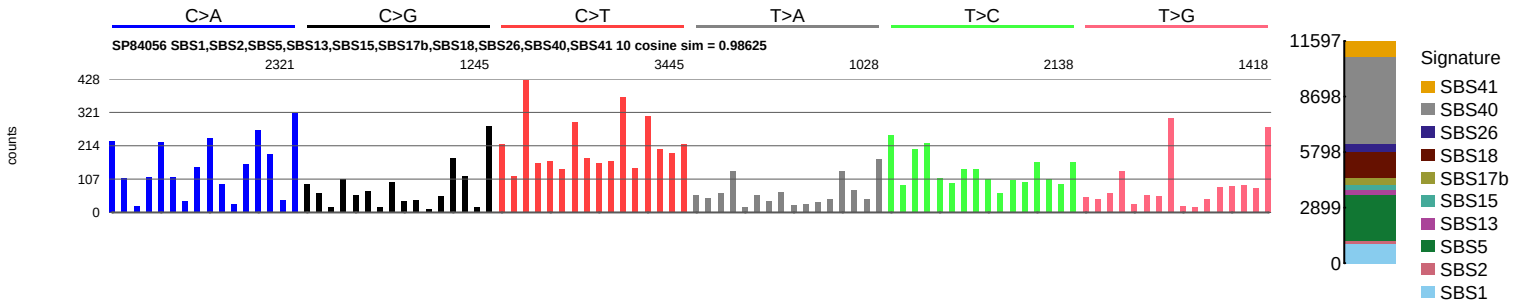
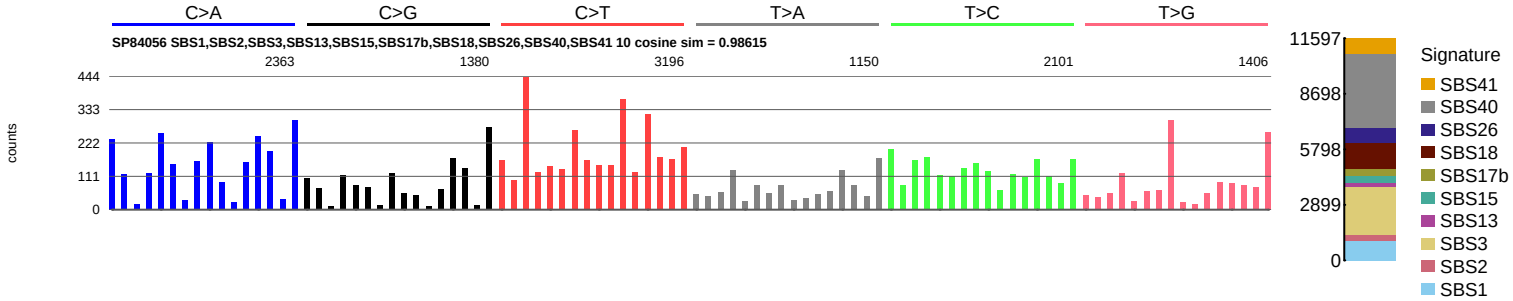
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98633
2300
1250
3429
1078
2143
1398
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.98631
2356
1368
3472
979
2121
1301
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.98629
2358
1368
3472
980
2117
1299
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98625
2321
1245
3445
1028
2138
1418
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98615
2363
1380
3196
1150
2101
1406
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
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SBS40
SBS26
SBS18
SBS17b
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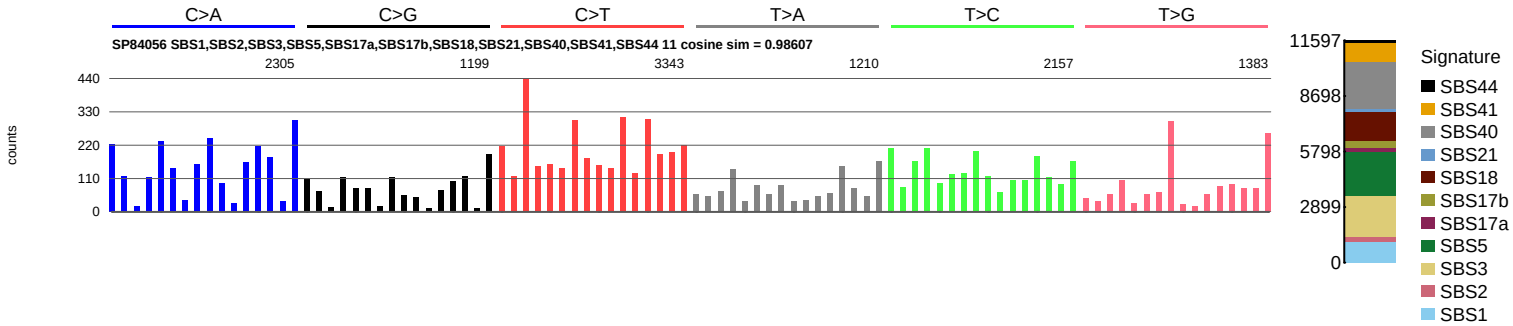
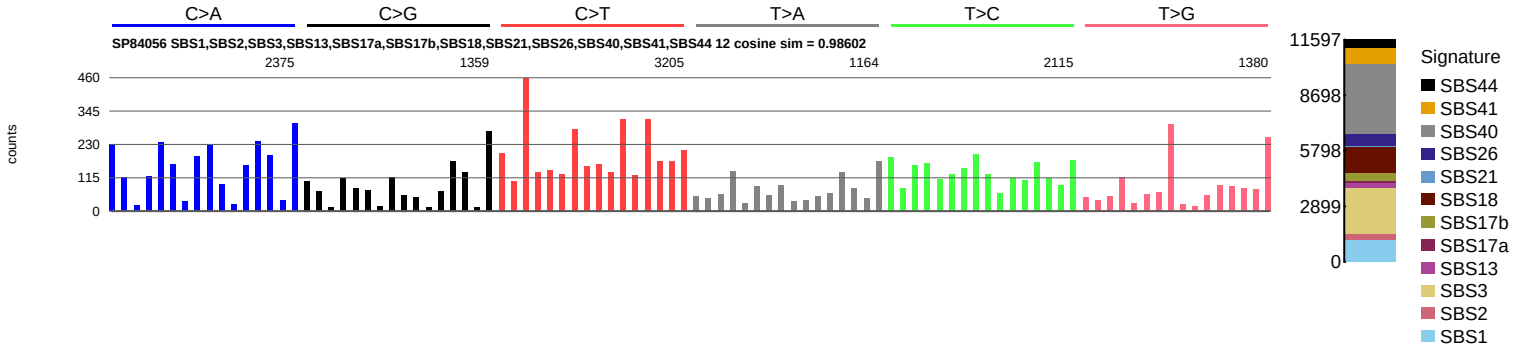
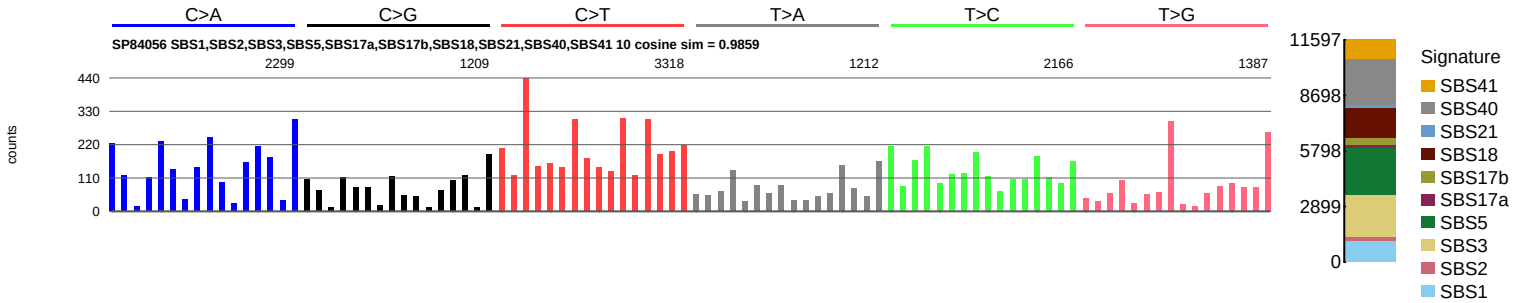
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98607
2305
1199
3343
1210
2157
1383
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98602
2375
1359
3205
1164
2115
1380
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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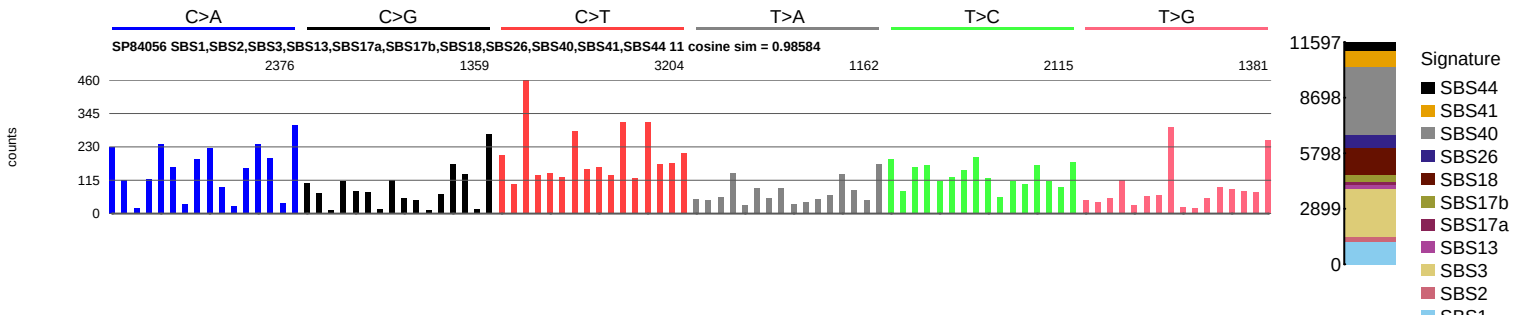
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9859
2299
1209
3318
1212
2166
1387
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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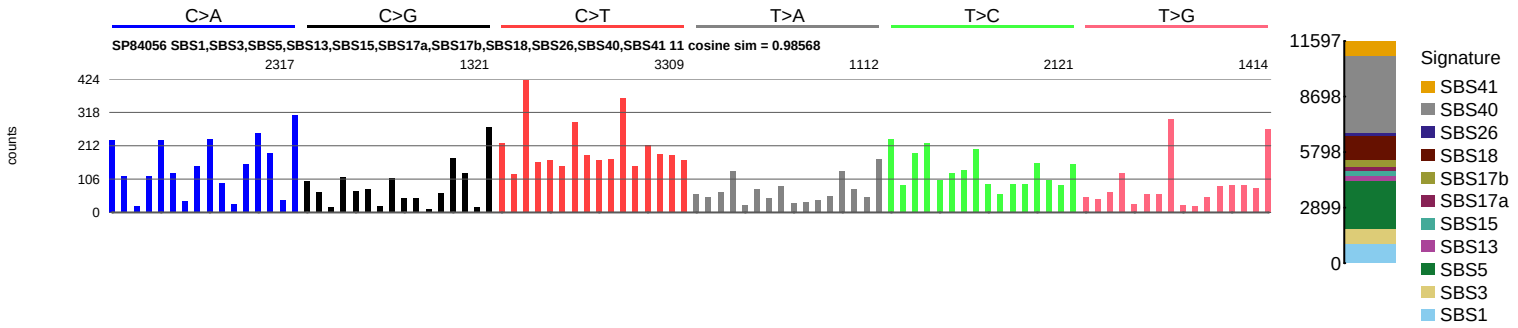
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98584
2376
1359
3204
1162
2115
1381
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
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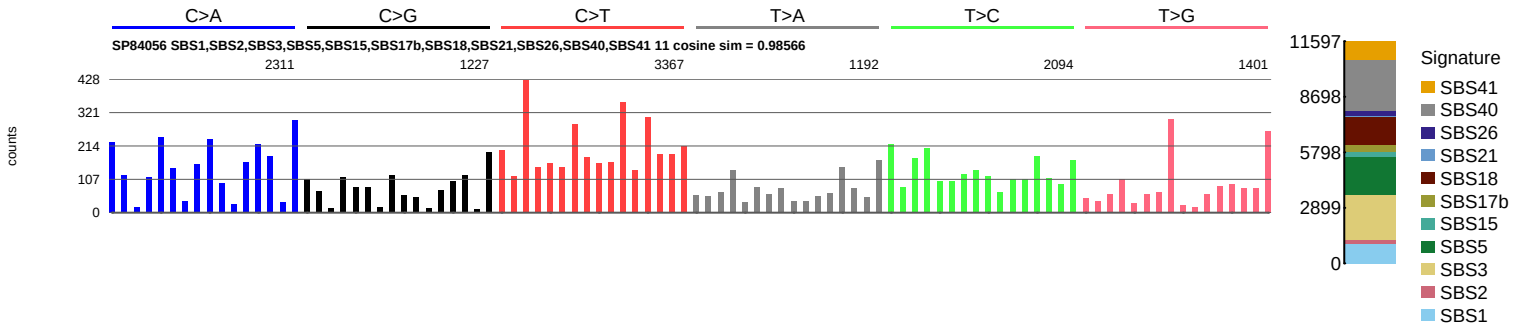
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98568
2317
1321
3309
1112
2121
1414
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
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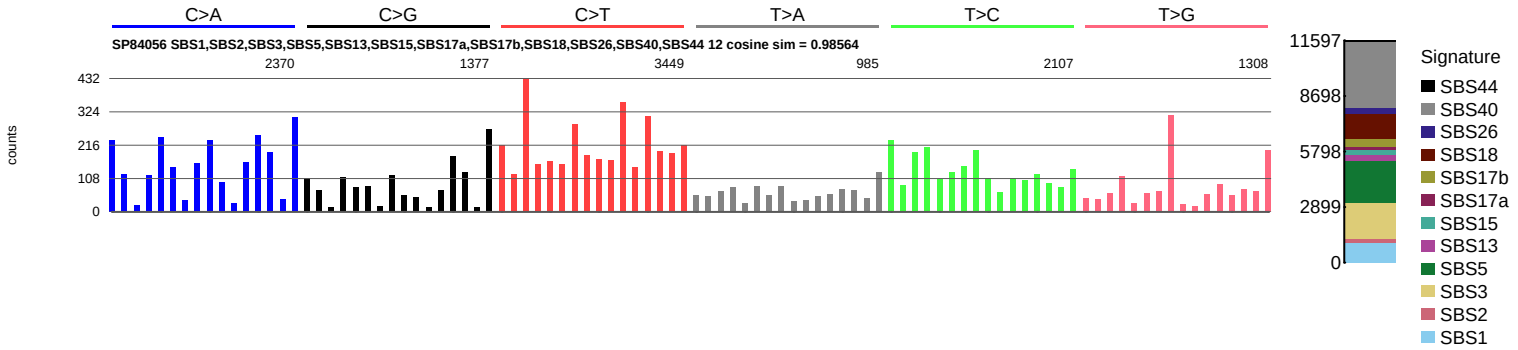
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98566
2311
1227
3367
1192
2094
1401
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 12 cosine sim = 0.98564
2370
1377
3449
985
2107
1308
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
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SBS18
SBS17b
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SBS15
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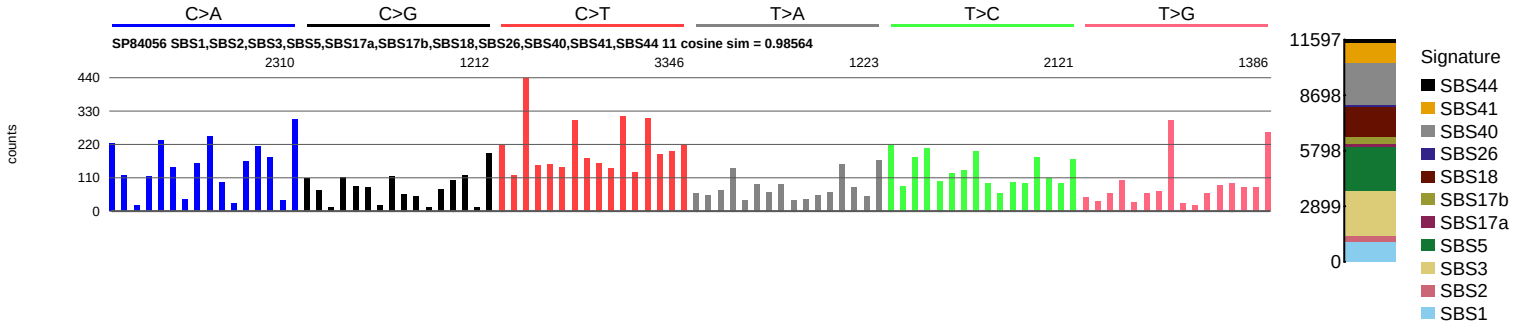
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98564
2310
1212
3346
1223
2121
1386
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
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SBS40
SBS26
SBS18
SBS17b
SBS17a
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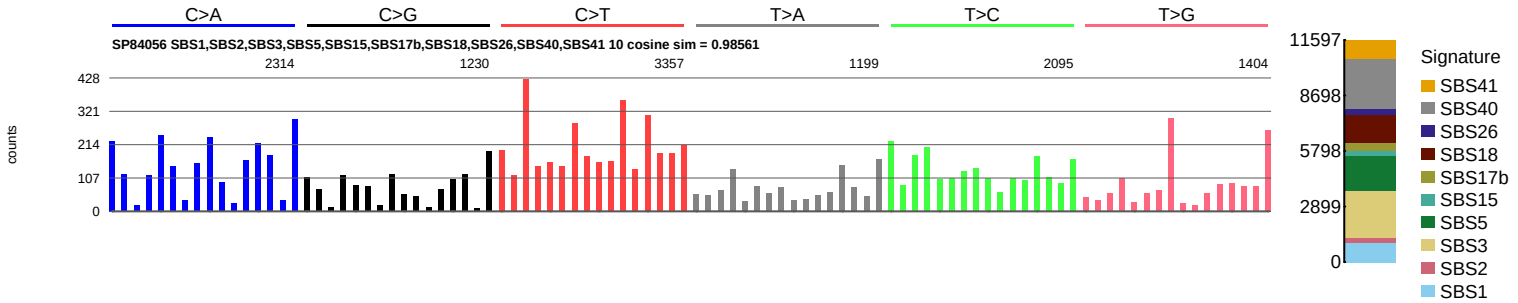
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98561
2314
1230
3357
1199
2095
1404
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
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SBS40
SBS26
SBS18
SBS17b
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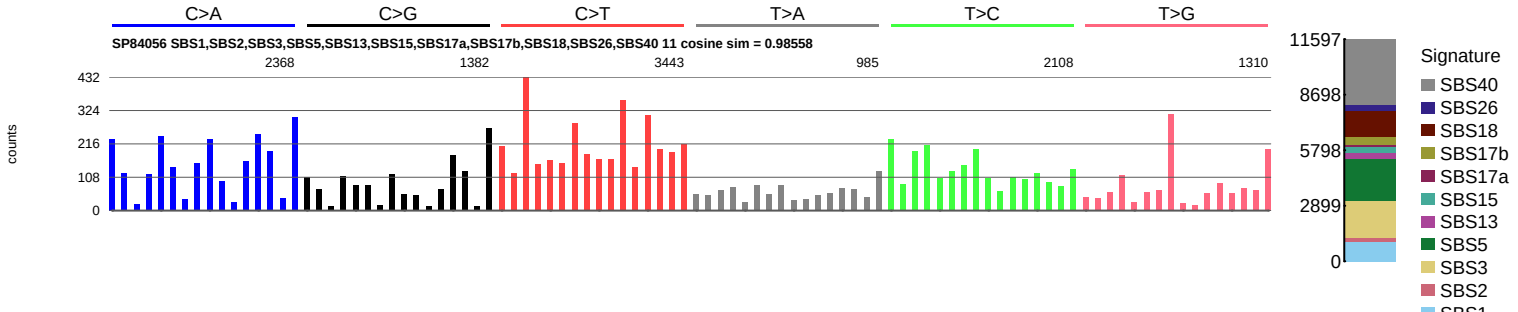
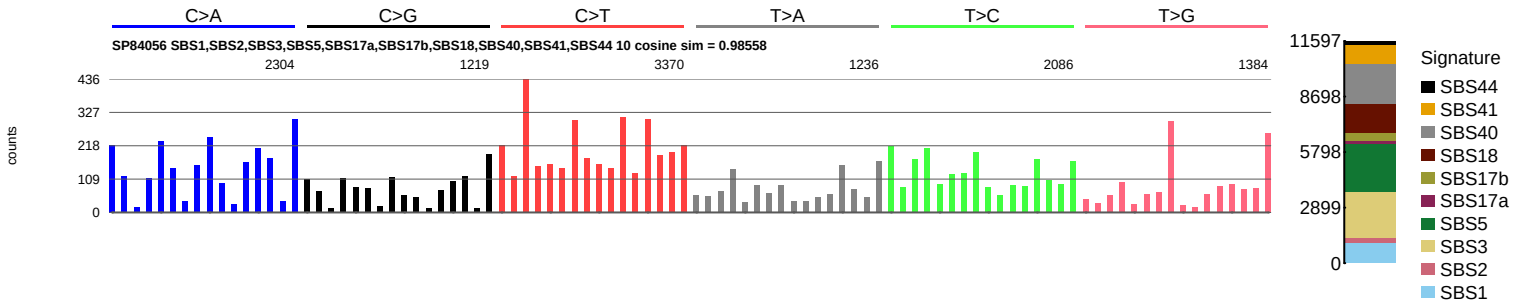
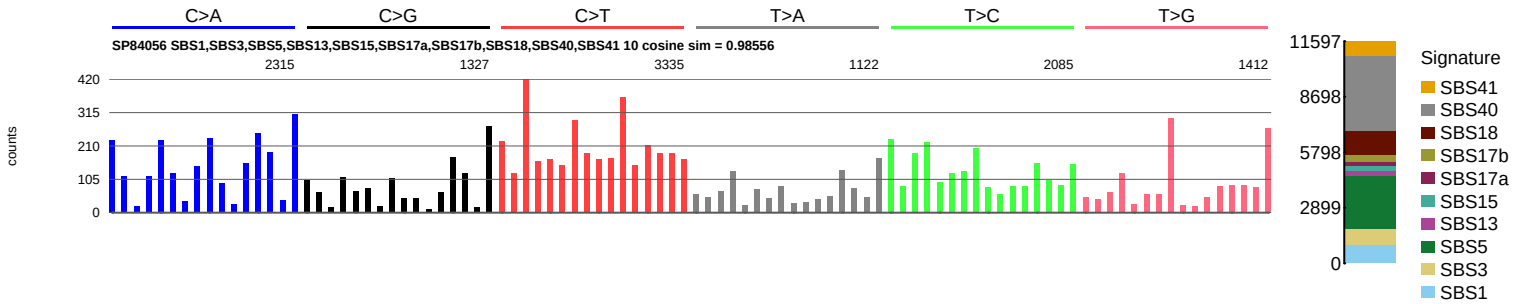
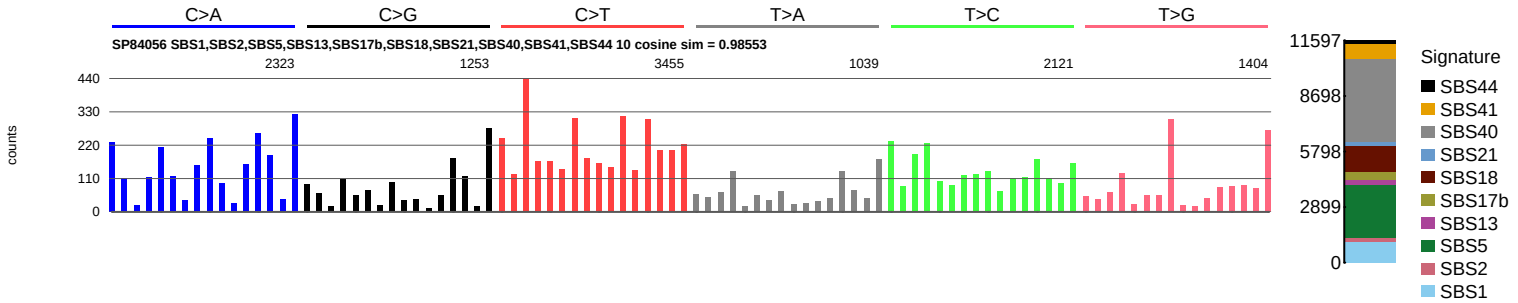
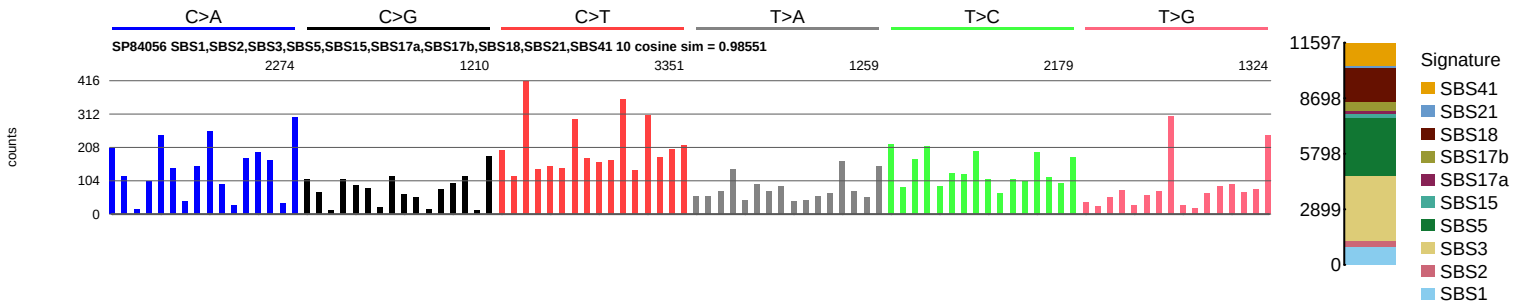
SBS5
SBS3
SBS2
SBS1

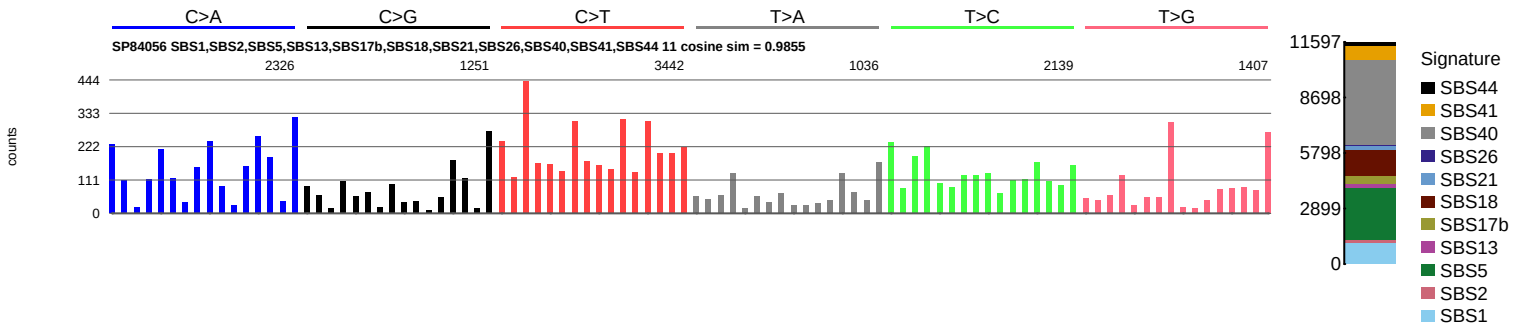

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 11 cosine sim = 0.98558
2368
1382
3443
985
2108
1310
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

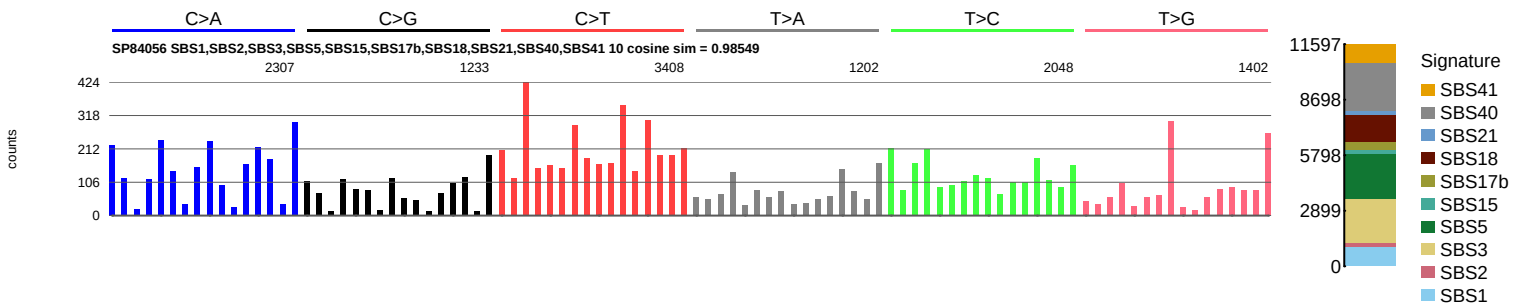

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98558
2304
1219
3370
1236
2086
1384
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

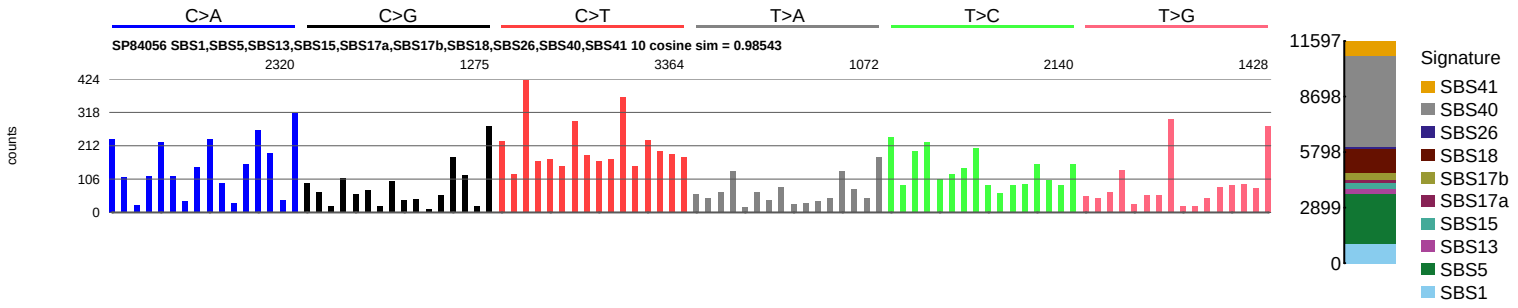

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98556
2315
1327
3335
1122
2085
1412
420
315
210
105
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

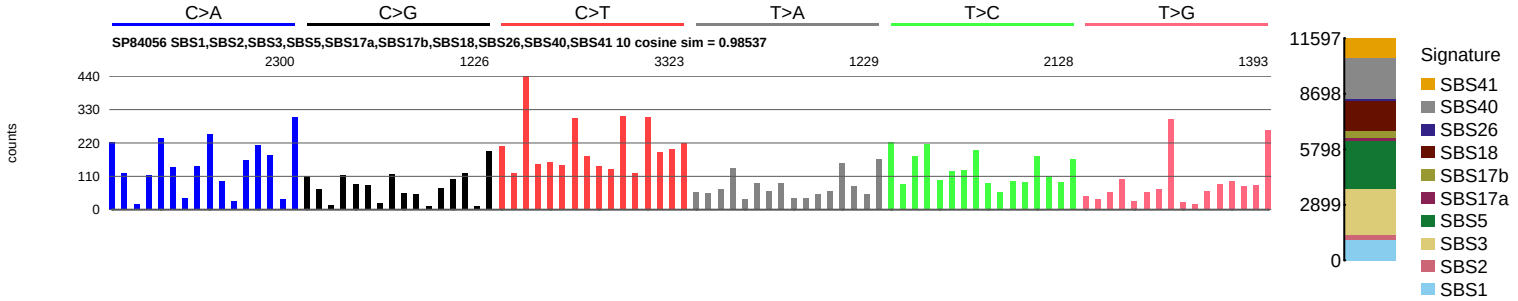

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98553
2323
1253
3455
1039
2121
1404
440
330
220
110
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

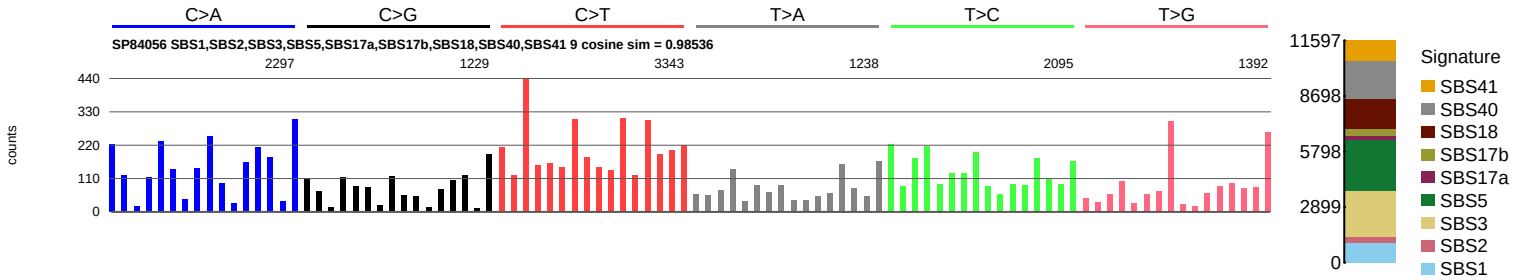

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98551
2274
1210
3351
1259
2179
1324
416
312
208
104
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

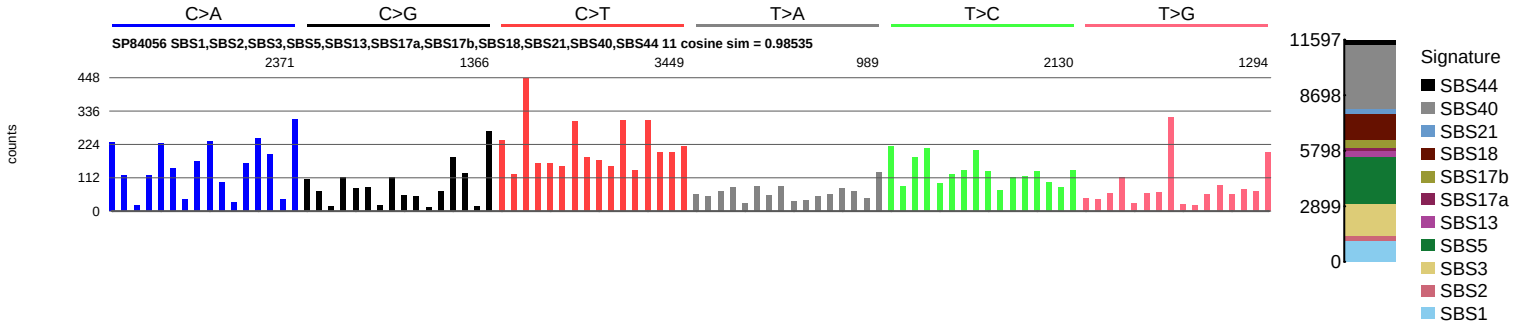

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9855
2326
1251
3442
1036
2139
1407
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

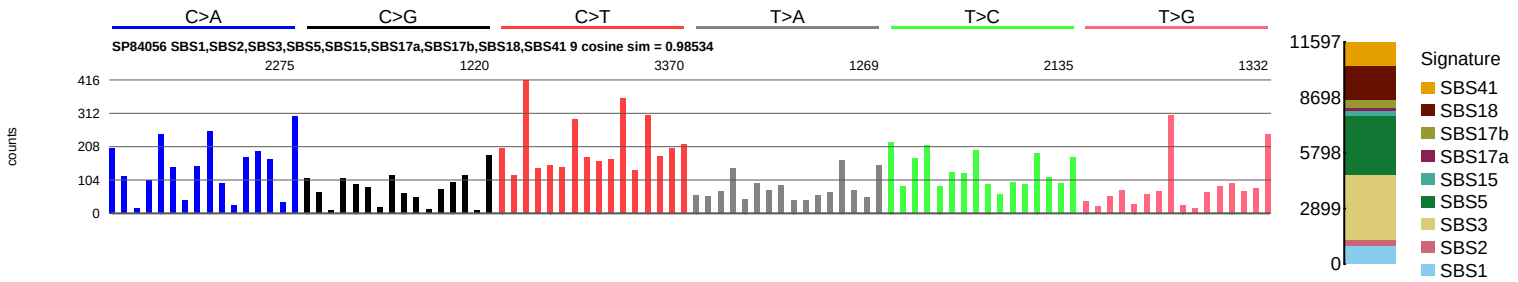

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98549
2307
1233
3408
1202
2048
1402
424
318
212
106
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98543
2320
1275
3364
1072
2140
1428
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98537
2300
1226
3323
1229
2128
1393
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98536
2297
1229
3343
1238
2095
1392
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98535
2371
1366
3449
989
2130
1294
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98534
2275
1220
3370
1269
2135
1332
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
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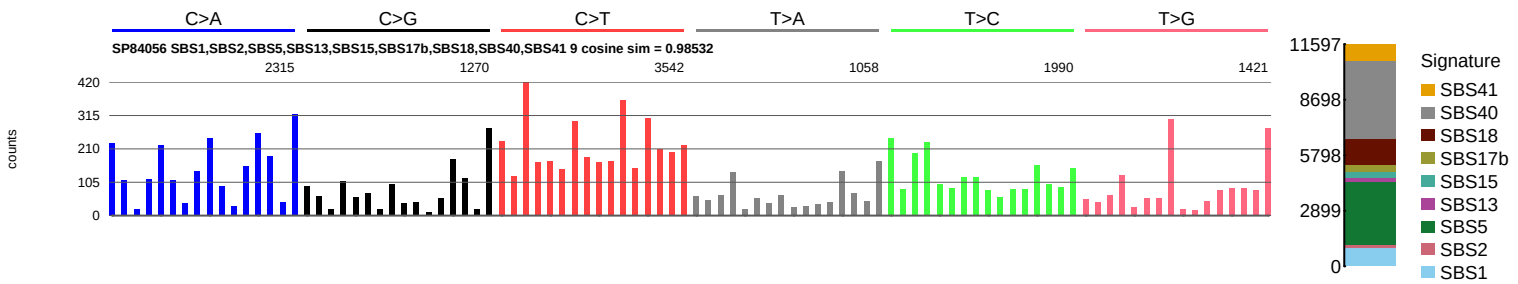
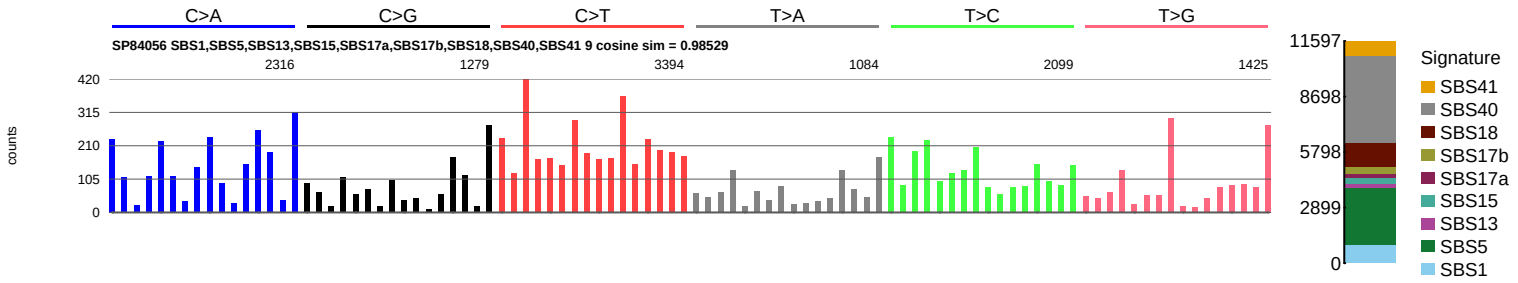
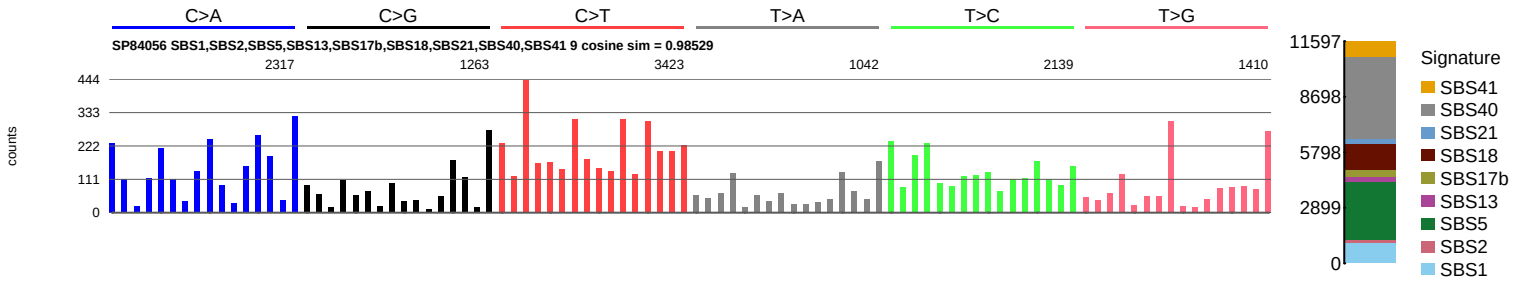
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98532
2315
1270
3542
1058
1990
1421
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98529
2316
1279
3394
1084
2099
1425
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98529
2317
1263
3423
1042
2139
1410
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
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SBS40
SBS21
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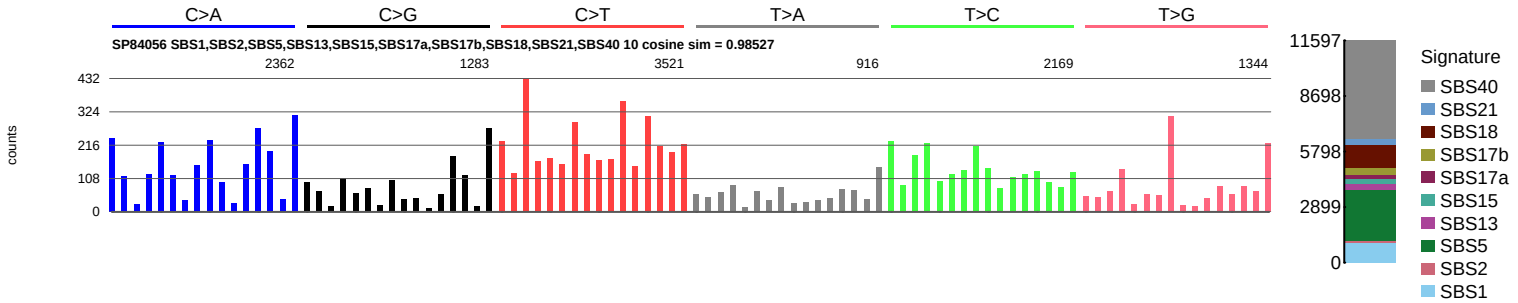
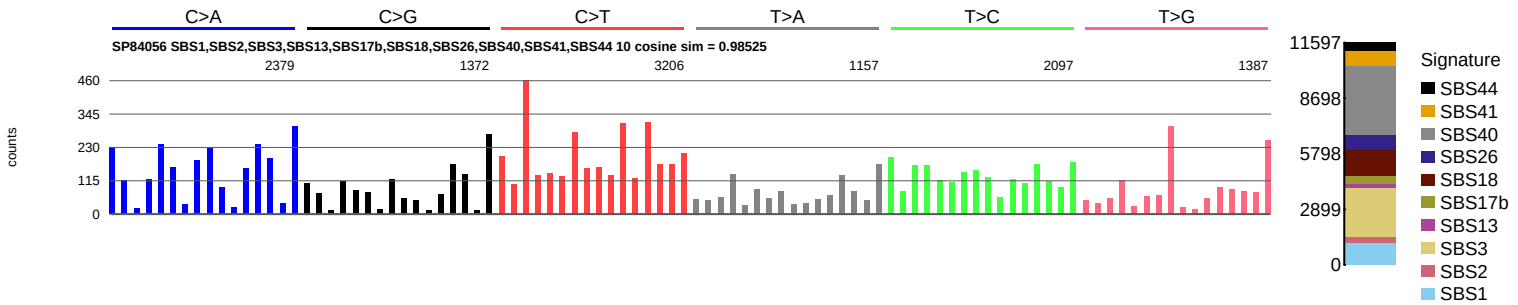
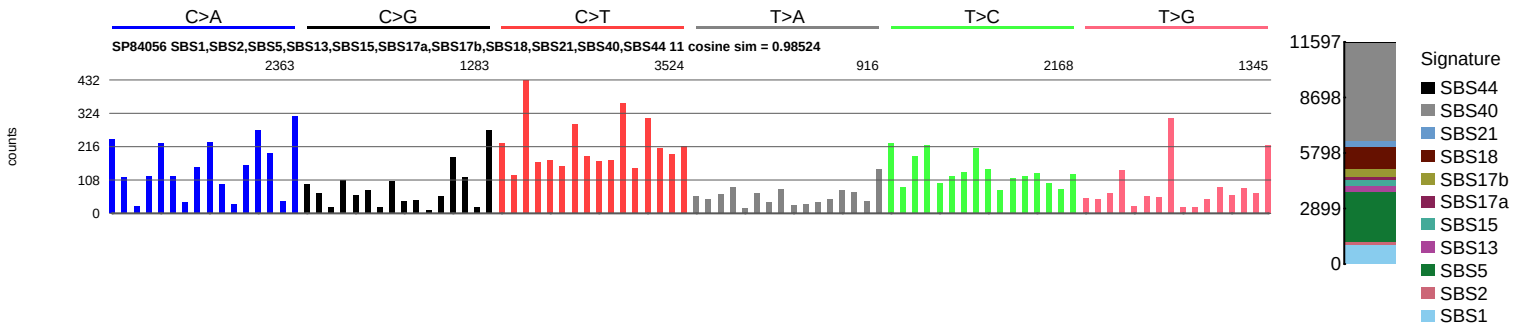
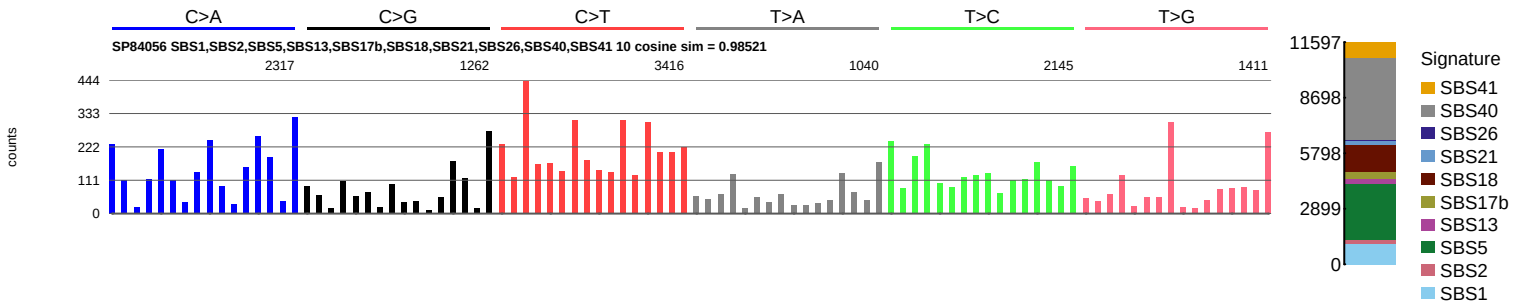
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98527
2362
1283
3521
916
2169
1344
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98525
2379
1372
3206
1157
2097
1387
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98524
2363
1283
3524
916
2168
1345
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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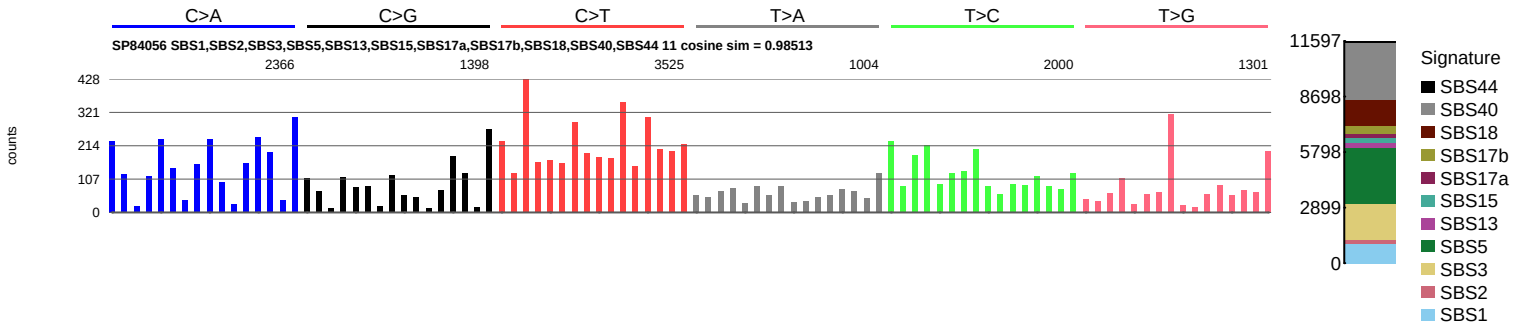
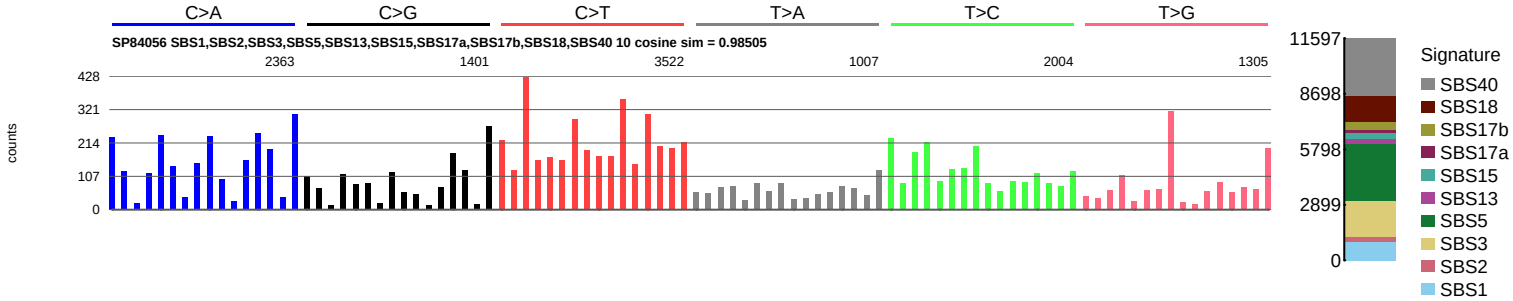
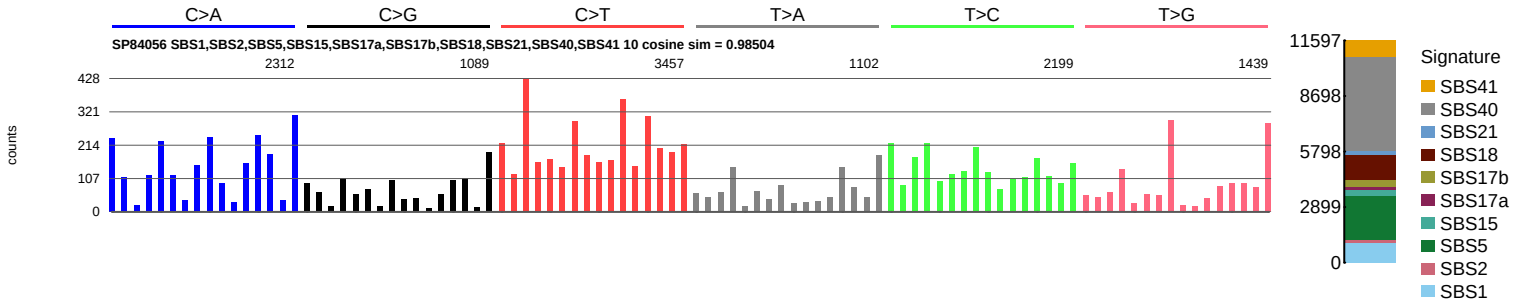
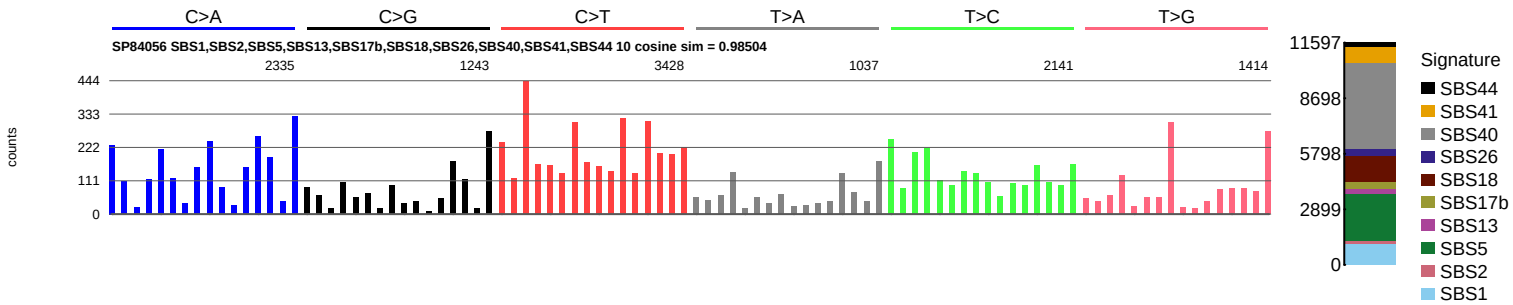
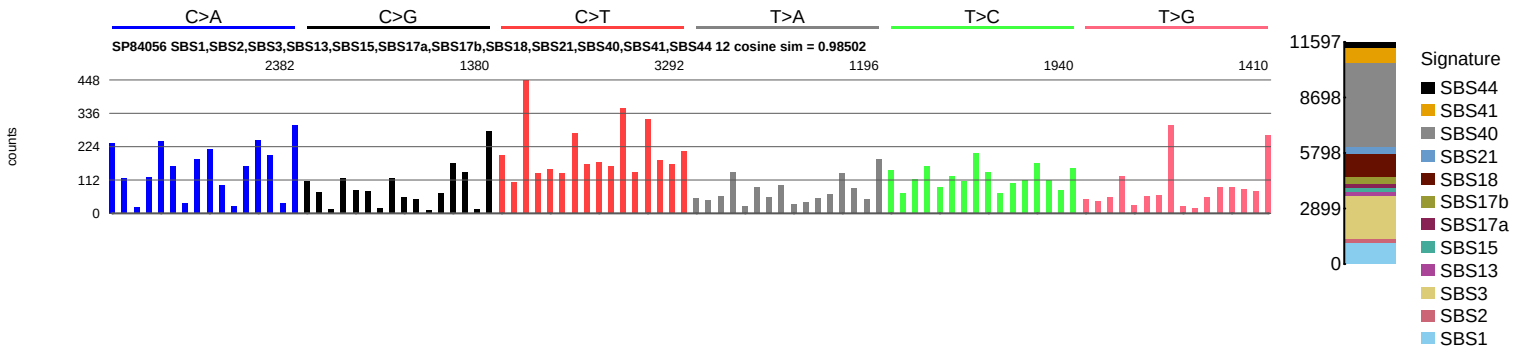
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98521
2317
1262
3416
1040
2145
1411
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

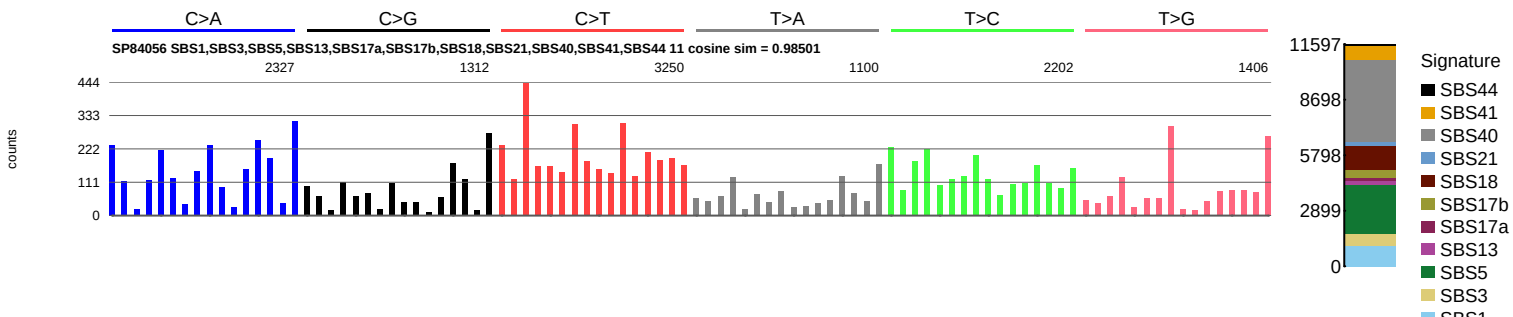

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98513
2366
1398
3525
1004
2000
1301
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

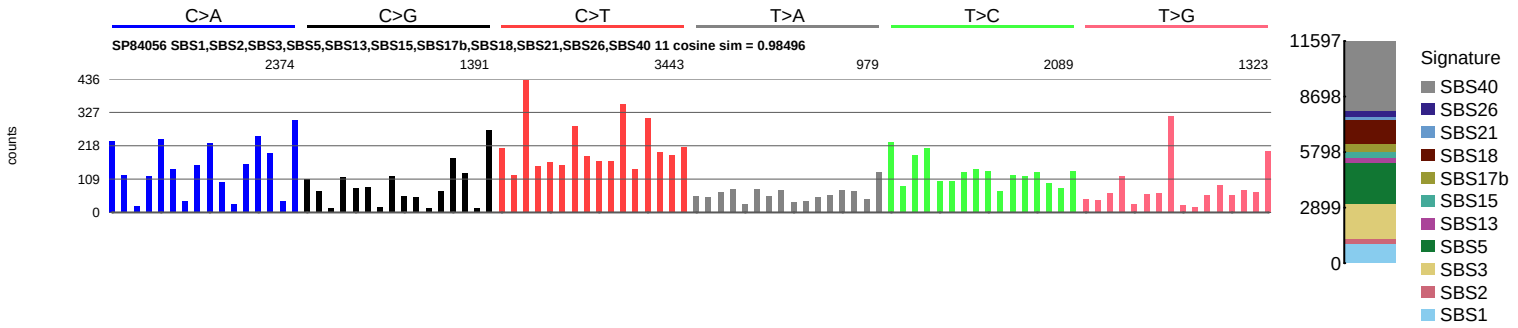

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98505
2363
1401
3522
1007
2004
1305
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

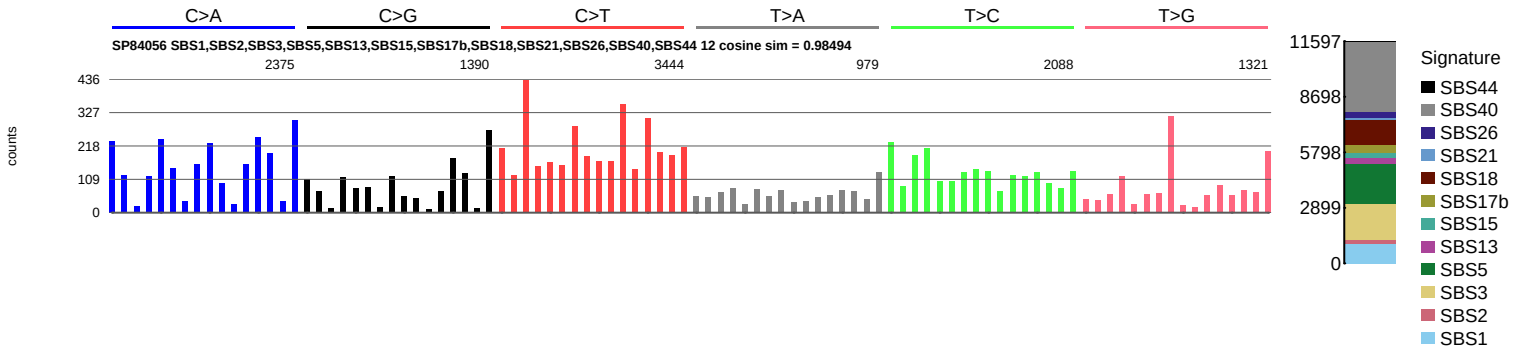

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98504
2312
1089
3457
1102
2199
1439
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

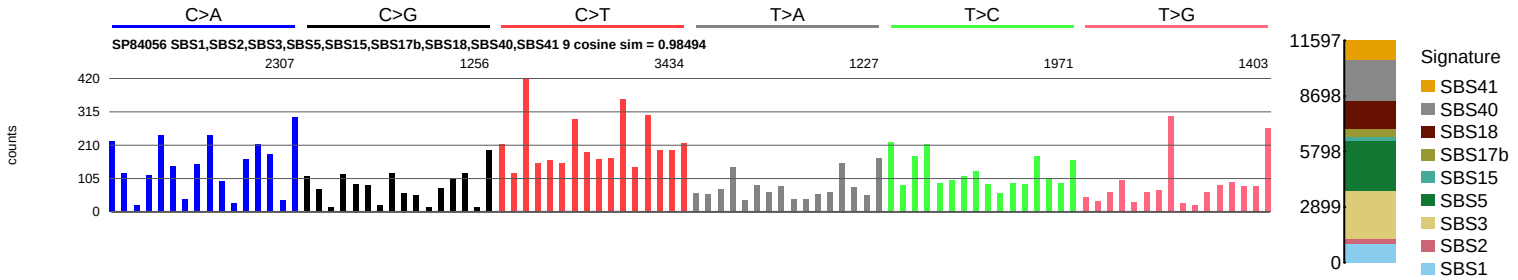

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98504
2335
1243
3428
1037
2141
1414
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

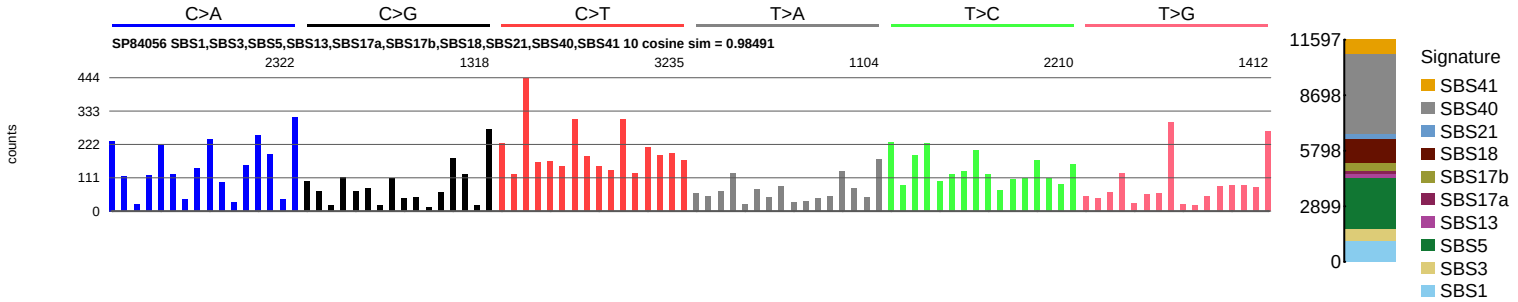

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98502
2382
1380
3292
1196
1940
1410
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

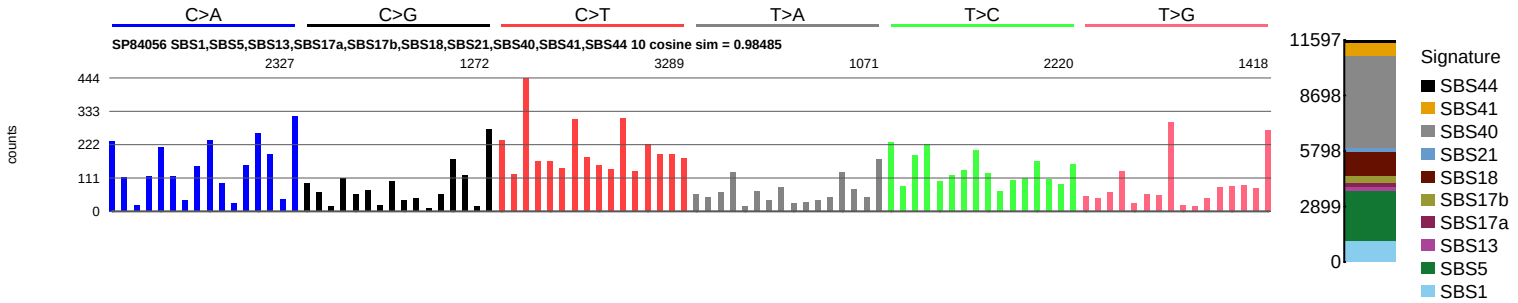

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98501
2327
1312
3250
1100
2202
1406
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

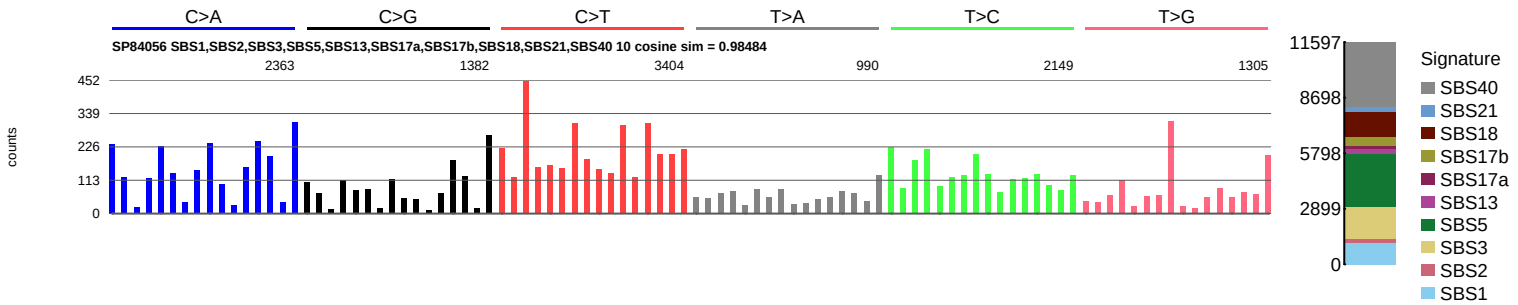

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98496
2374
1391
3443
979
2089
1323
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

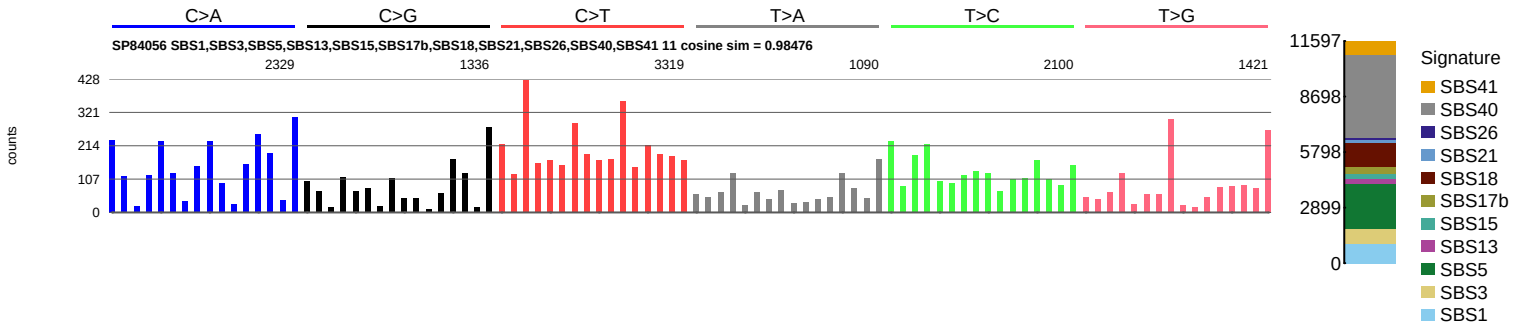

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98494
2375
1390
3444
979
2088
1321
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

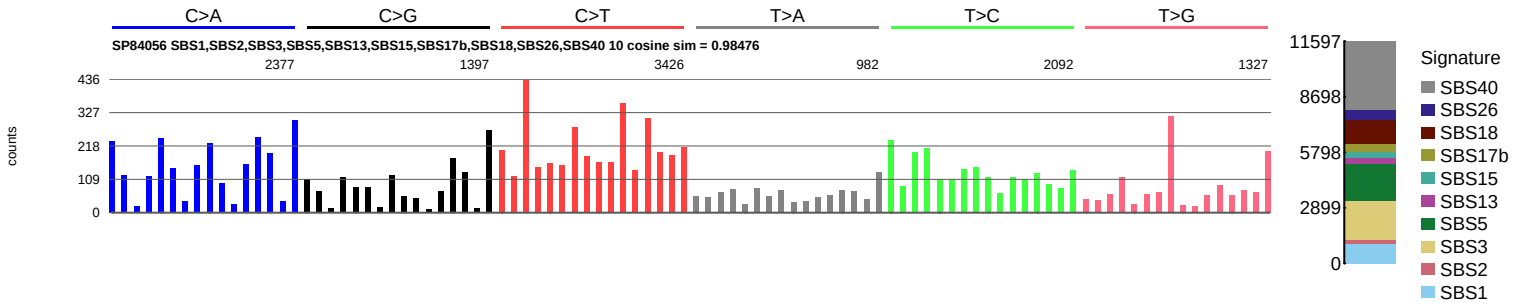

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98494
2307
1256
3434
1227
1971
1403
420
315
210
105
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

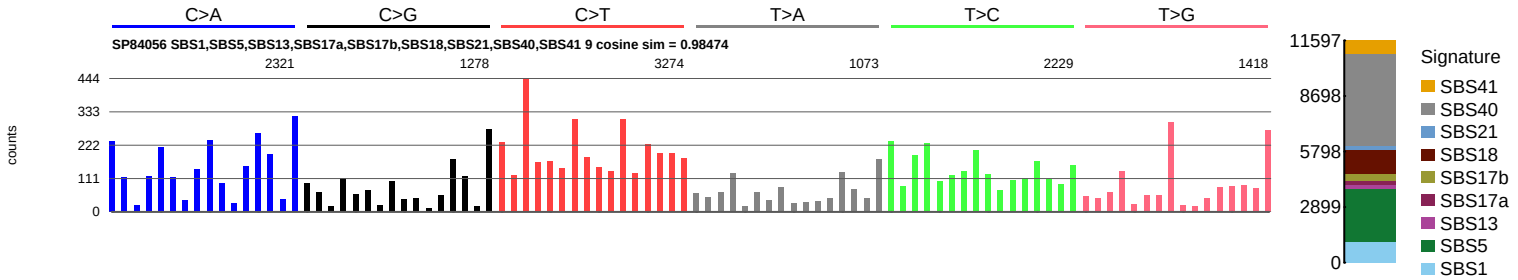

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98491
2322
1318
3235
1104
2210
1412
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

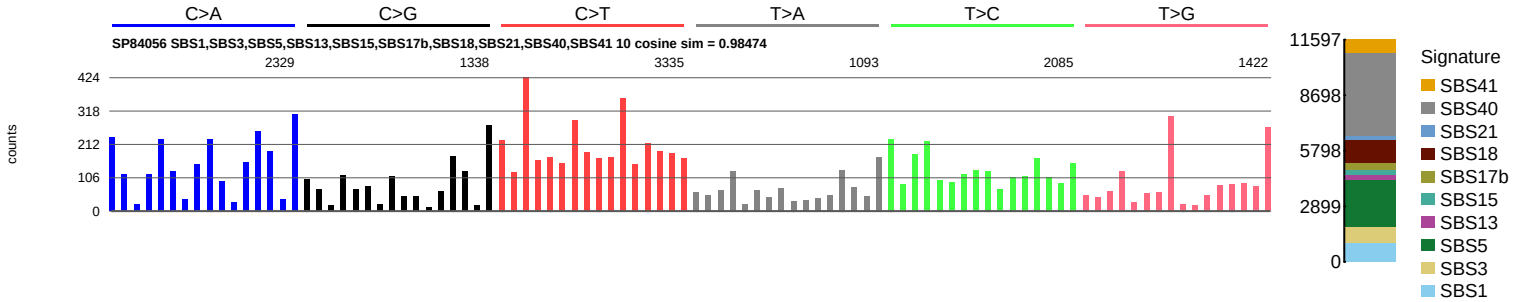

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98485
2327
1272
3289
1071
2220
1418
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

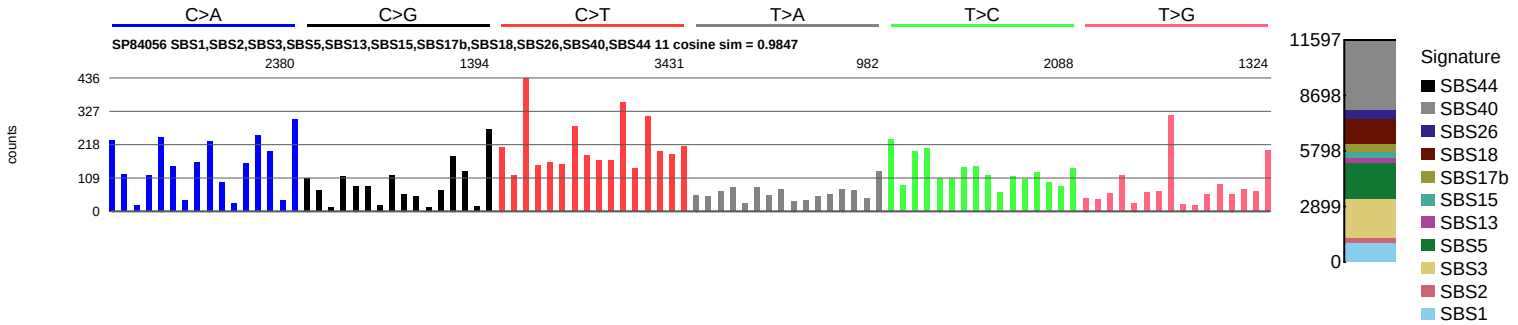

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98484
2363
1382
3404
990
2149
1305
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

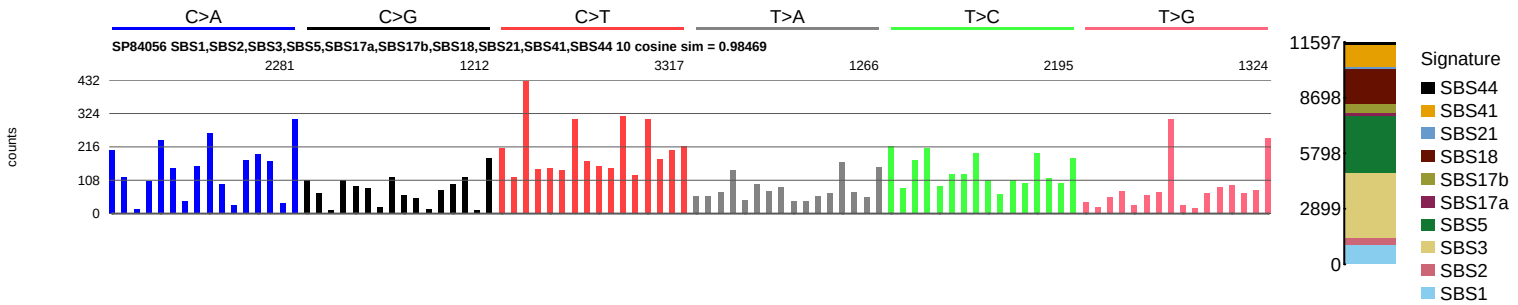

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98476
2329
1336
3319
1090
2100
1421
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

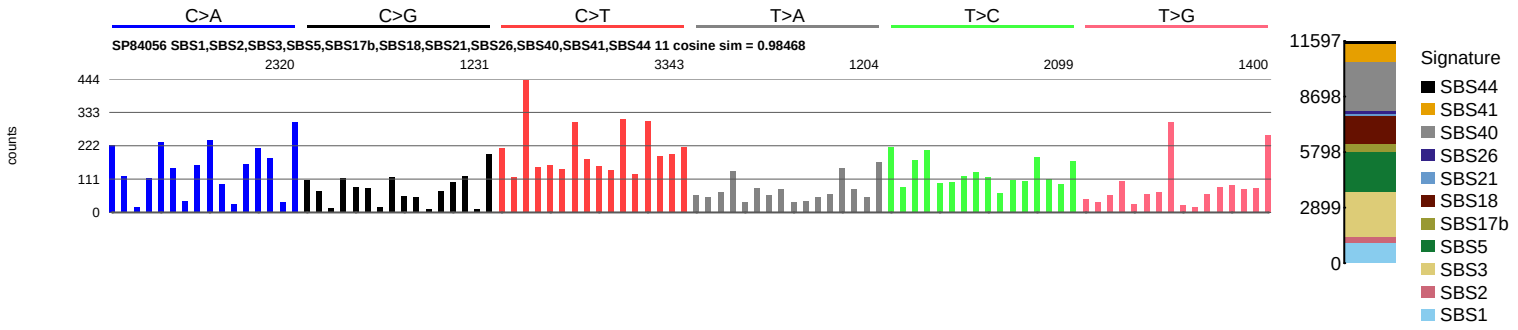

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98476
2377
1397
3426
982
2092
1327
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

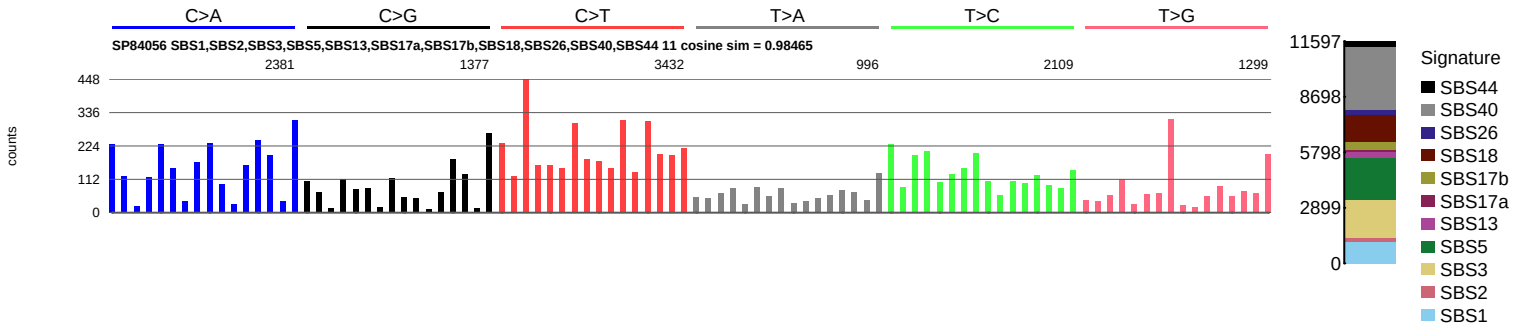

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98474
2321
1278
3274
1073
2229
1418
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

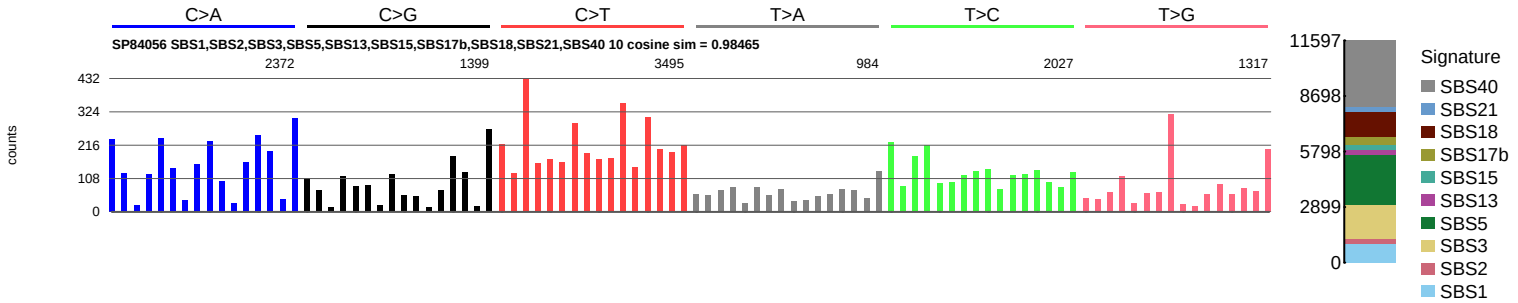

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98474
2329
1338
3335
1093
2085
1422
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

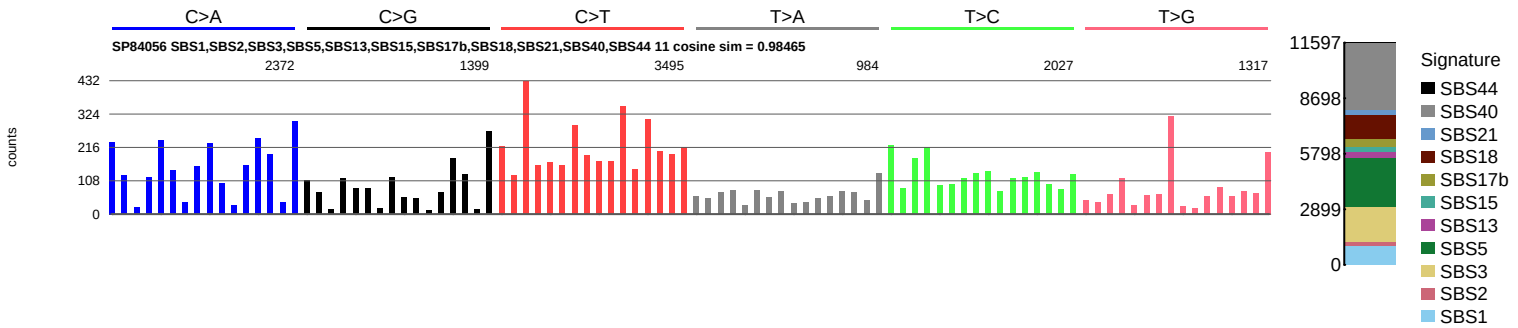

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9847
2380
1394
3431
982
2088
1324
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98469
2281
1212
3317
1266
2195
1324
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98468
2320
1231
3343
1204
2099
1400
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98465
2381
1377
3432
996
2109
1299
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98465
2372
1399
3495
984
2027
1317
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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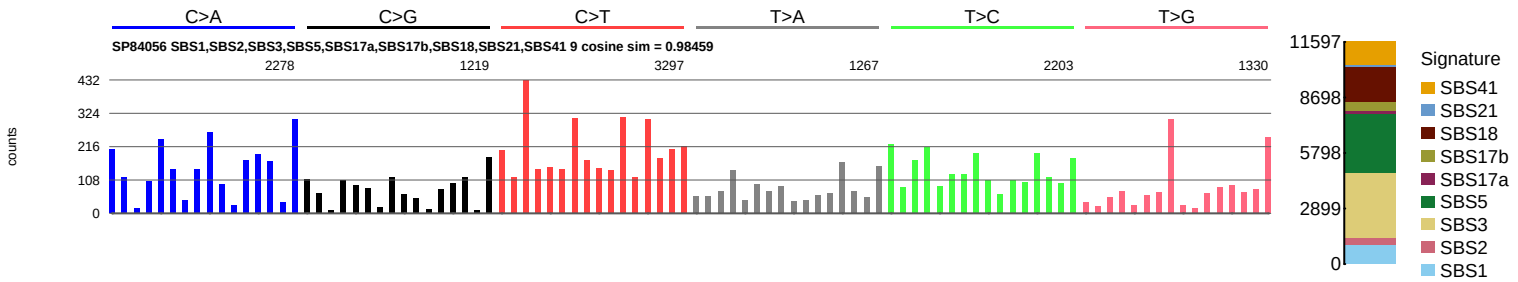
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98465
2372
1399
3495
984
2027
1317
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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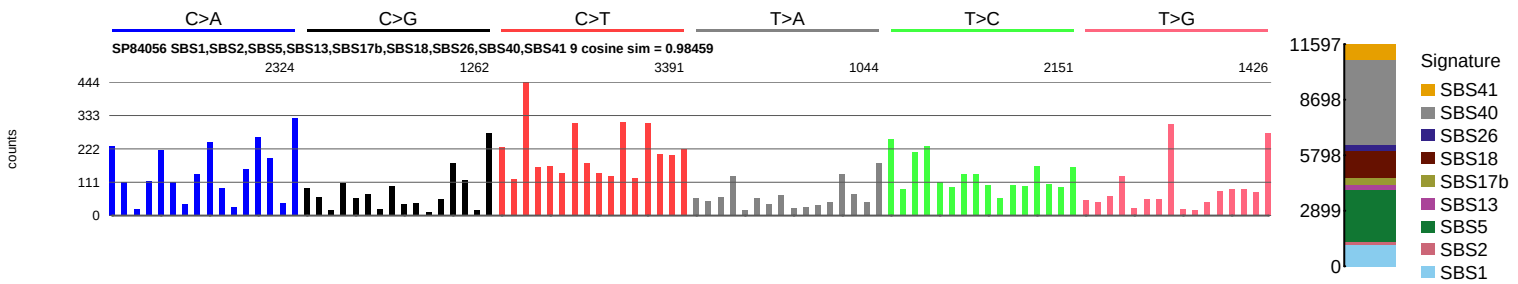
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98459
2278
1219
3297
1267
2203
1330
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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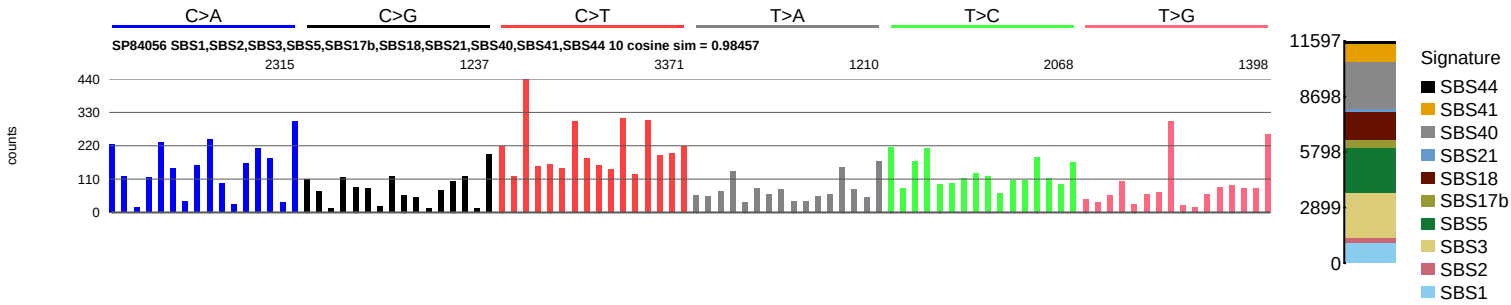
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98459
2324
1262
3391
1044
2151
1426
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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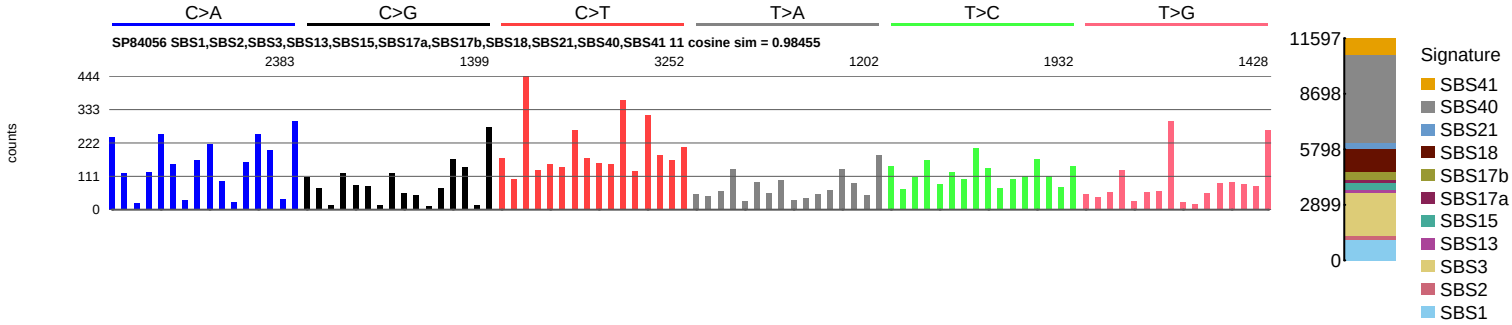
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98457
2315
1237
3371
1210
2068
1398
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
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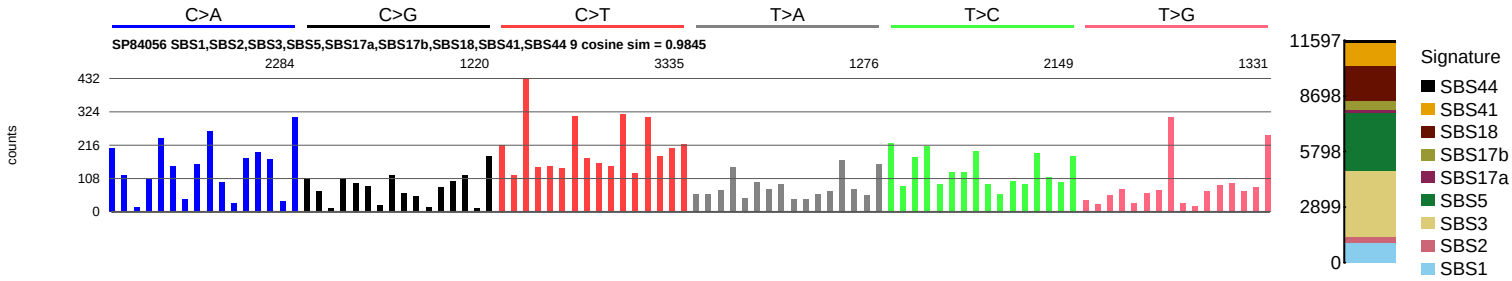
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98455
2383
1399
3252
1202
1932
1428
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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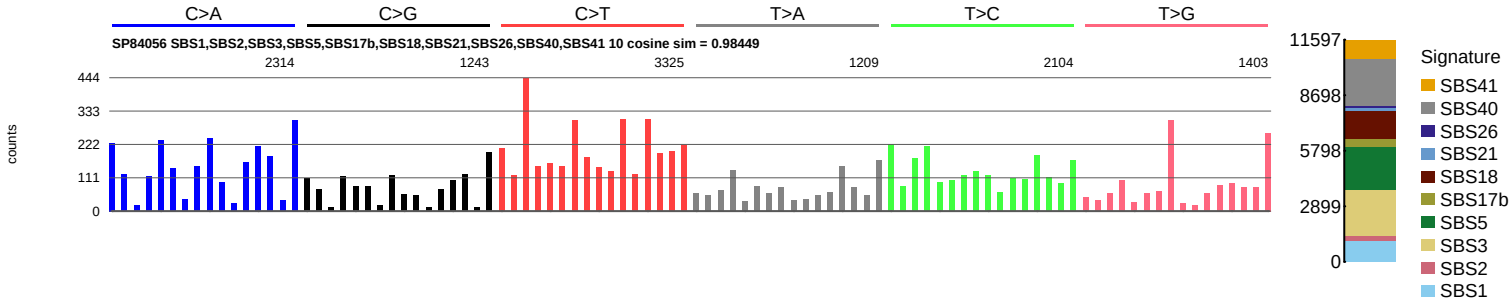
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9845
2284
1220
3335
1276
2149
1331
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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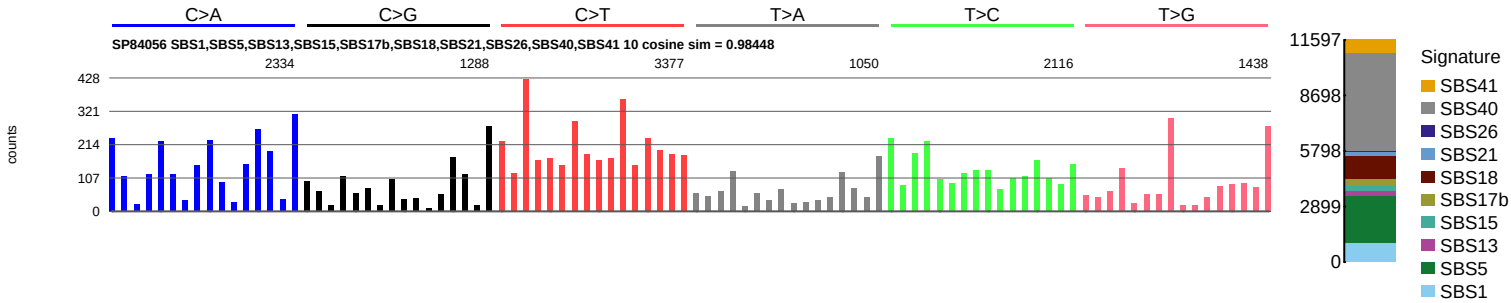
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98449
2314
1243
3325
1209
2104
1403
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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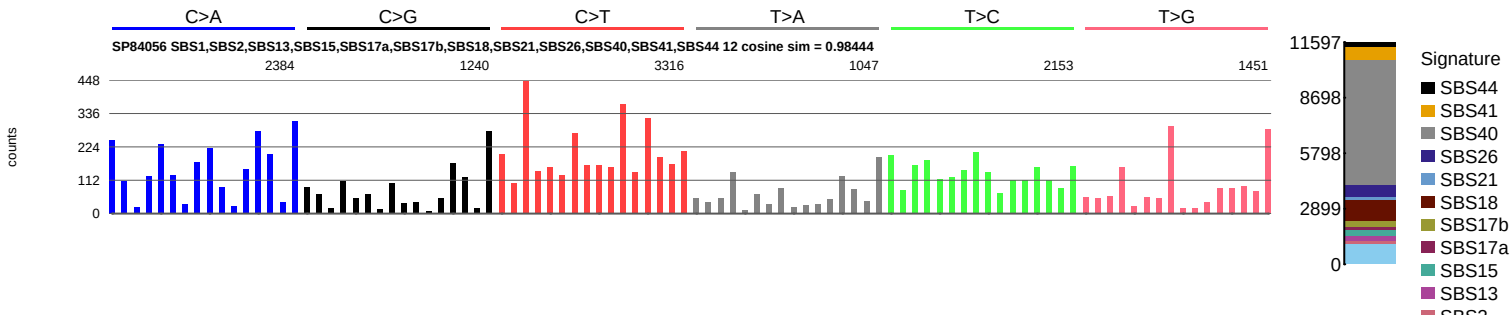
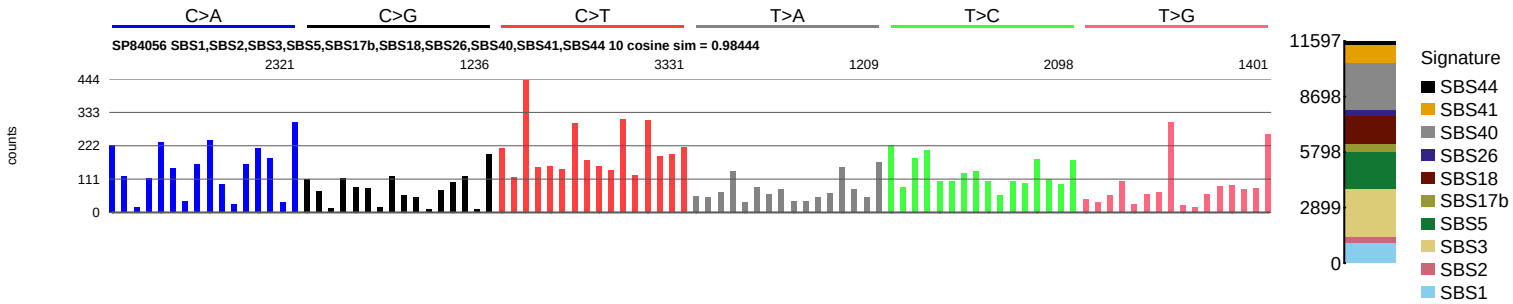
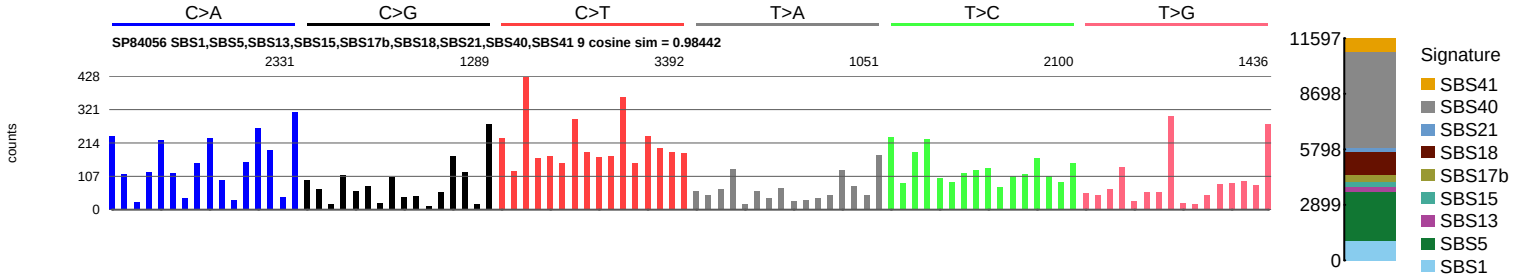
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98448
2334
1288
3377
1050
2116
1438
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98444
2384
1240
3316
1047
2153
1451
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98444
2321
1236
3331
1209
2098
1401
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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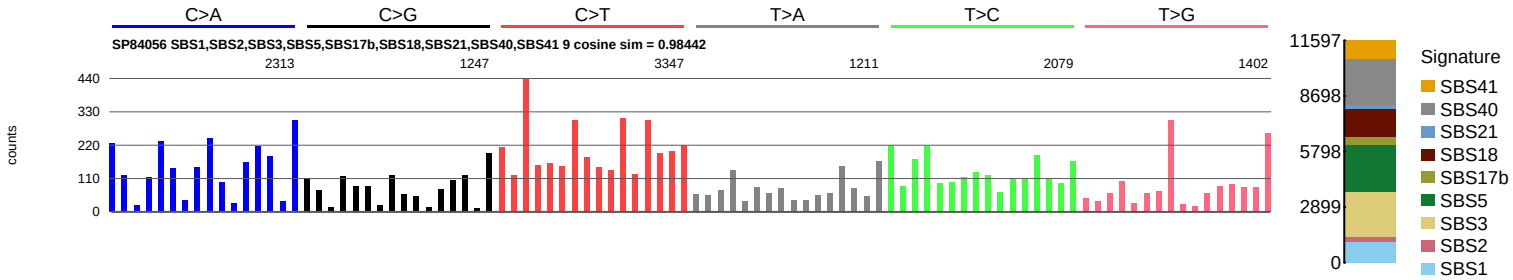
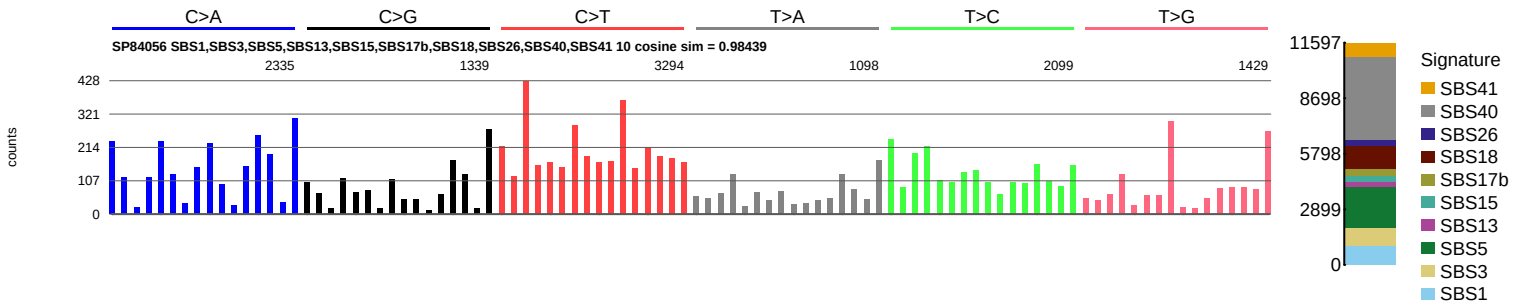
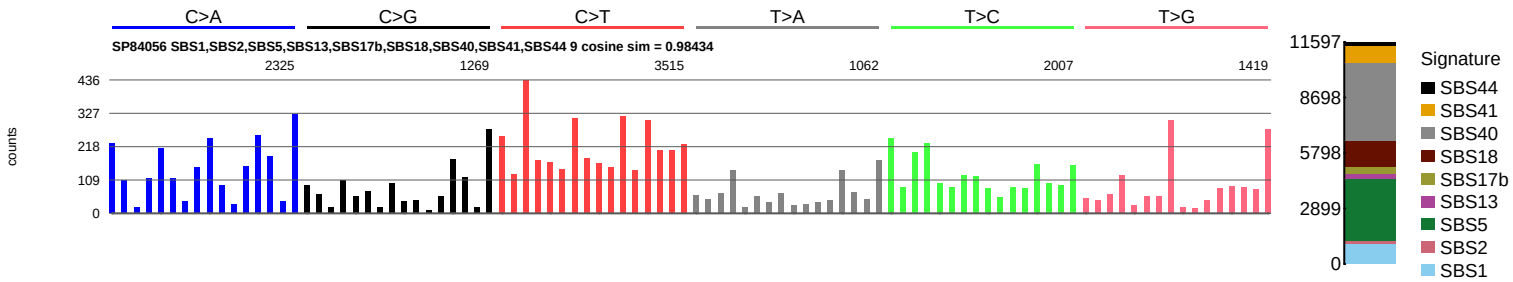
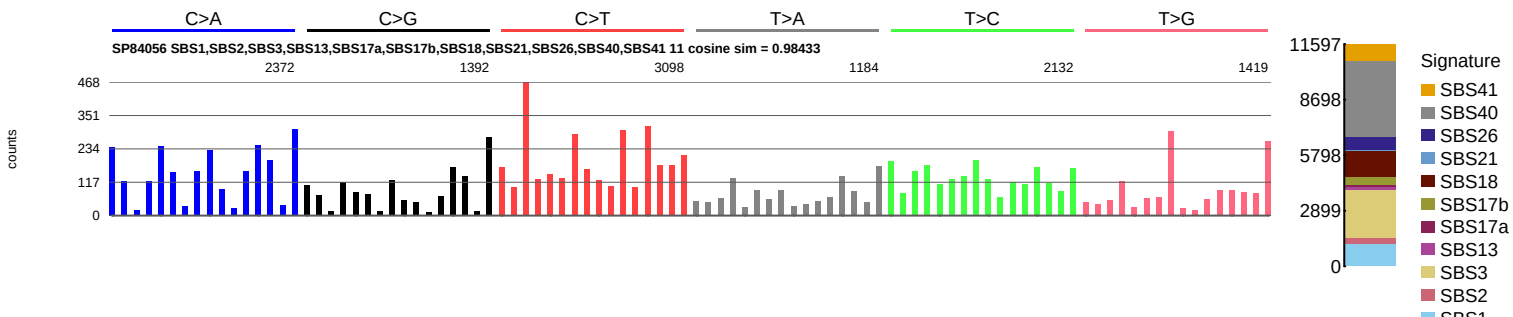
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98442
2331
1289
3392
1051
2100
1436
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98442
2313
1247
3347
1211
2079
1402
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98439
2335
1339
3294
1098
2099
1429
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98434
2325
1269
3515
1062
2007
1419
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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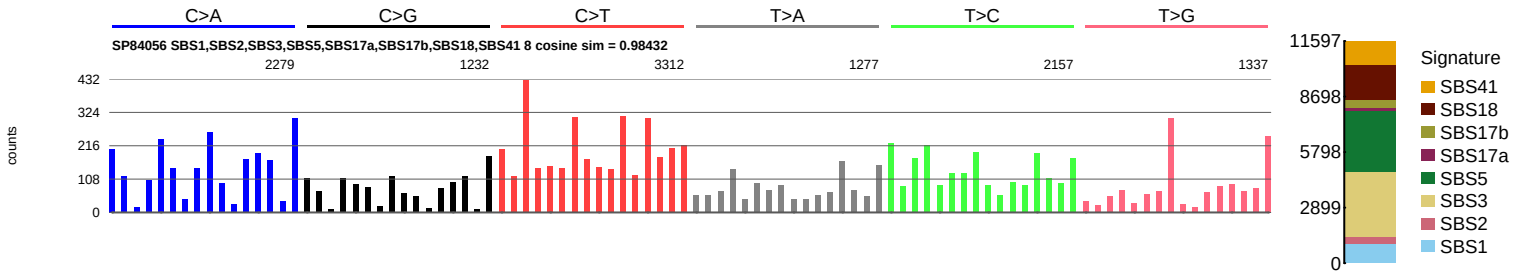
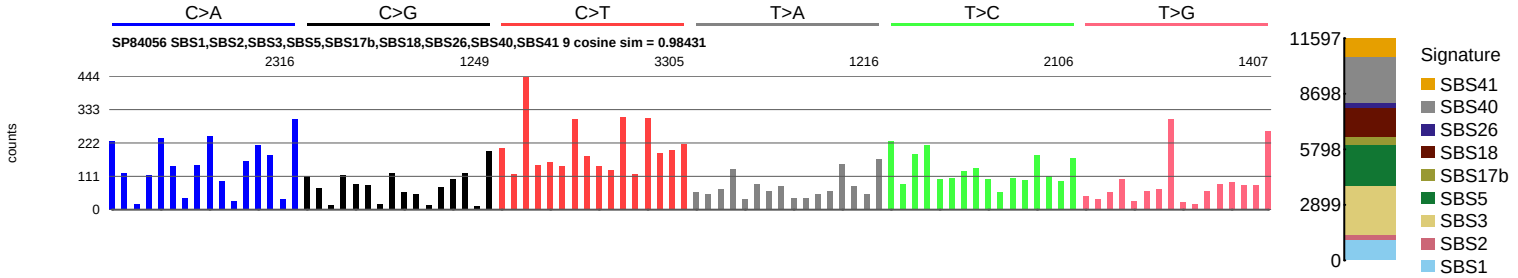
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98433
2372
1392
3098
1184
2132
1419
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98432
2279
1232
3312
1277
2157
1337
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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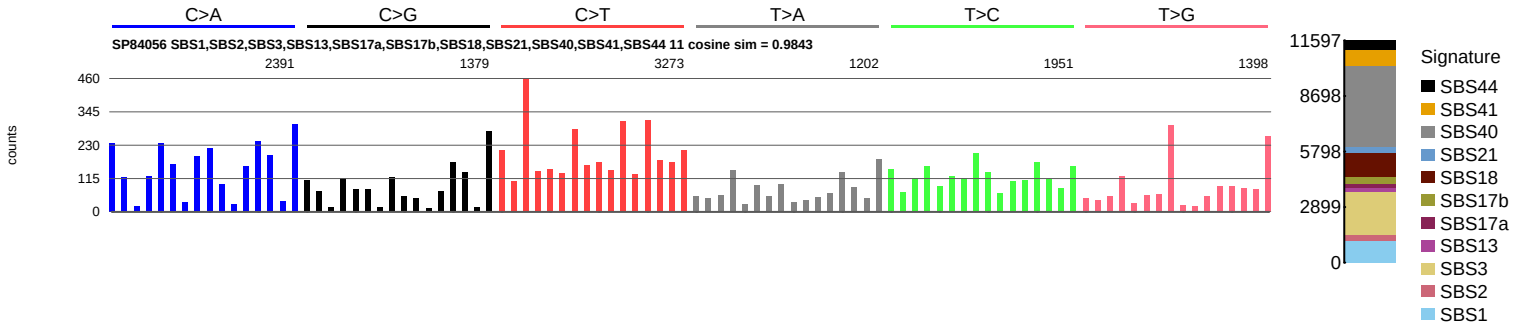
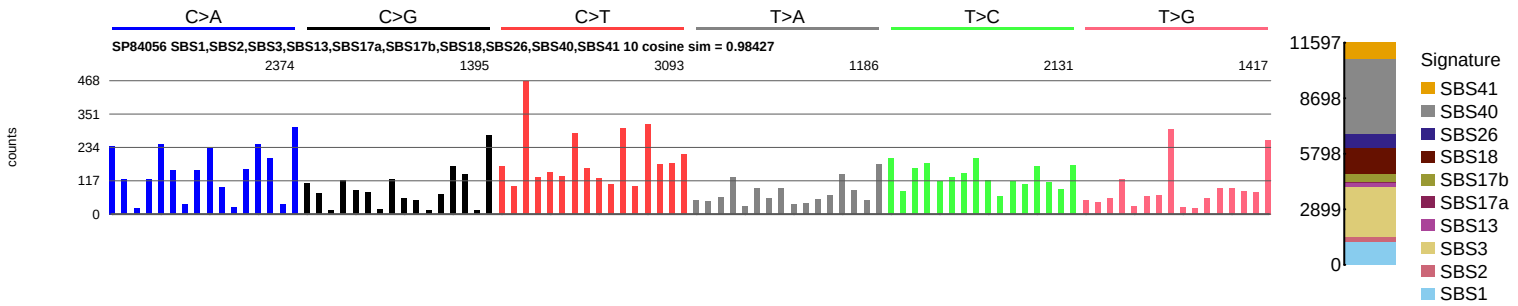
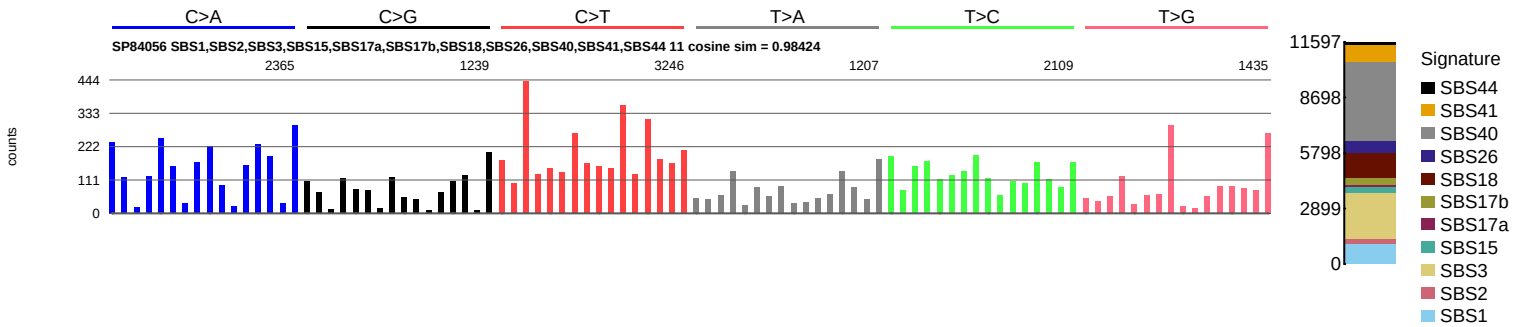
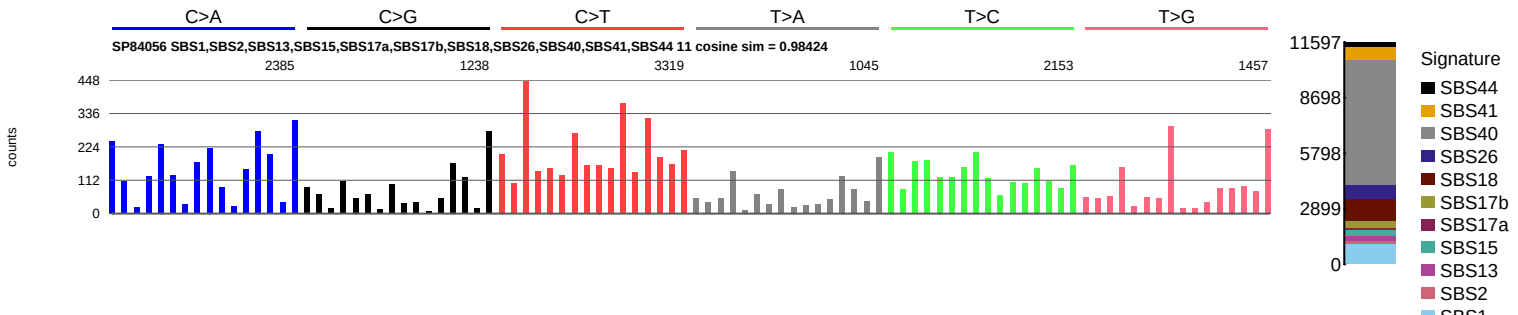
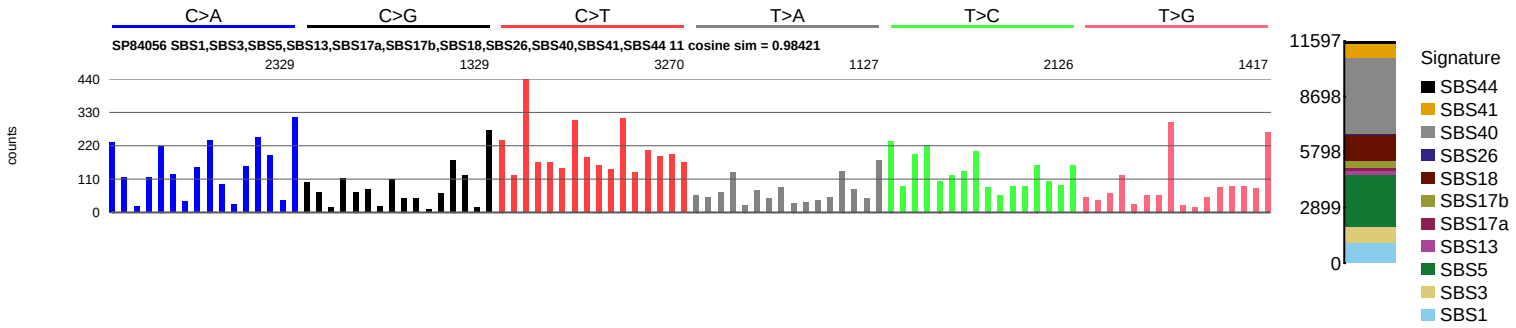
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98431
2316
1249
3305
1216
2106
1407
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

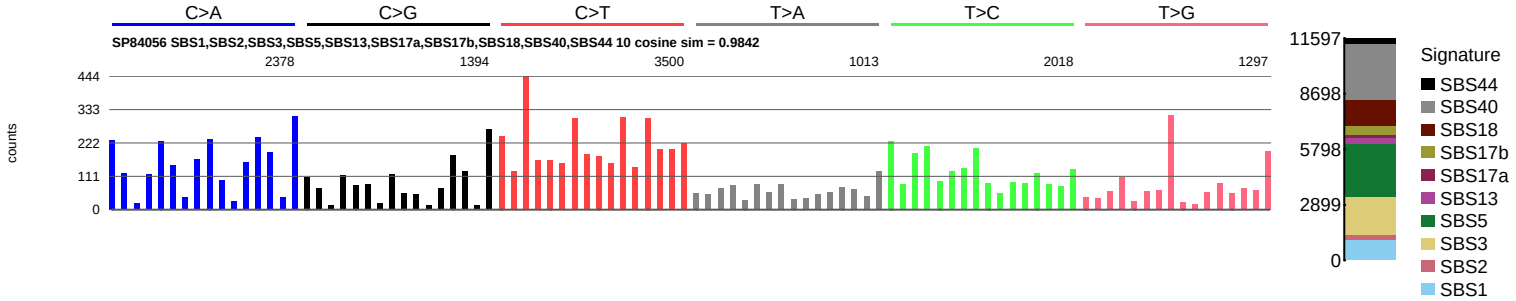

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9843
2391
1379
3273
1202
1951
1398
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

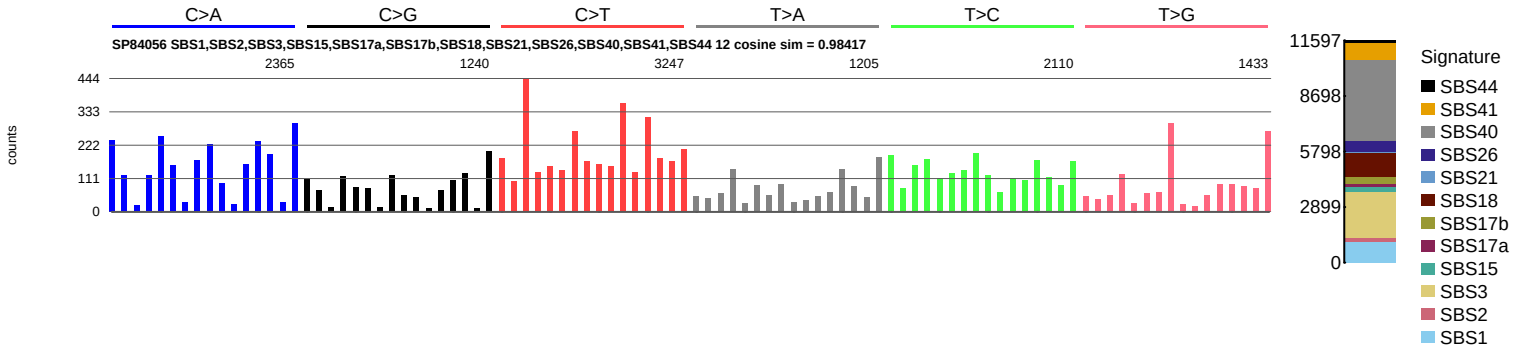

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98427
2374
1395
3093
1186
2131
1417
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98424
2365
1239
3246
1207
2109
1435
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98424
2385
1238
3319
1045
2153
1457
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98421
2329
1329
3270
1127
2126
1417
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9842
2378
1394
3500
1013
2018
1297
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98417
2365
1240
3247
1205
2110
1433
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
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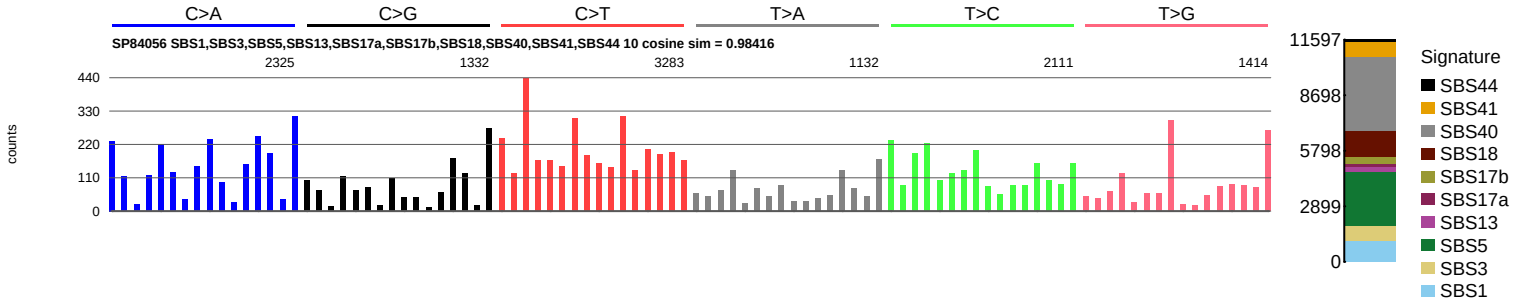
SBS1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SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98416
2325
1332
3283
1132
2111
1414
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
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SBS40
SBS18
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SBS13
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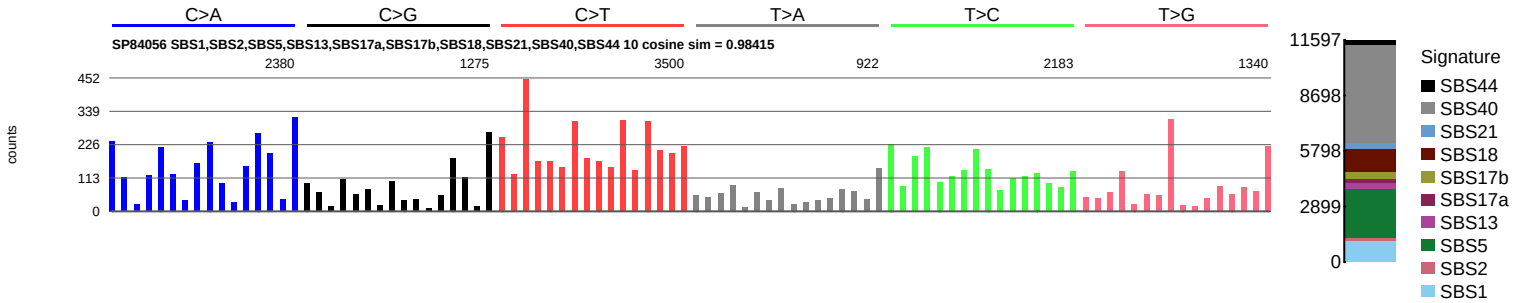
SBS1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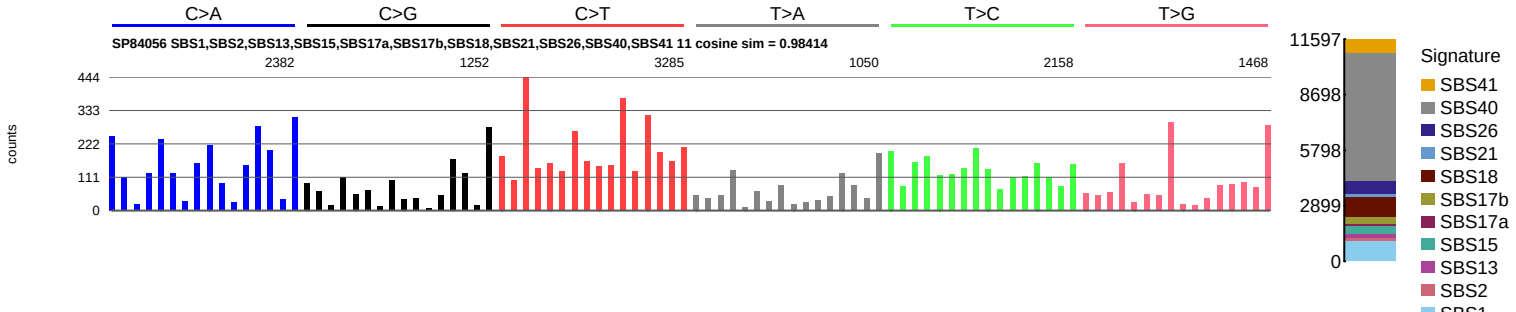
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98415
2380
1275
3500
922
2183
1340
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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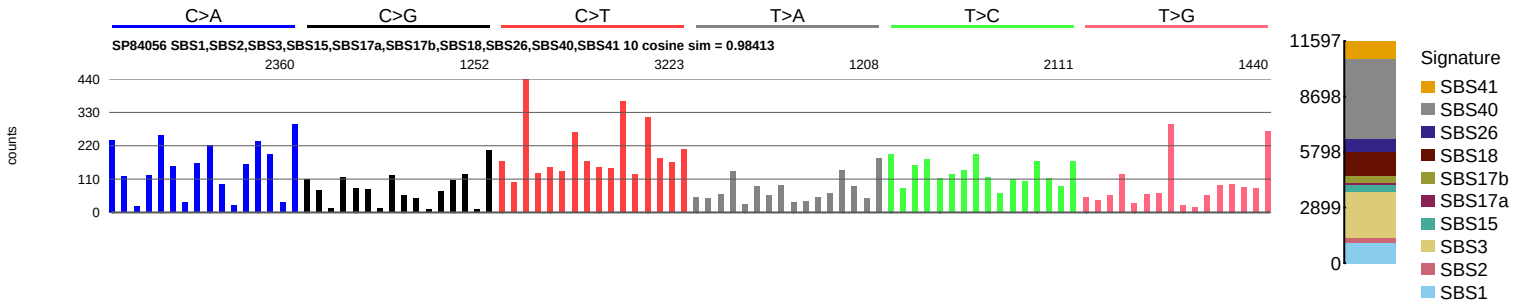
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98414
2382
1252
3285
1050
2158
1468
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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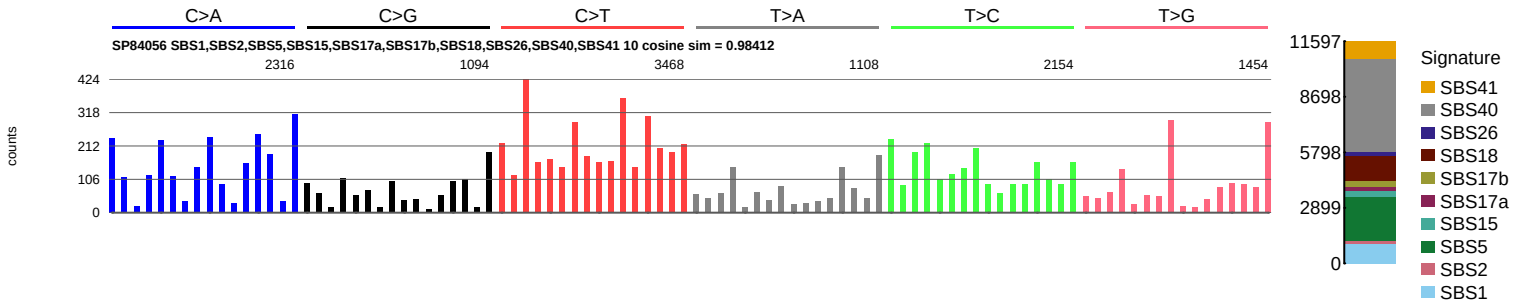
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98413
2360
1252
3223
1208
2111
1440
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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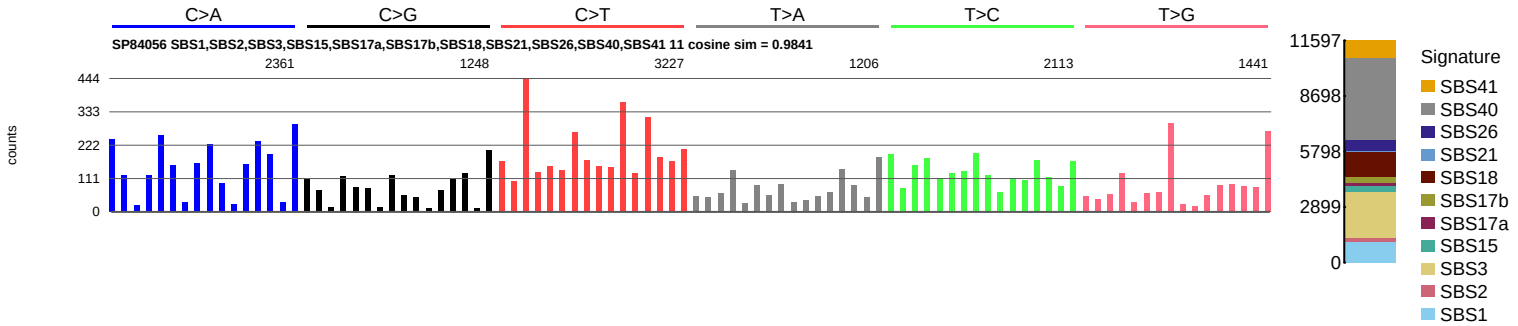
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98412
2316
1094
3468
1108
2154
1454
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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T>C
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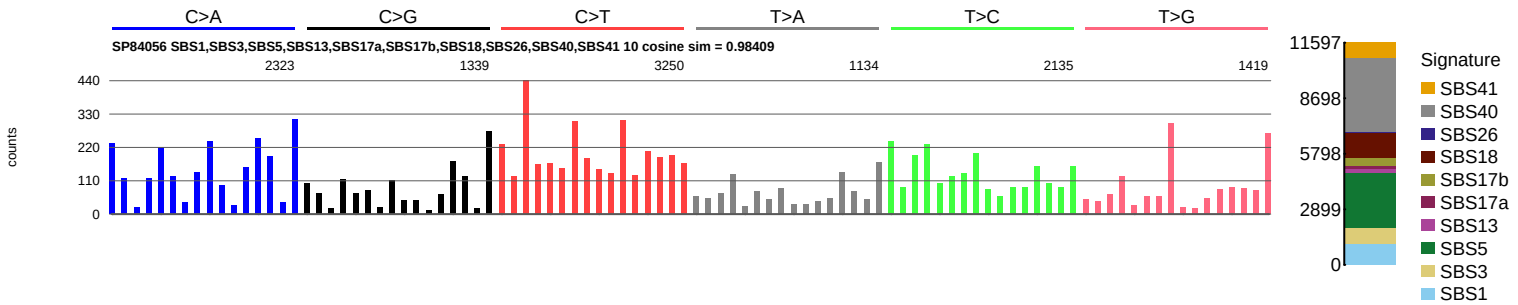
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9841
2361
1248
3227
1206
2113
1441
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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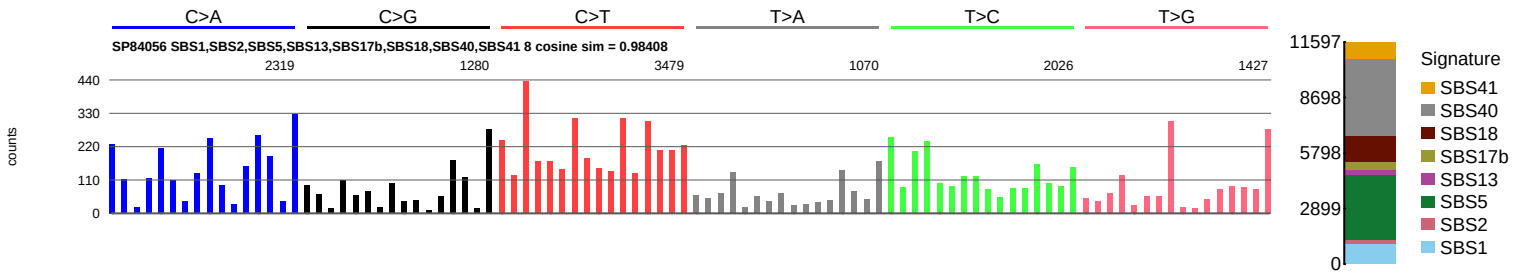
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98409
2323
1339
3250
1134
2135
1419
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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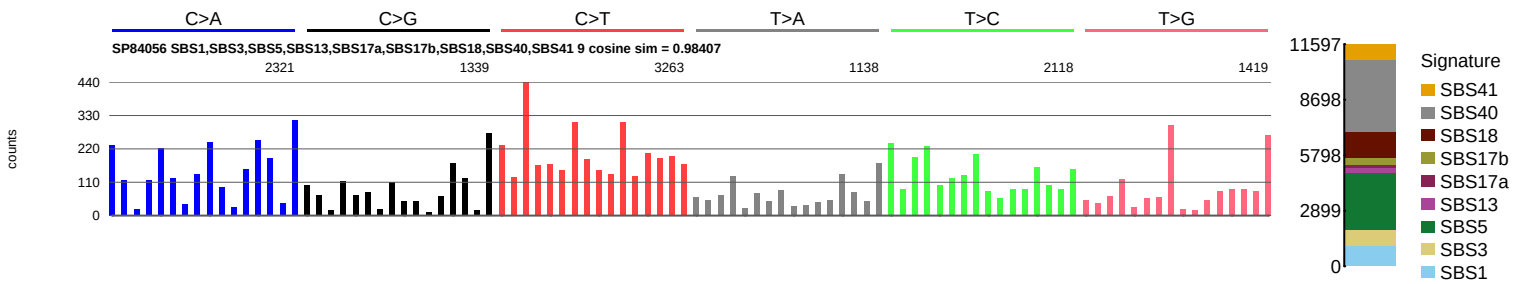
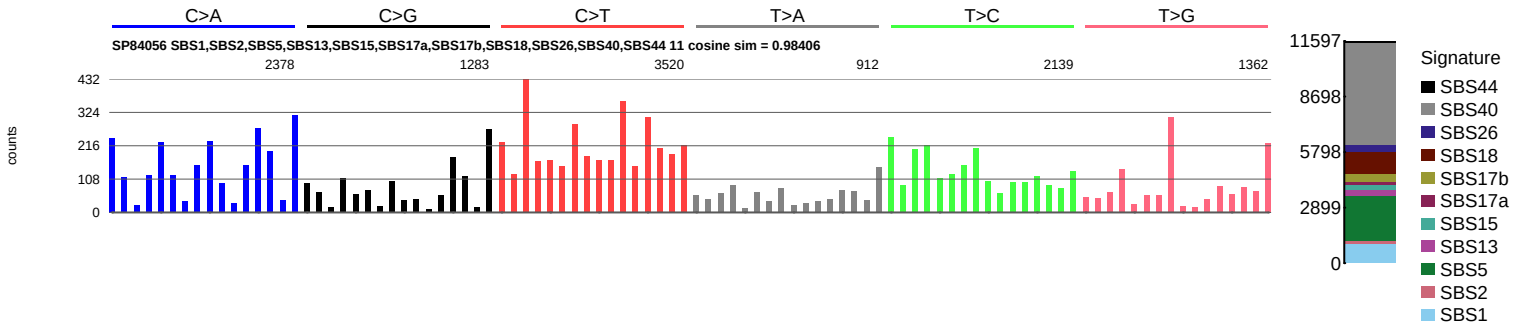
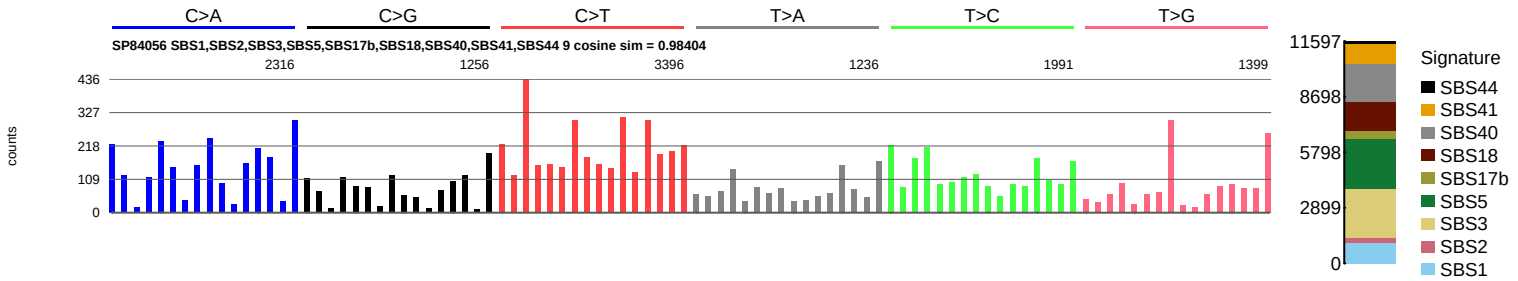
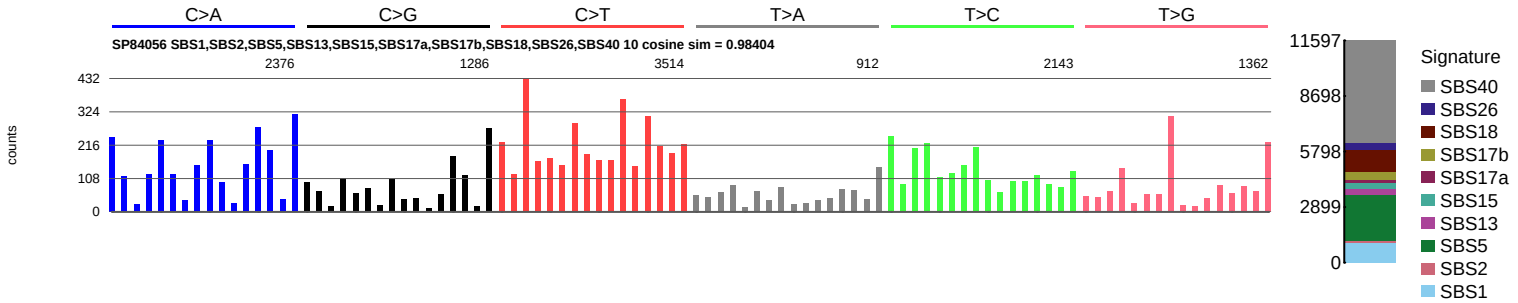
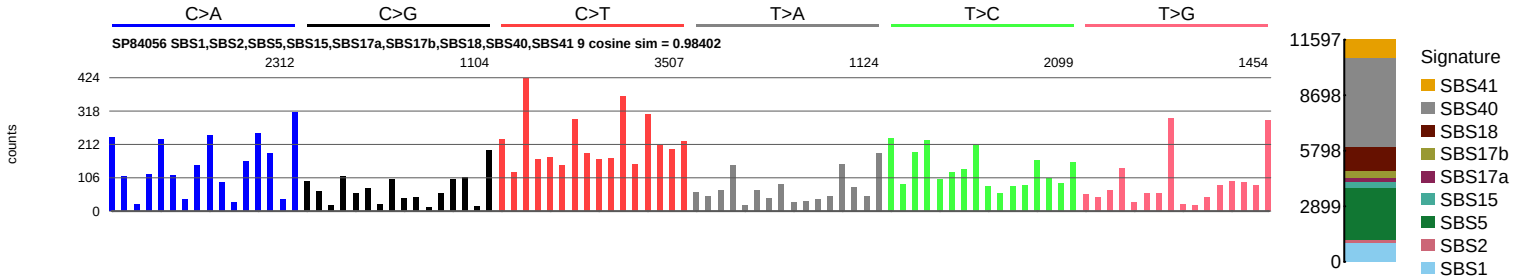
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98408
2319
1280
3479
1070
2026
1427
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98407
2321
1339
3263
1138
2118
1419
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98406
2378
1283
3520
912
2139
1362
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98404
2316
1256
3396
1236
1991
1399
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98404
2376
1286
3514
912
2143
1362
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98402
2312
1104
3507
1124
2099
1454
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
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SBS17b
SBS17a
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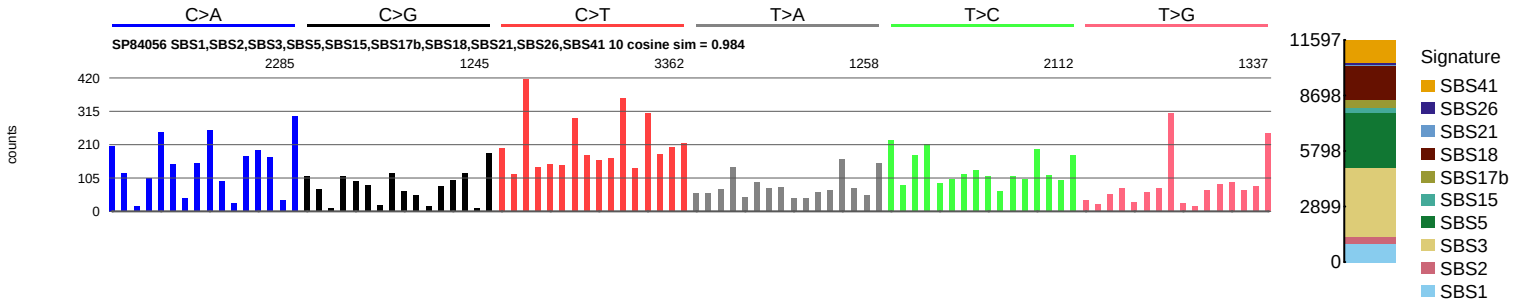
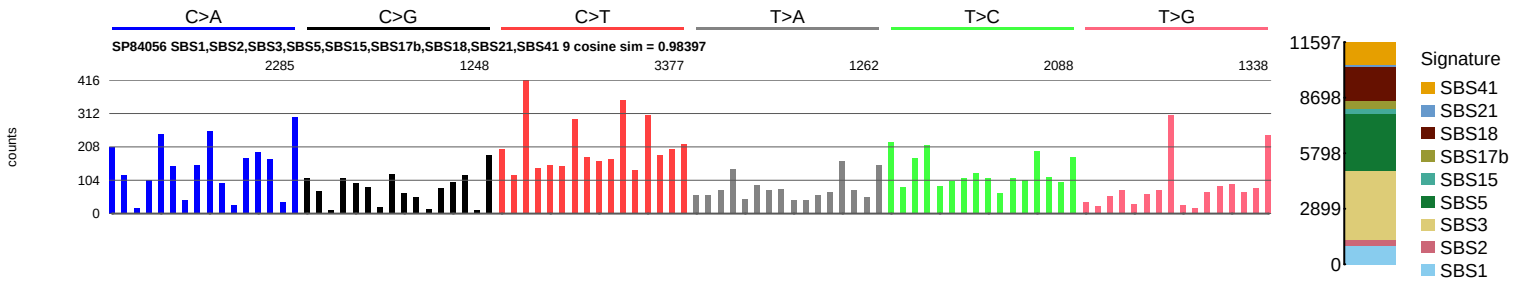
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.984
2285
1245
3362
1258
2112
1337
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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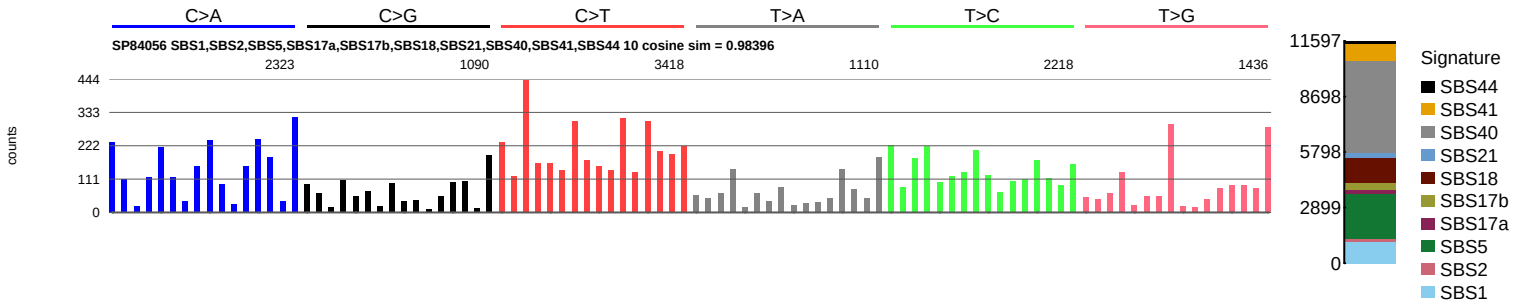
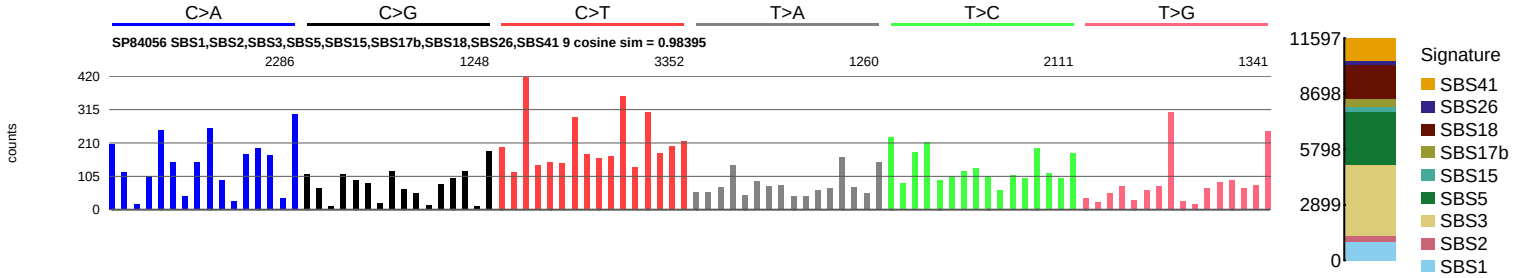
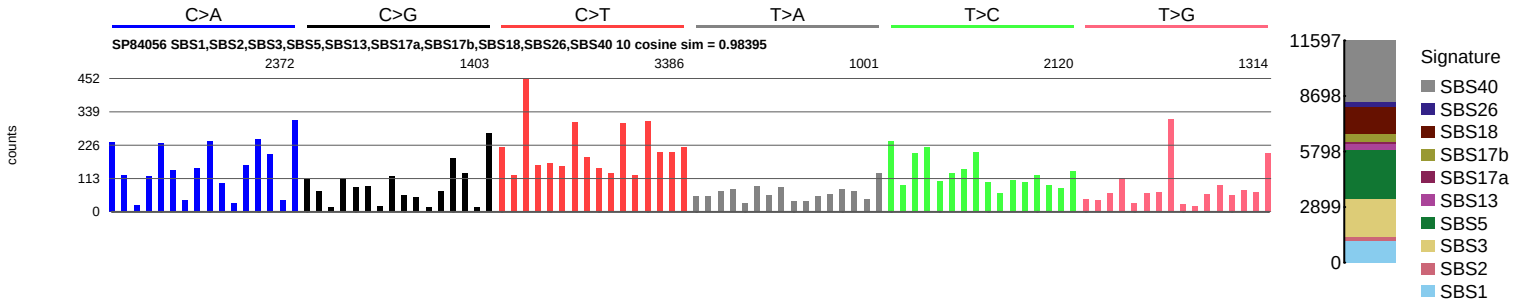
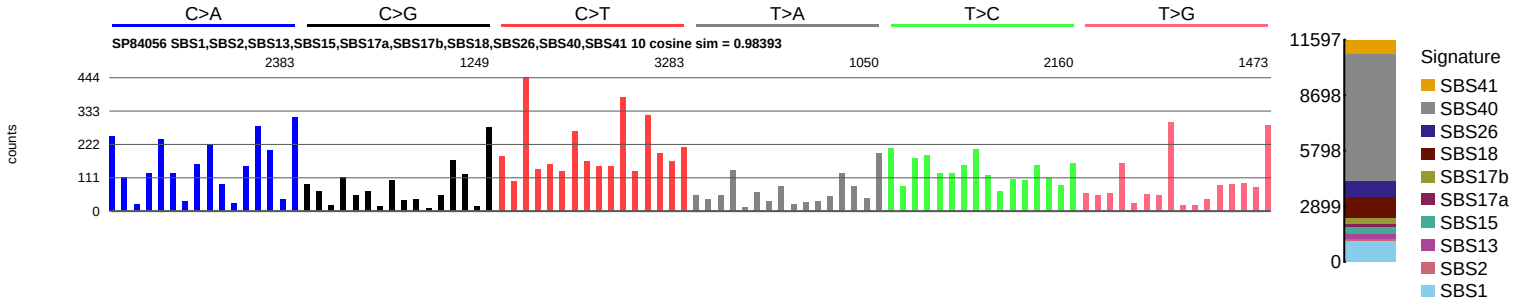
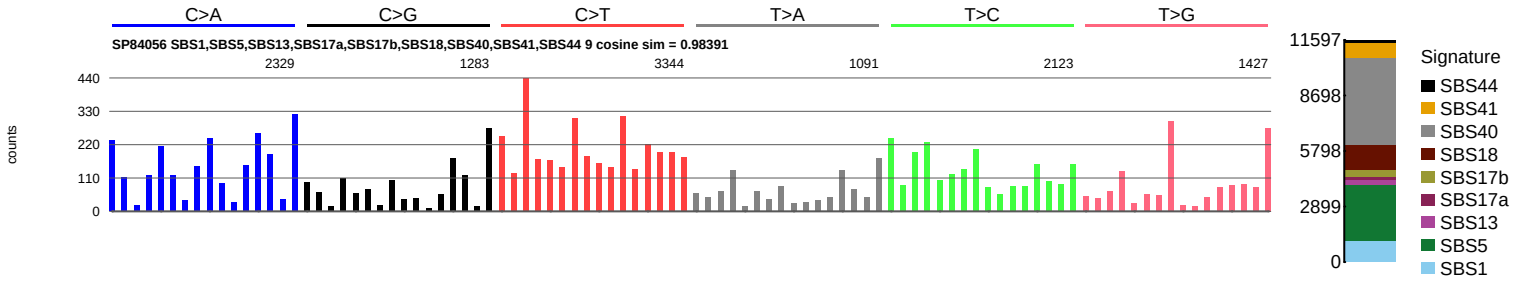
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98397
2285
1248
3377
1262
2088
1338
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

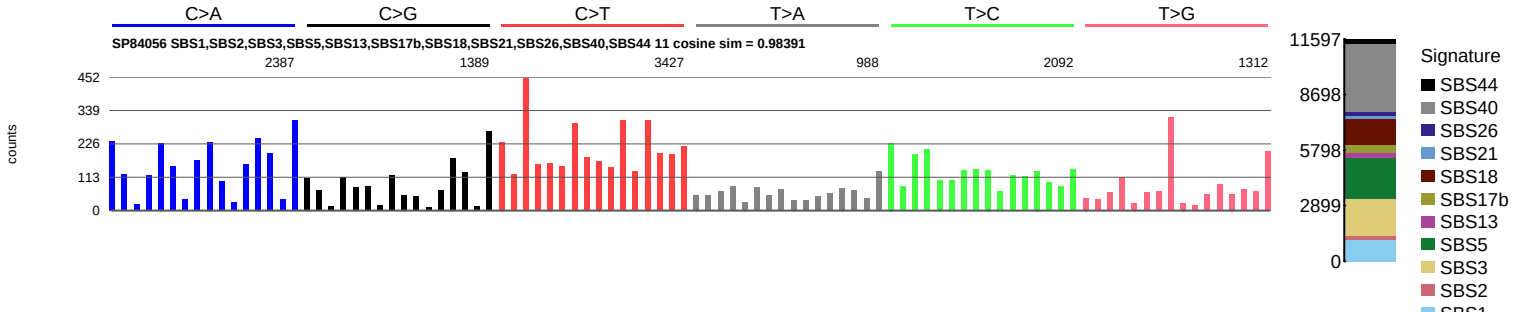

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98396
2323
1090
3418
1110
2218
1436
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

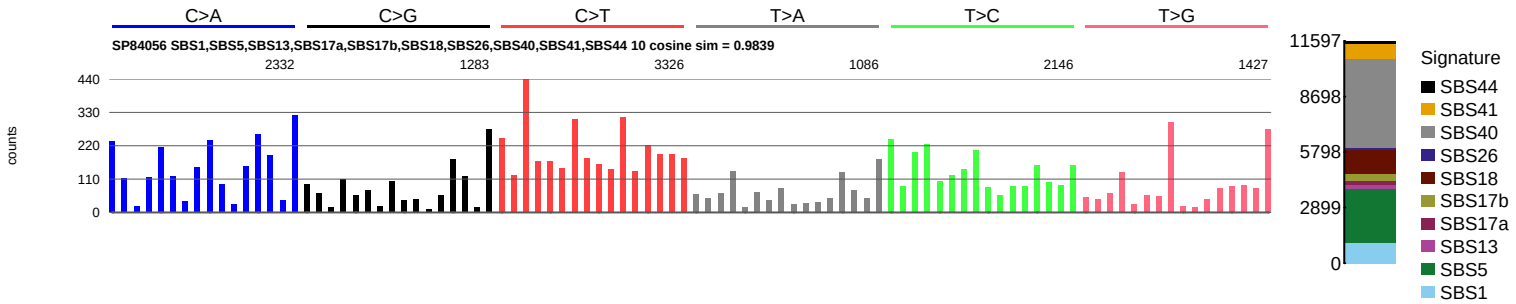

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98395
2286
1248
3352
1260
2111
1341
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

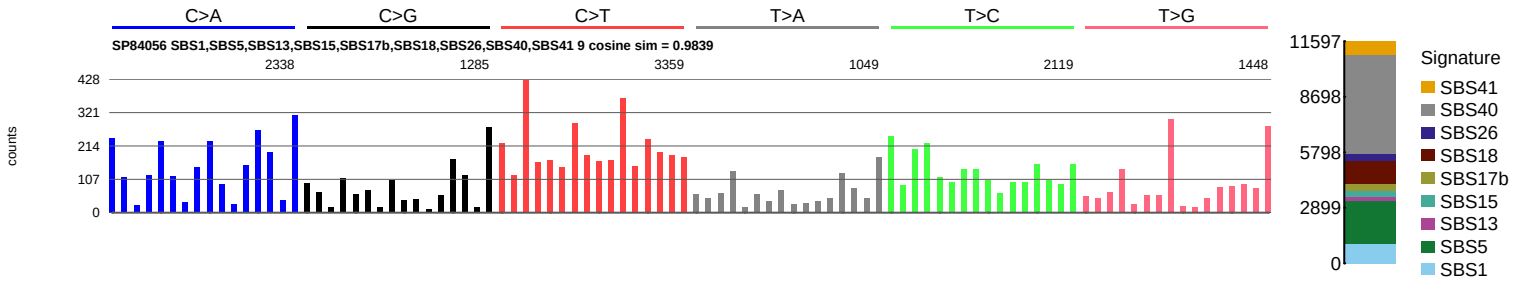

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98395
2372
1403
3386
1001
2120
1314
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

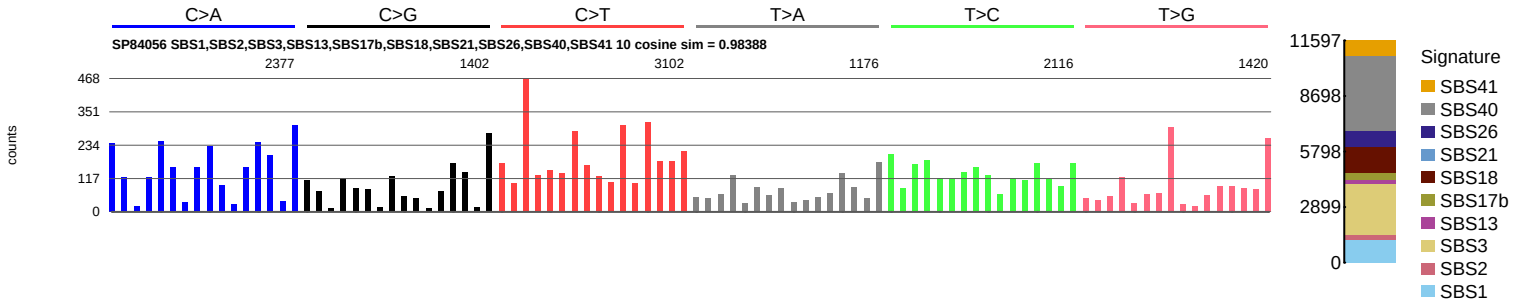

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98393
2383
1249
3283
1050
2160
1473
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

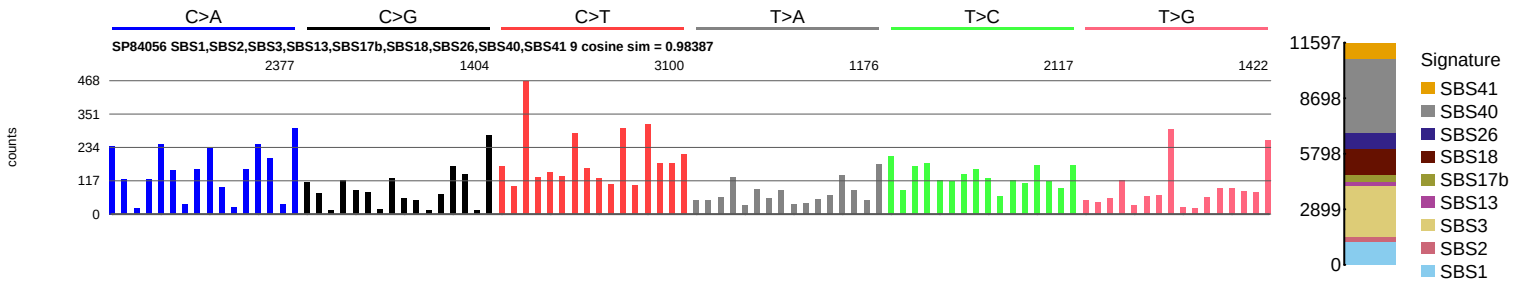

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98391
2329
1283
3344
1091
2123
1427
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98391
2387
1389
3427
988
2092
1312
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9839
2332
1283
3326
1086
2146
1427
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9839
2338
1285
3359
1049
2119
1448
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98388
2377
1402
3102
1176
2116
1420
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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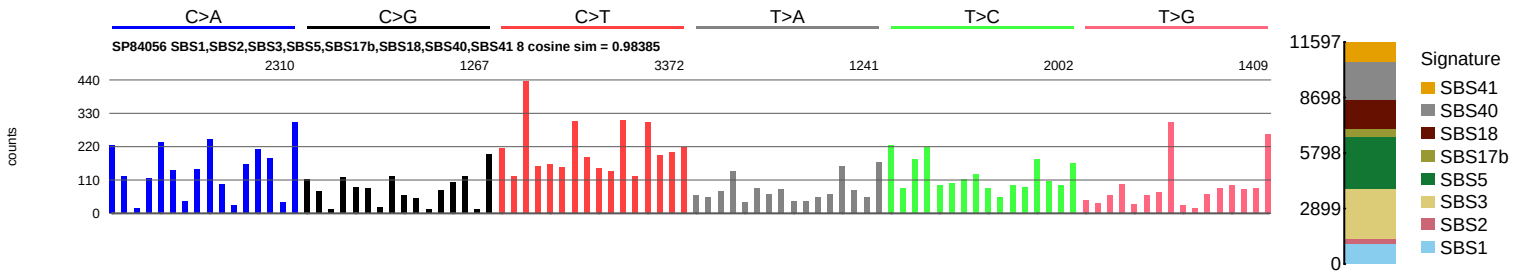
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98387
2377
1404
3100
1176
2117
1422
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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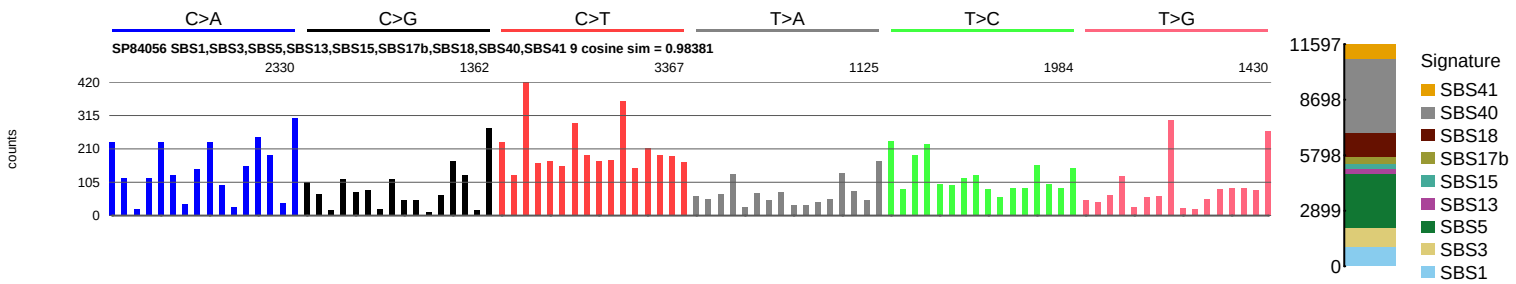
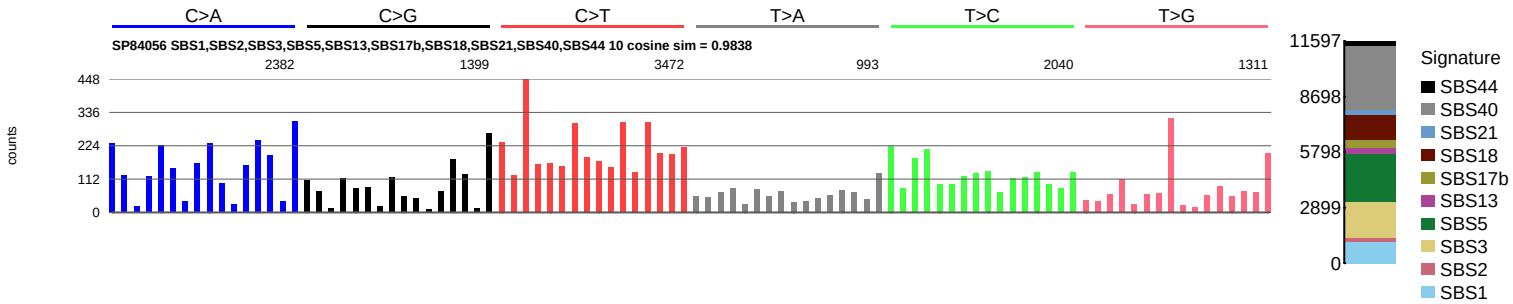
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98385
2310
1267
3372
1241
2002
1409
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98381
2330
1362
3367
1125
1984
1430
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1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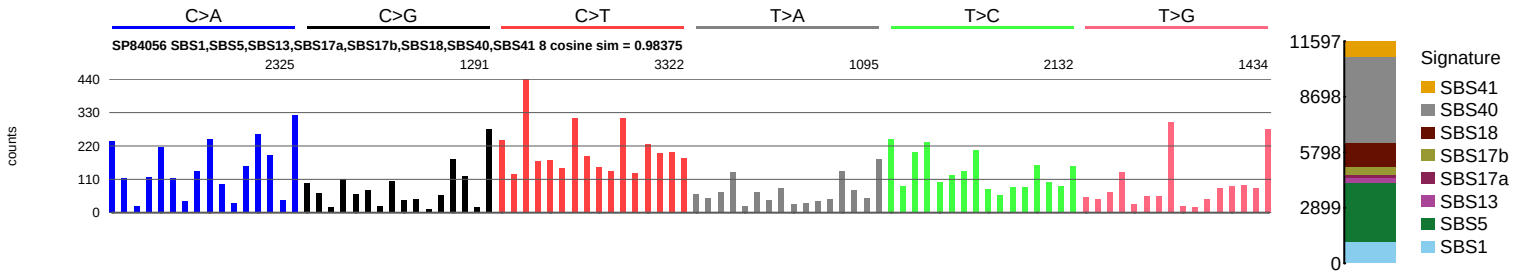
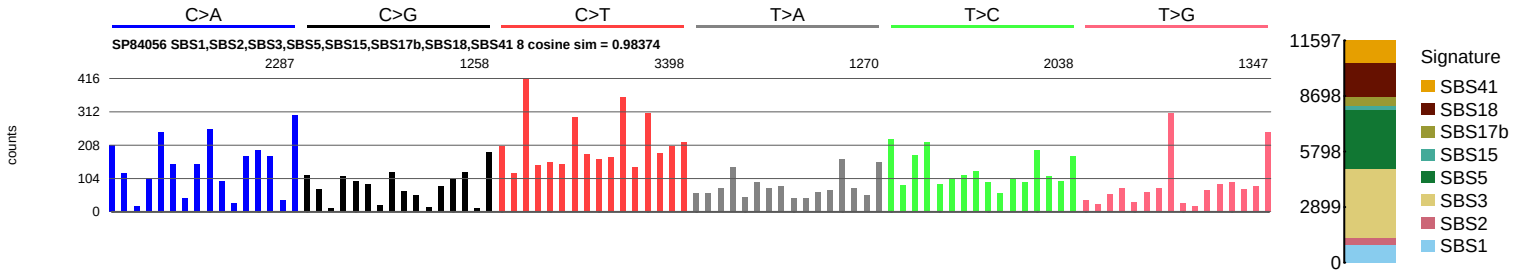
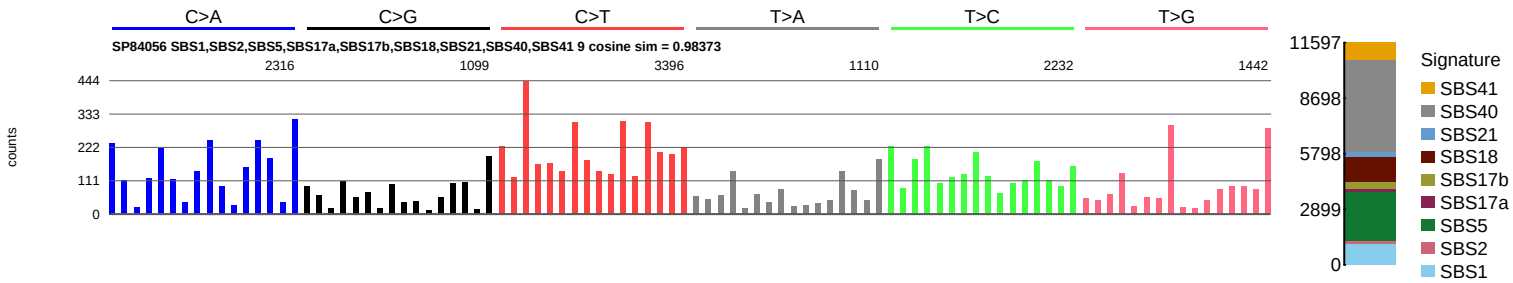
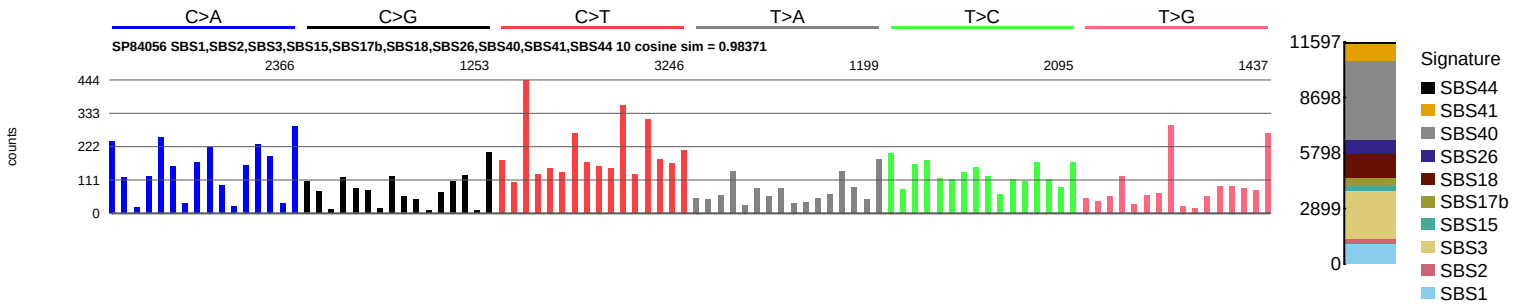
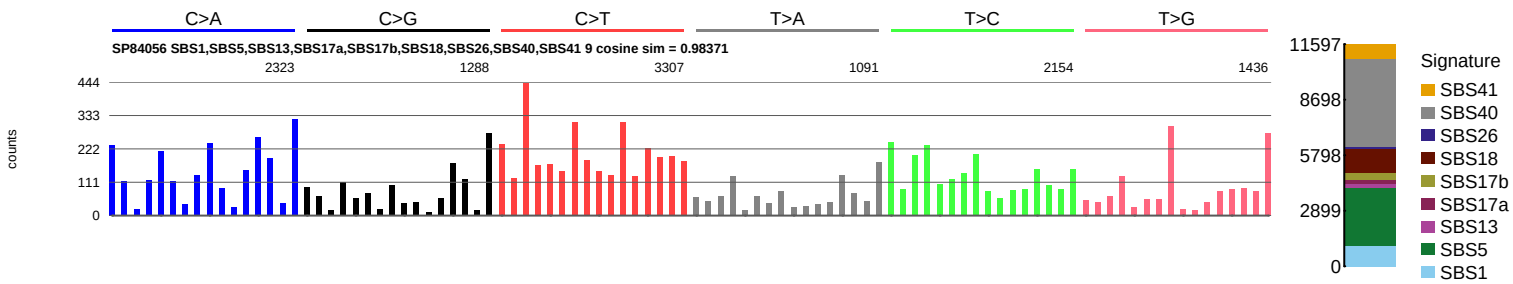
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9838
2382
1399
3472
993
2040
1311
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

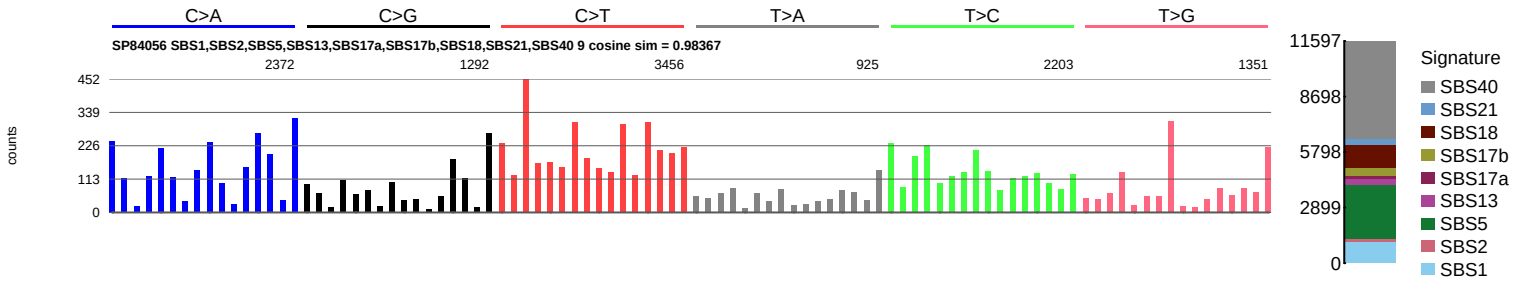

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98375
2325
1291
3322
1095
2132
1434
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

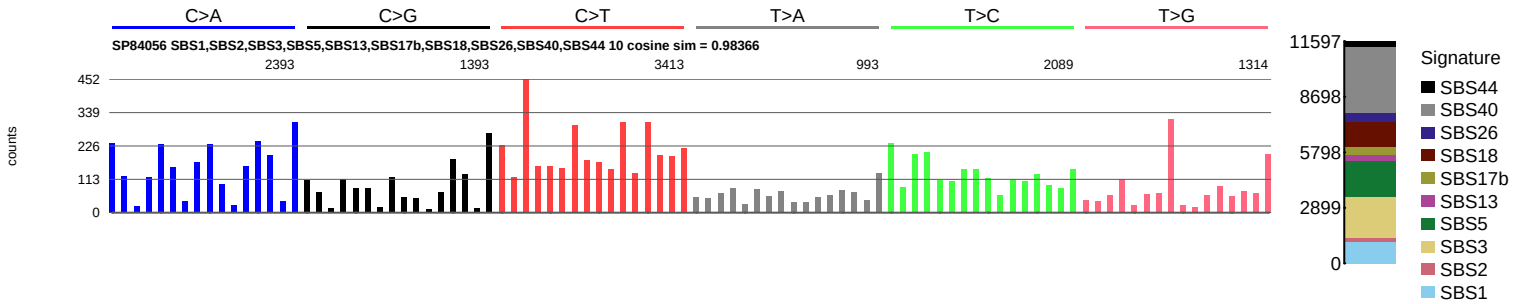

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98374
2287
1258
3398
1270
2038
1347
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

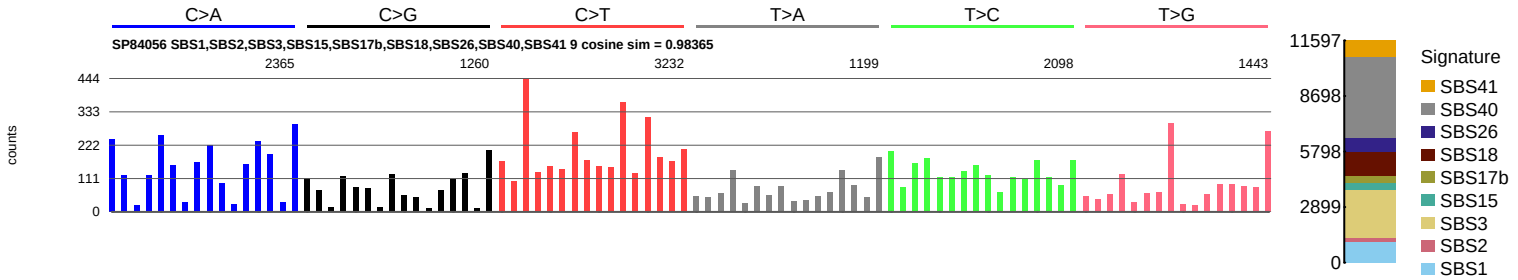

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98373
2316
1099
3396
1110
2232
1442
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

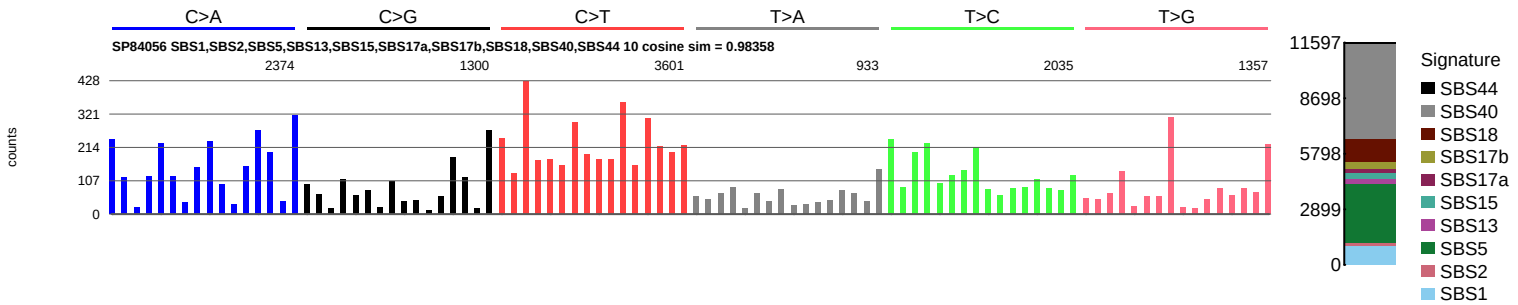

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98371
2366
1253
3246
1199
2095
1437
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

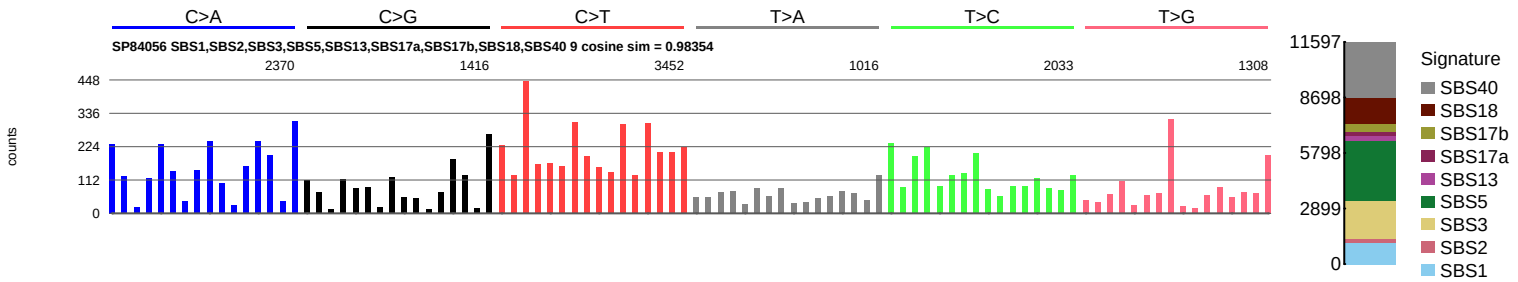

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98371
2323
1288
3307
1091
2154
1436
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

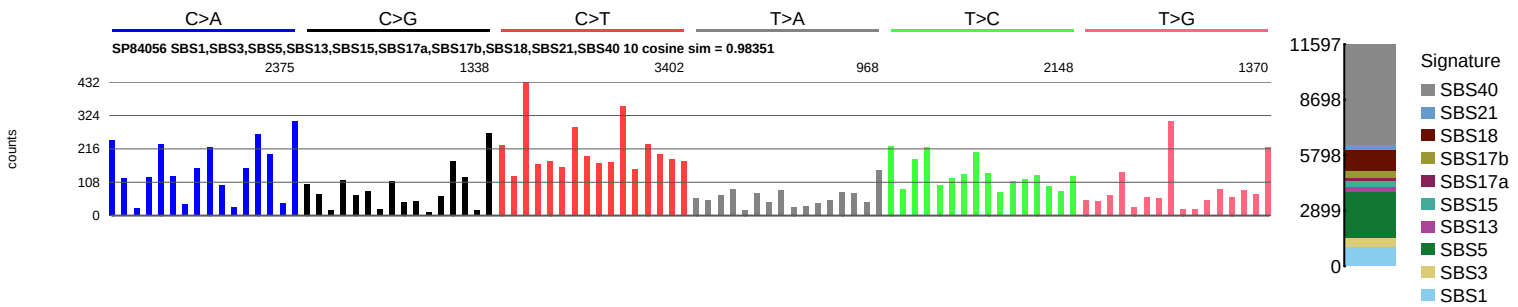

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98367
2372
1292
3456
925
2203
1351
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

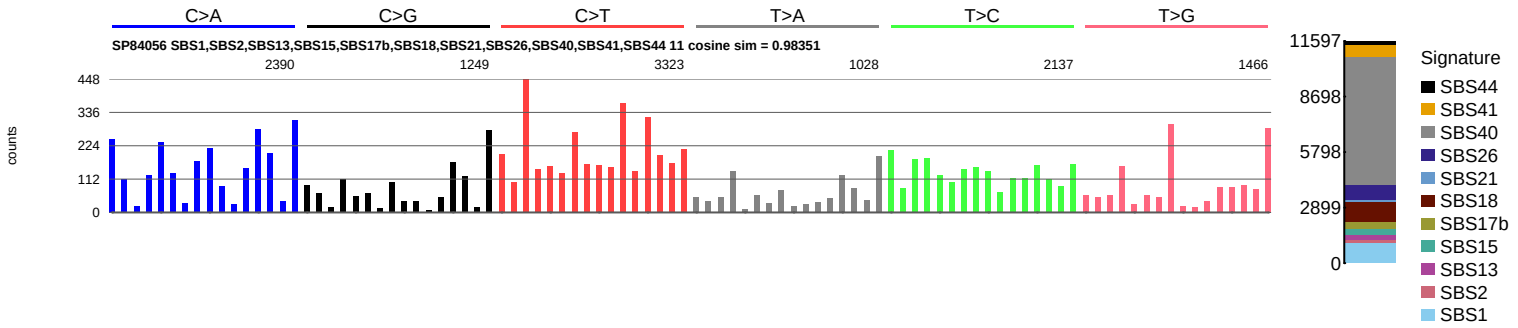

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98366
2393
1393
3413
993
2089
1314
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

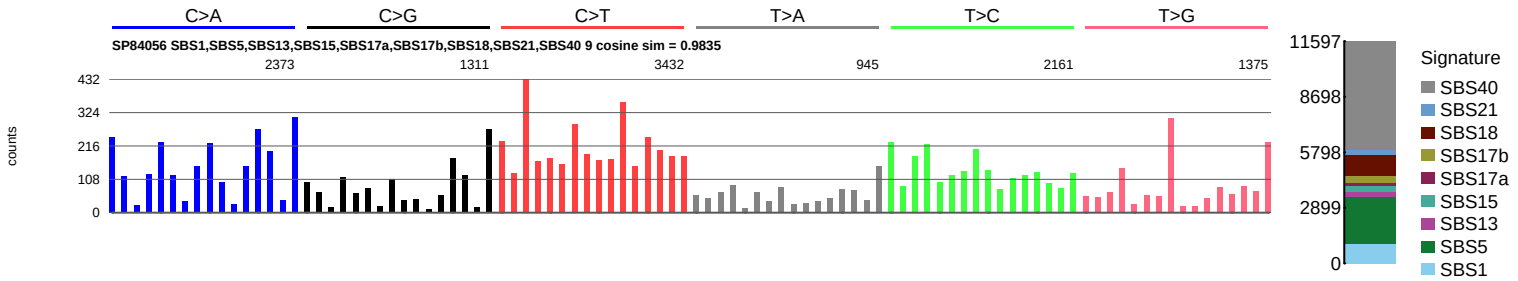

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98365
2365
1260
3232
1199
2098
1443
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

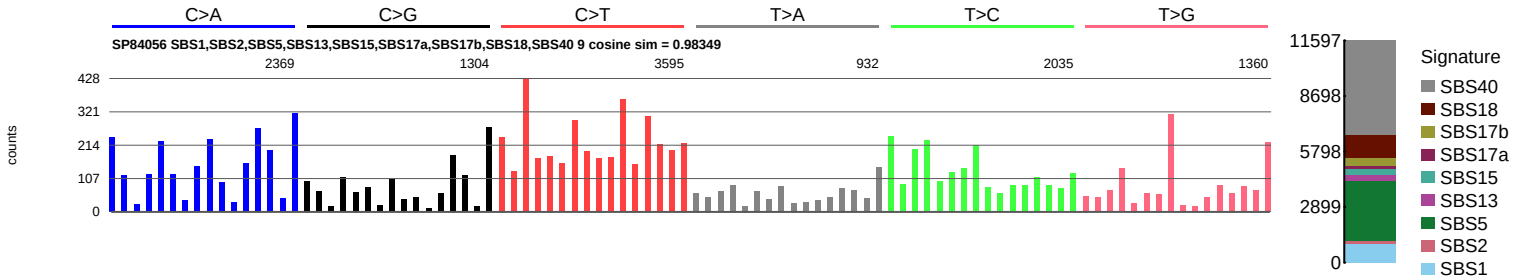

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98358
2374
1300
3601
933
2035
1357
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

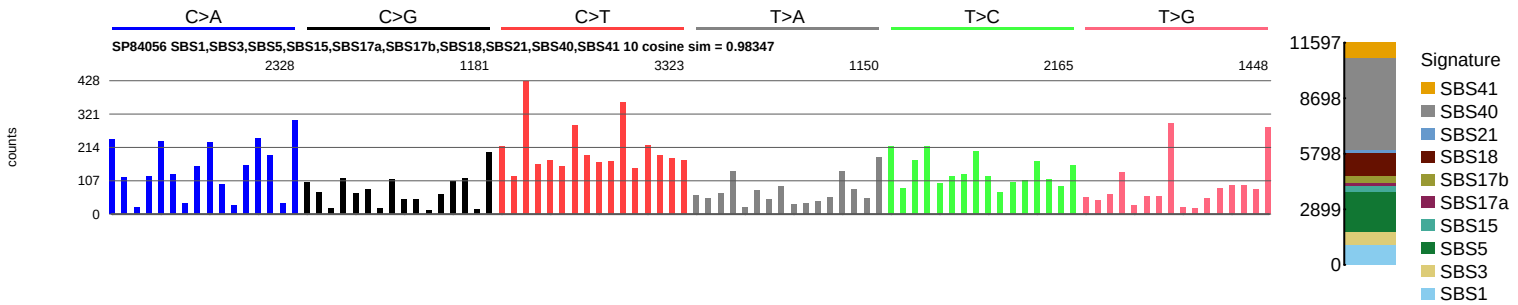

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98354
2370
1416
3452
1016
2033
1308
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

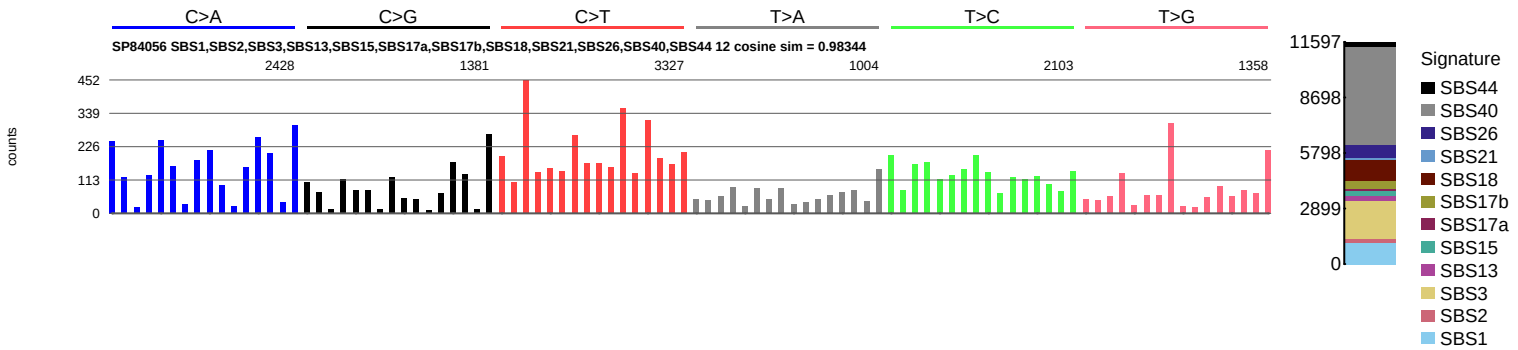

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98351
2375
1338
3402
968
2148
1370
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

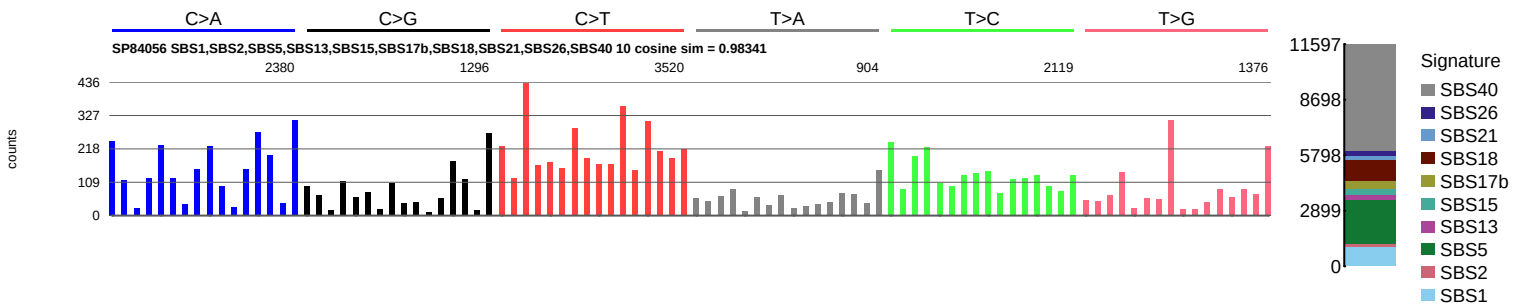

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98351
2390
1249
3323
1028
2137
1466
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

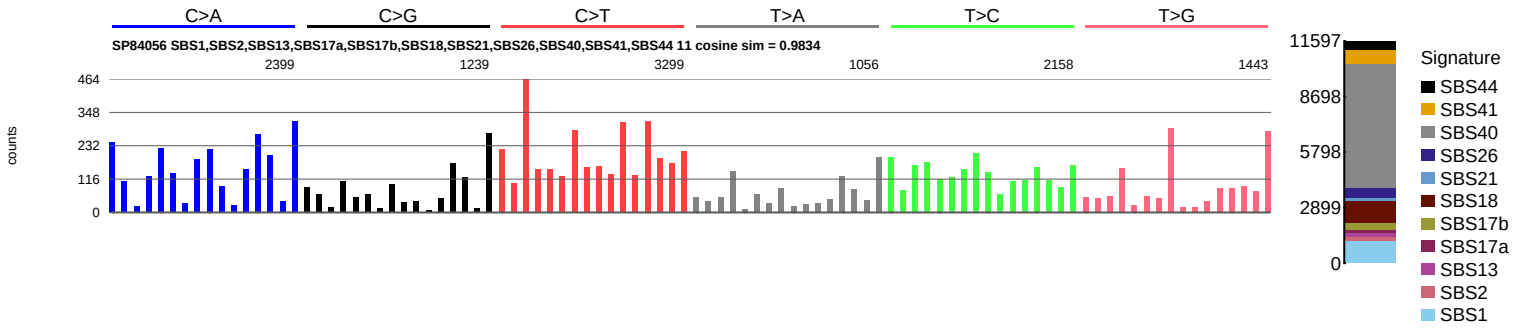

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.9835
2373
1311
3432
945
2161
1375
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

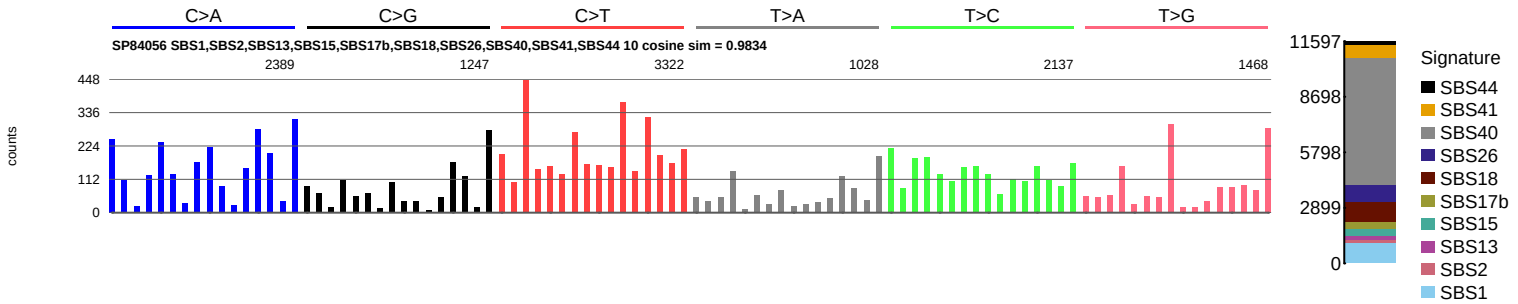

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98349
2369
1304
3595
932
2035
1360
428
321
214
107
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

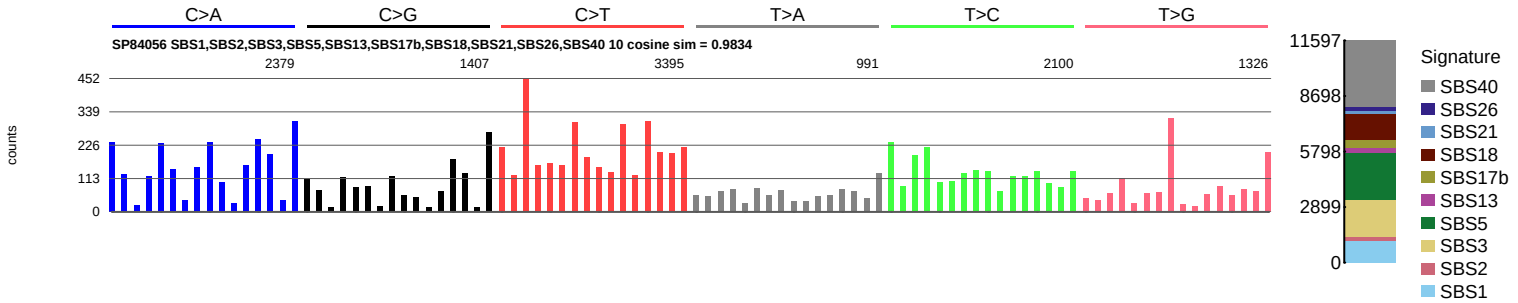

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98347
2328
1181
3323
1150
2165
1448
428
321
214
107
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

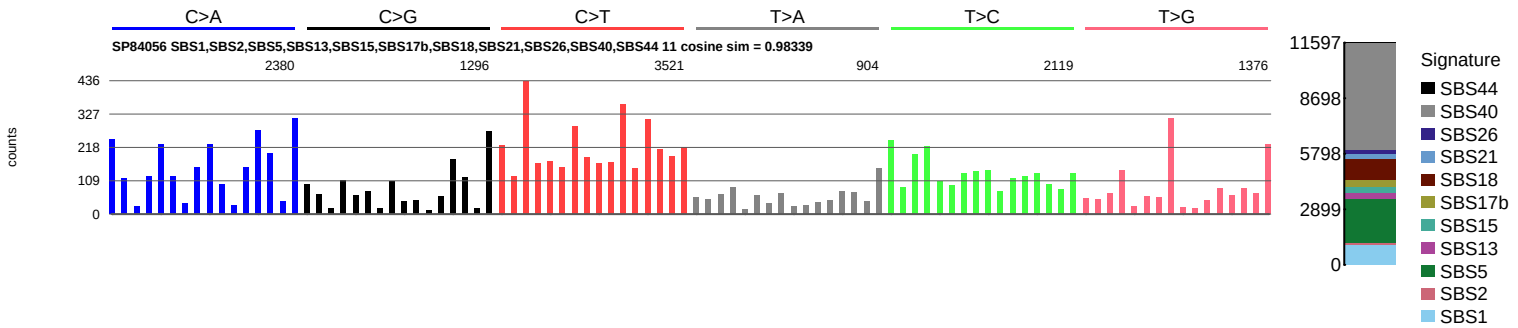

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98344
2428
1381
3327
1004
2103
1358
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

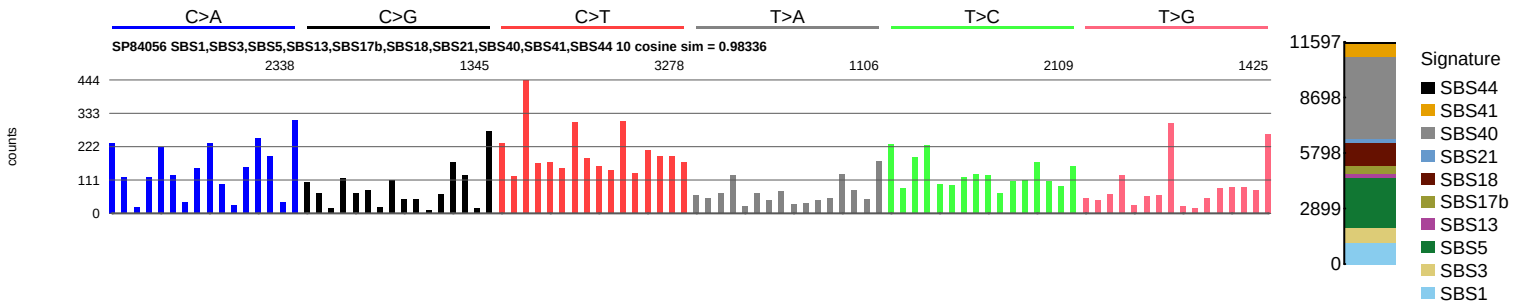

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98341
2380
1296
3520
904
2119
1376
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9834
2399
1239
3299
1056
2158
1443
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9834
2389
1247
3322
1028
2137
1468
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.9834
2379
1407
3395
991
2100
1326
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98339
2380
1296
3521
904
2119
1376
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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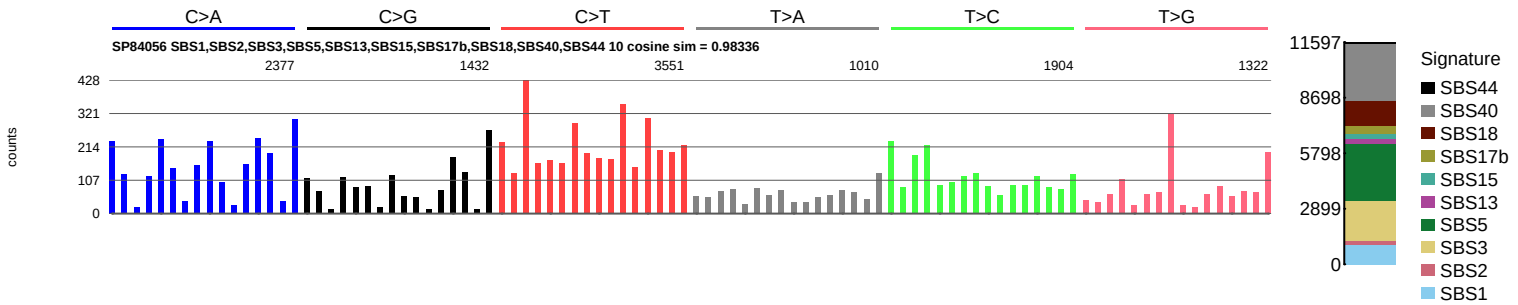
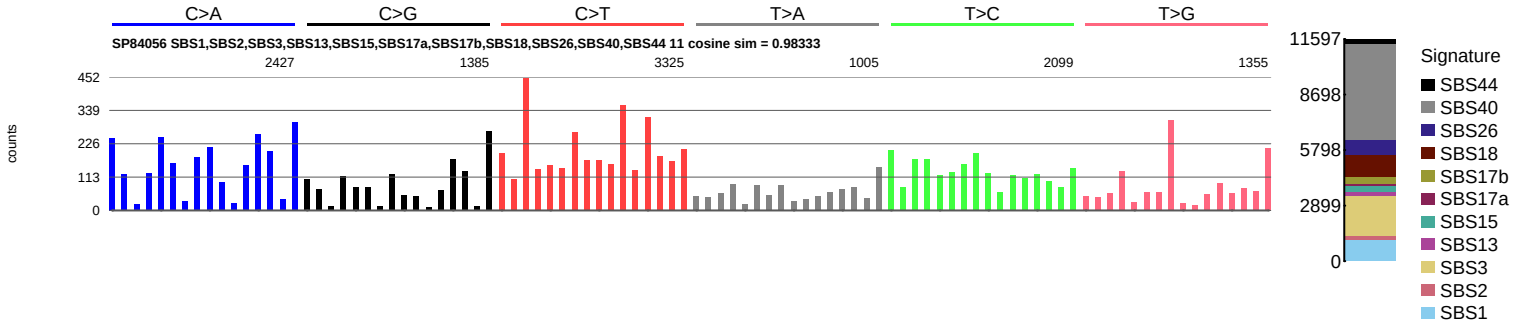
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98336
2338
1345
3278
1106
2109
1425
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98336
2377
1432
3551
1010
1904
1322
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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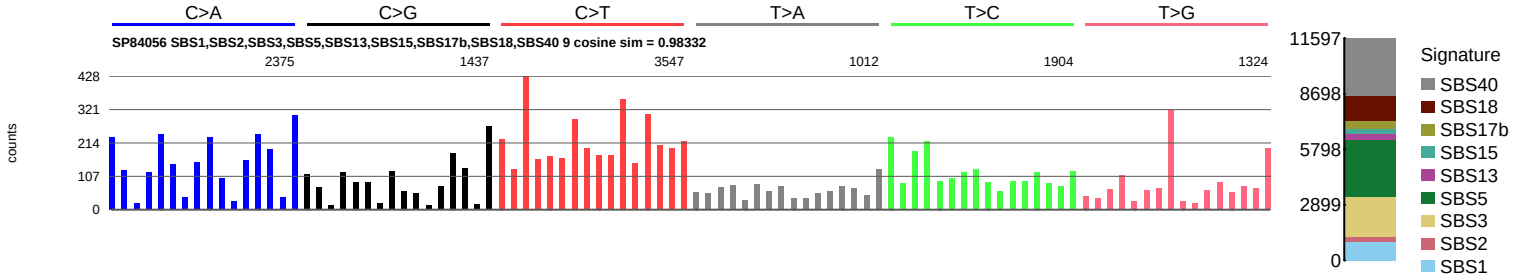
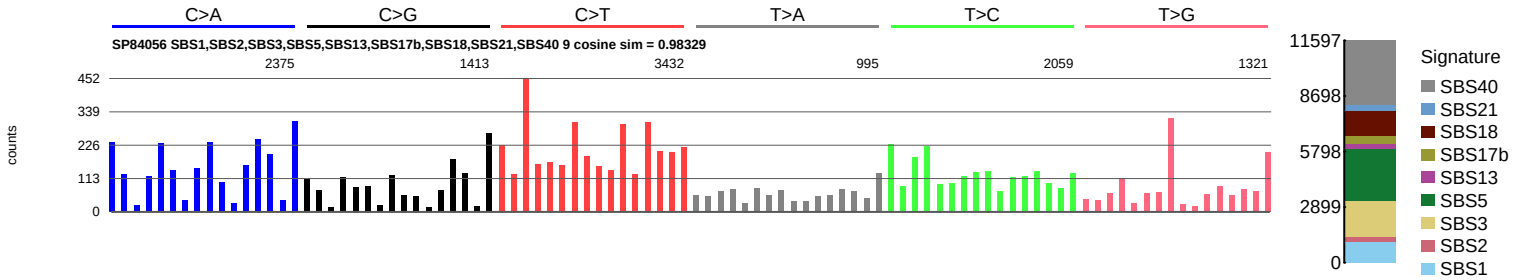
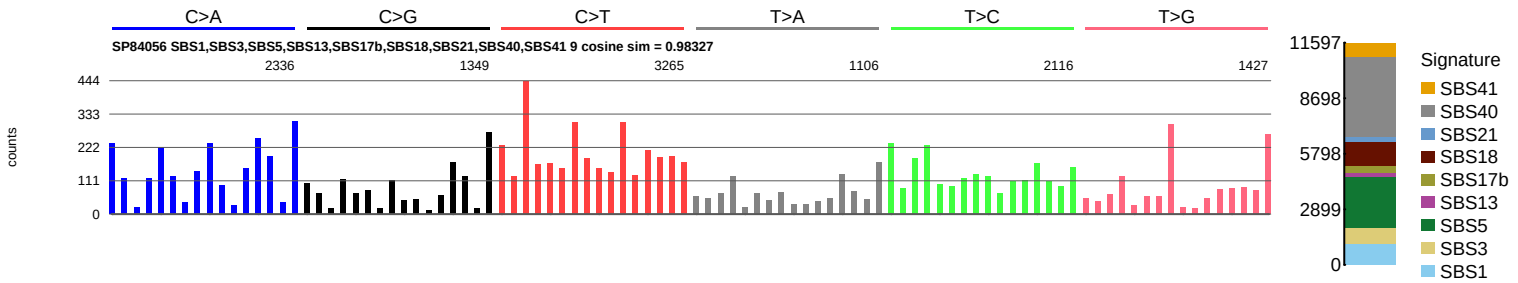
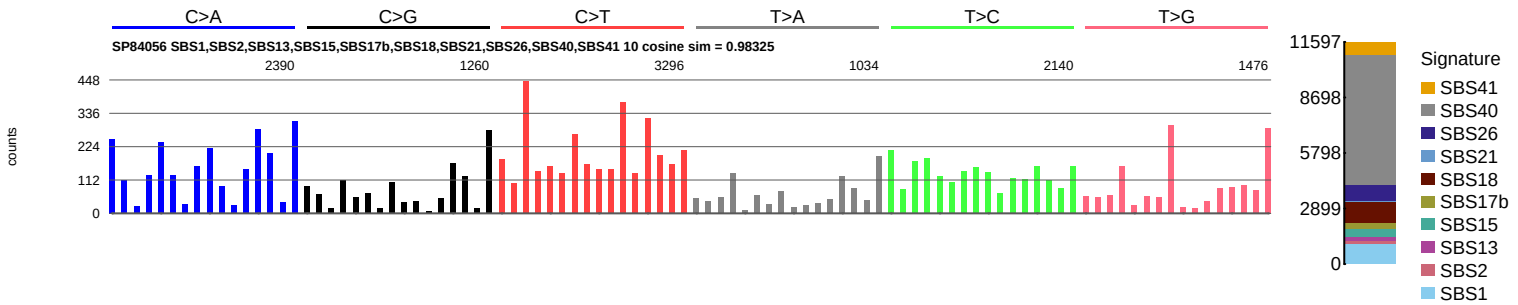
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98333
2427
1385
3325
1005
2099
1355
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.98332
2375
1437
3547
1012
1904
1324
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98329
2375
1413
3432
995
2059
1321
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98327
2336
1349
3265
1106
2116
1427
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98325
2390
1260
3296
1034
2140
1476
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
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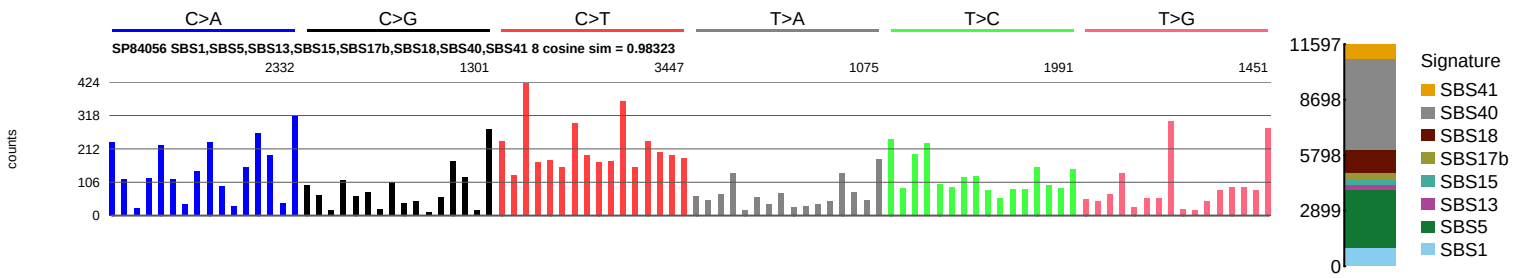
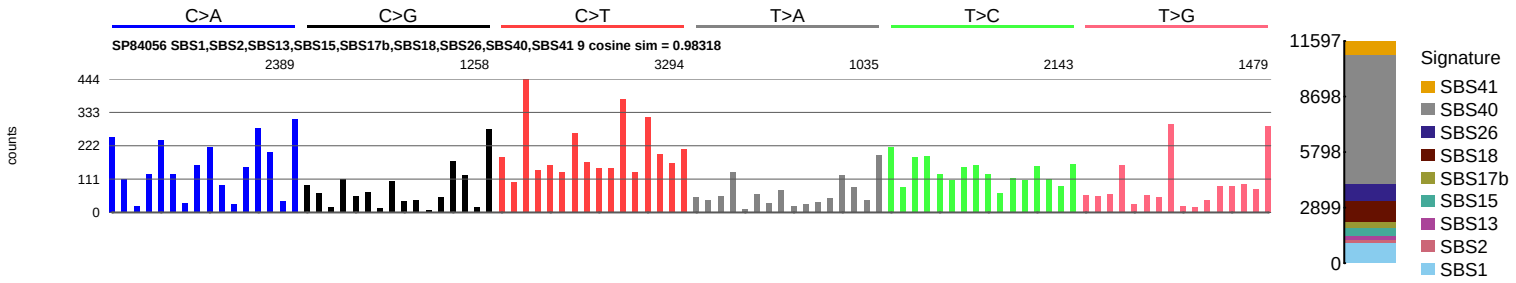
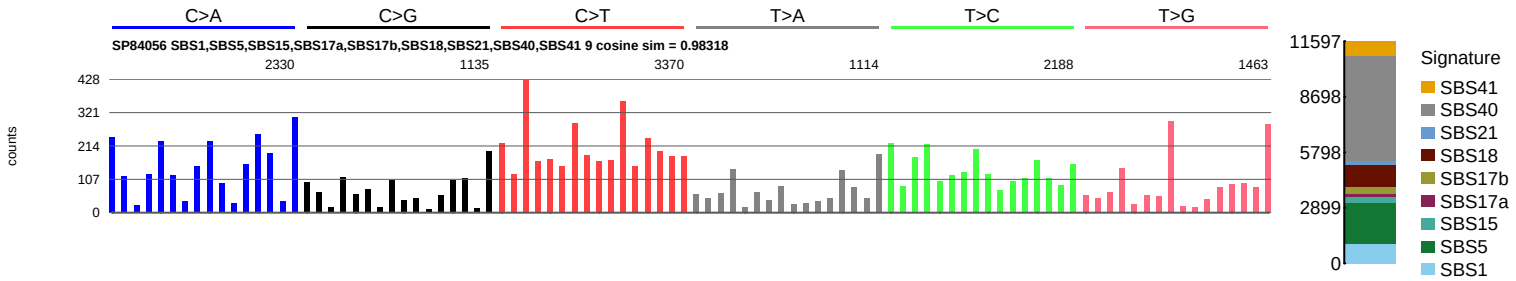
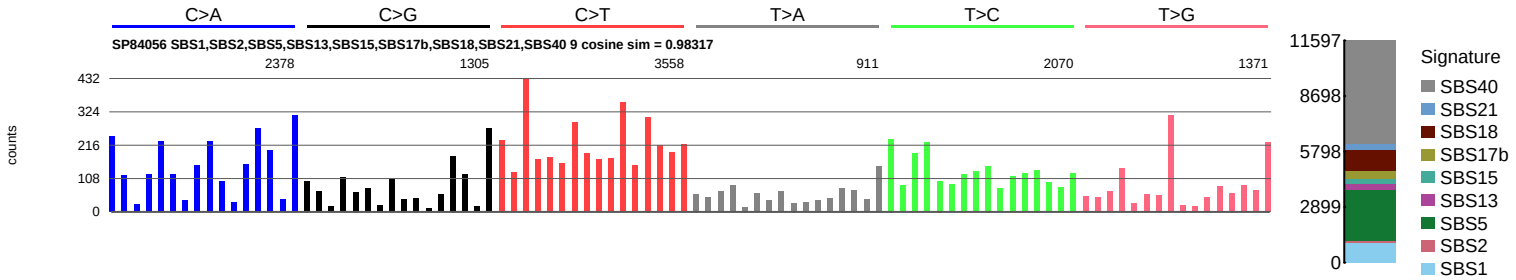
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98323
2332
1301
3447
1075
1991
1451
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98318
2389
1258
3294
1035
2143
1479
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98318
2330
1135
3370
1114
2188
1463
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98317
2378
1305
3558
911
2070
1371
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
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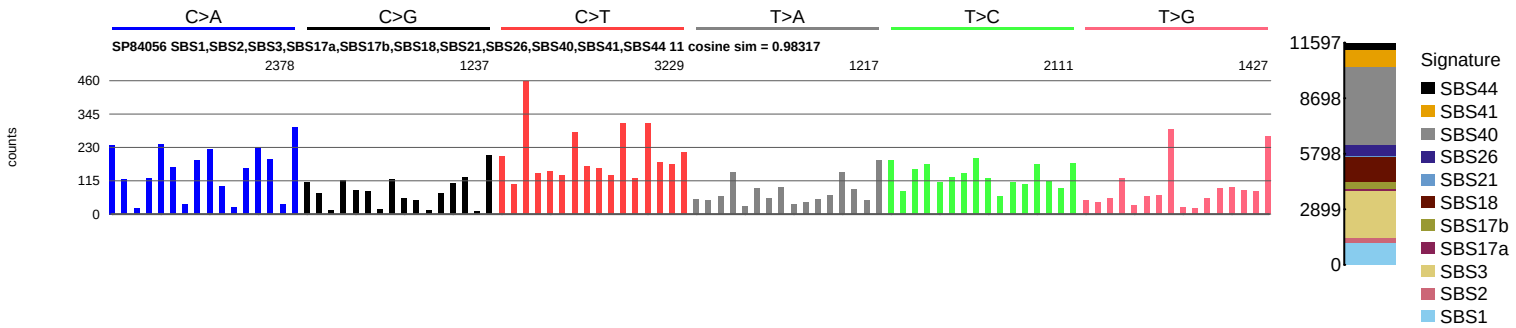
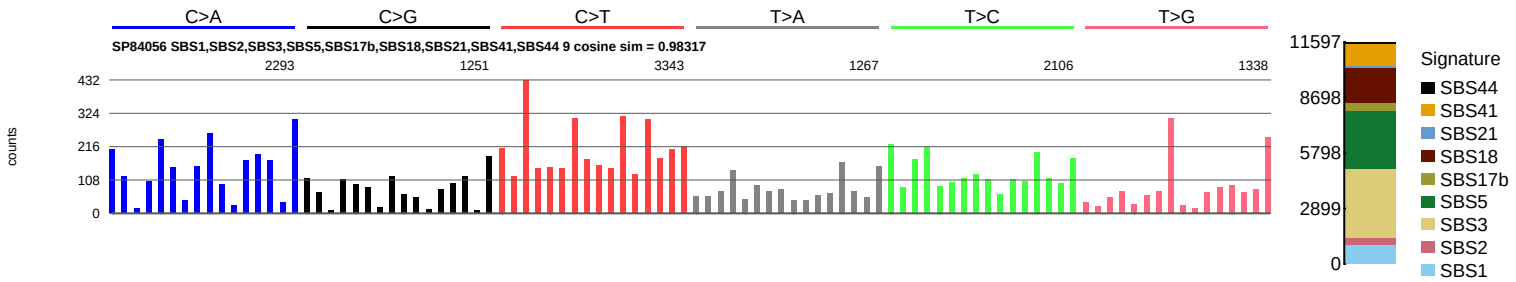
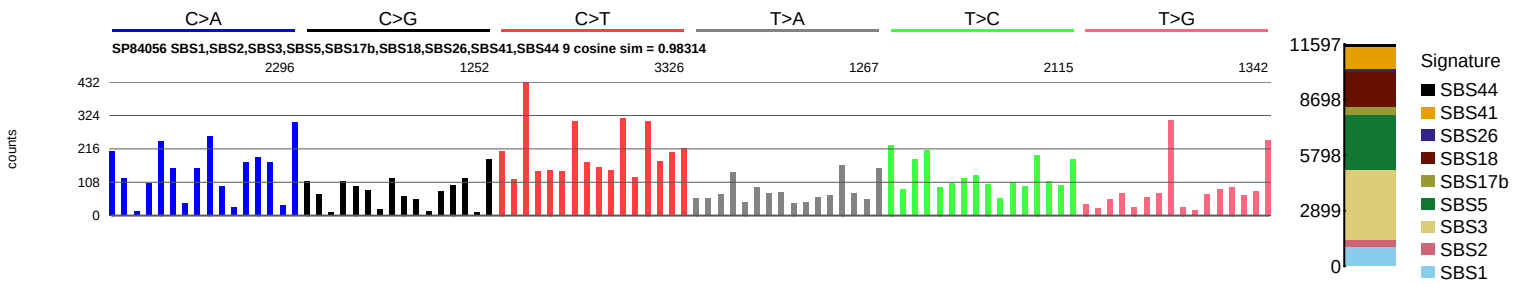
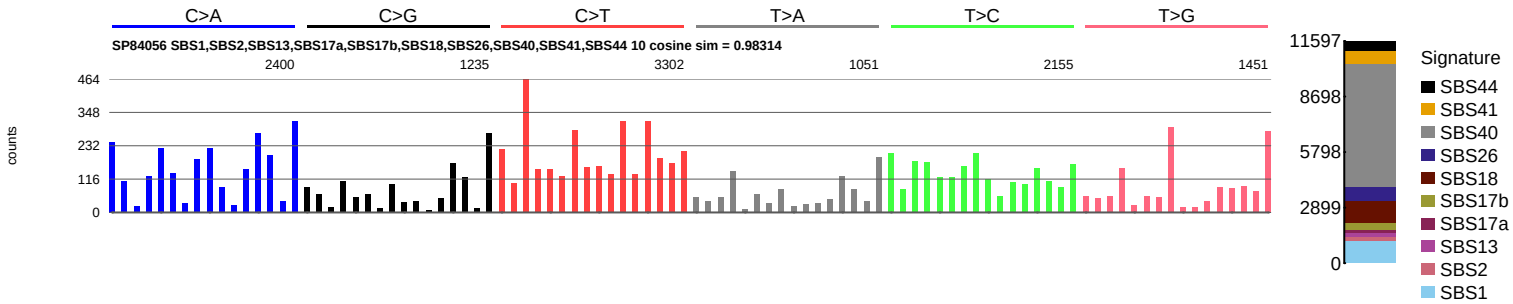
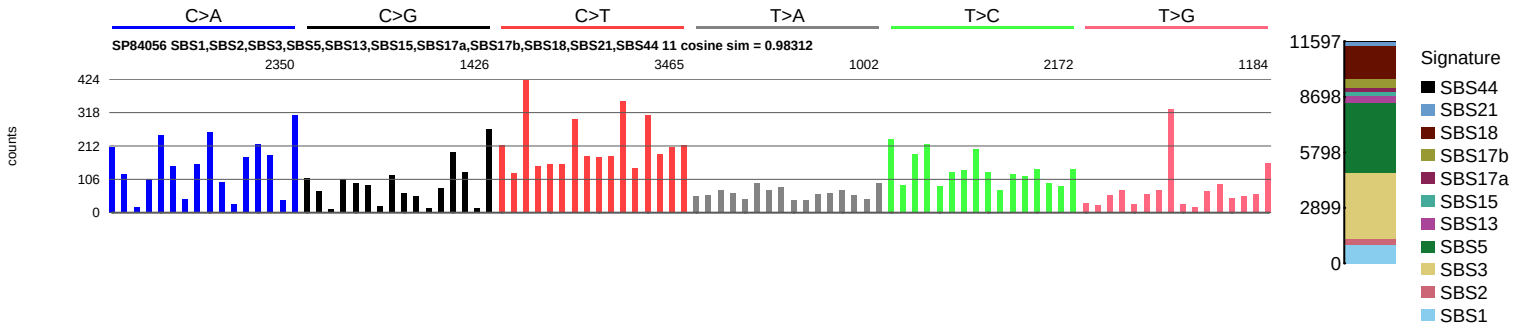
SBS13
SBS5
SBS2
SBS1

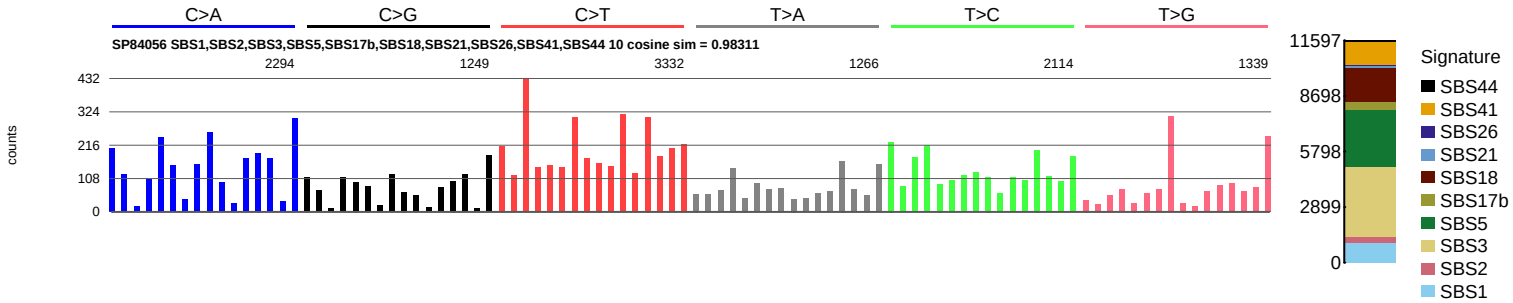

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98317
2378
1237
3229
1217
2111
1427
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

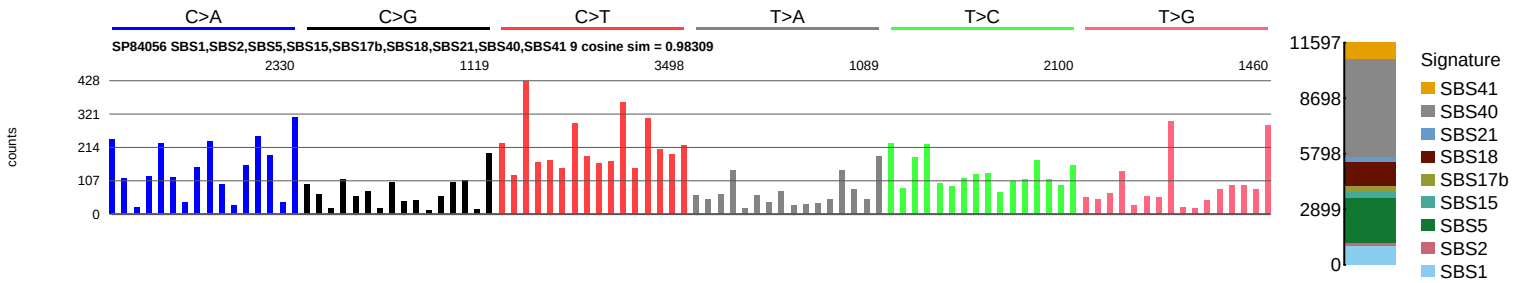

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98317
2293
1251
3343
1267
2106
1338
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

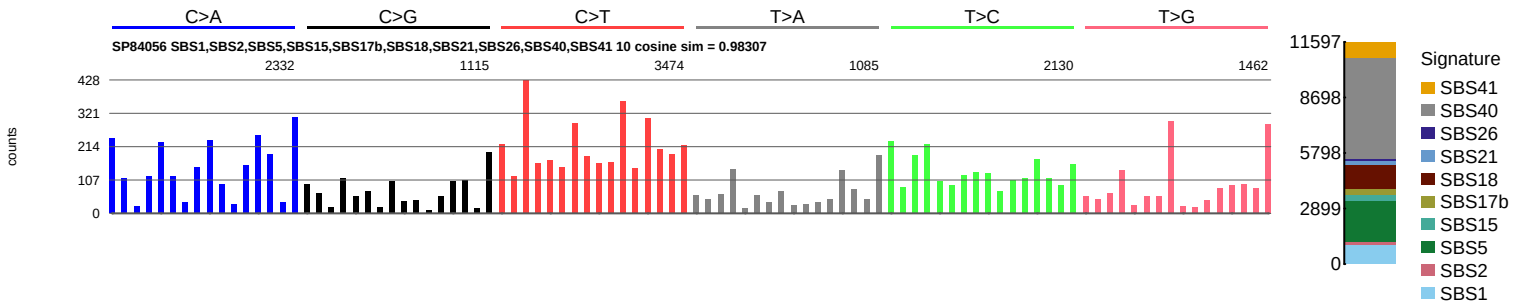

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98314
2296
1252
3326
1267
2115
1342
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

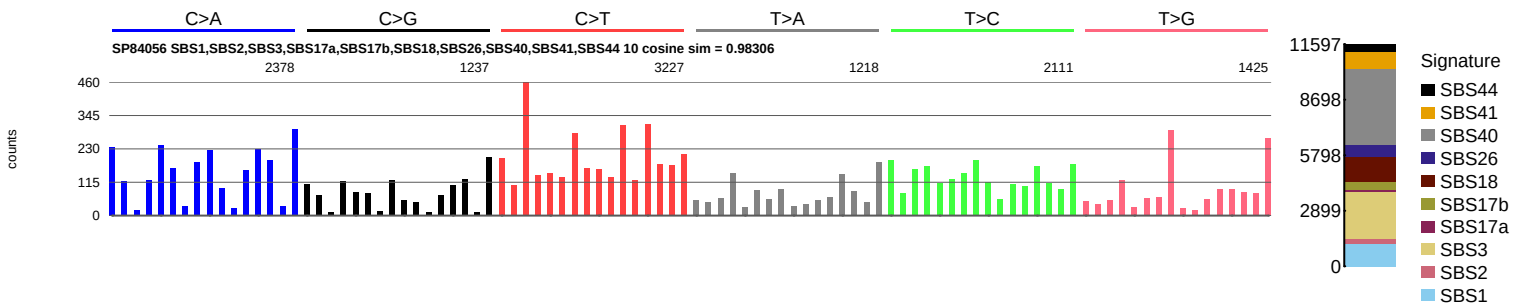

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98314
2400
1235
3302
1051
2155
1451
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

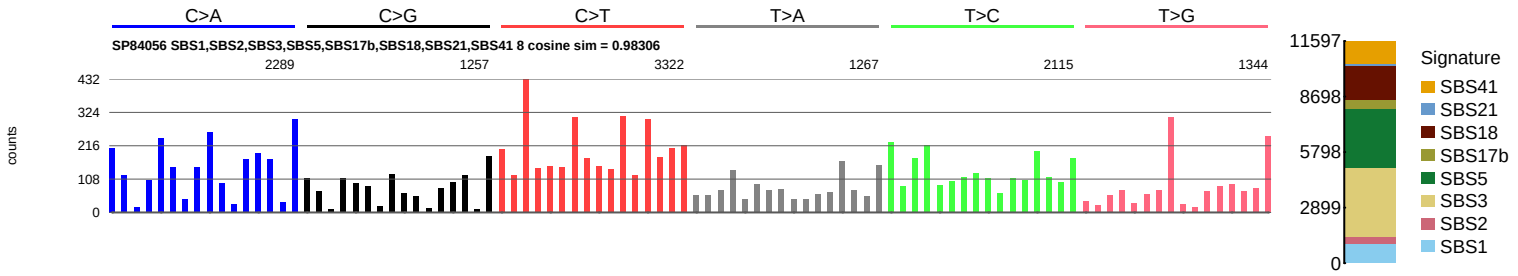

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98312
2350
1426
3465
1002
2172
1184
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98311
2294
1249
3332
1266
2114
1339
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98309
2330
1119
3498
1089
2100
1460
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98307
2332
1115
3474
1085
2130
1462
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98306
2378
1237
3227
1218
2111
1425
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98306
2289
1257
3322
1267
2115
1344
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
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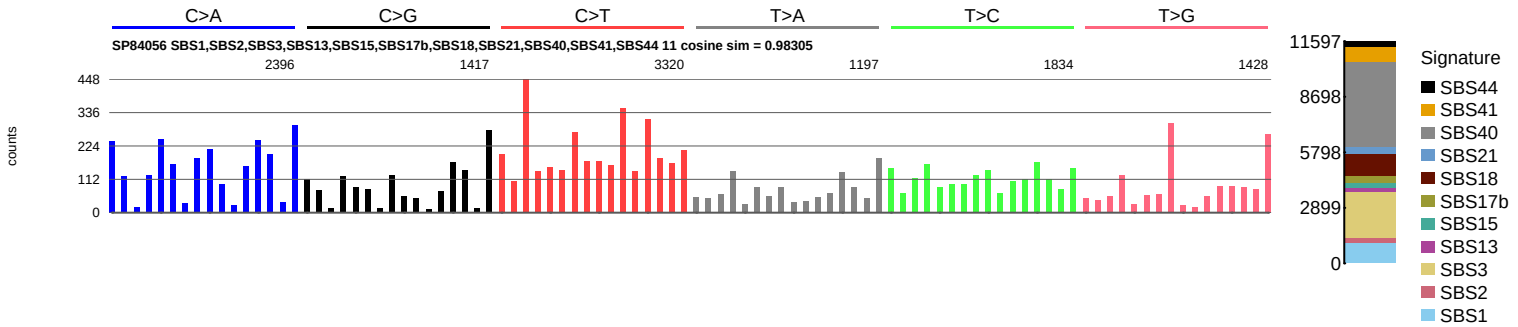
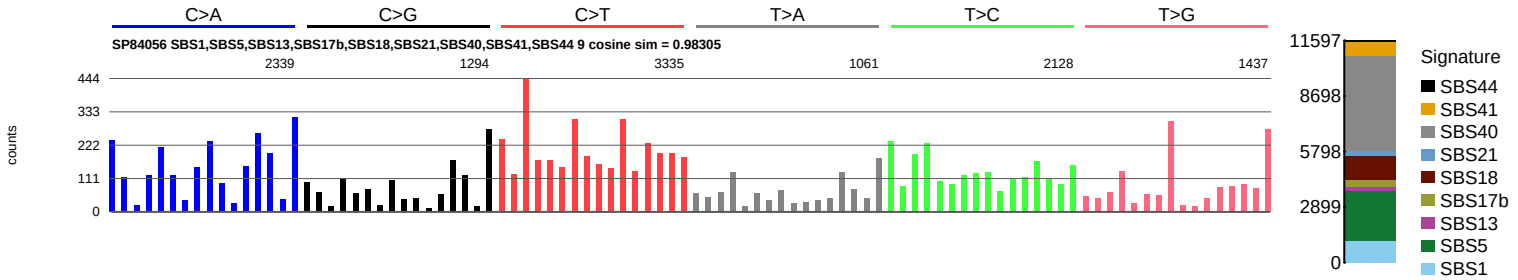
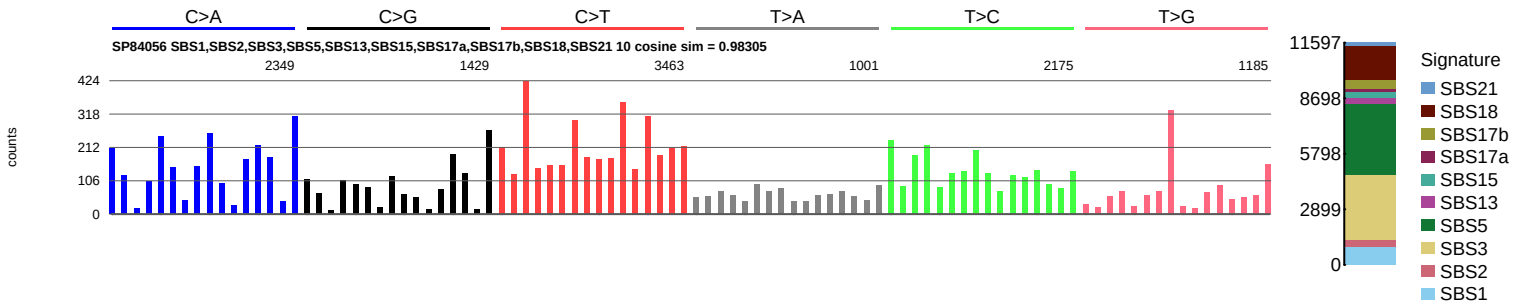
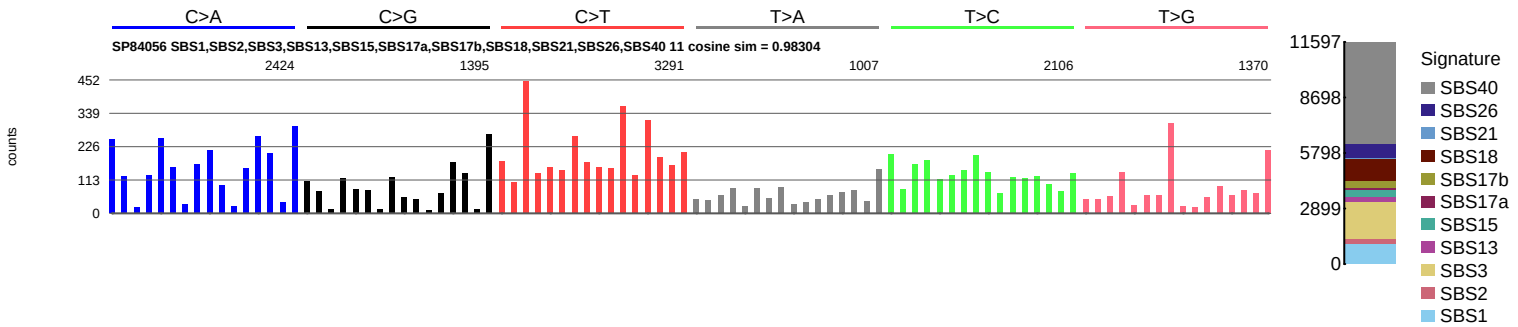
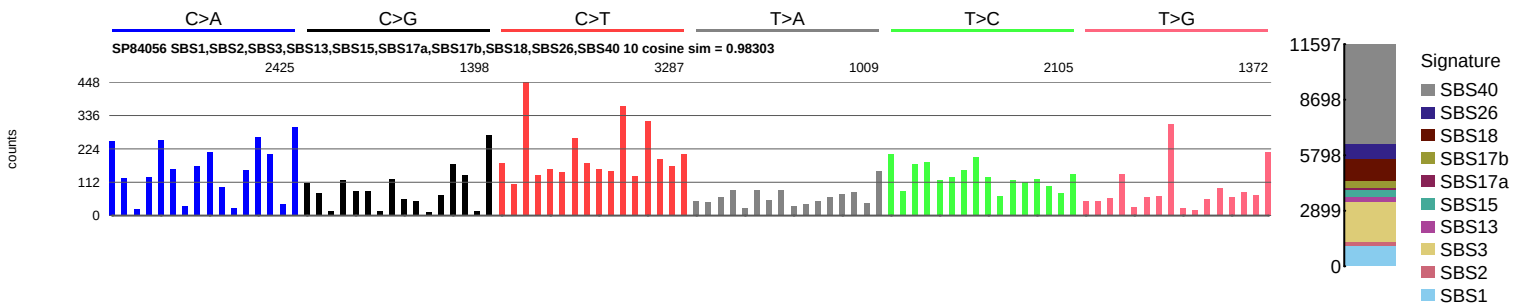
SBS3
SBS2
SBS1

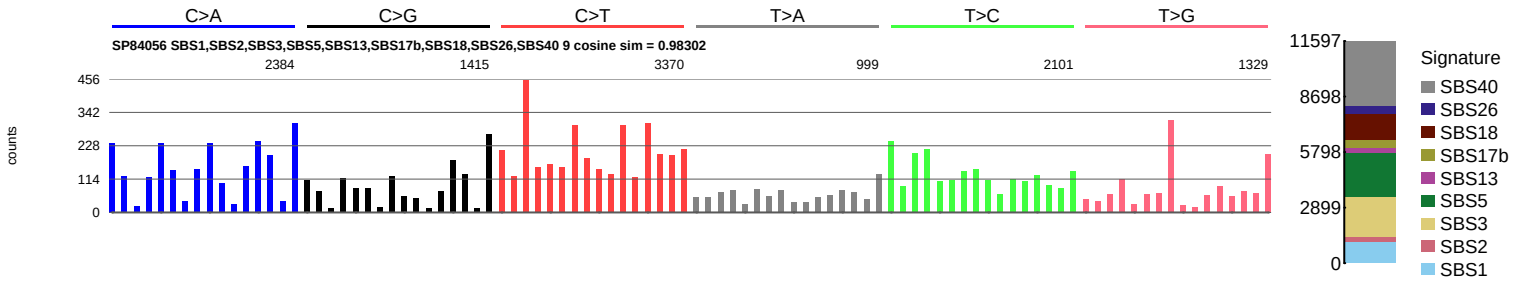

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98305
2396
1417
3320
1197
1834
1428
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

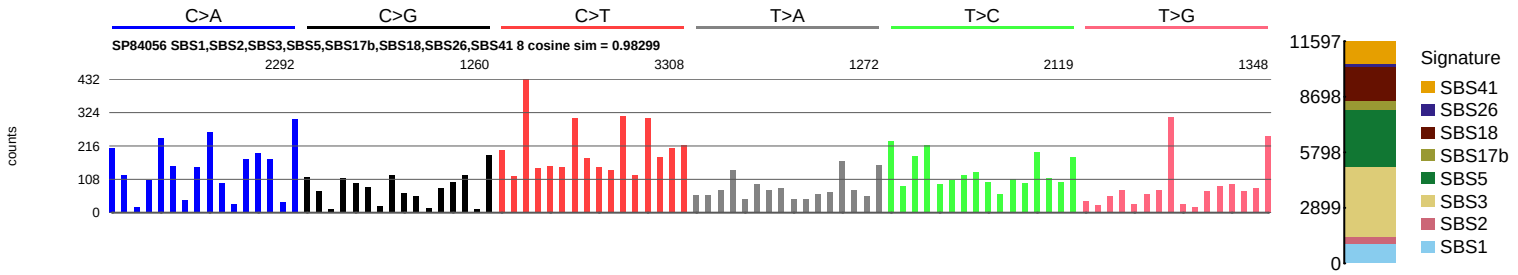

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98305
2339
1294
3335
1061
2128
1437
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

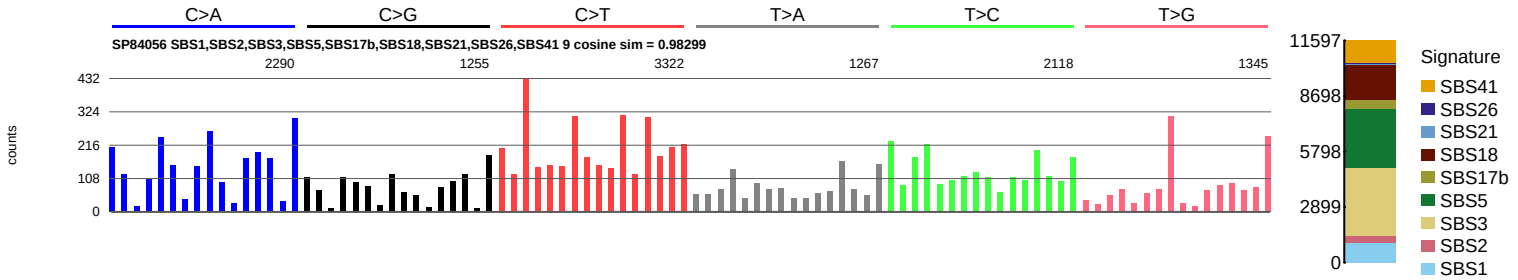

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98305
2349
1429
3463
1001
2175
1185
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

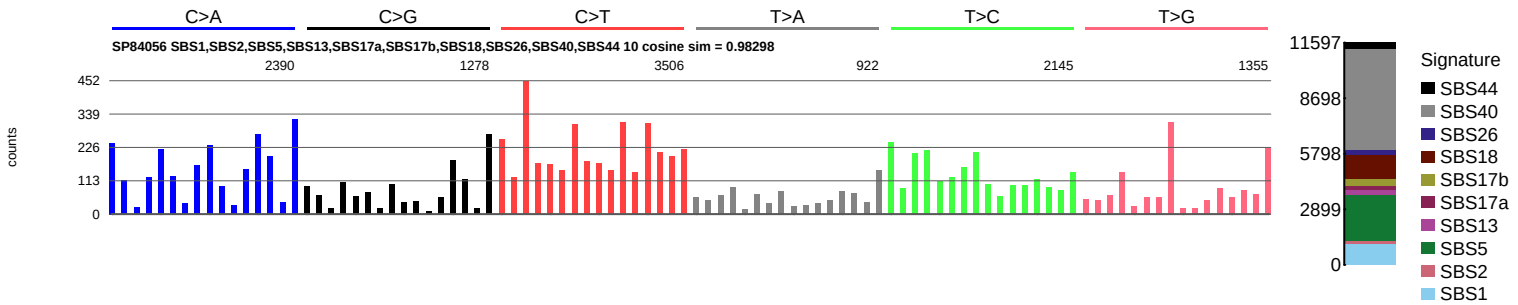

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98304
2424
1395
3291
1007
2106
1370
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

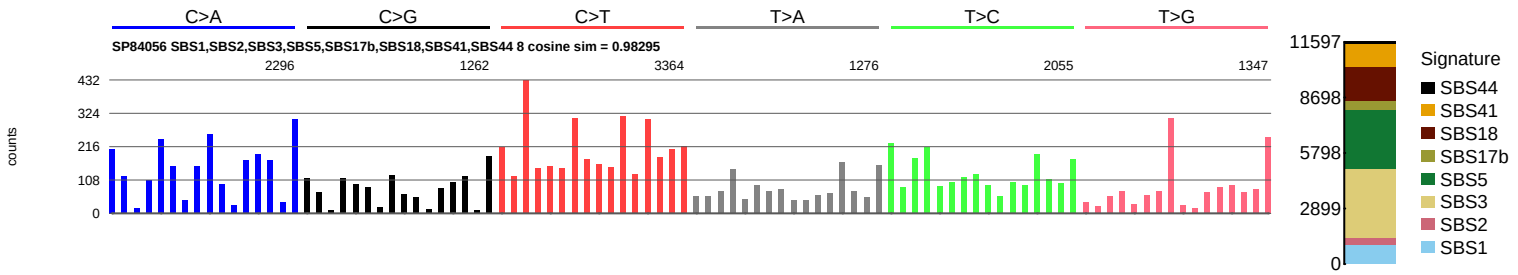

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98303
2425
1398
3287
1009
2105
1372
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

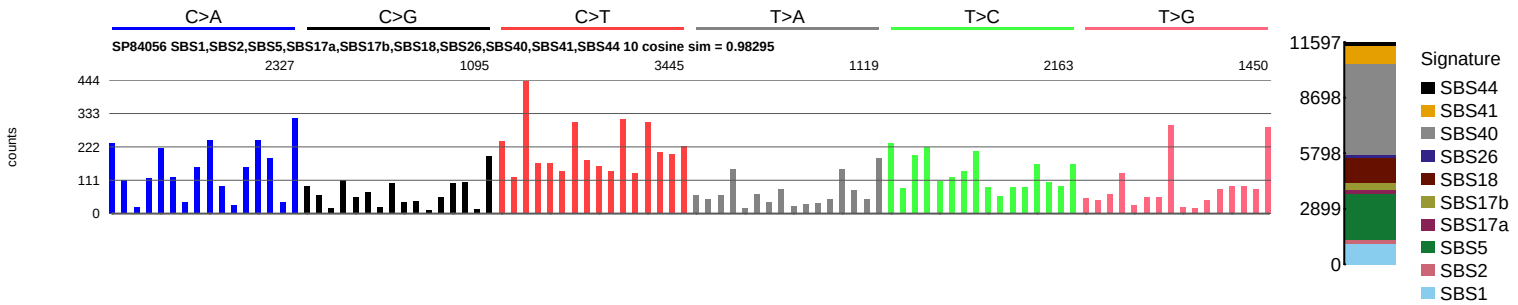

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98302
2384
1415
3370
999
2101
1329
456
342
228
114
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

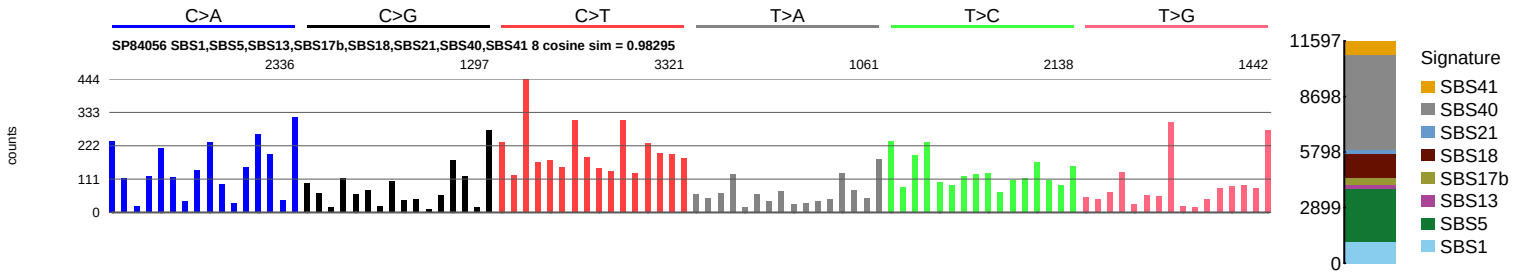

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98299
2292
1260
3308
1272
2119
1348
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

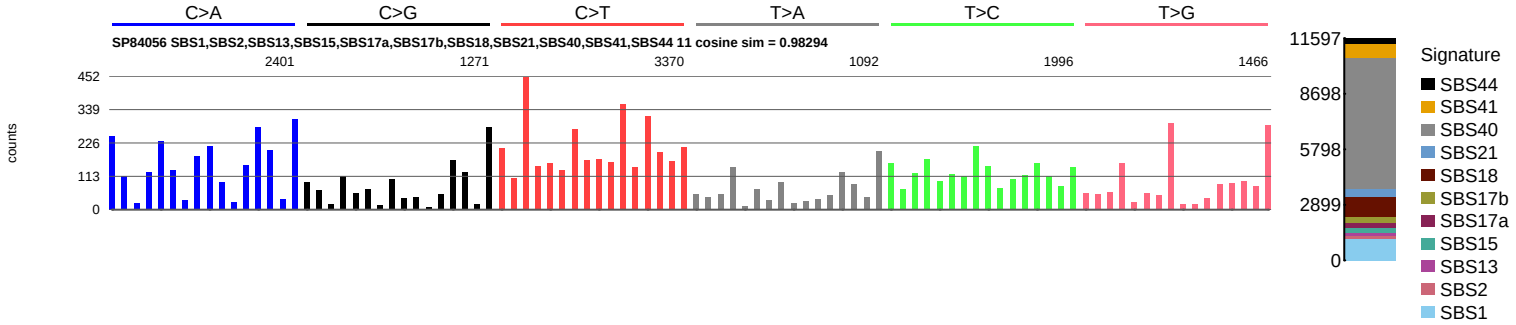

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98299
2290
1255
3322
1267
2118
1345
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

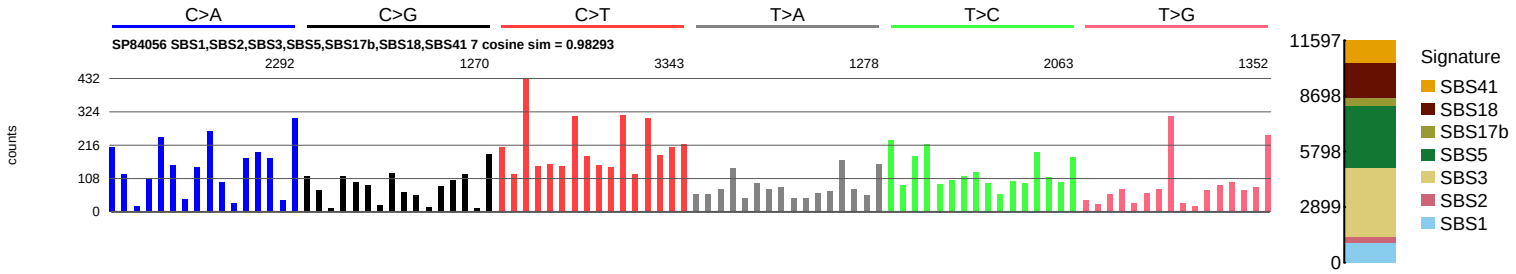

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98298
2390
1278
3506
922
2145
1355
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

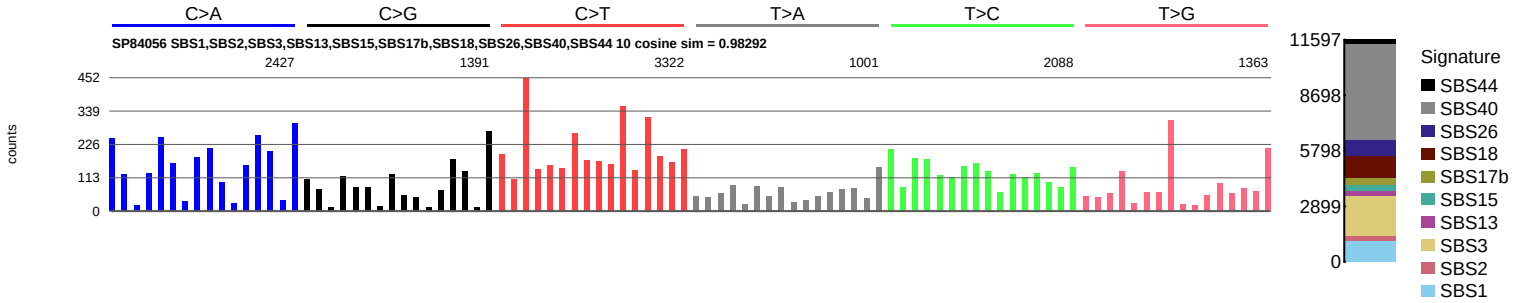

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98295
2296
1262
3364
1276
2055
1347
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

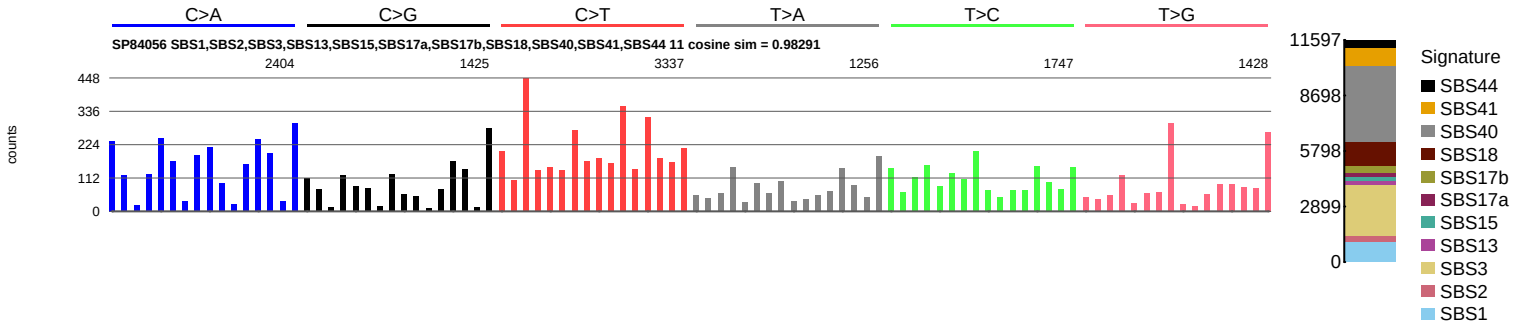

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98295
2327
1095
3445
1119
2163
1450
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

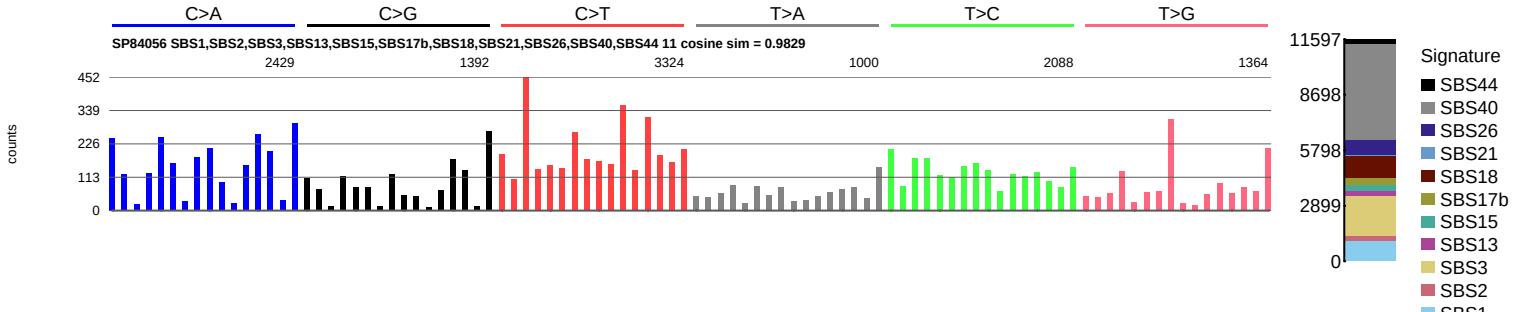

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98295
2336
1297
3321
1061
2138
1442
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

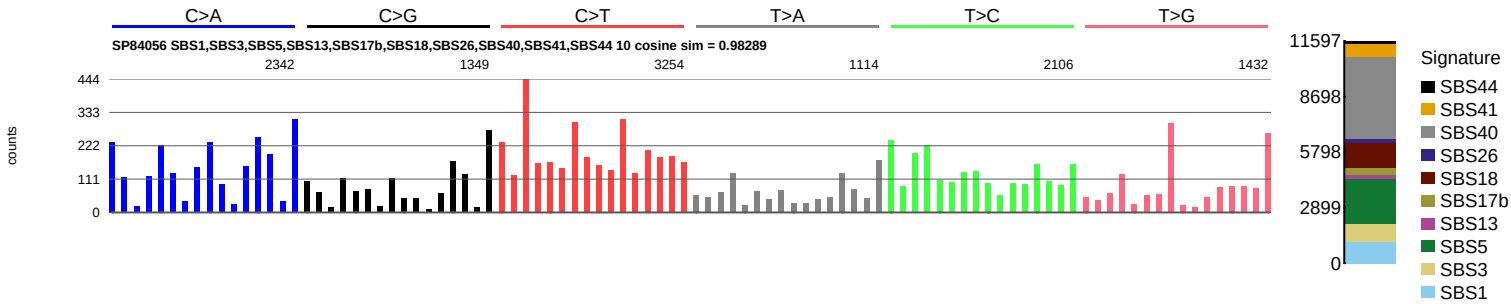

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98294
2401
1271
3370
1092
1996
1466
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

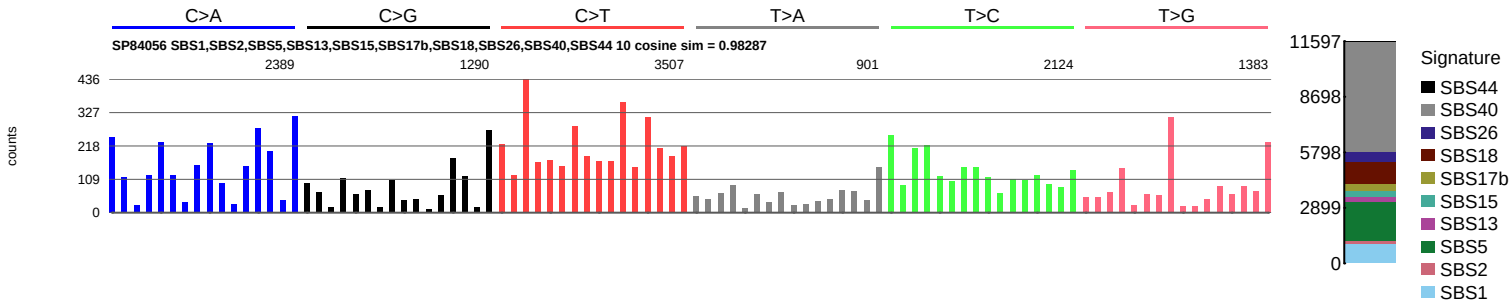

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98293
2292
1270
3343
1278
2063
1352
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

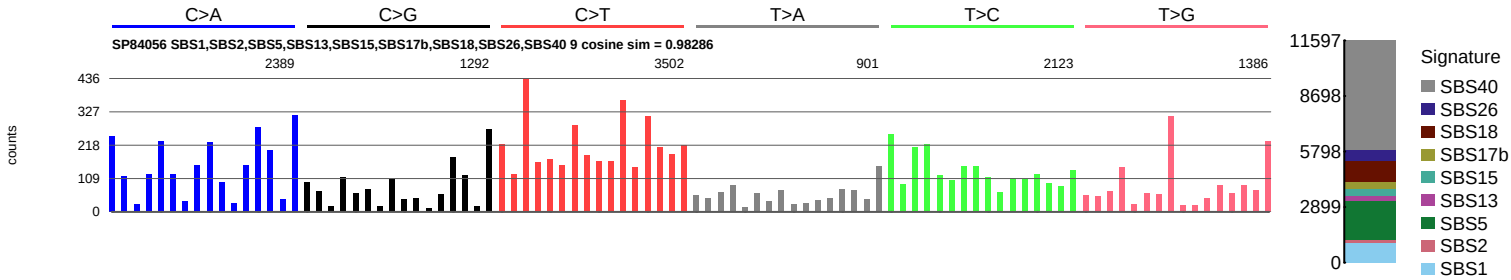

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98292
2427
1391
3322
1001
2088
1363
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

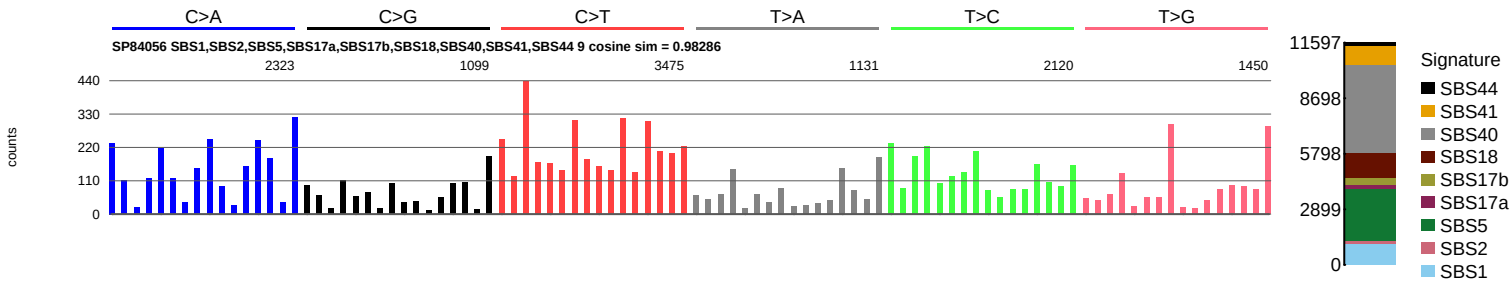

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98291
2404
1425
3337
1256
1747
1428
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

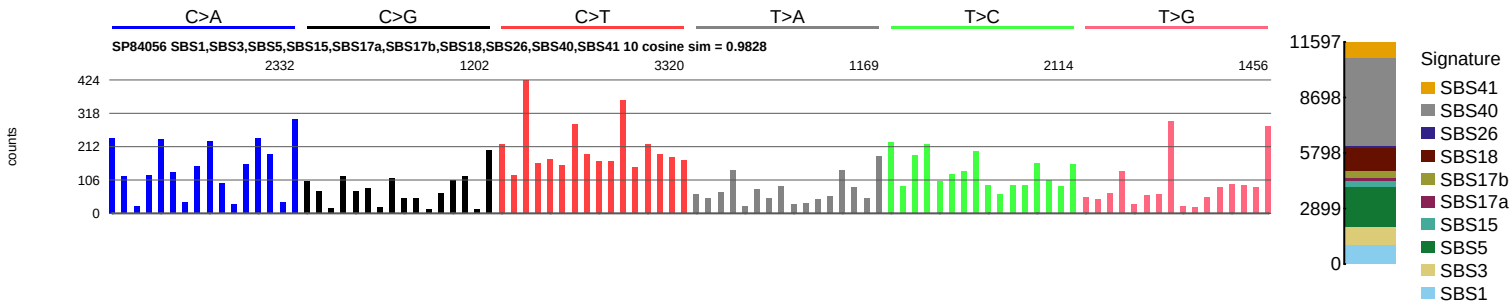

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9829
2429
1392
3324
1000
2088
1364
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2

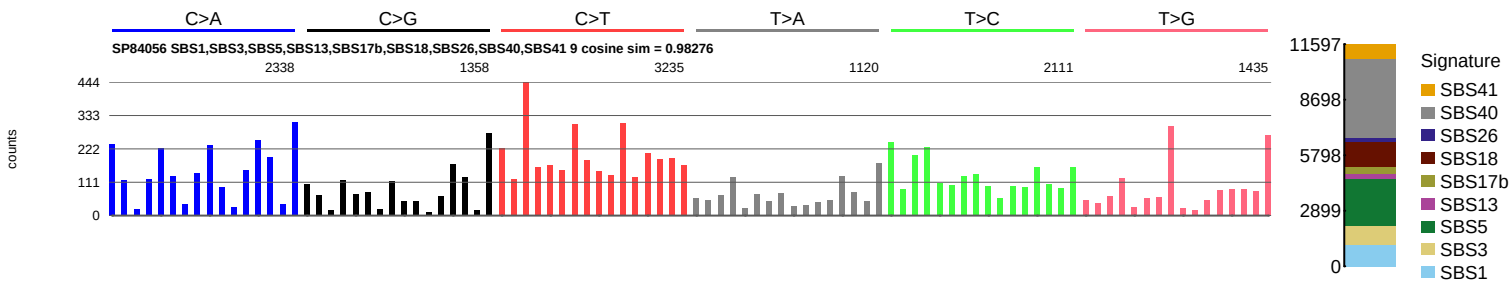

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98289
2342
1349
3254
1114
2106
1432
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1

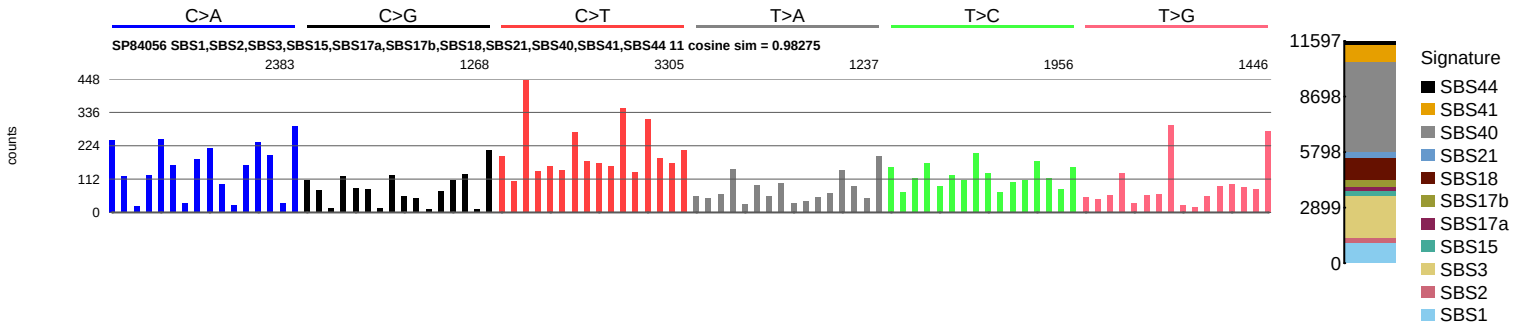

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98287
2389
1290
3507
901
2124
1383
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98286
2389
1292
3502
901
2123
1386
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98286
2323
1099
3475
1131
2120
1450
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.9828
2332
1202
3320
1169
2114
1456
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98276
2338
1358
3235
1120
2111
1435
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98275
2383
1268
3305
1237
1956
1446
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
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SBS40
SBS21
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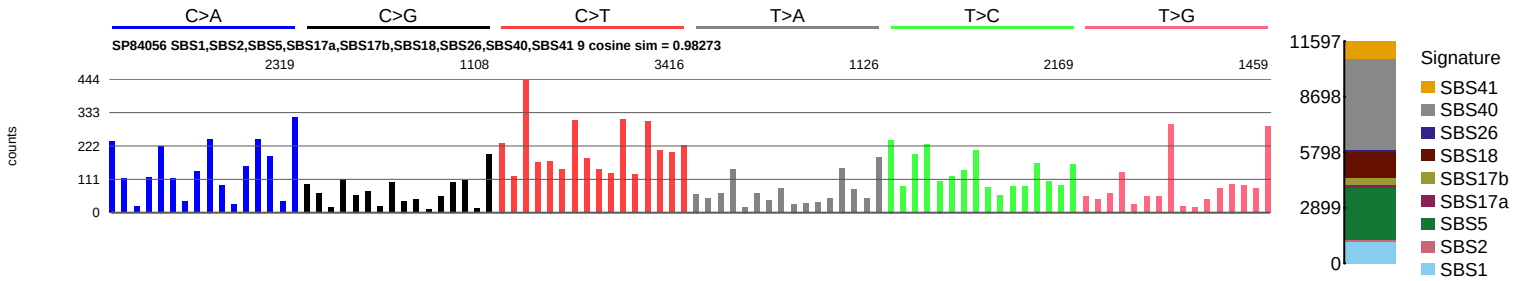
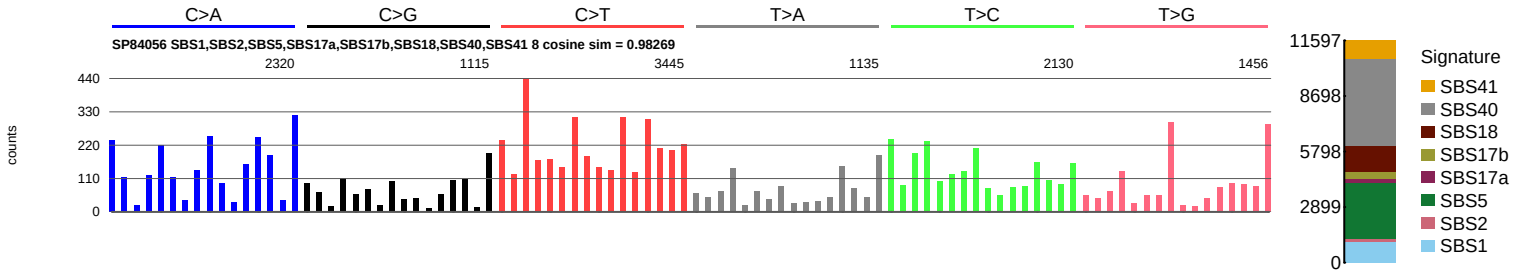
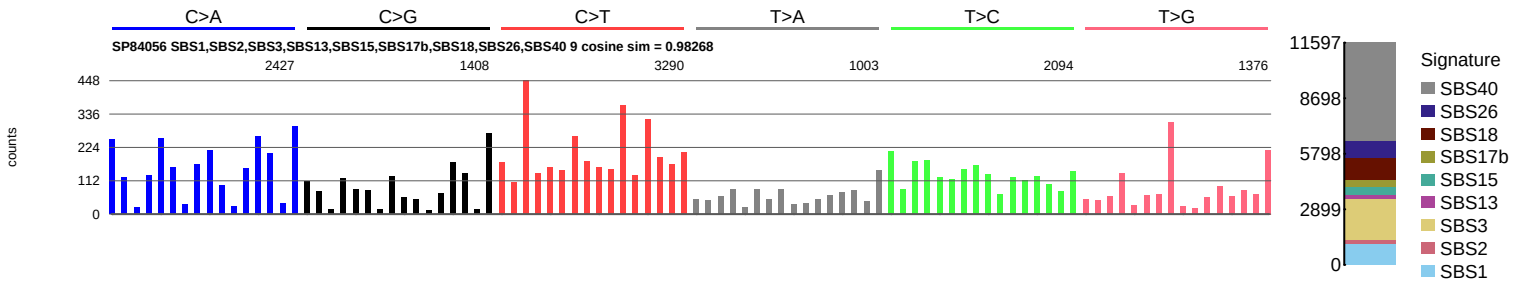
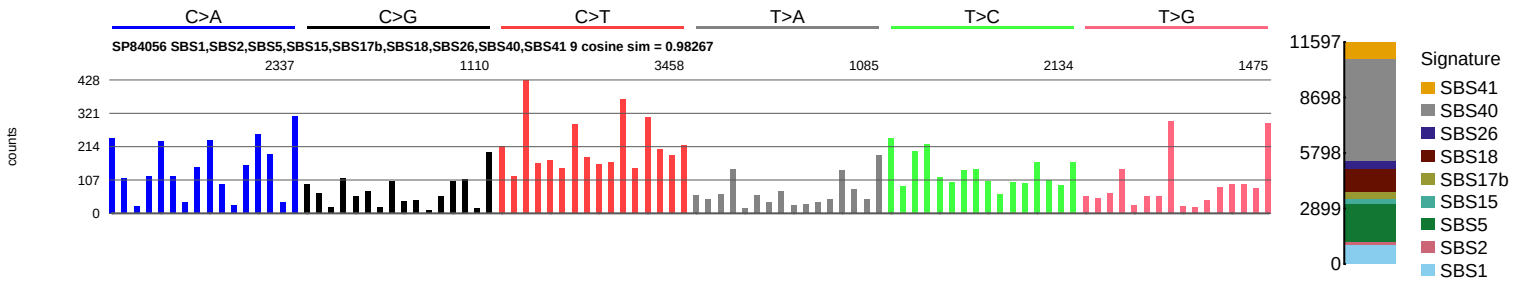
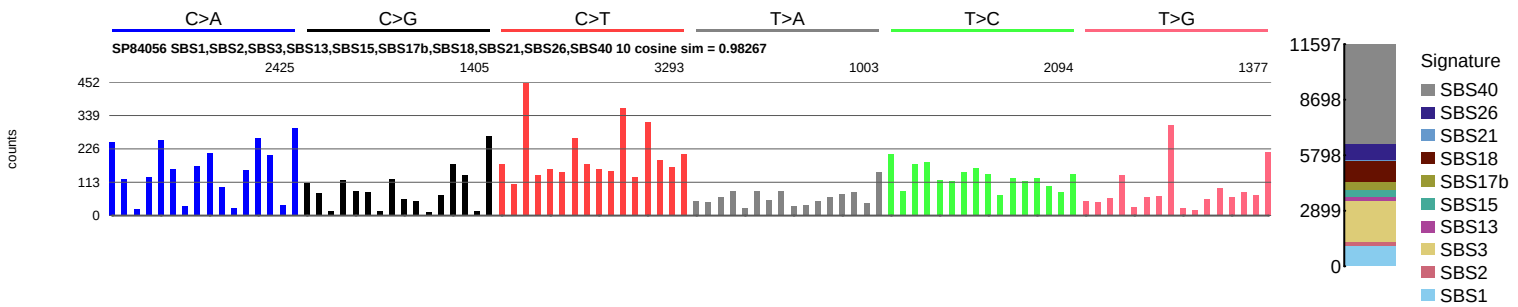
SBS3
SBS2
SBS1

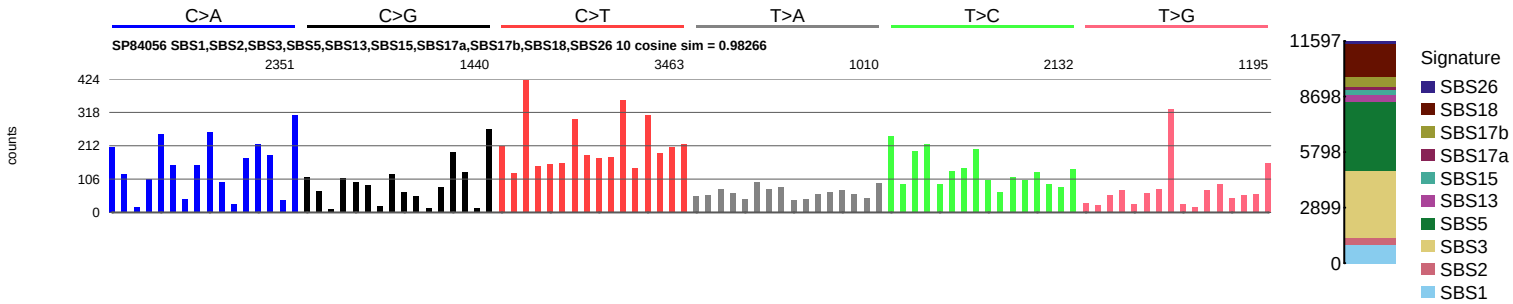

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98273
2319
1108
3416
1126
2169
1459
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

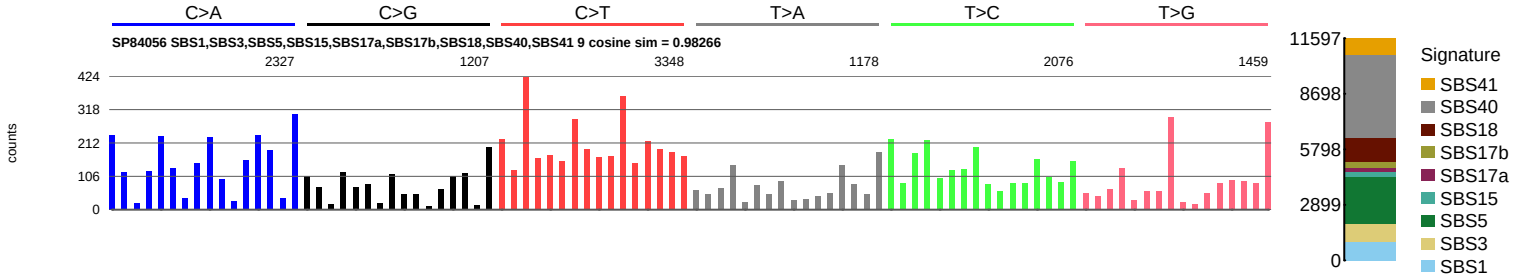

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98269
2320
1115
3445
1135
2130
1456
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

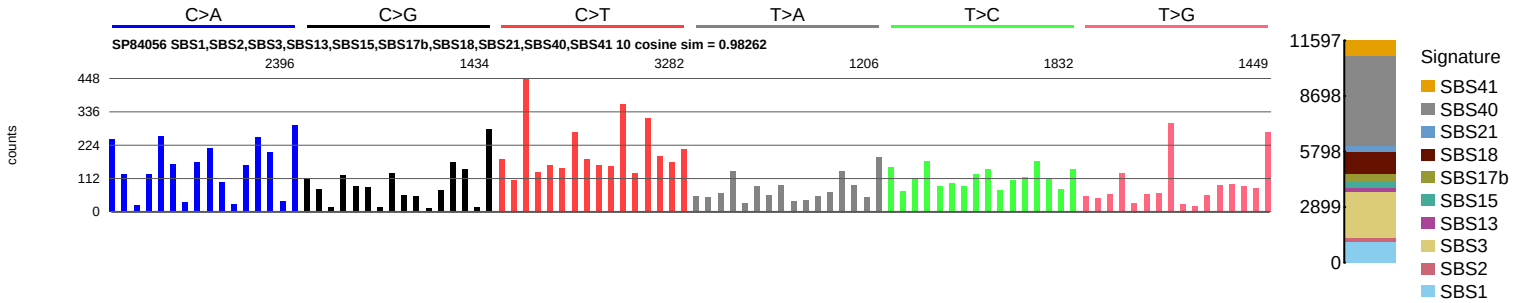

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98268
2427
1408
3290
1003
2094
1376
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

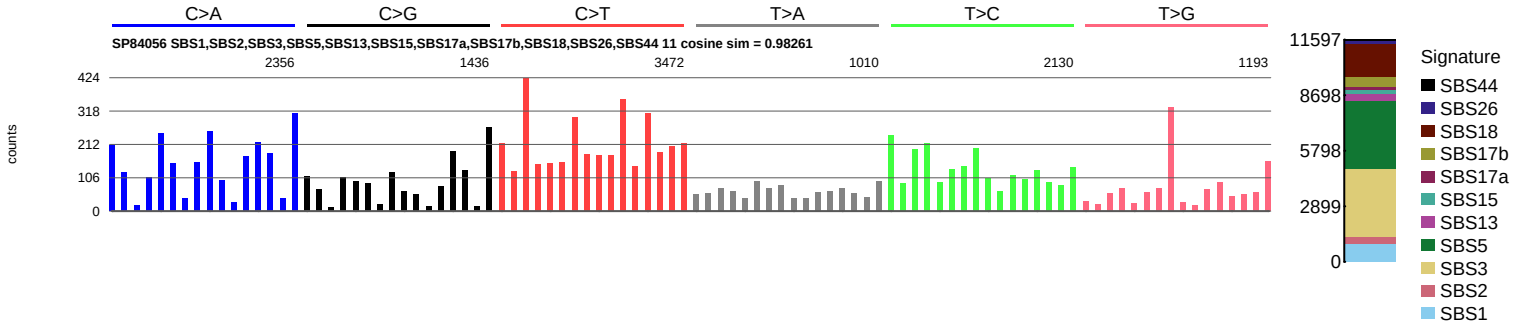

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98267
2337
1110
3458
1085
2134
1475
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

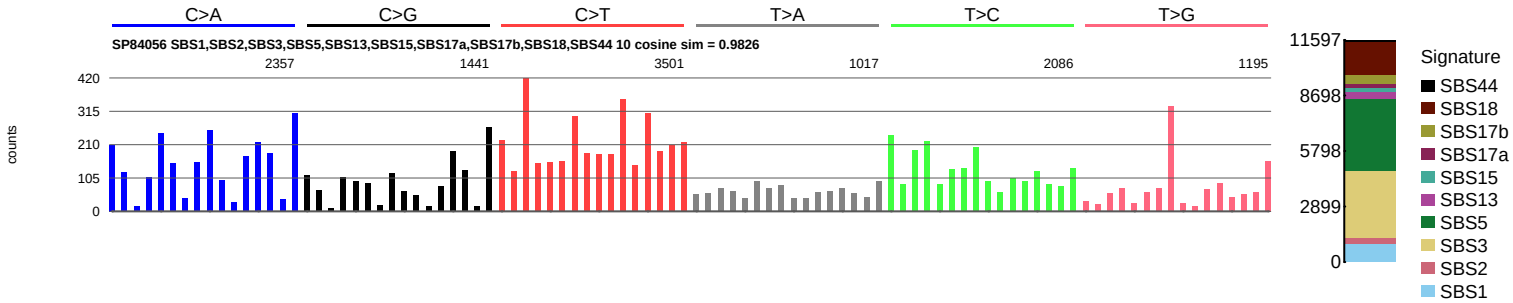

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98267
2425
1405
3293
1003
2094
1377
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

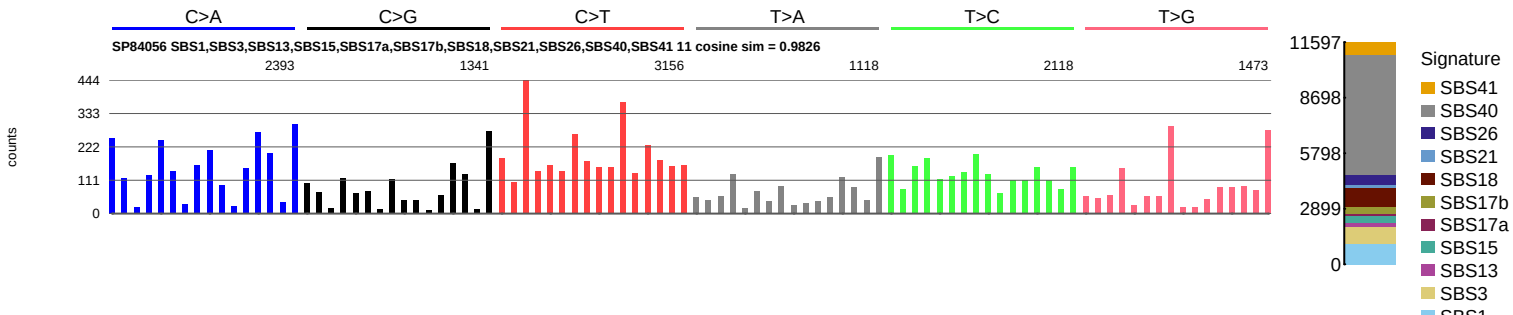

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98266
2351
1440
3463
1010
2132
1195
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

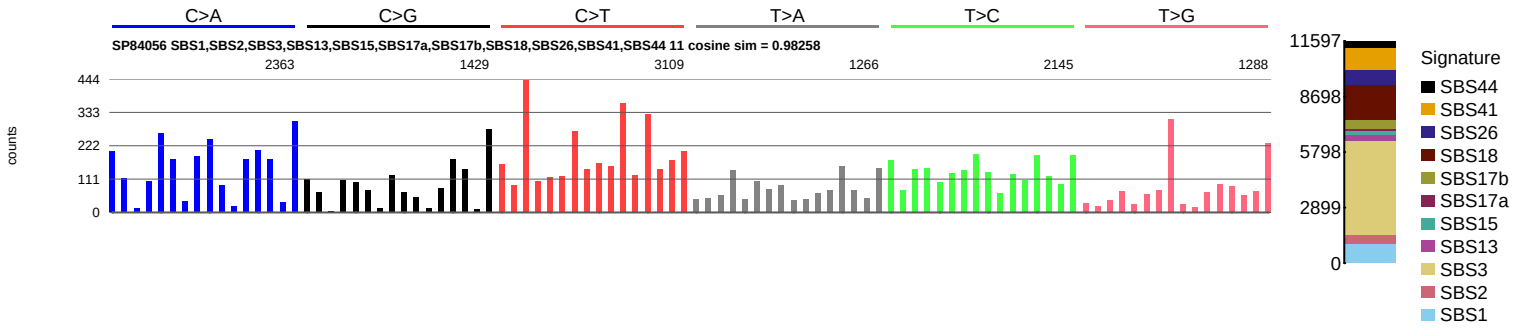

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98266
2327
1207
3348
1178
2076
1459
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

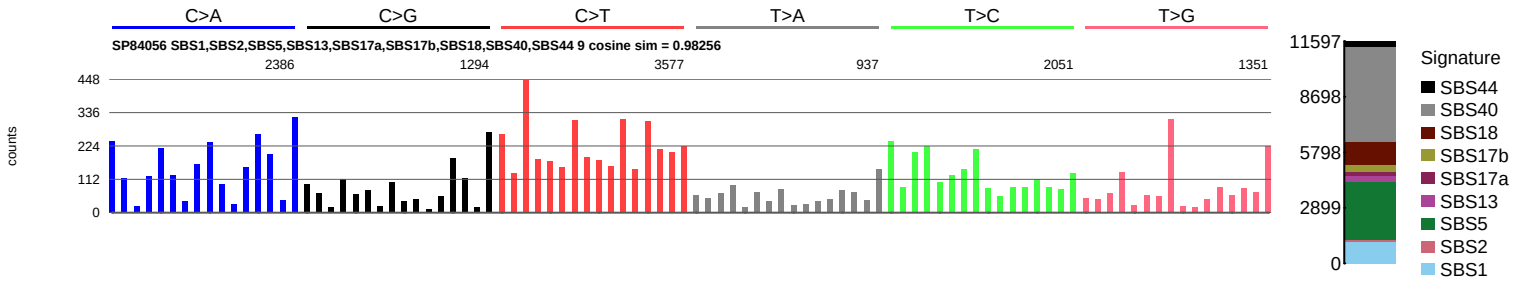

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98262
2396
1434
3282
1206
1832
1449
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

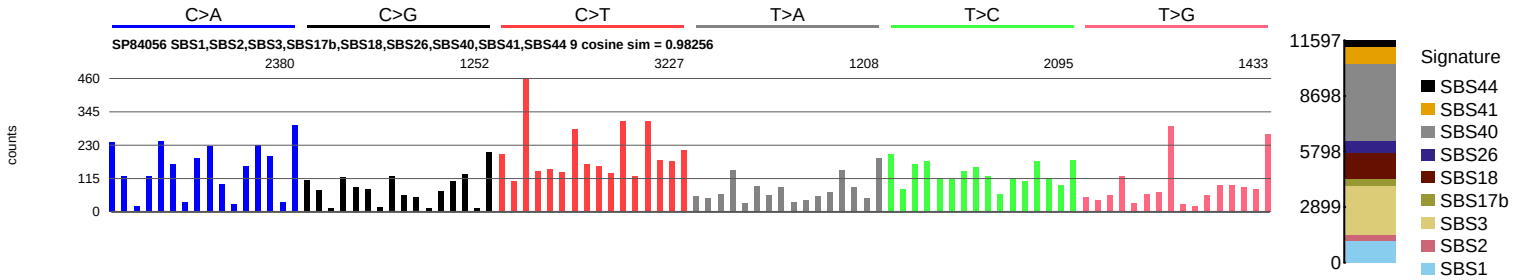

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 11 cosine sim = 0.98261
2356
1436
3472
1010
2130
1193
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

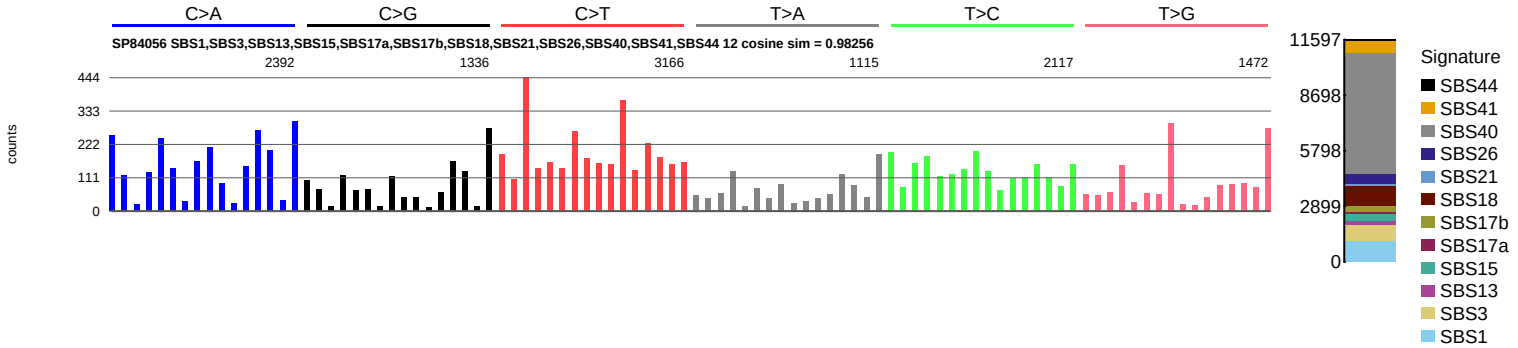

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.9826
2357
1441
3501
1017
2086
1195
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

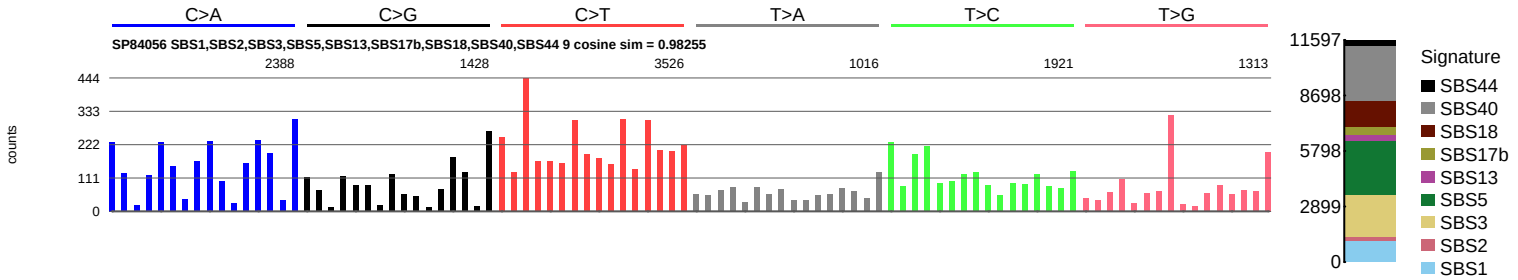

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9826
2393
1341
3156
1118
2118
1473
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

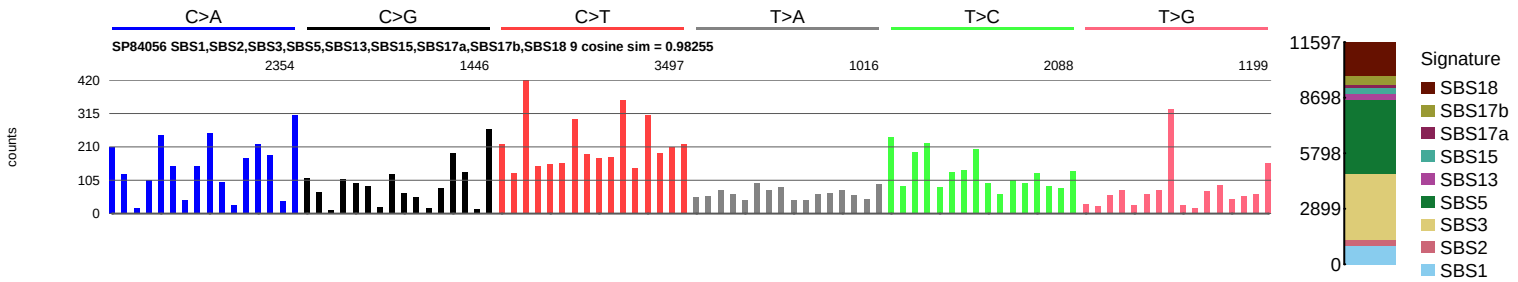

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98258
2363
1429
3109
1266
2145
1288
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

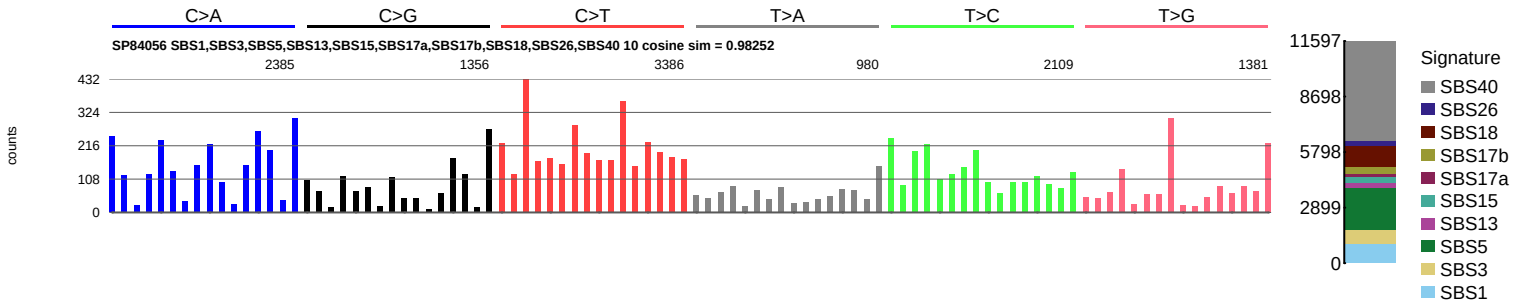

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98256
2386
1294
3577
937
2051
1351
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

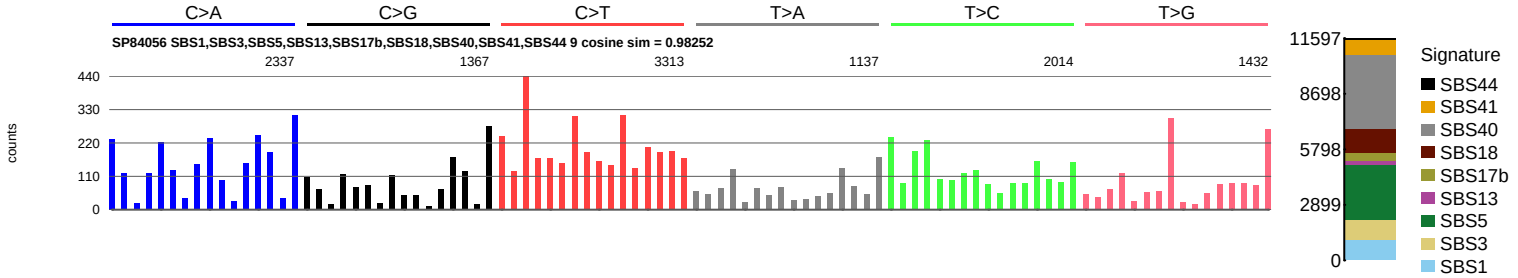

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98256
2380
1252
3227
1208
2095
1433
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

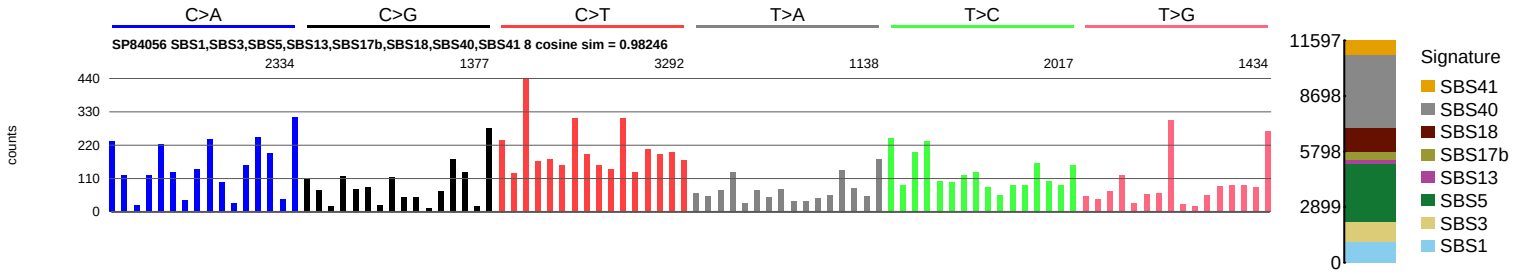

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98256
2392
1336
3166
1115
2117
1472
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

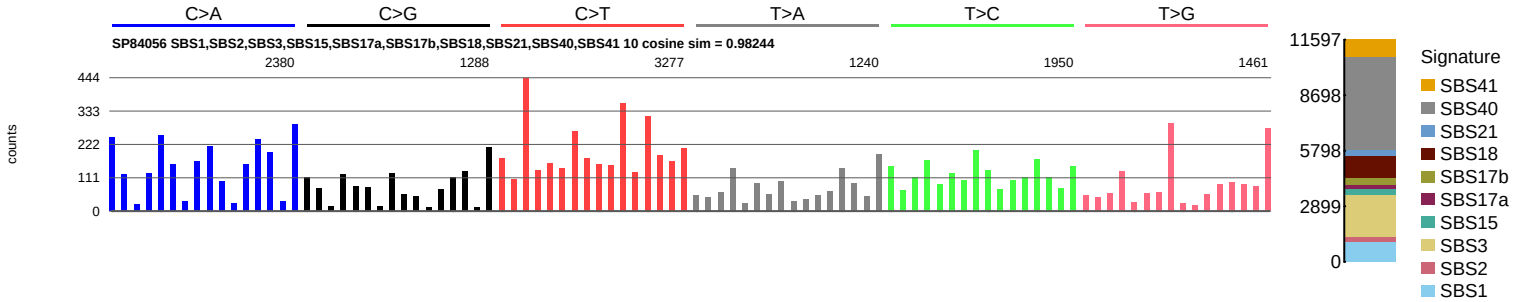

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98255
2388
1428
3526
1016
1921
1313
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

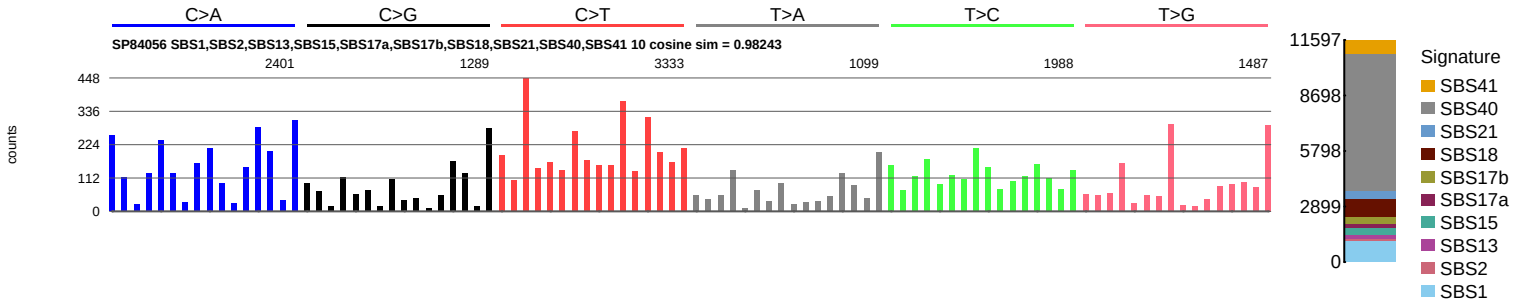

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98255
2354
1446
3497
1016
2088
1199
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

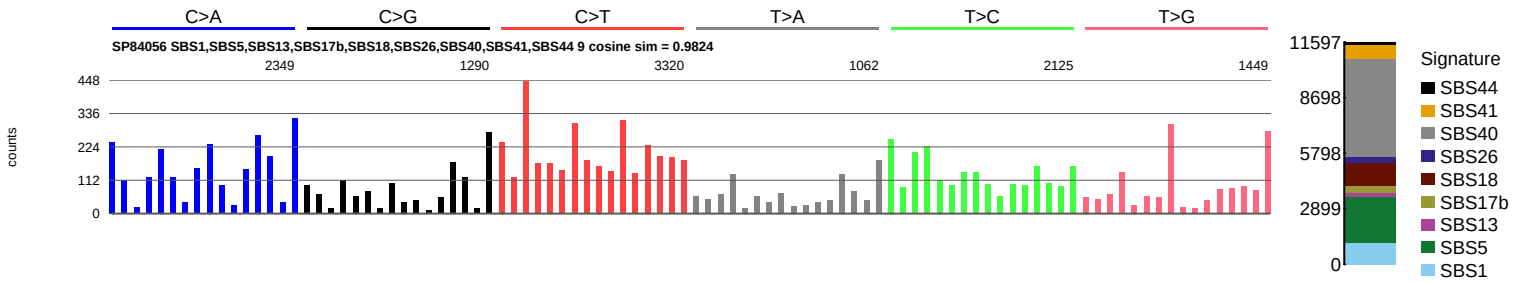

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98252
2385
1356
3386
980
2109
1381
432
324
216
108
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

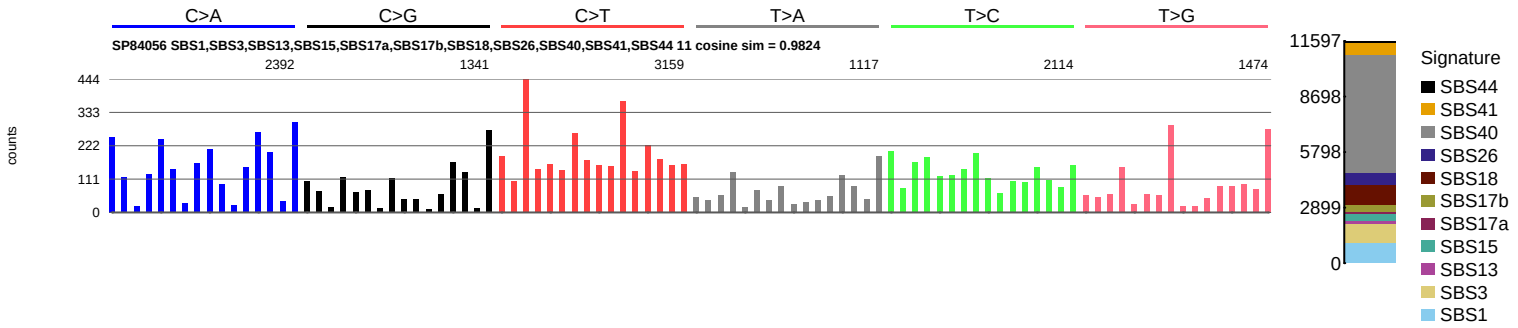

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98252
2337
1367
3313
1137
2014
1432
440
330
220
110
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1

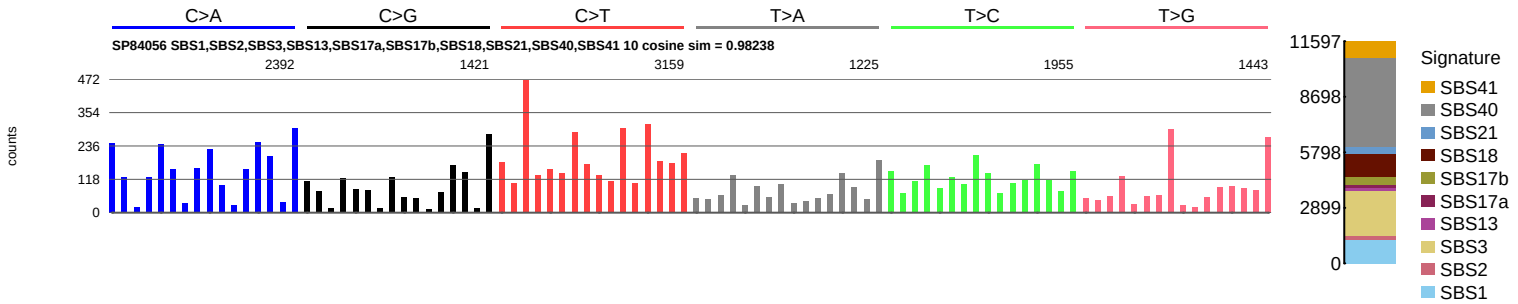

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98246
2334
1377
3292
1138
2017
1434
440
330
220
110
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1

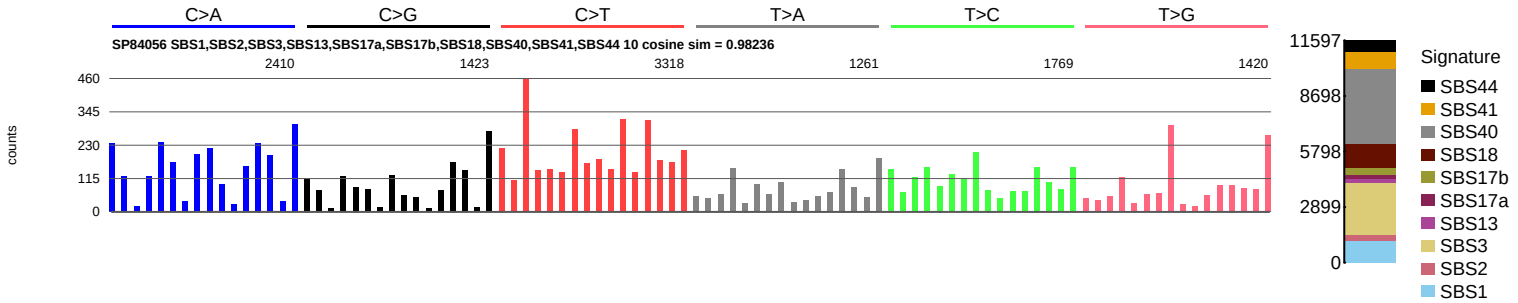

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98244
2380
1288
3277
1240
1950
1461
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98243
2401
1289
3333
1099
1988
1487
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9824
2349
1290
3320
1062
2125
1449
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9824
2392
1341
3159
1117
2114
1474
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98238
2392
1421
3159
1225
1955
1443
counts
472
354
236
118
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
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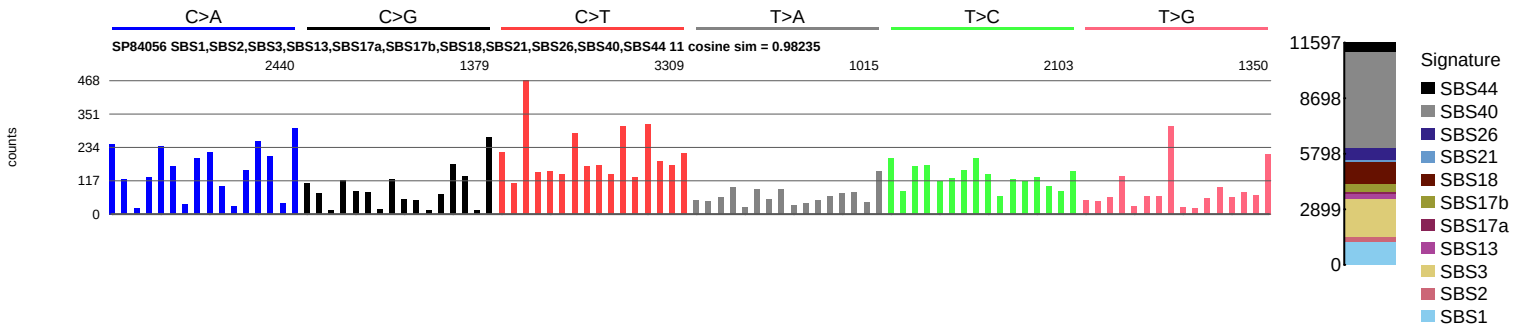
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98236
2410
1423
3318
1261
1769
1420
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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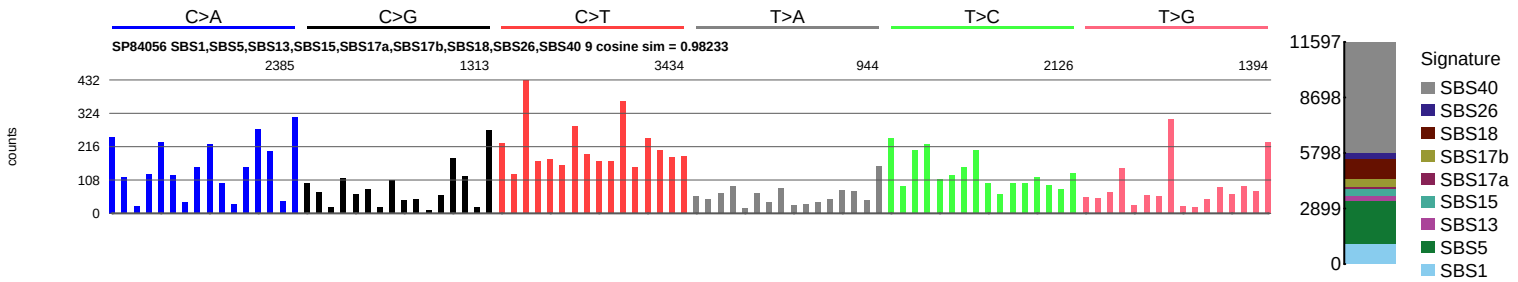
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98235
2440
1379
3309
1015
2103
1350
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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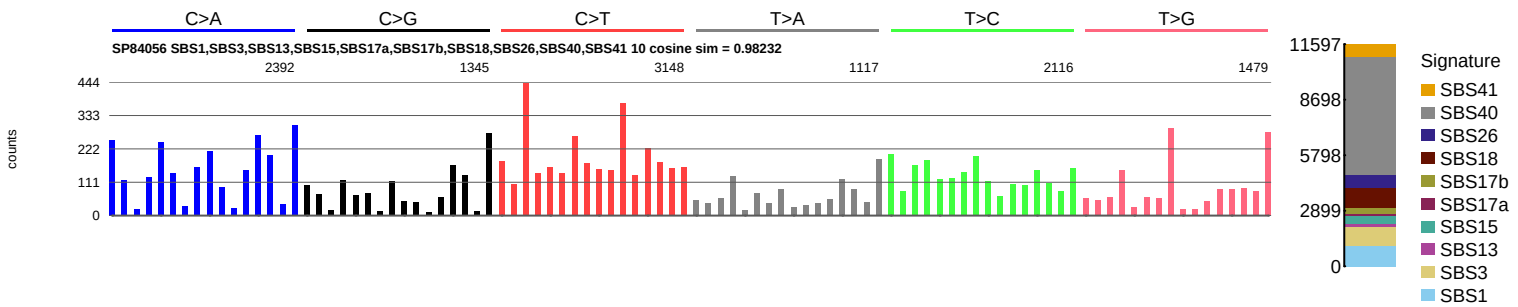
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98233
2385
1313
3434
944
2126
1394
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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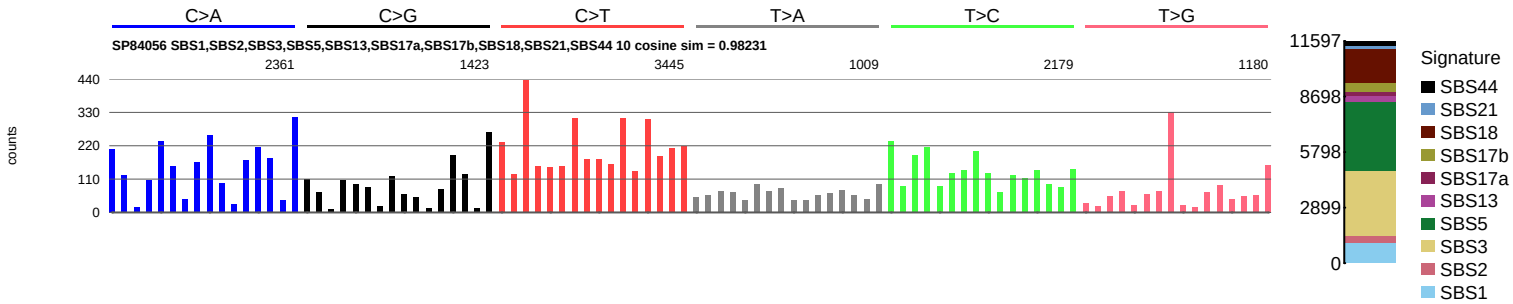
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98232
2392
1345
3148
1117
2116
1479
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
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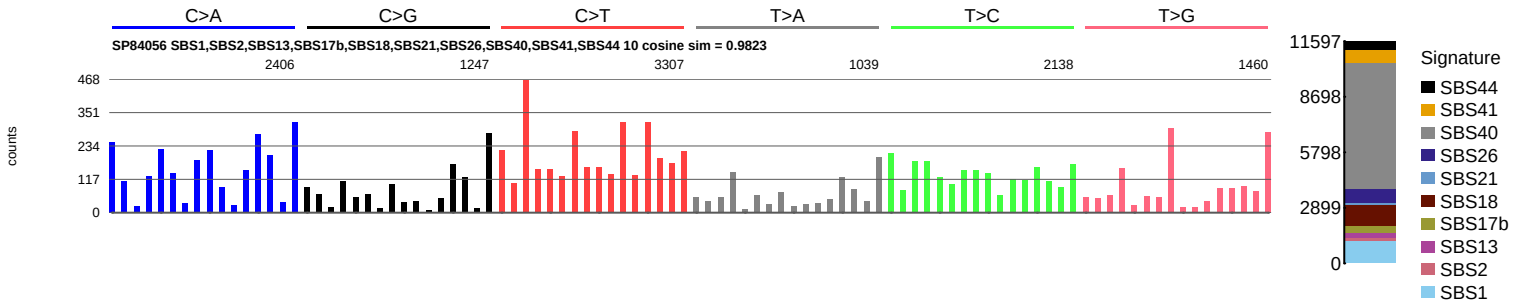
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98231
2361
1423
3445
1009
2179
1180
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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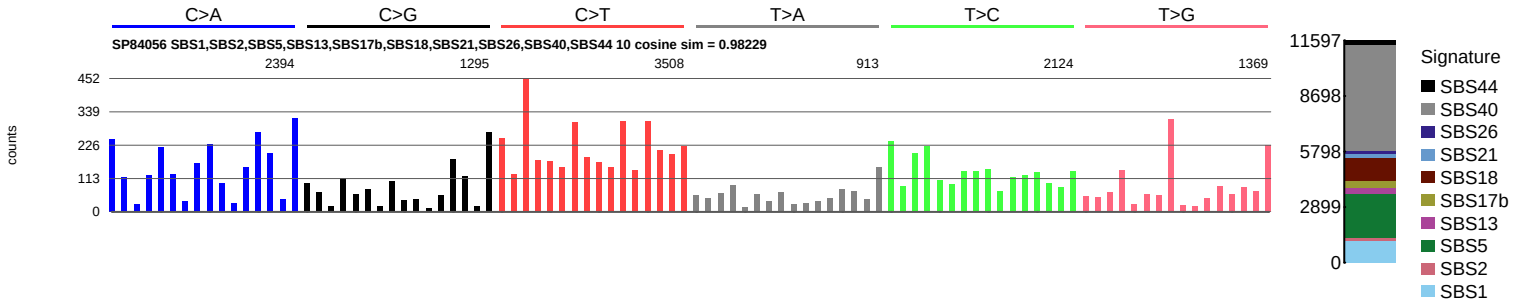
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9823
2406
1247
3307
1039
2138
1460
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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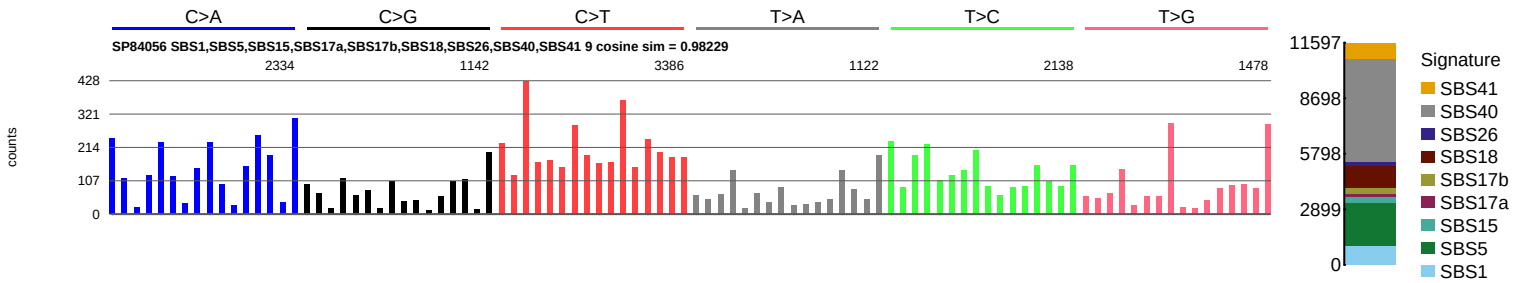
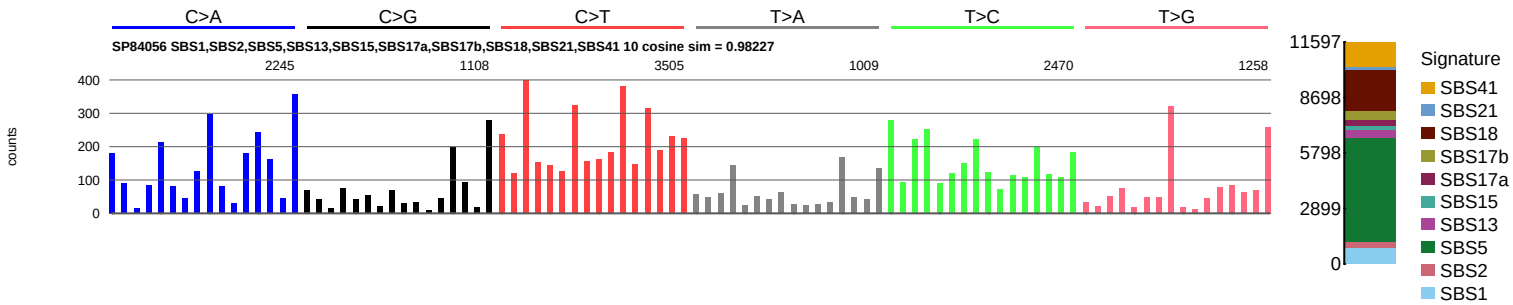
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98229
2394
1295
3508
913
2124
1369
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98229
2334
1142
3386
1122
2138
1478
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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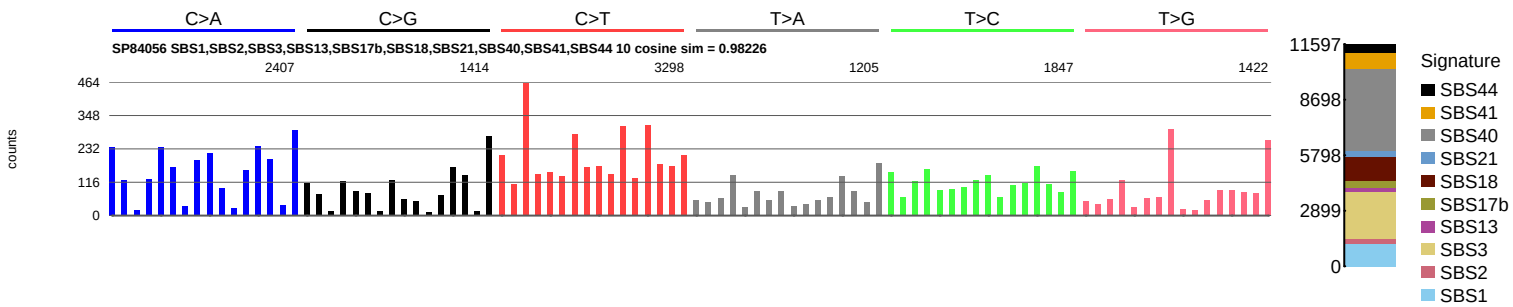
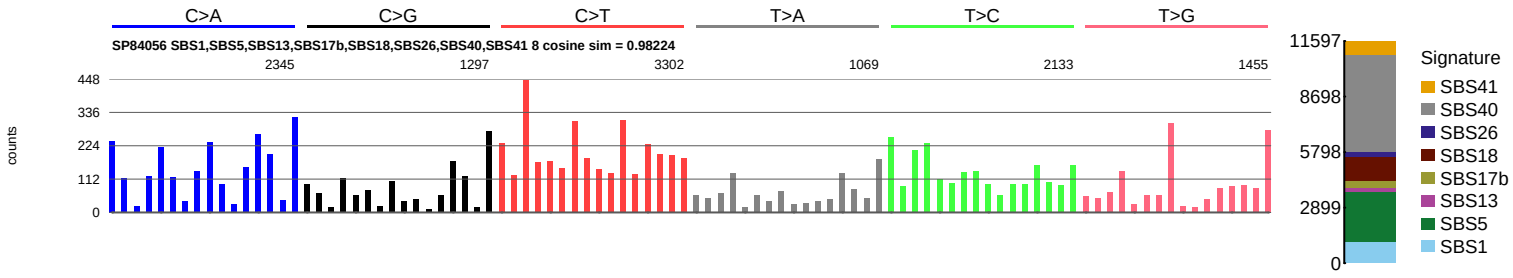
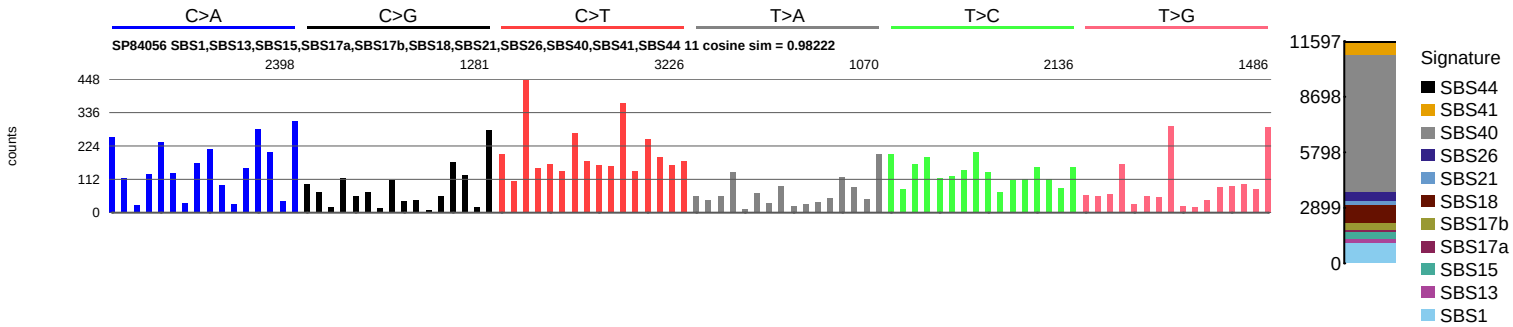
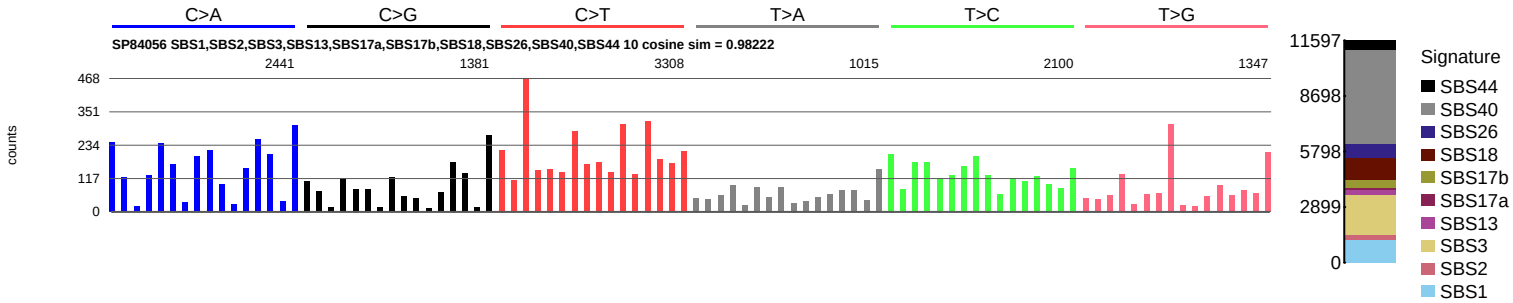
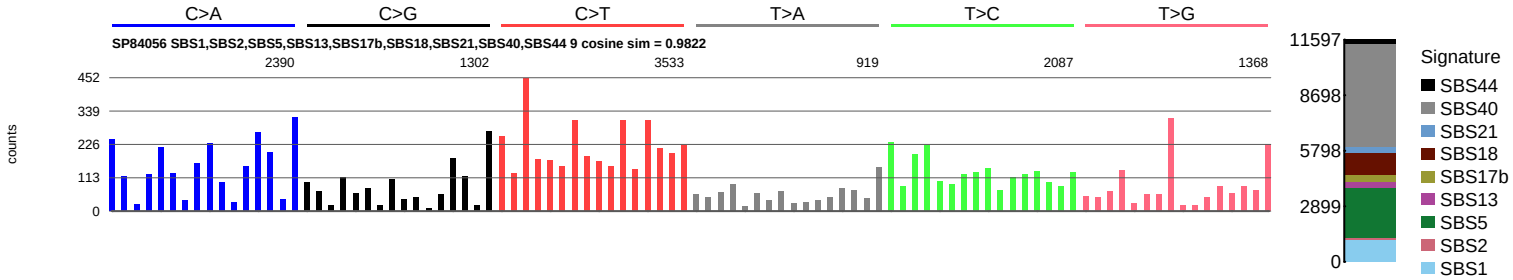
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98227
2245
1108
3505
1009
2470
1258
counts
400
300
200
100
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

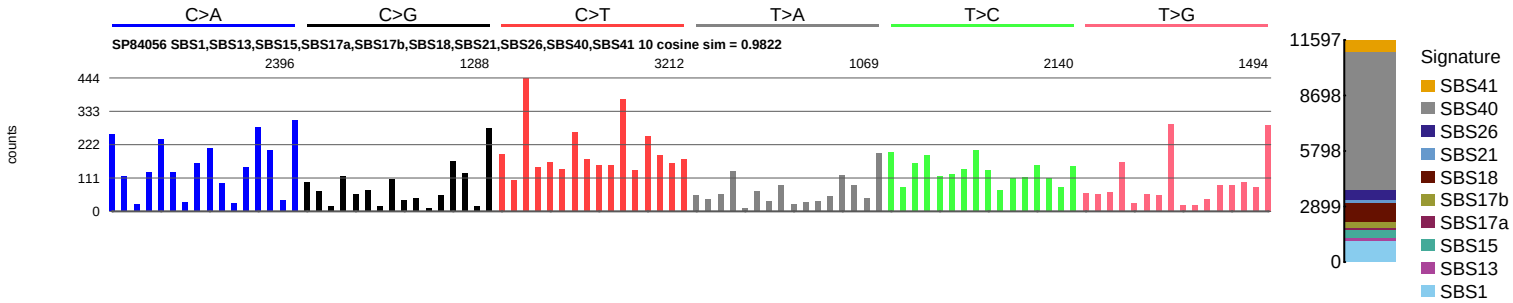

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98226
2407
1414
3298
1205
1847
1422
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98224
2345
1297
3302
1069
2133
1455
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98222
2398
1281
3226
1070
2136
1486
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98222
2441
1381
3308
1015
2100
1347
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9822
2390
1302
3533
919
2087
1368
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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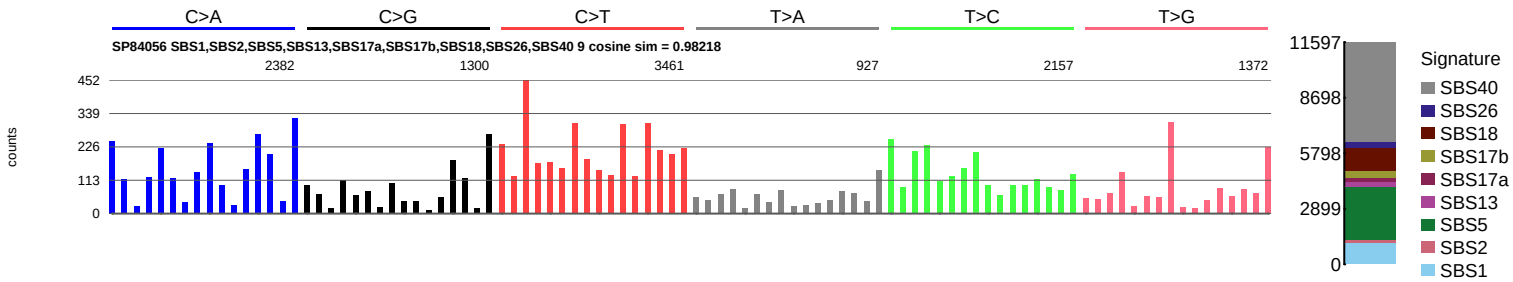
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9822
2396
1288
3212
1069
2140
1494
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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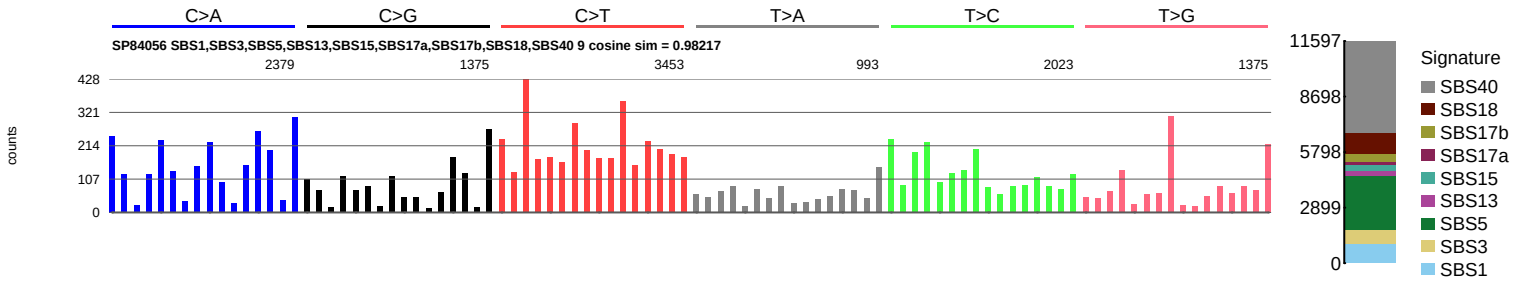
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98218
2382
1300
3461
927
2157
1372
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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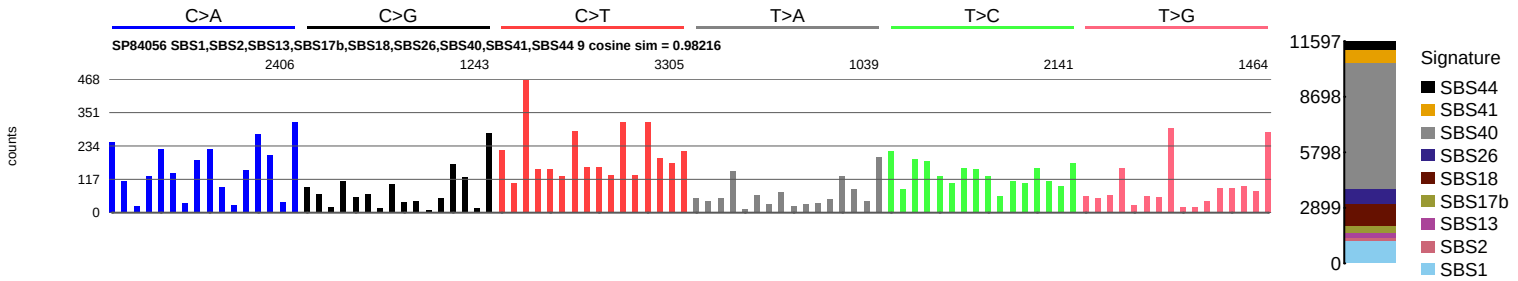
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98217
2379
1375
3453
993
2023
1375
428
321
214
107
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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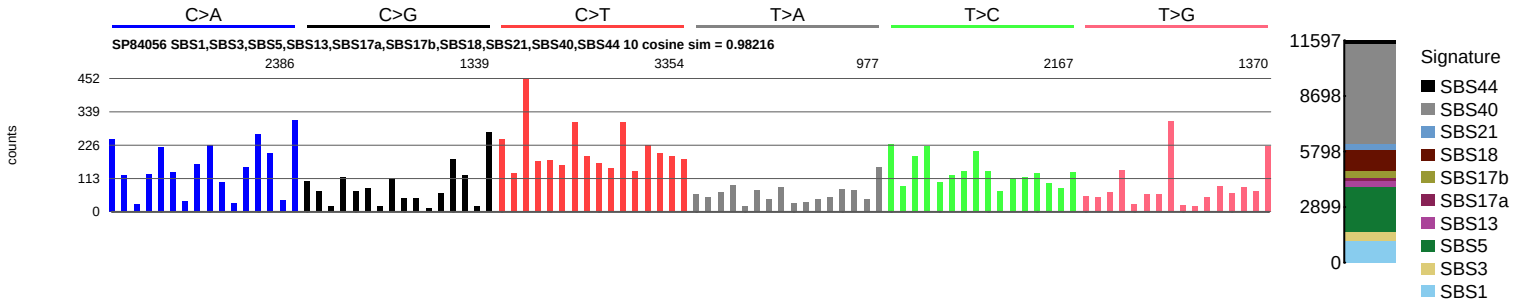
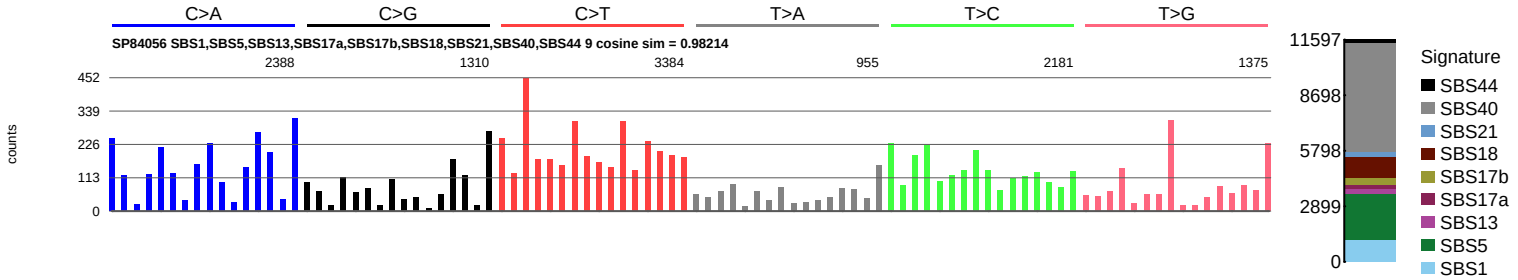
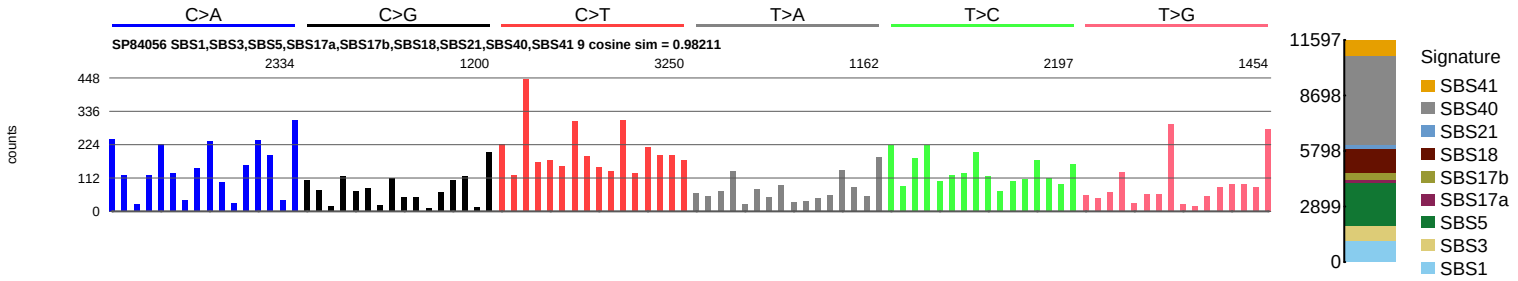
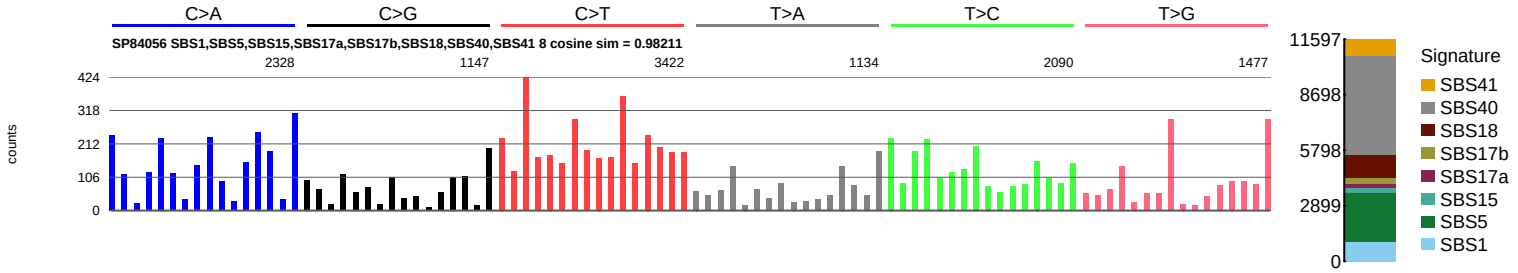
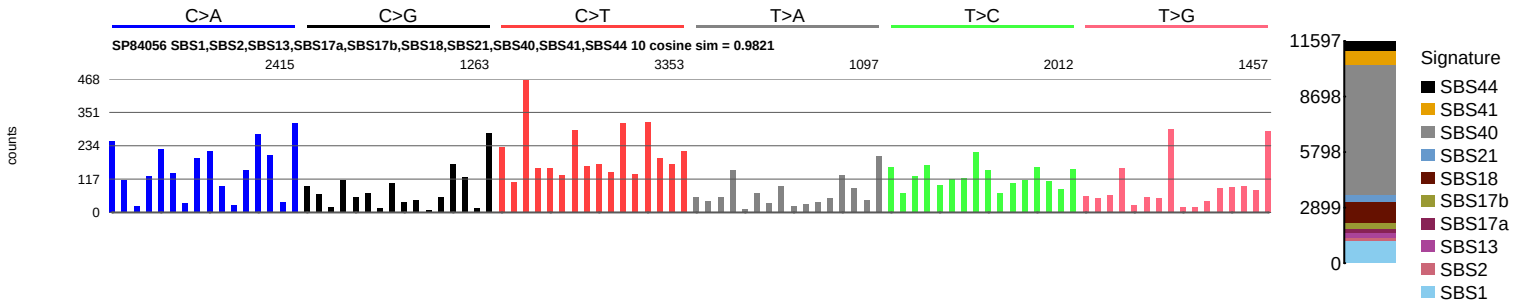
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98216
2406
1243
3305
1039
2141
1464
468
351
234
117
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

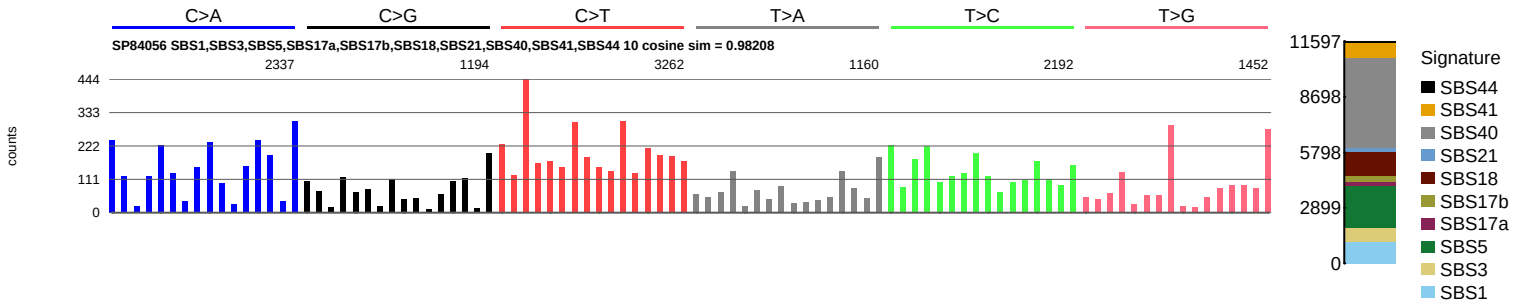

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98216
2386
1339
3354
977
2167
1370
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

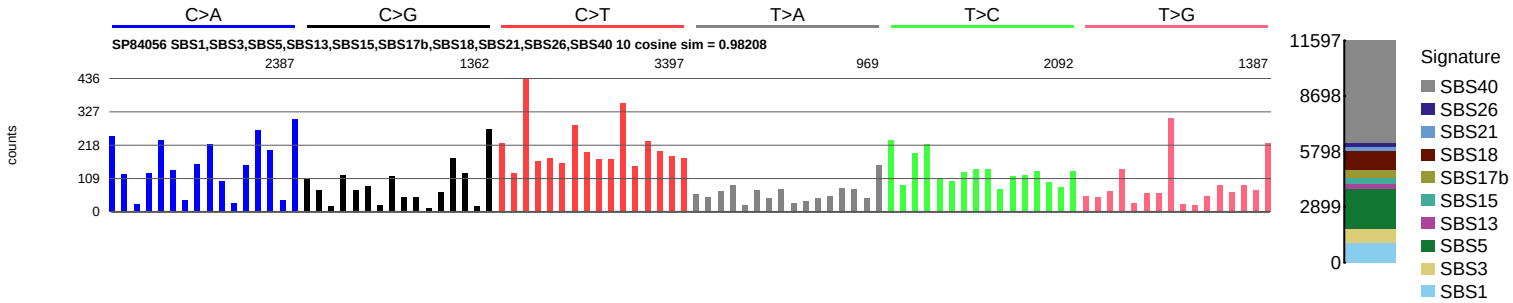

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98214
2388
1310
3384
955
2181
1375
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98211
2334
1200
3250
1162
2197
1454
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98211
2328
1147
3422
1134
2090
1477
424
318
212
106
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9821
2415
1263
3353
1097
2012
1457
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98208
2337
1194
3262
1160
2192
1452
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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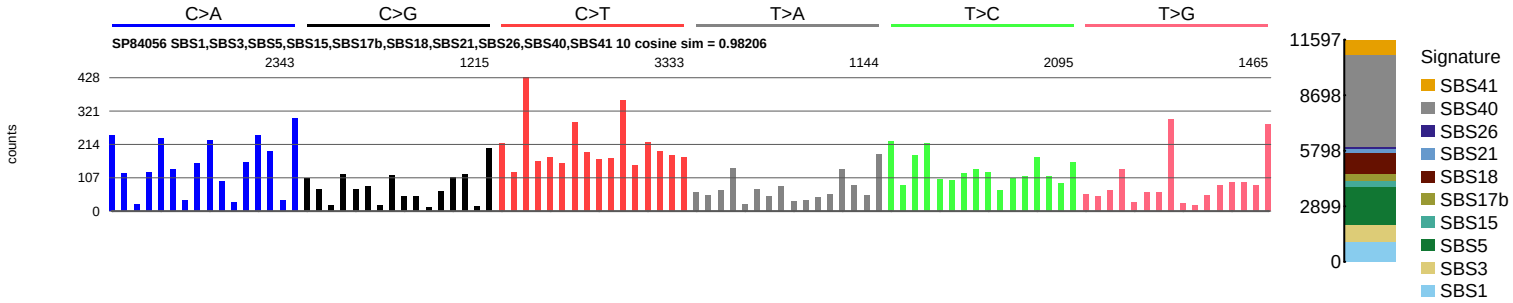
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98208
2387
1362
3397
969
2092
1387
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
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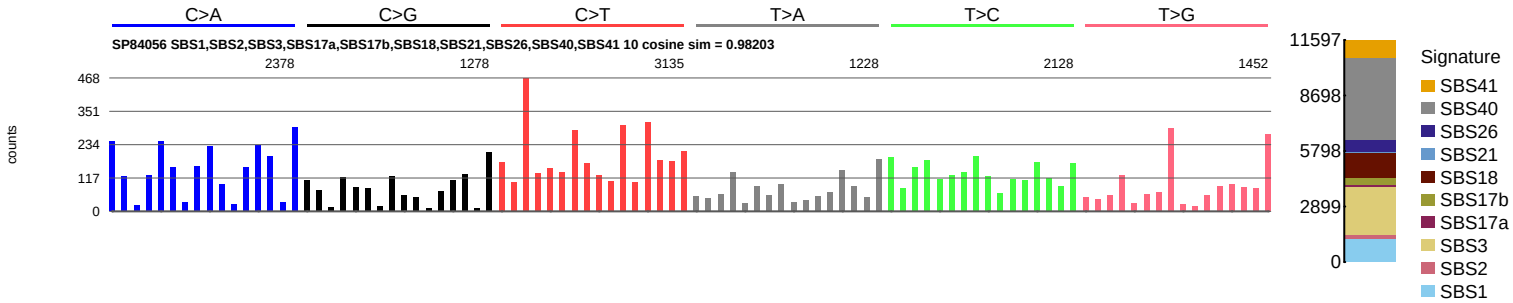
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98206
2343
1215
3333
1144
2095
1465
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1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C>G
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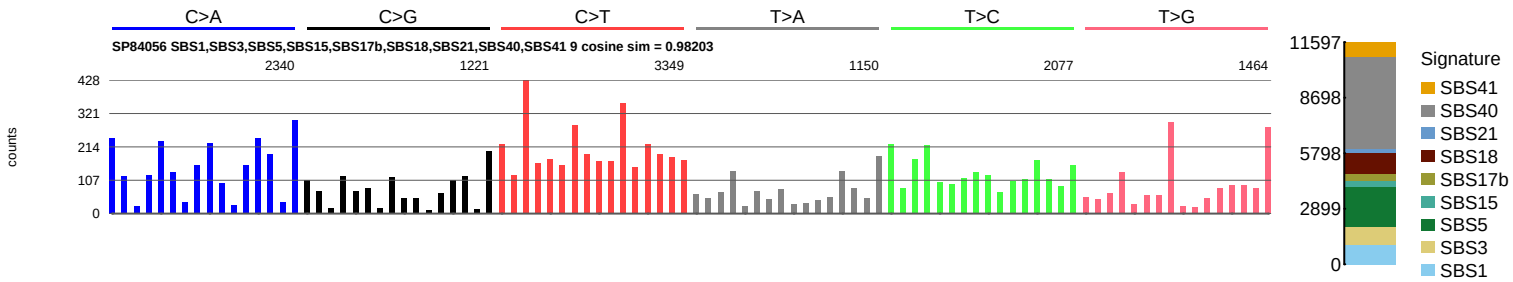
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98203
2378
1278
3135
1228
2128
1452
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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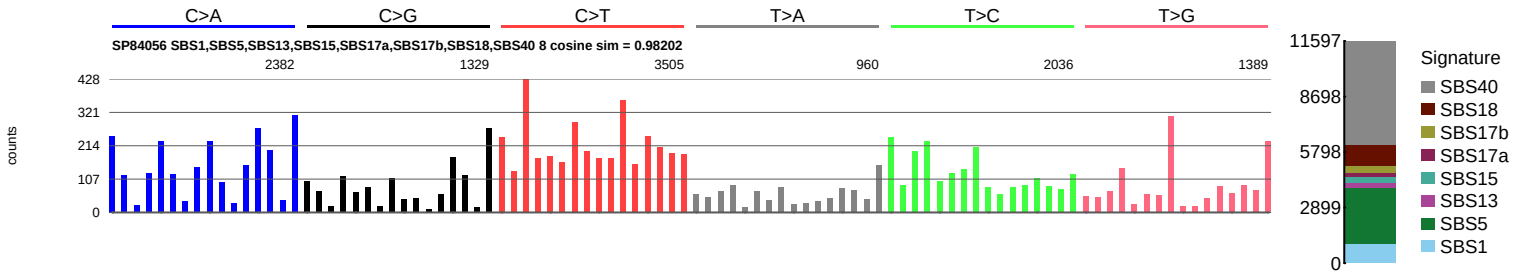
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98203
2340
1221
3349
1150
2077
1464
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1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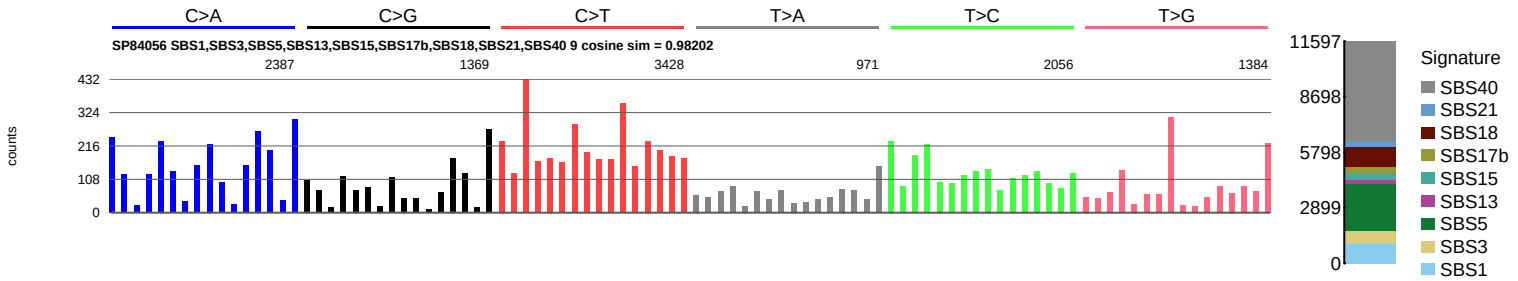
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98202
2382
1329
3505
960
2036
1389
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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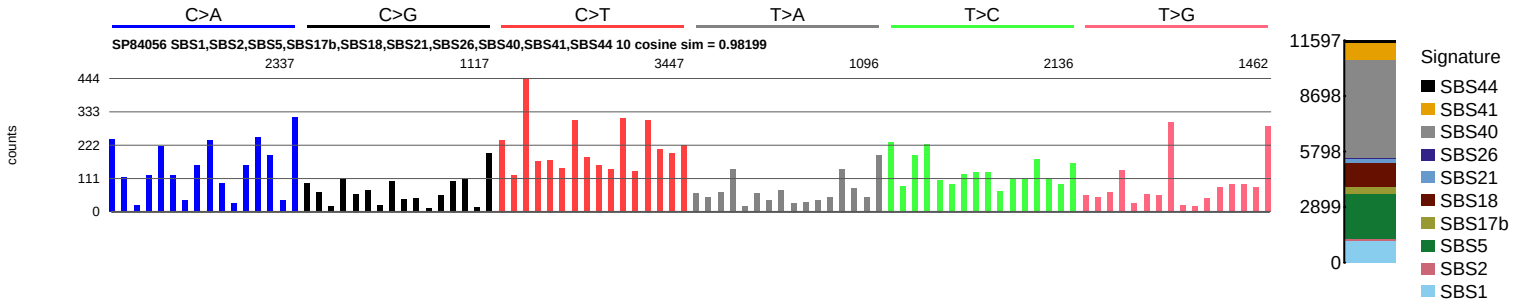
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98202
2387
1369
3428
971
2056
1384
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1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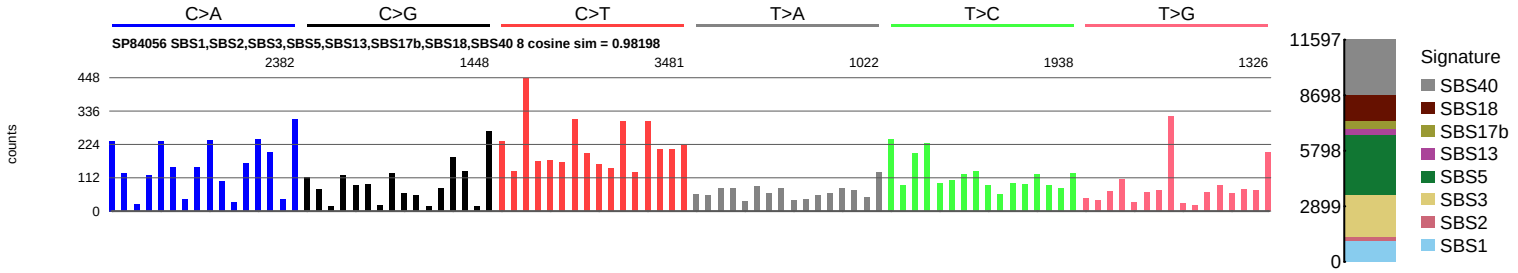
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98199
2337
1117
3447
1096
2136
1462
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98198
2382
1448
3481
1022
1938
1326
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
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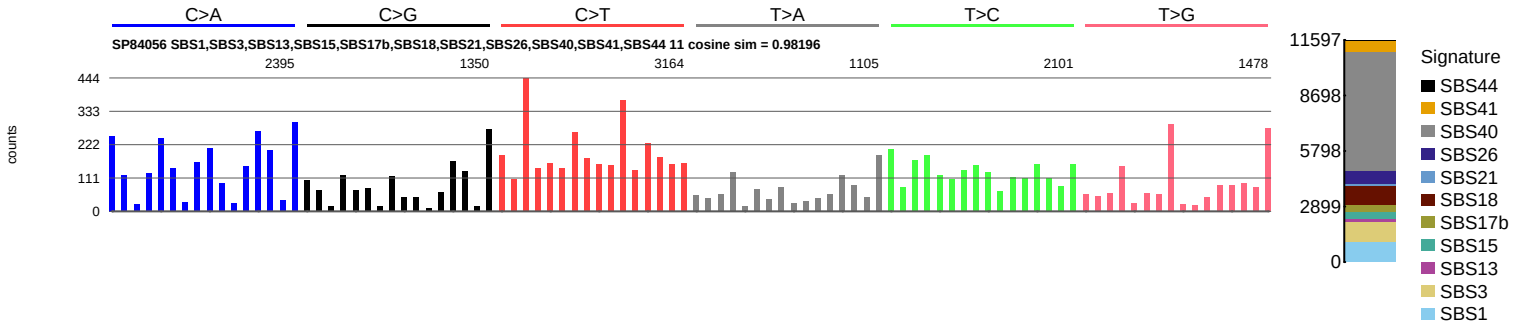
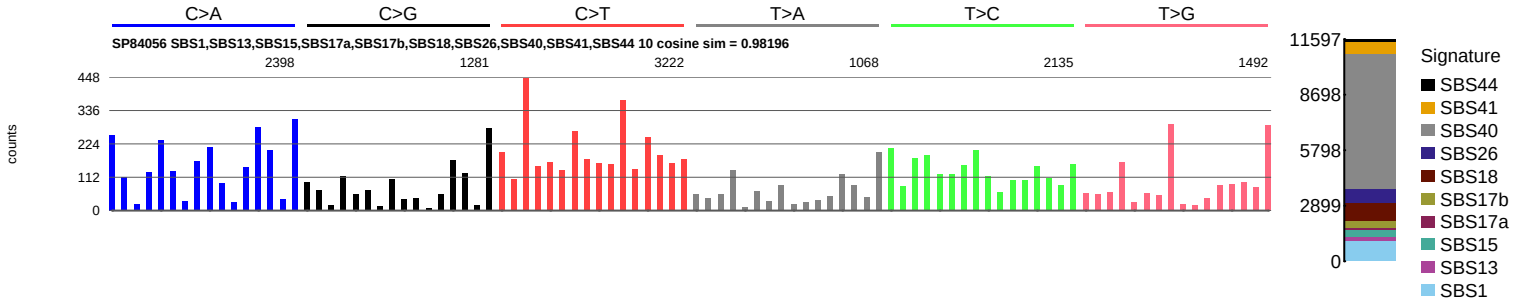
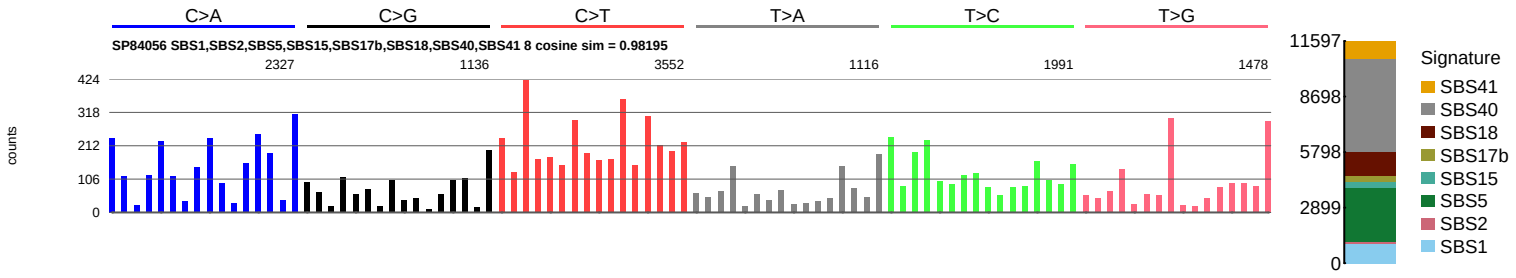
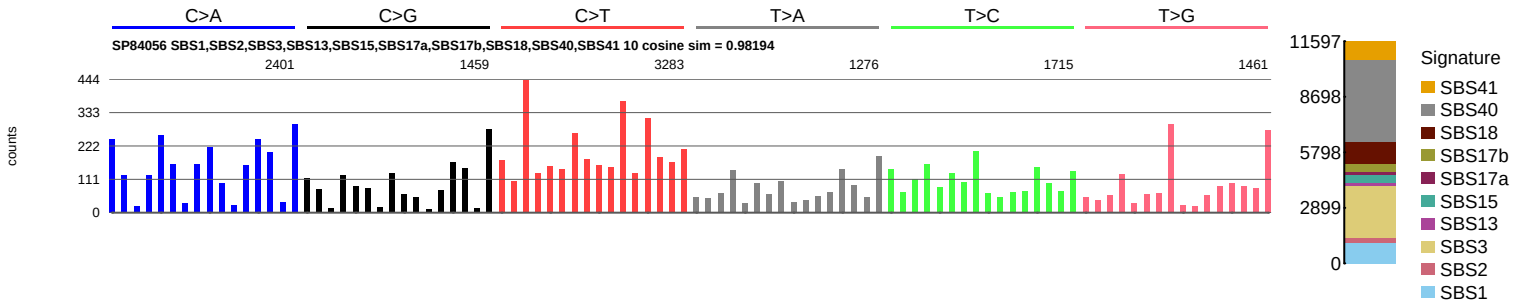
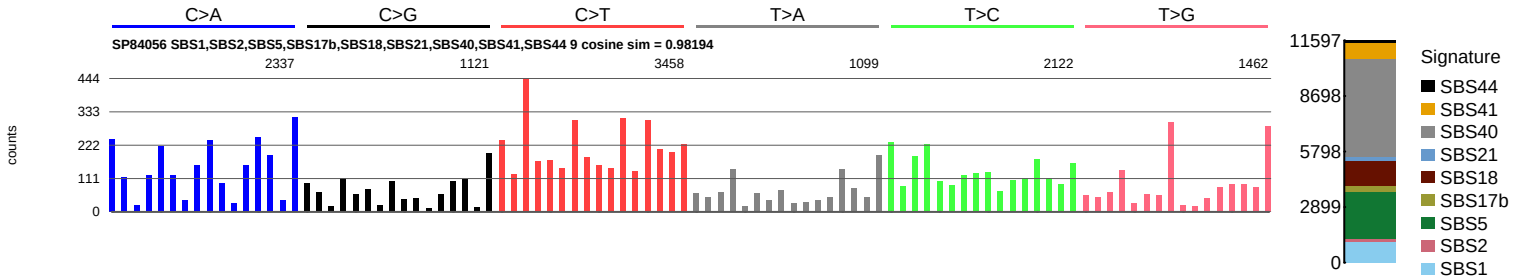
SBS3
SBS2
SBS1

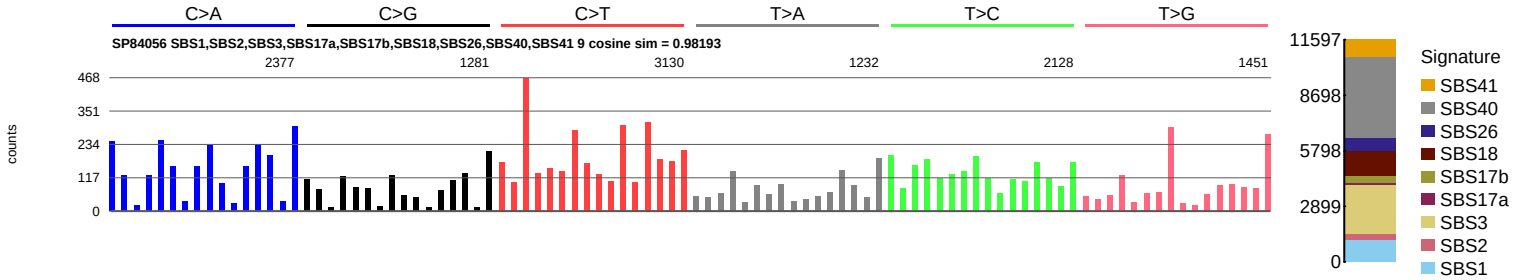

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98196
2395
1350
3164
1105
2101
1478
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS1

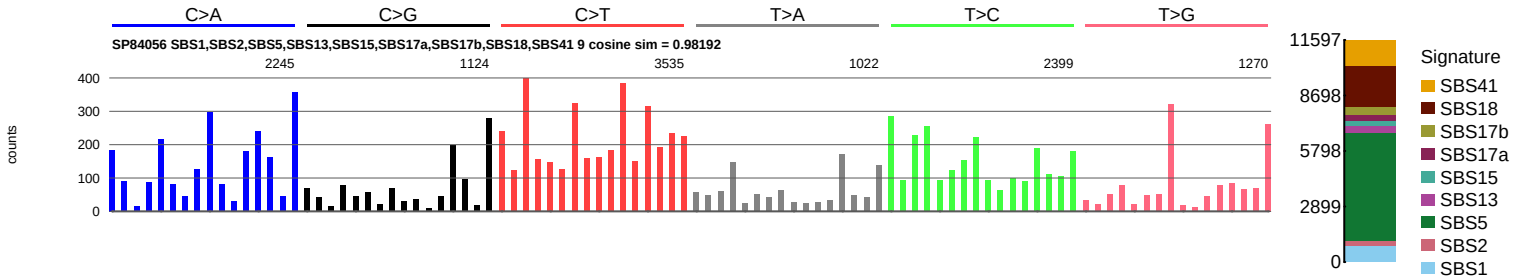

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98196
2398
1281
3222
1068
2135
1492
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

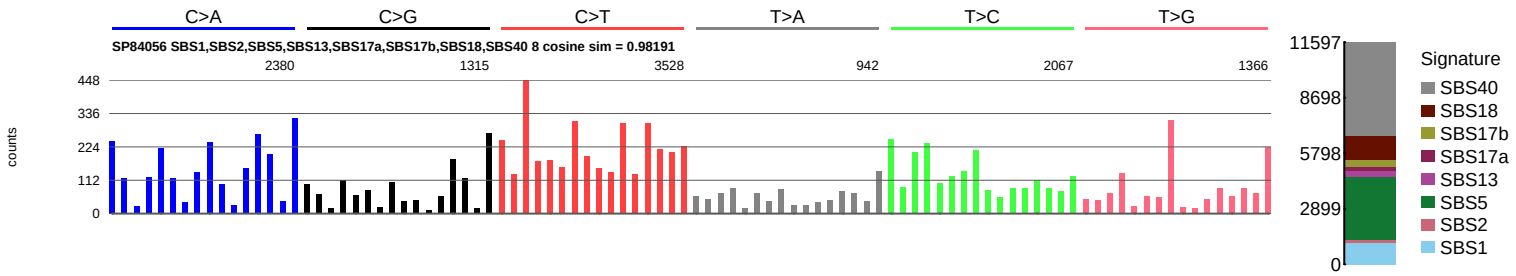

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98195
2327
1136
3552
1116
1991
1478
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

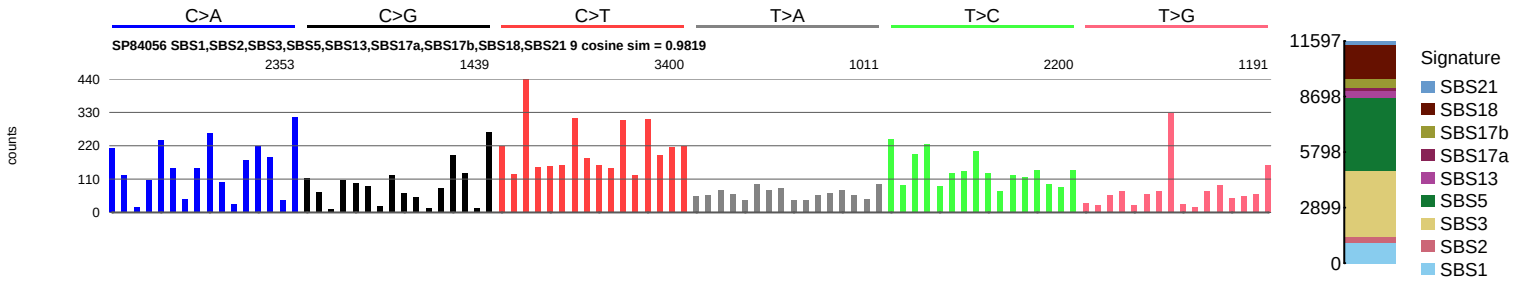

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98194
2401
1459
3283
1276
1715
1461
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

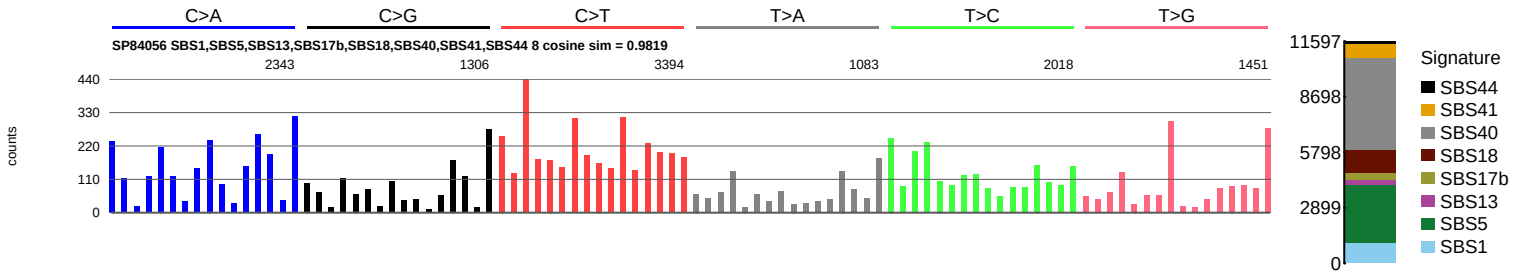

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98194
2337
1121
3458
1099
2122
1462
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

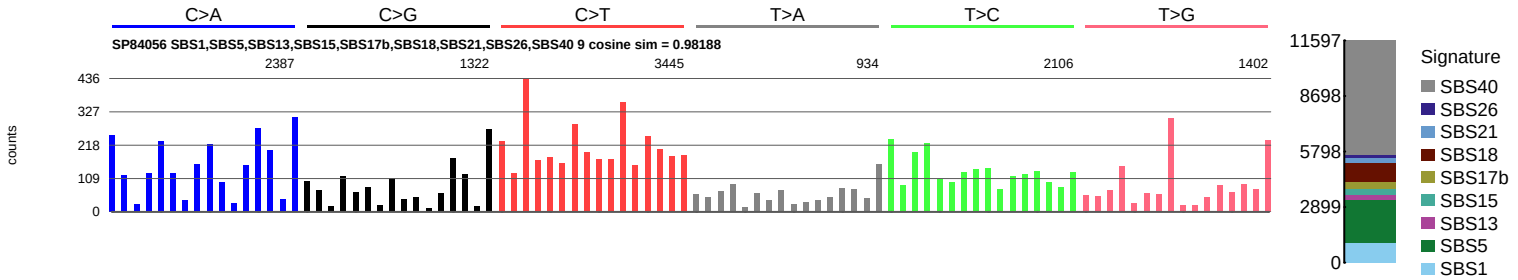

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98193
2377
1281
3130
1232
2128
1451
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

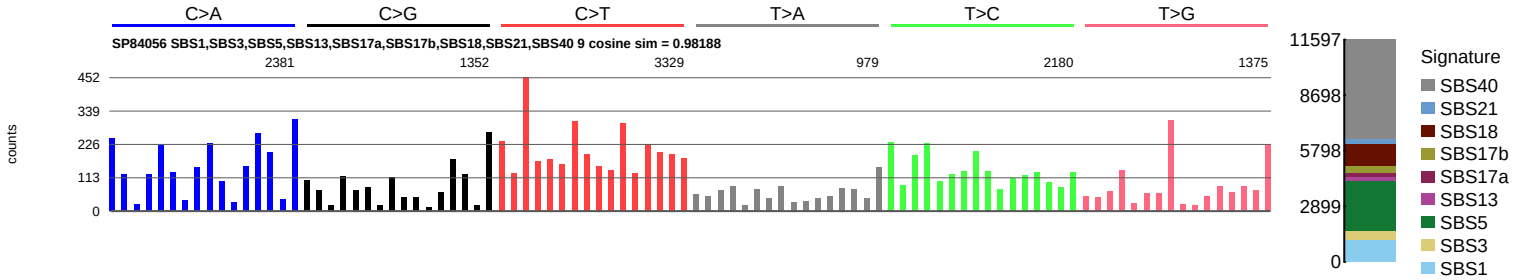

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98192
2245
1124
3535
1022
2399
1270
counts
400
300
200
100
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

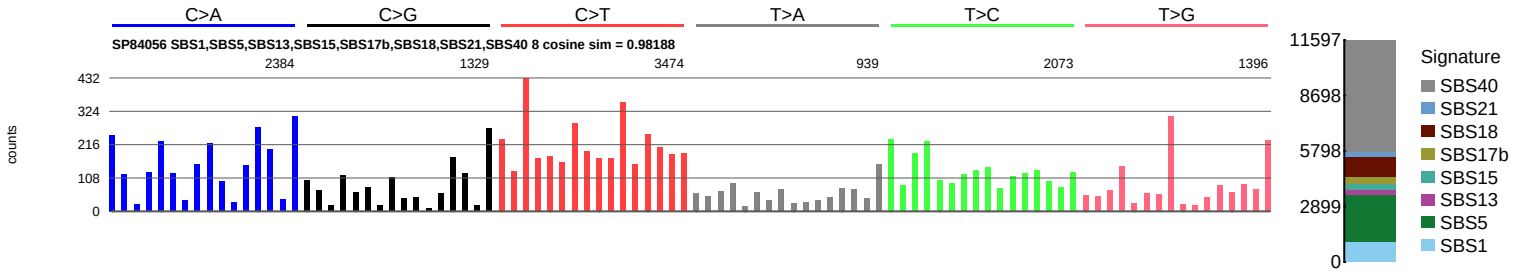

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98191
2380
1315
3528
942
2067
1366
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9819
2353
1439
3400
1011
2200
1191
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9819
2343
1306
3394
1083
2018
1451
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98188
2387
1322
3445
934
2106
1402
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98188
2381
1352
3329
979
2180
1375
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98188
2384
1329
3474
939
2073
1396
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
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SBS17b
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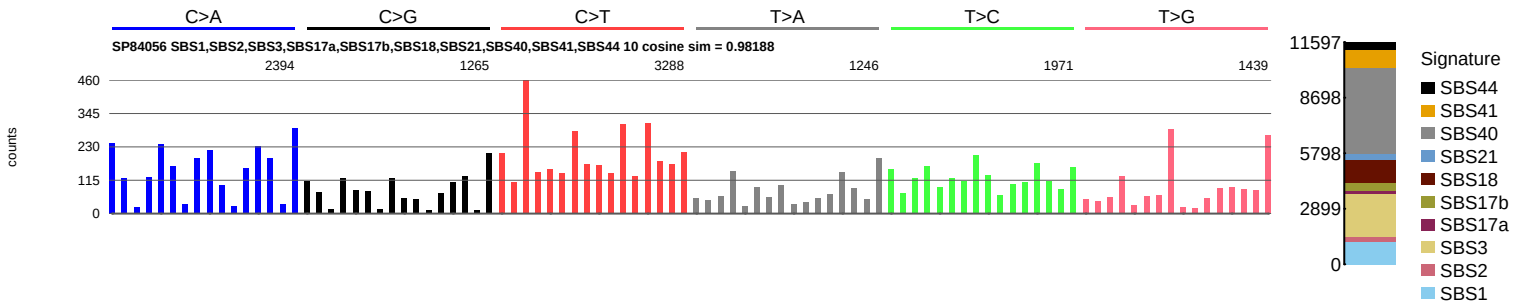
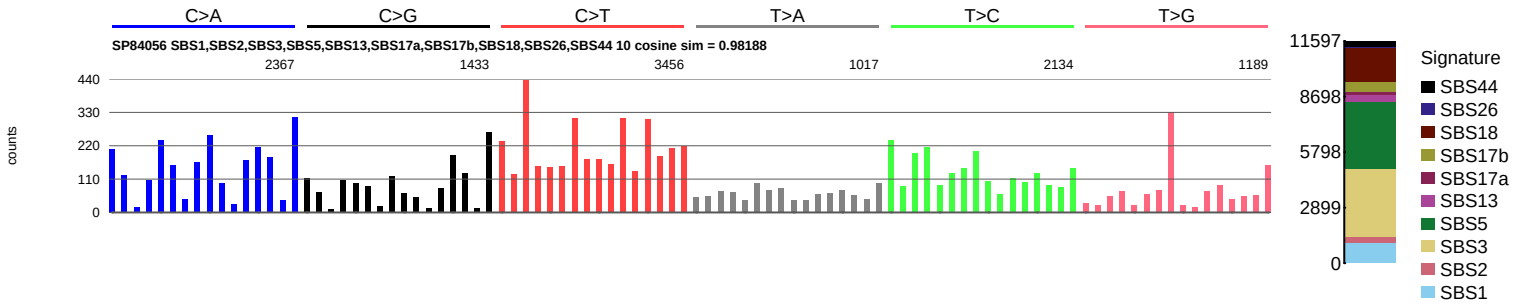
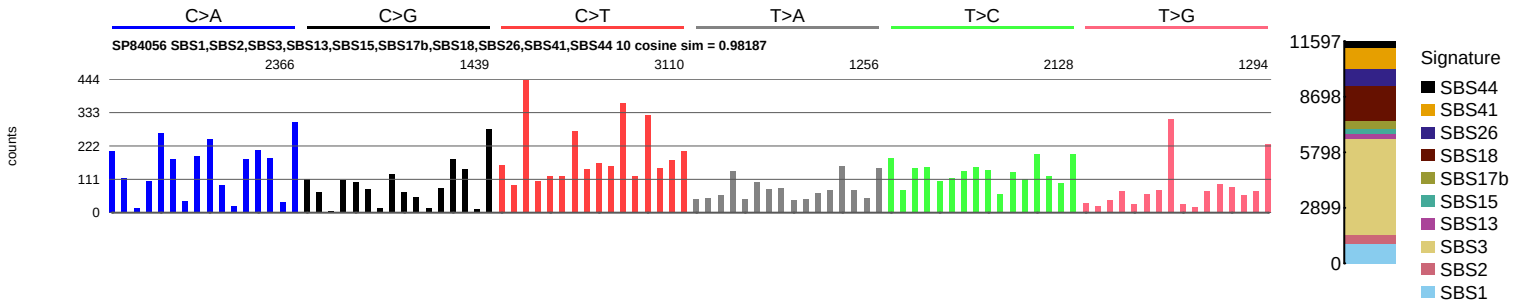
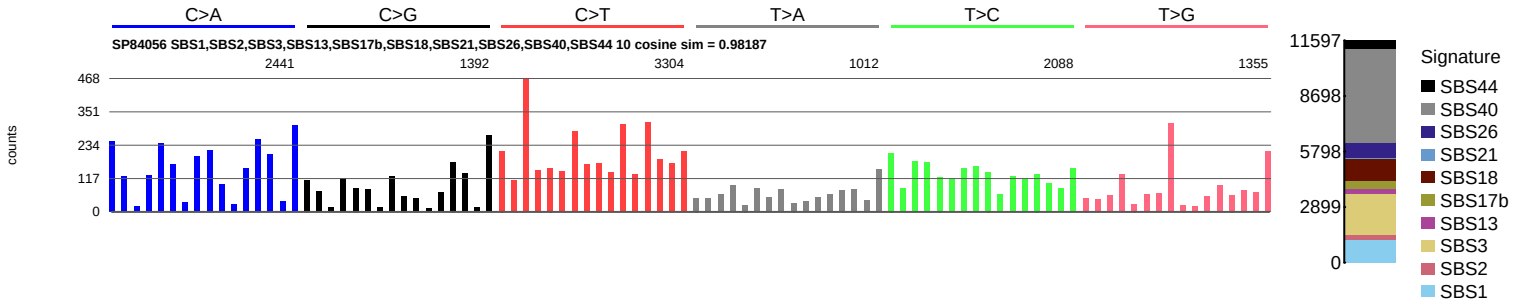
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98188
2394
1265
3288
1246
1971
1439
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98188
2367
1433
3456
1017
2134
1189
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98187
2366
1439
3110
1256
2128
1294
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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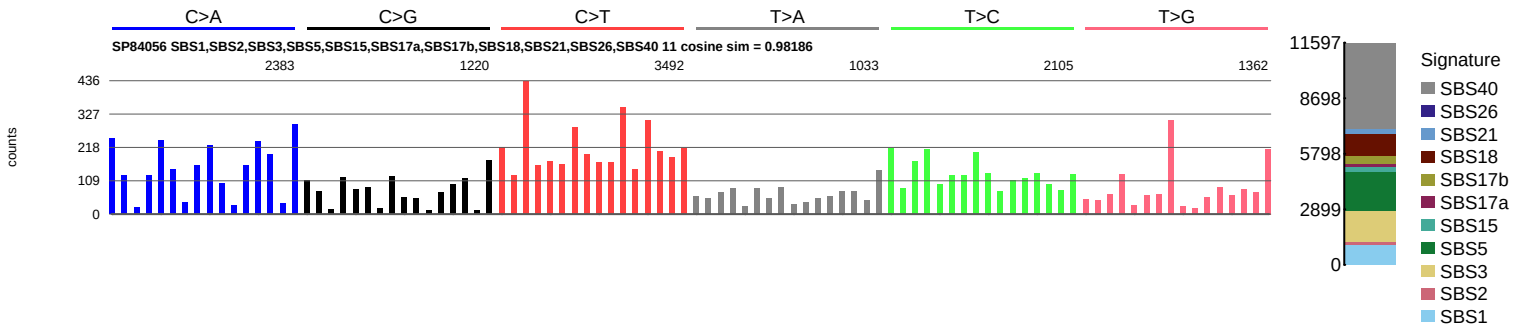
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98187
2441
1392
3304
1012
2088
1355
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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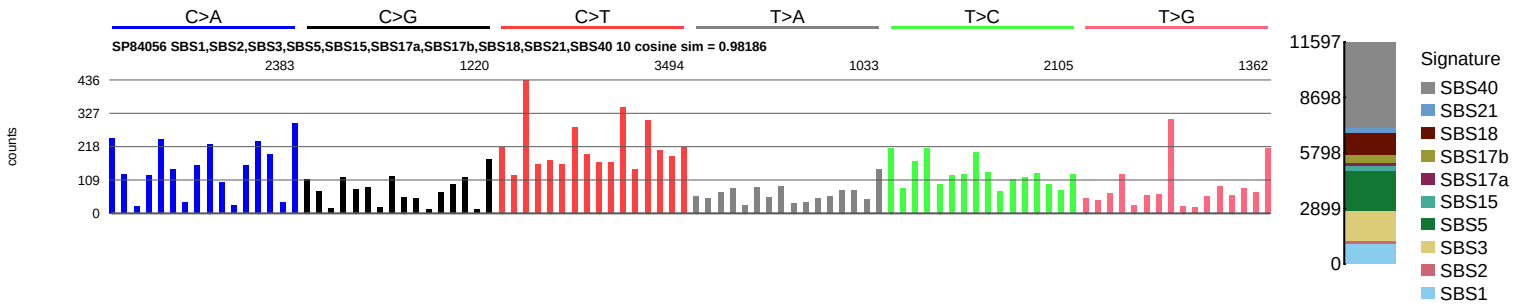
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98186
2383
1220
3492
1033
2105
1362
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>T
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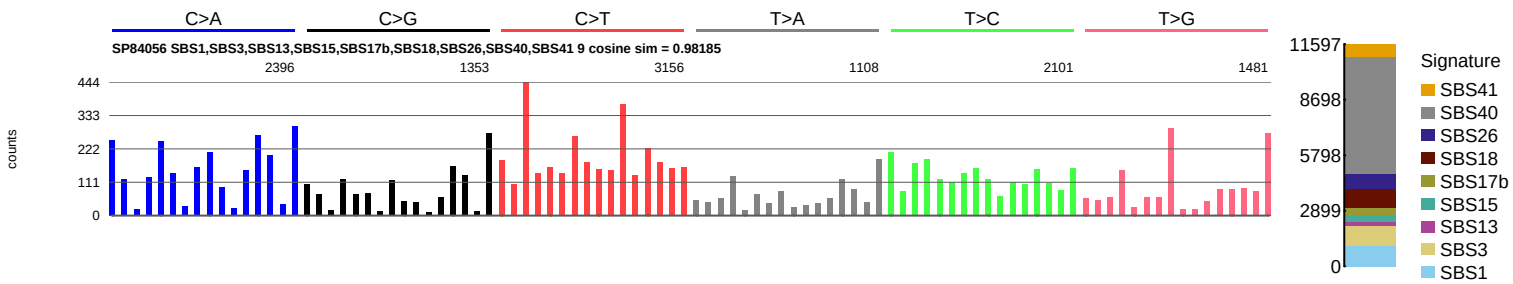
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98186
2383
1220
3494
1033
2105
1362
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
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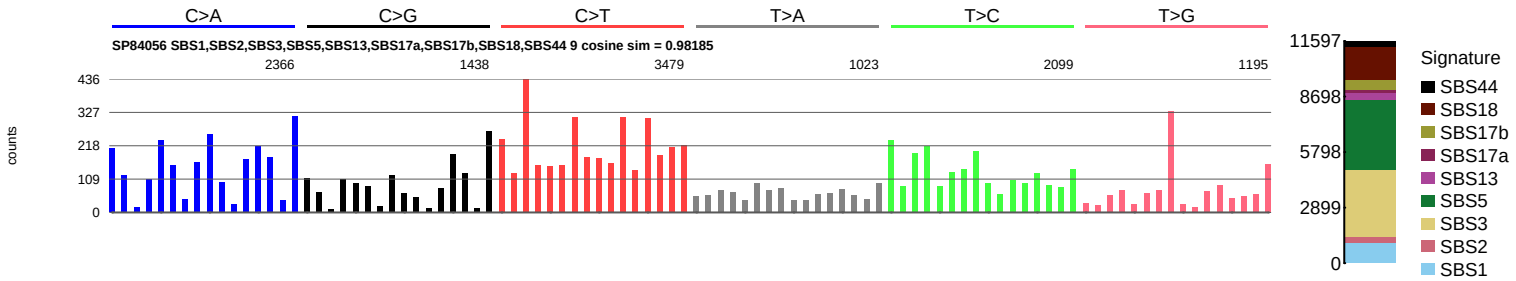
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98185
2396
1353
3156
1108
2101
1481
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS1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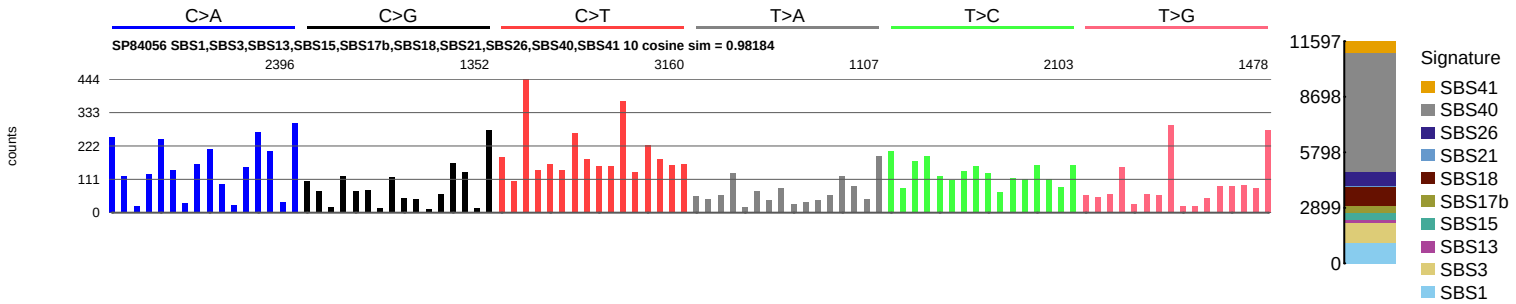
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98185
2366
1438
3479
1023
2099
1195
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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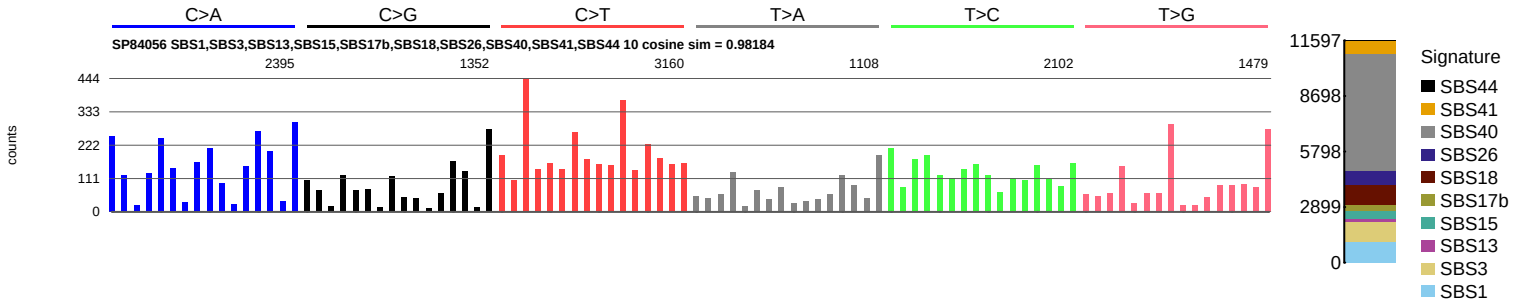
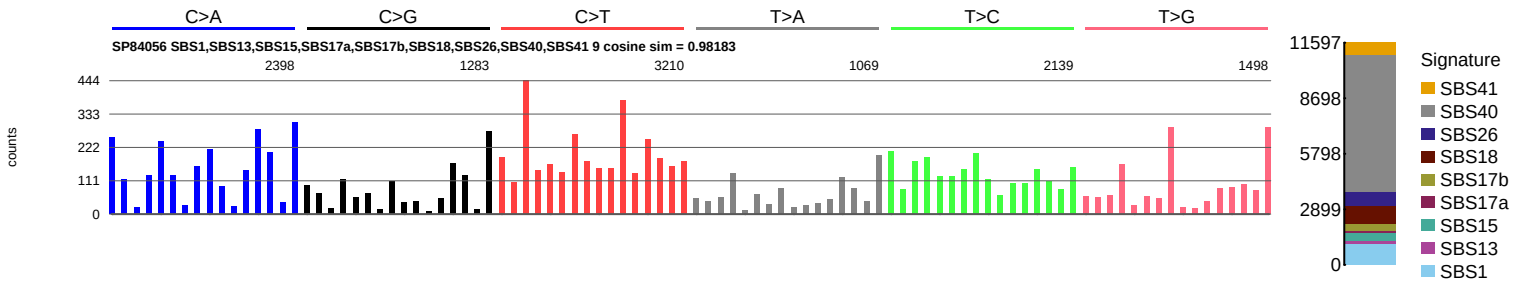
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98184
2396
1352
3160
1107
2103
1478
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98184
2395
1352
3160
1108
2102
1479
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS1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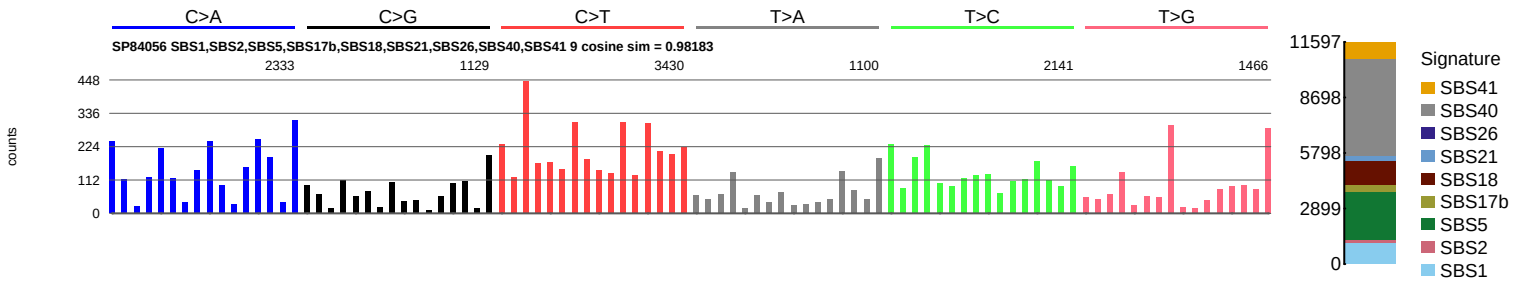
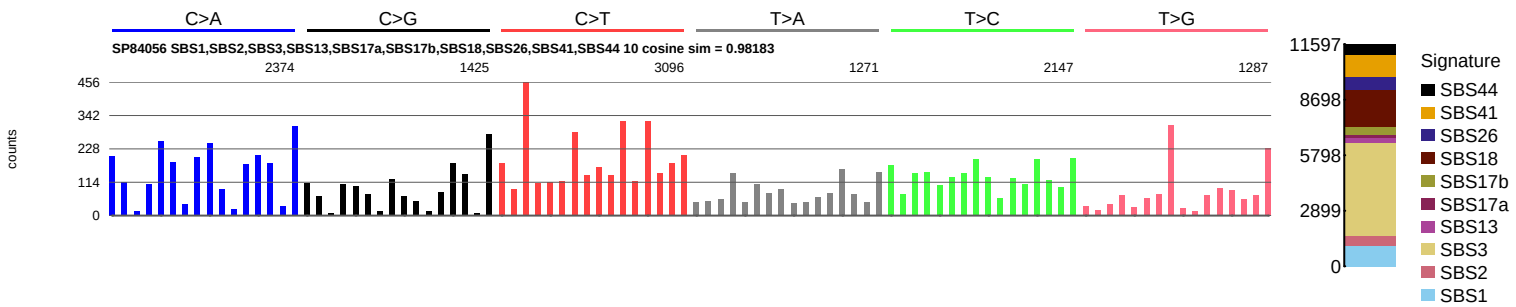
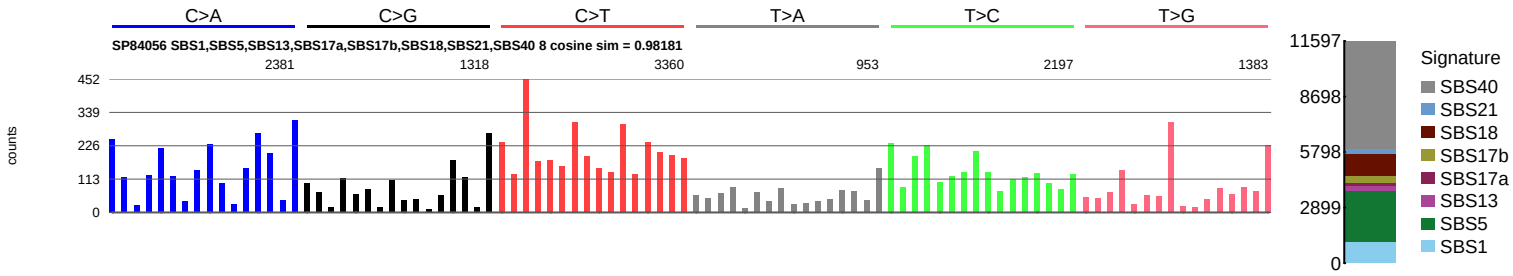
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98183
2398
1283
3210
1069
2139
1498
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98183
2333
1129
3430
1100
2141
1466
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98183
2374
1425
3096
1271
2147
1287
456
342
228
114
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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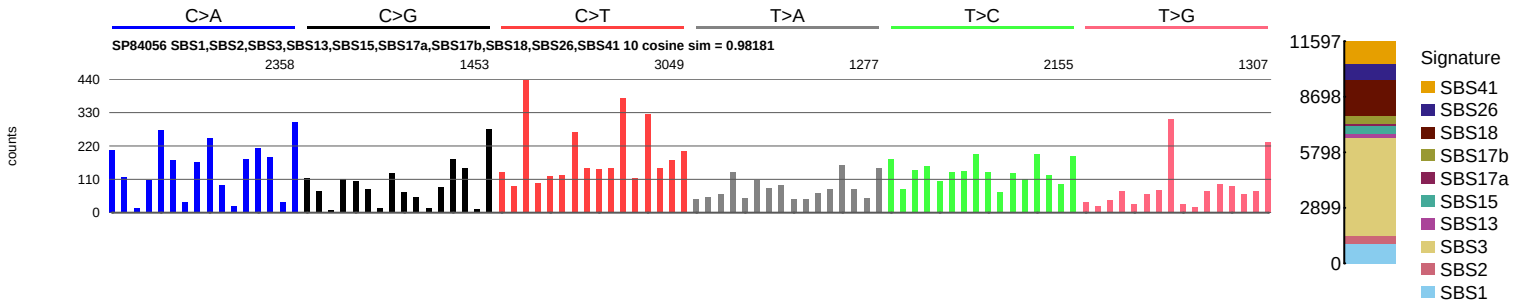
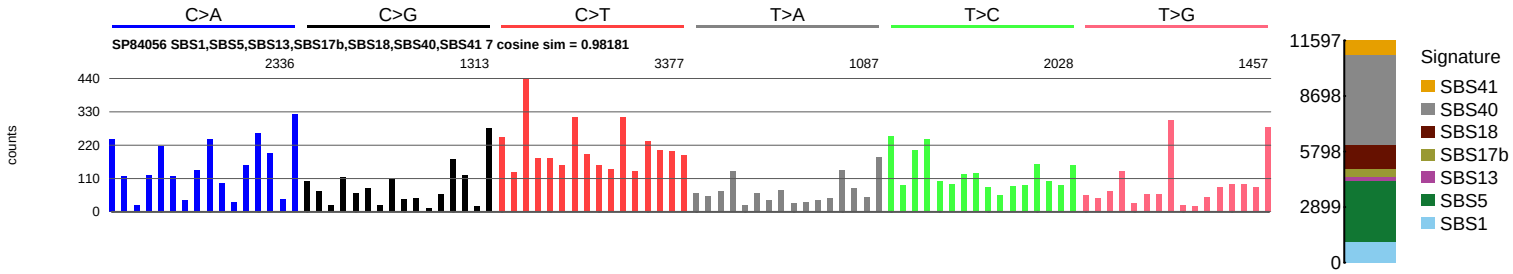
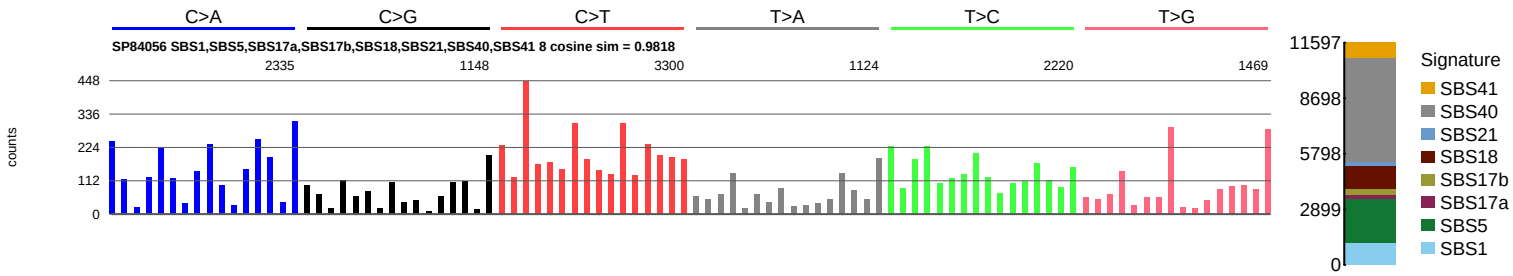
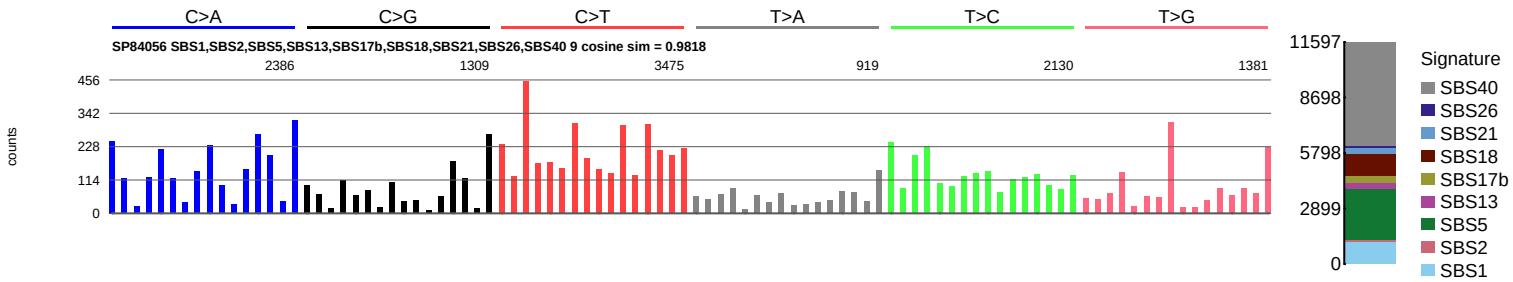
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98181
2381
1318
3360
953
2197
1383
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98181
2358
1453
3049
1277
2155
1307
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98181
2336
1313
3377
1087
2028
1457
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9818
2335
1148
3300
1124
2220
1469
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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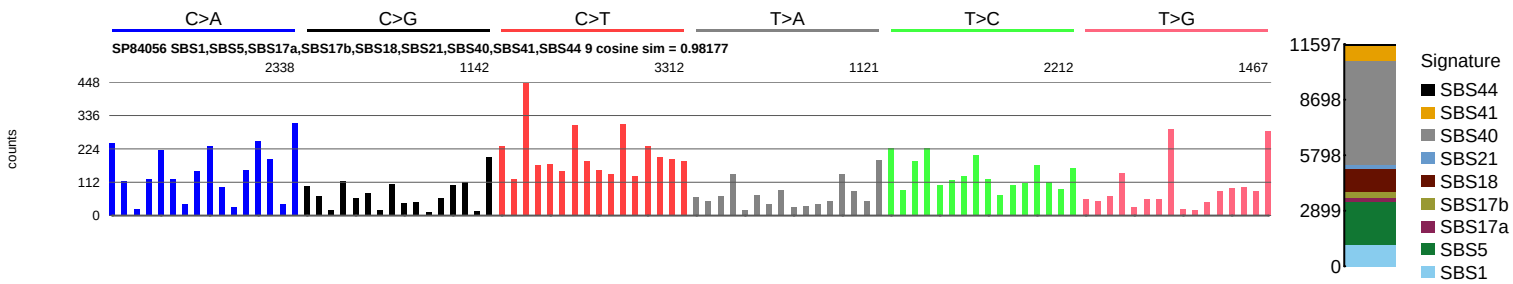
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.9818
2386
1309
3475
919
2130
1381
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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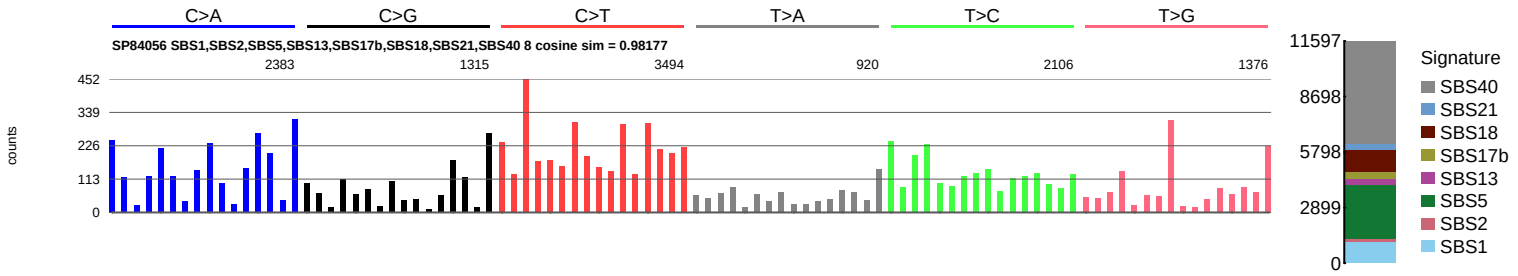
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98177
2338
1142
3312
1121
2212
1467
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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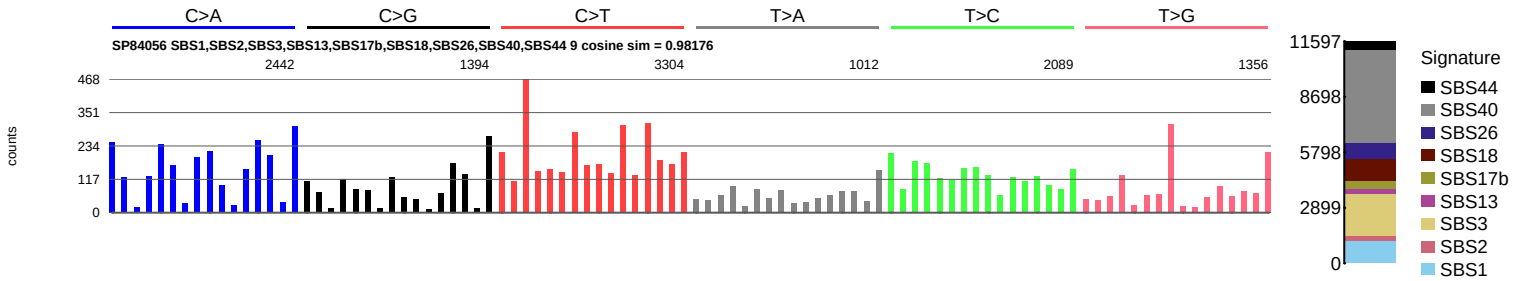
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98177
2383
1315
3494
920
2106
1376
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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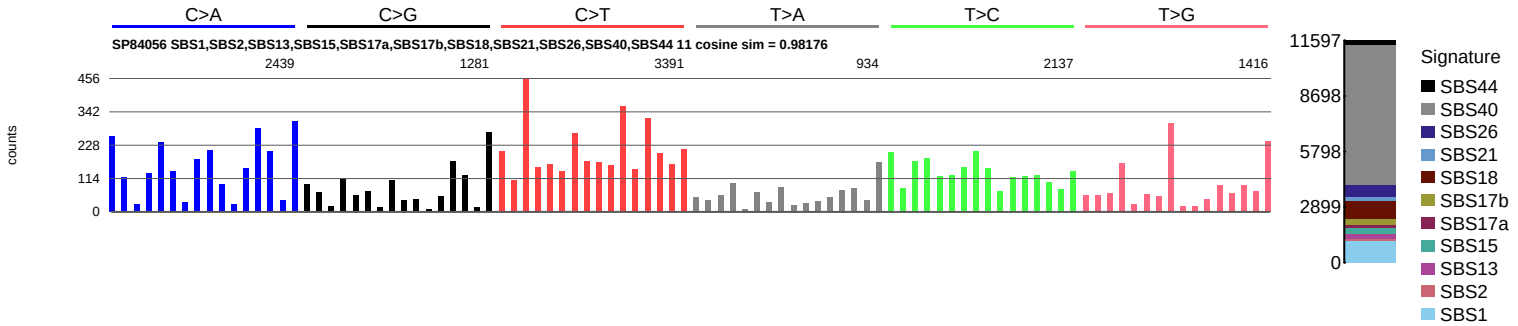
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98176
2442
1394
3304
1012
2089
1356
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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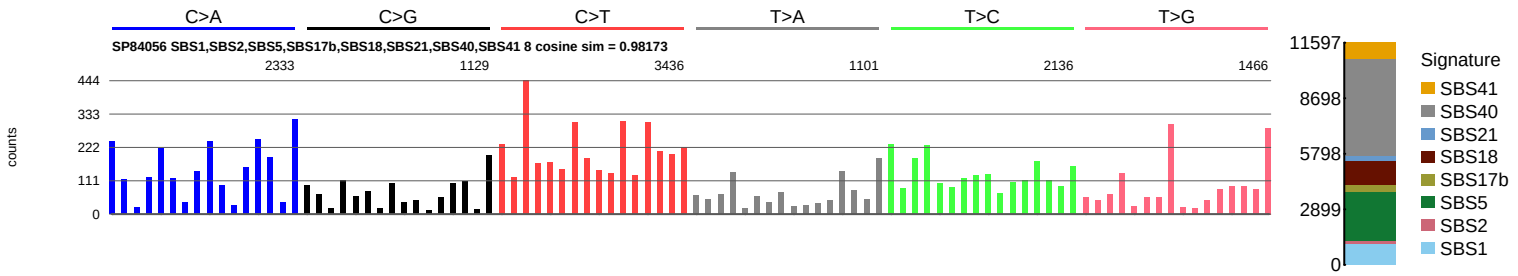
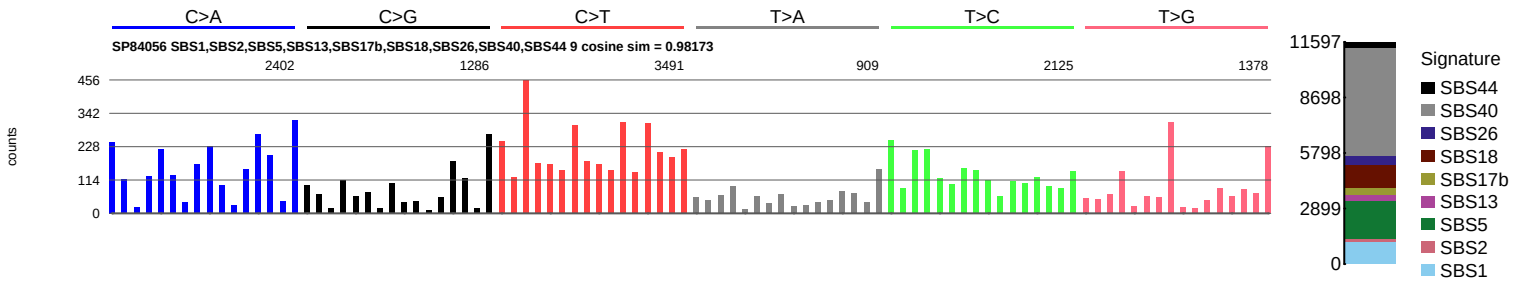
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98176
2439
1281
3391
934
2137
1416
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98173
2333
1129
3436
1101
2136
1466
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98173
2402
1286
3491
909
2125
1378
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
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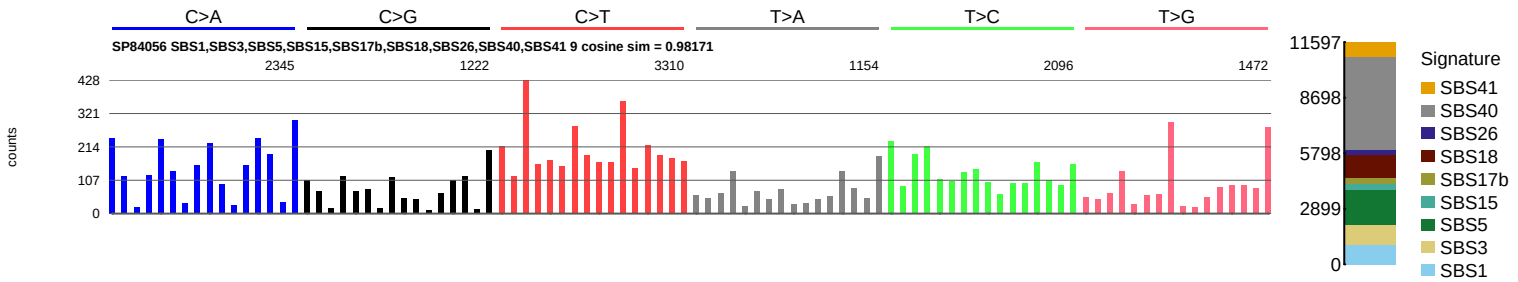
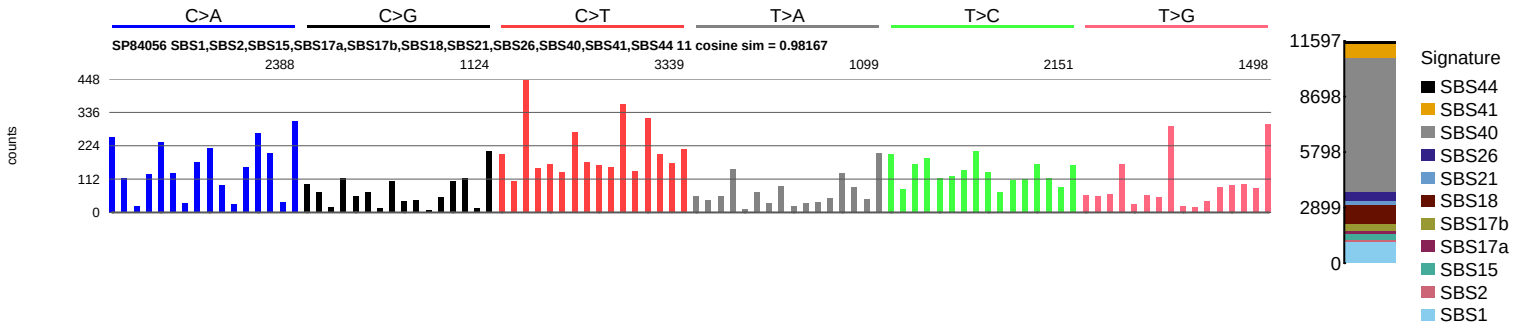
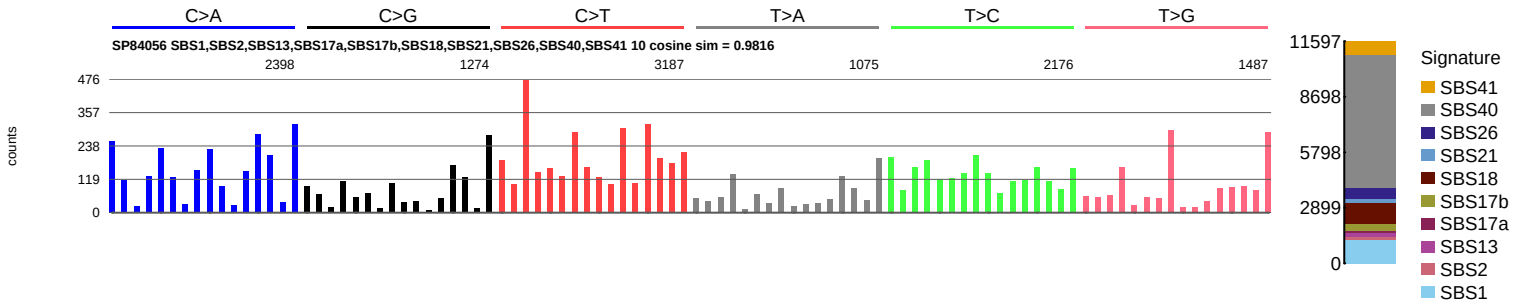
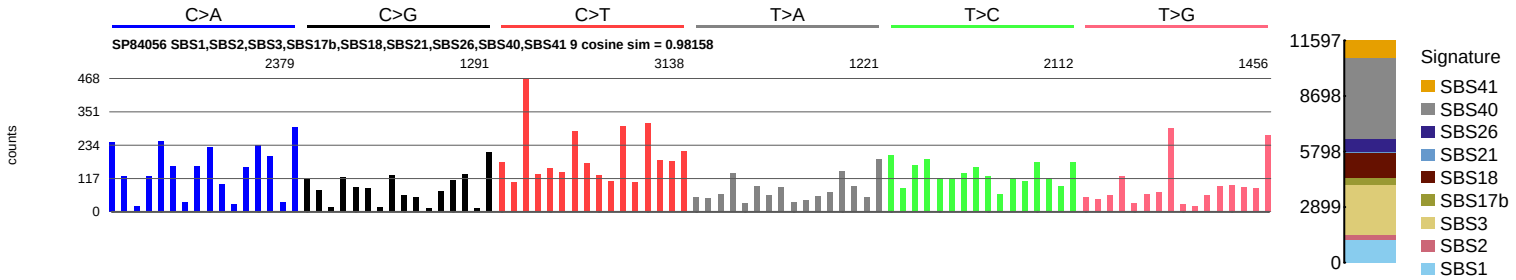
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98171
2345
1222
3310
1154
2096
1472
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98167
2388
1124
3339
1099
2151
1498
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9816
2398
1274
3187
1075
2176
1487
counts
476
357
238
119
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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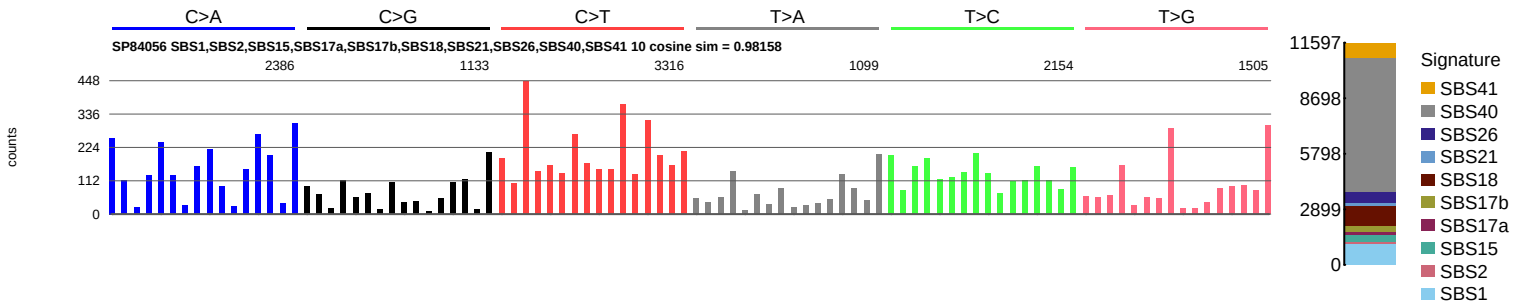
SP84056 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98158
2379
1291
3138
1221
2112
1456
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1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C>G
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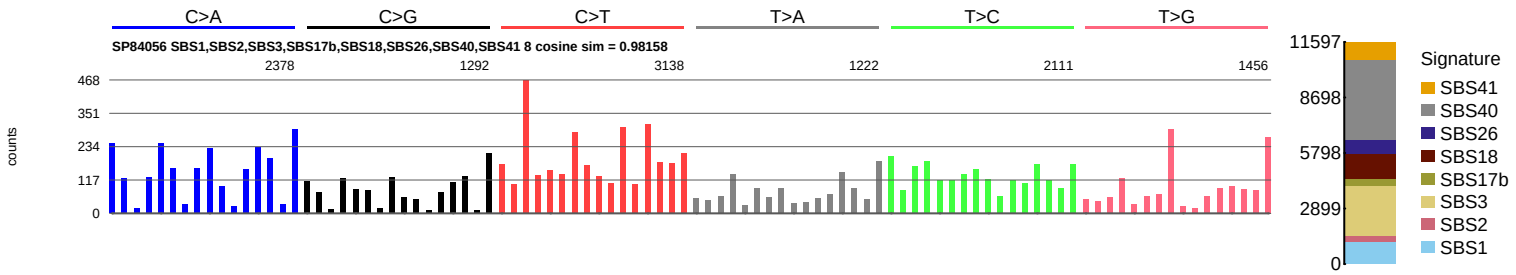
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98158
2386
1133
3316
1099
2154
1505
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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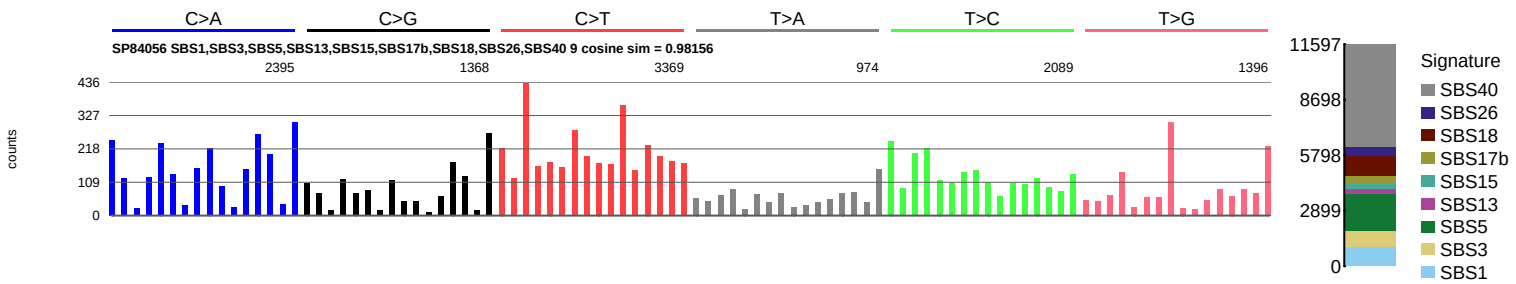
T>G
SP84056 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98158
2378
1292
3138
1222
2111
1456
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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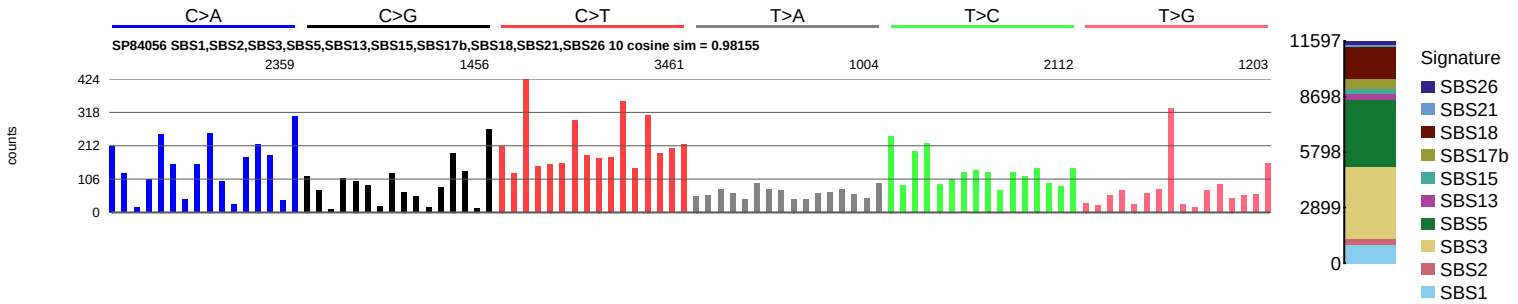
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98156
2395
1368
3369
974
2089
1396
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98155
2359
1456
3461
1004
2112
1203
counts
424
318
212
106
0
11597
8698
5798
2899
0
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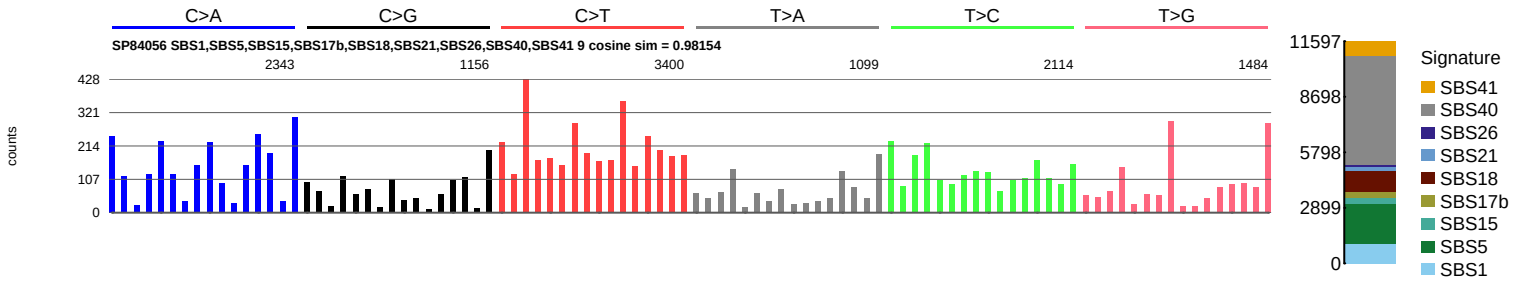
SBS2
SBS1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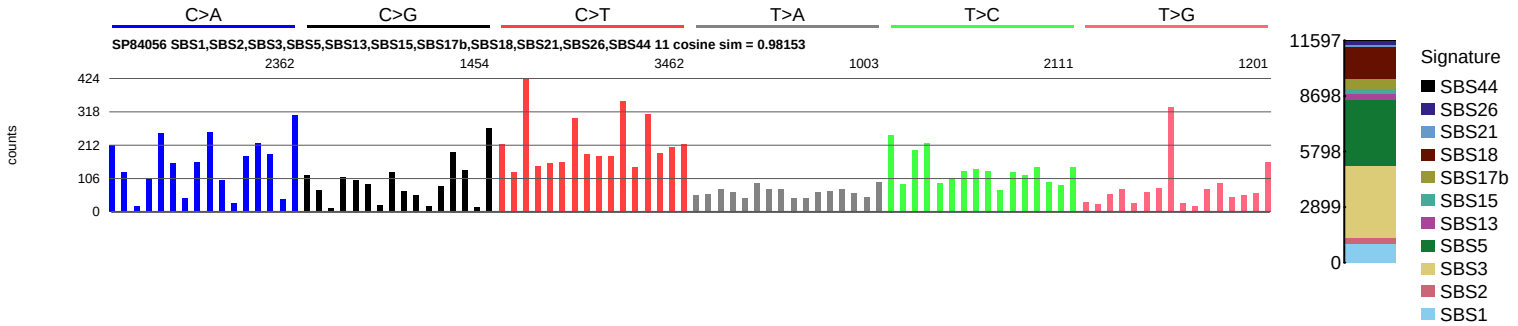
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98154
2343
1156
3400
1099
2114
1484
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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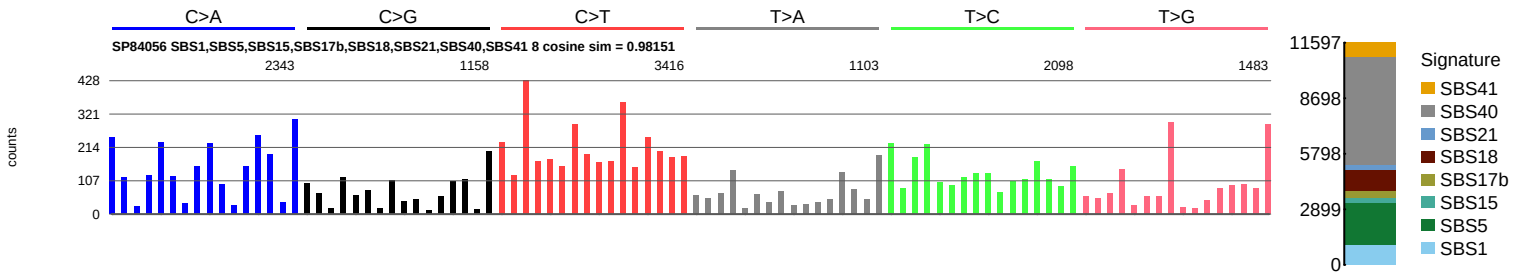
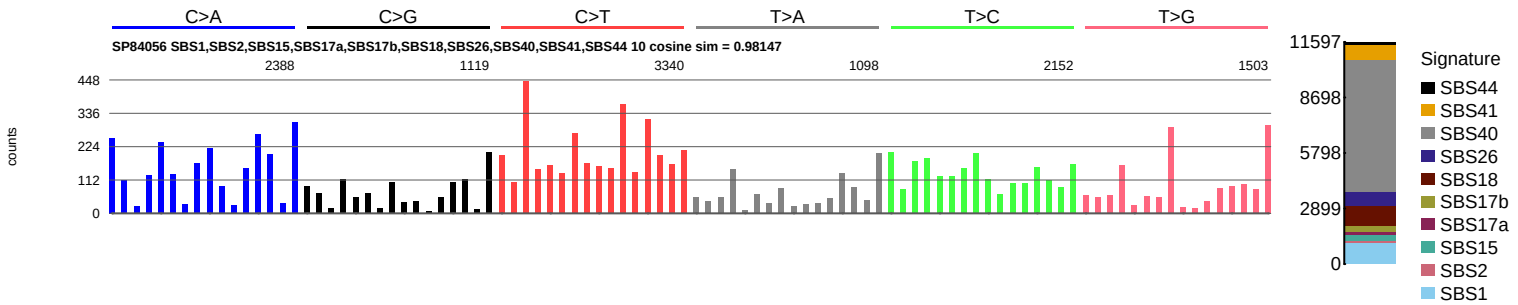
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98153
2362
1454
3462
1003
2111
1201
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98151
2343
1158
3416
1103
2098
1483
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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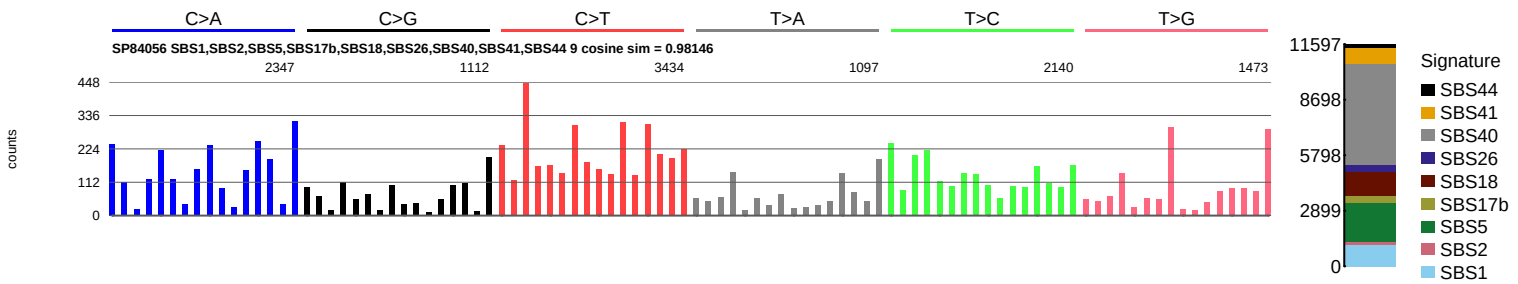
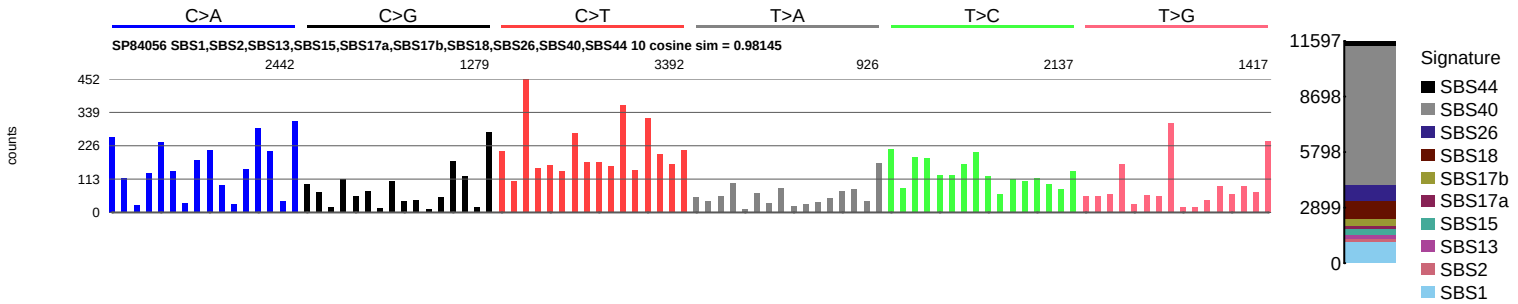
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98147
2388
1119
3340
1098
2152
1503
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98146
2347
1112
3434
1097
2140
1473
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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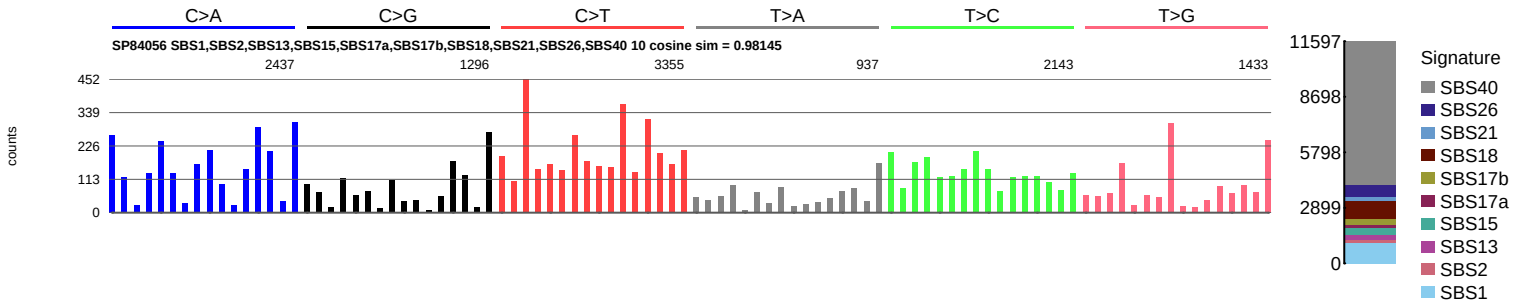
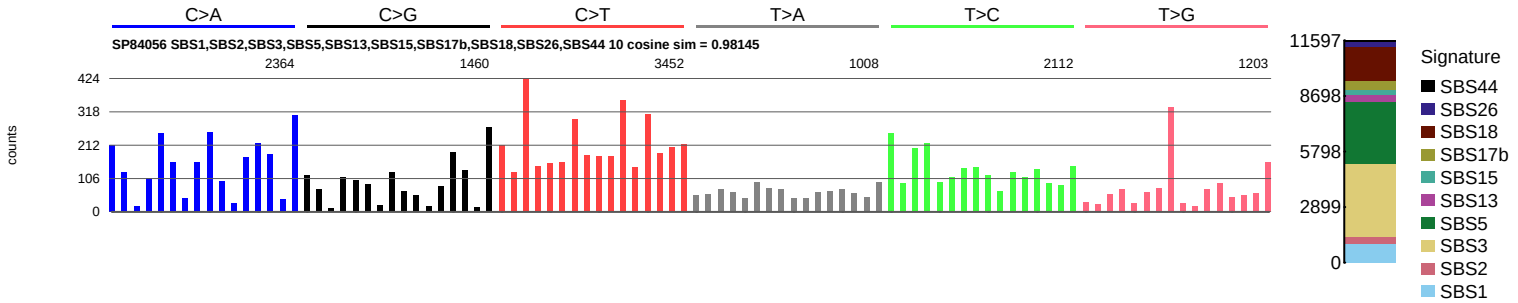
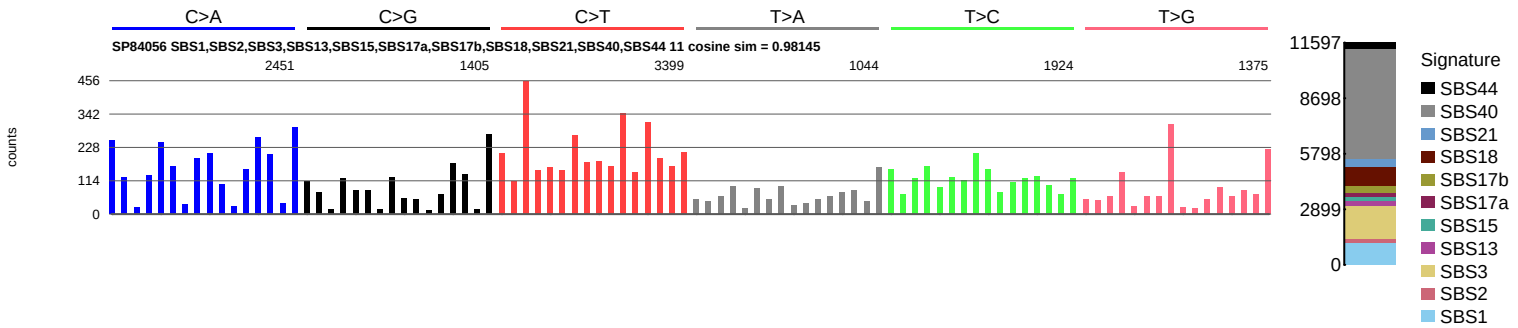
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98145
2442
1279
3392
926
2137
1417
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98145
2437
1296
3355
937
2143
1433
452
339
226
113
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98145
2364
1460
3452
1008
2112
1203
424
318
212
106
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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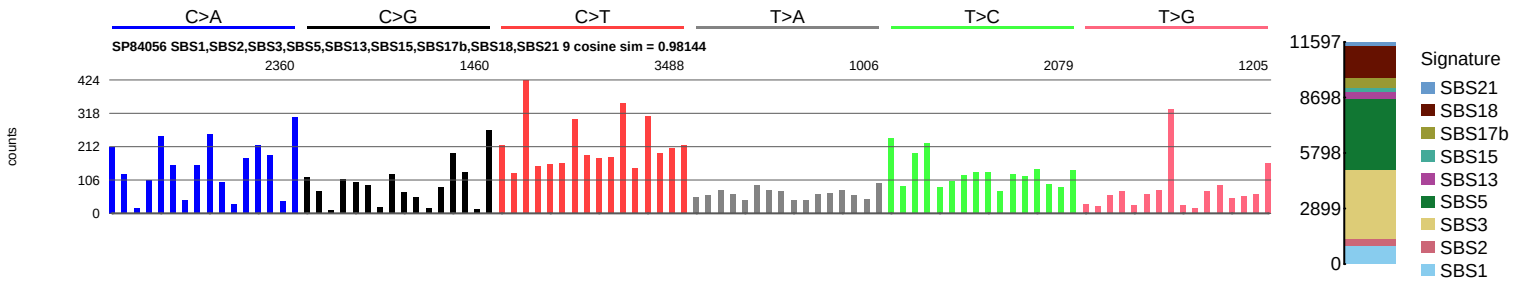
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98145
2451
1405
3399
1044
1924
1375
456
342
228
114
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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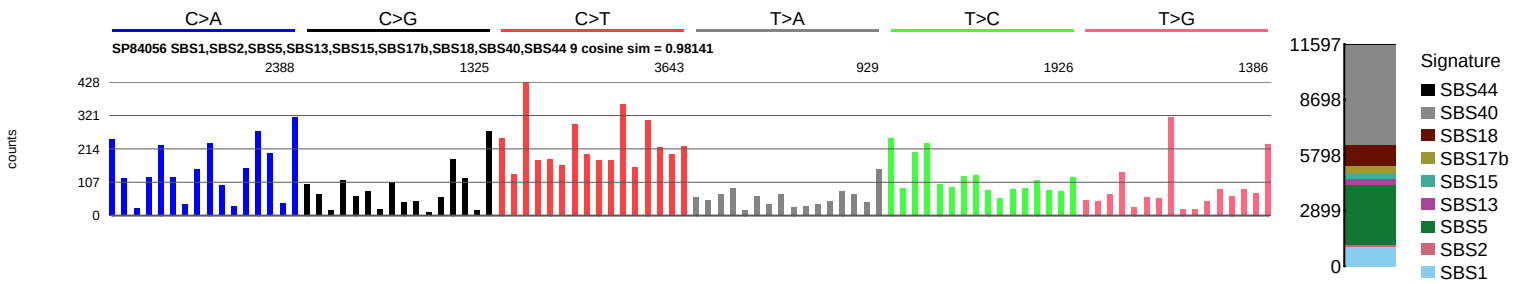
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98144
2360
1460
3488
1006
2079
1205
424
318
212
106
0
counts
11597
8698
5798
2899
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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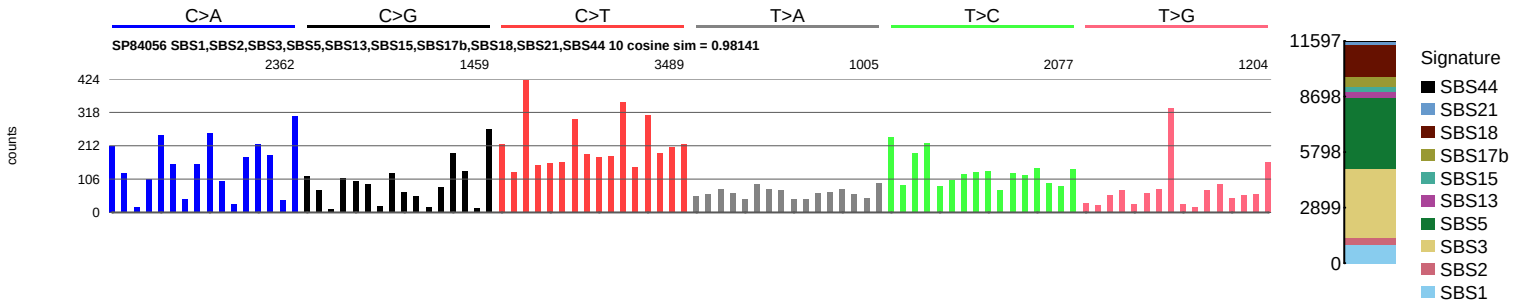
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98141
2388
1325
3643
929
1926
1386
428
321
214
107
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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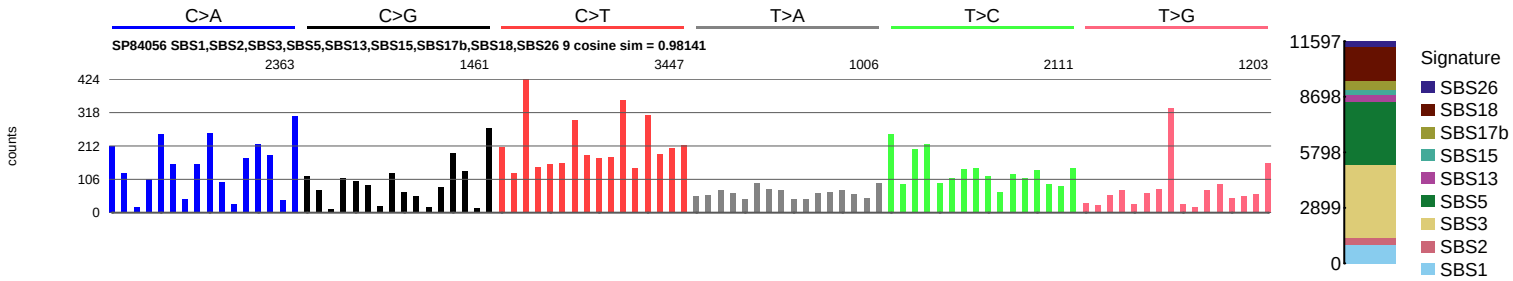
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98141
2362
1459
3489
1005
2077
1204
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 9 cosine sim = 0.98141
2363
1461
3447
1006
2111
1203
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
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SBS18
SBS17b
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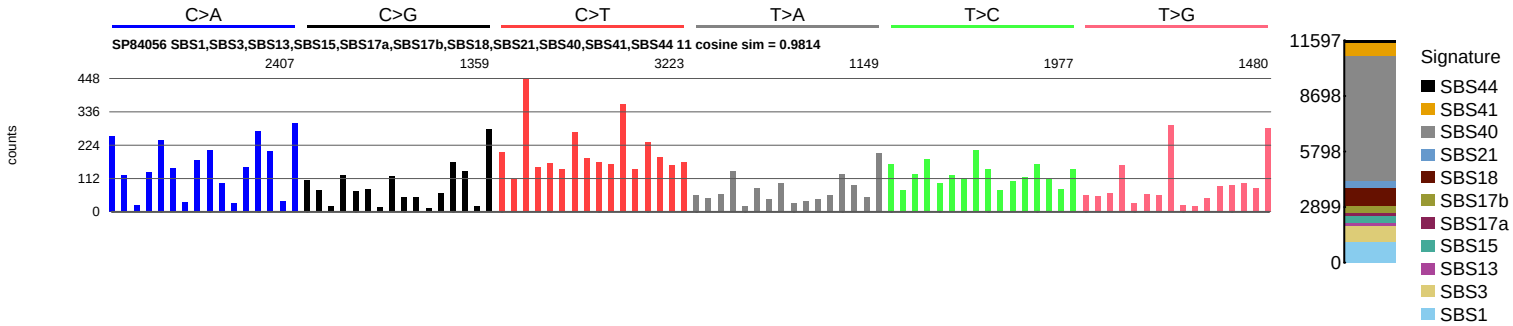
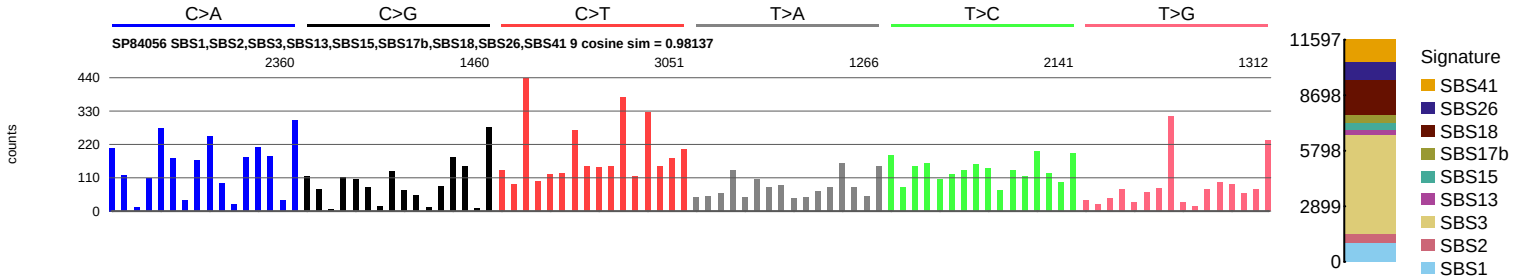
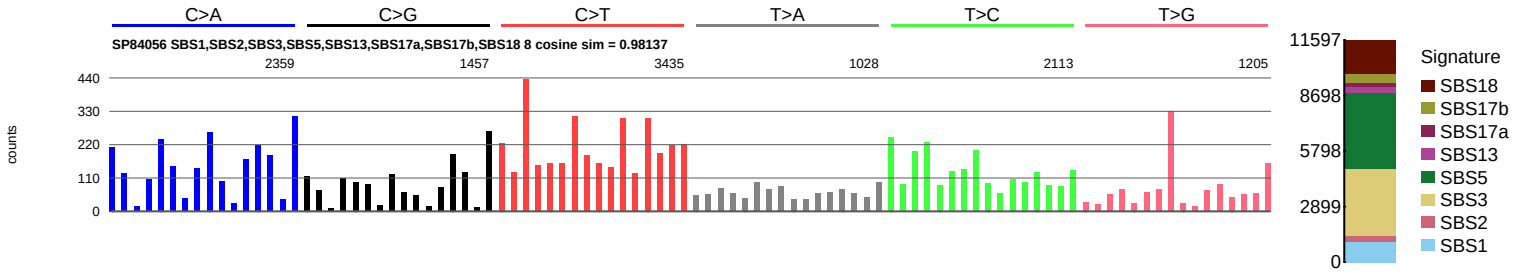
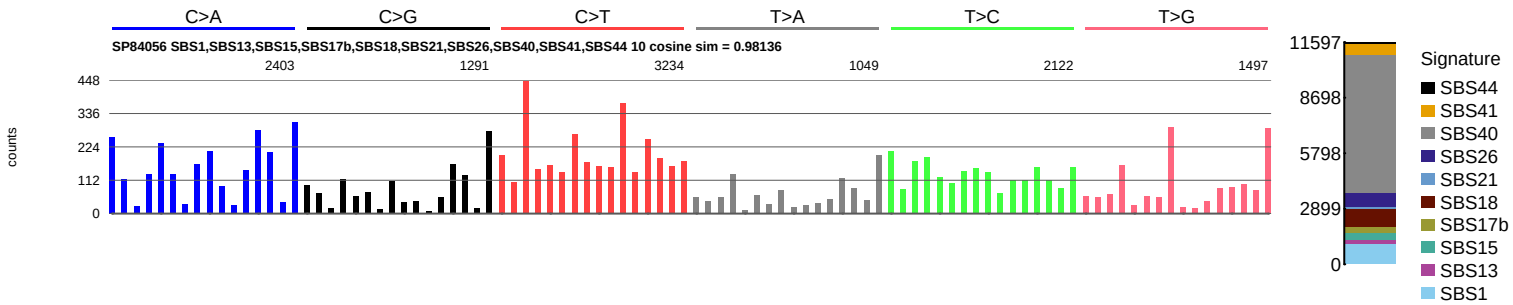
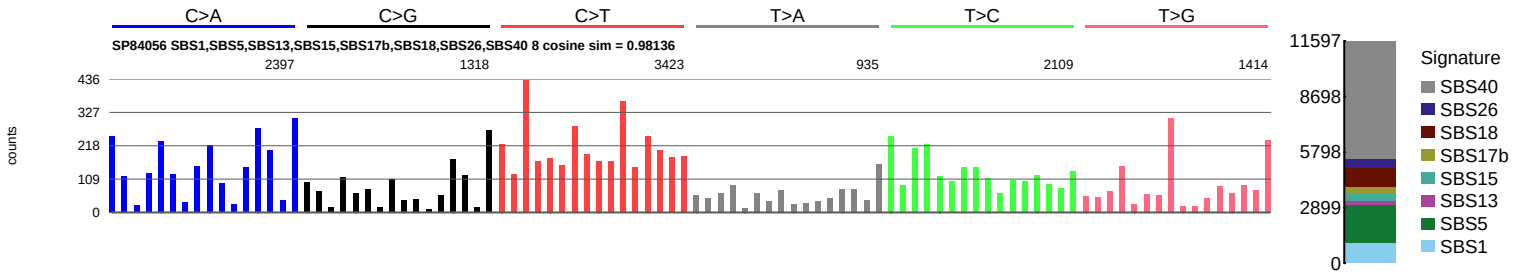
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9814
2407
1359
3223
1149
1977
1480
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98137
2360
1460
3051
1266
2141
1312
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98137
2359
1457
3435
1028
2113
1205
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98136
2403
1291
3234
1049
2122
1497
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98136
2397
1318
3423
935
2109
1414
436
327
218
109
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
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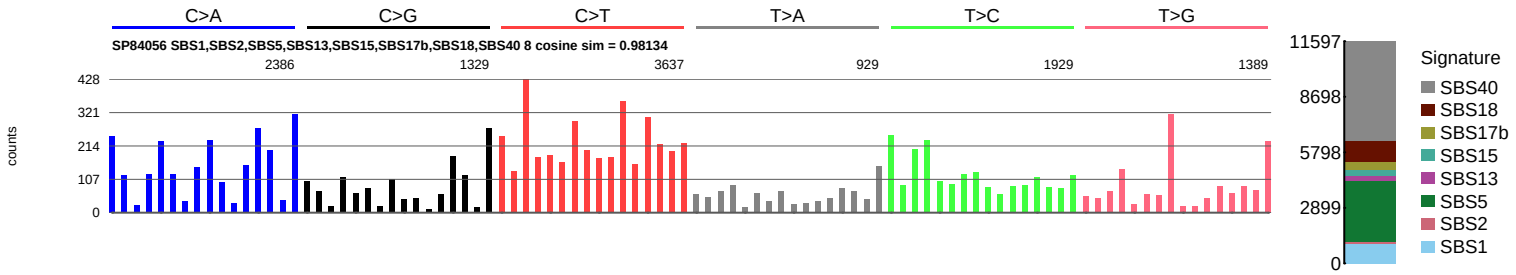
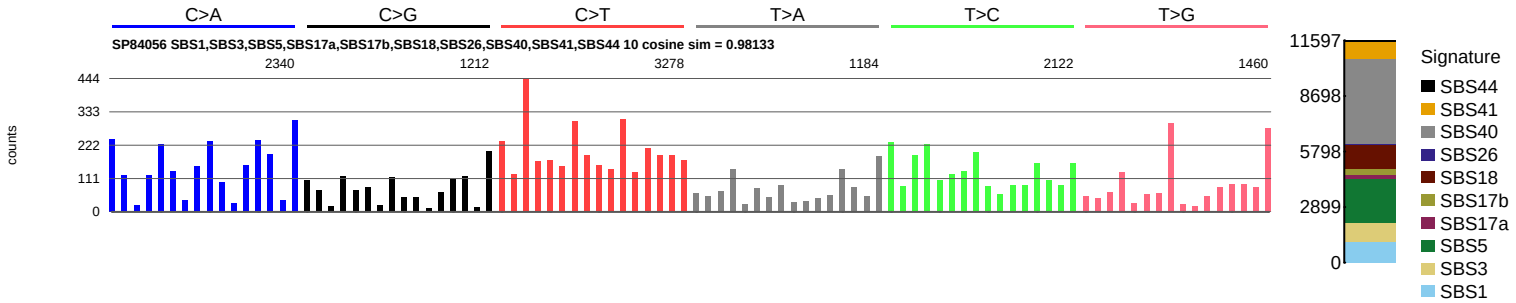
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98134
2386
1329
3637
929
1929
1389
428
321
214
107
0
counts
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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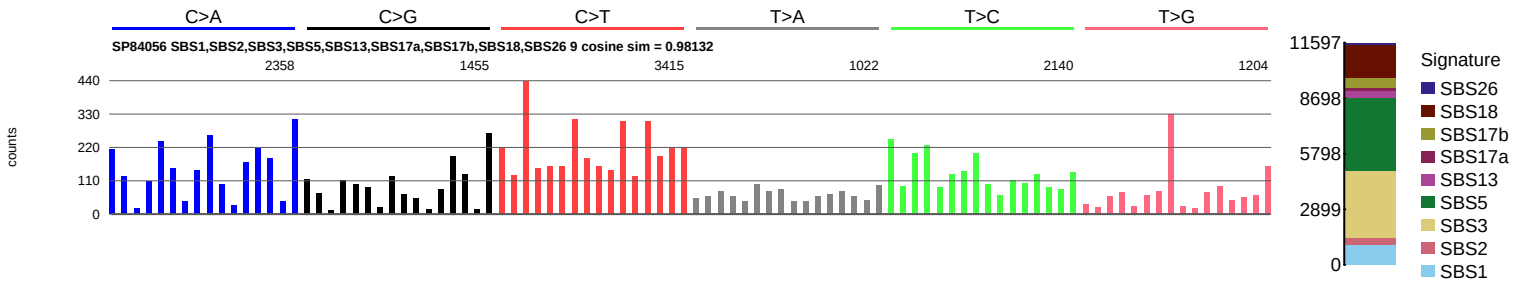
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98133
2340
1212
3278
1184
2122
1460
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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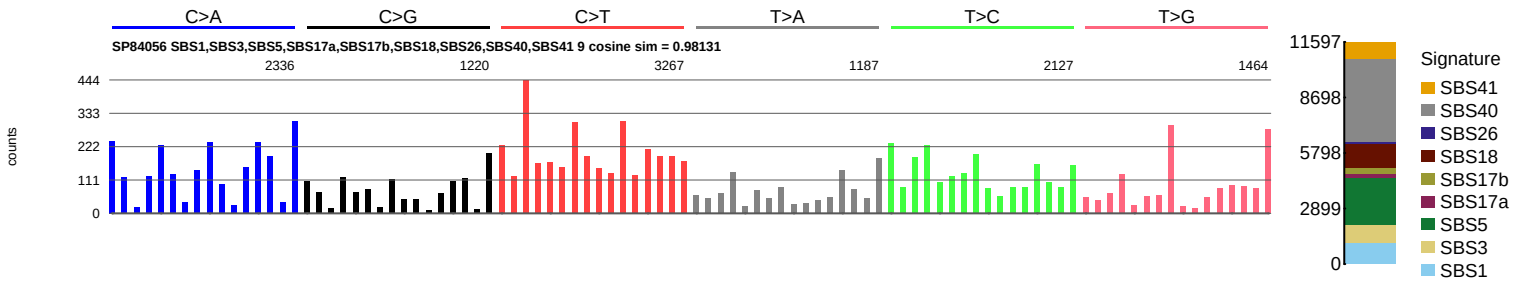
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98132
2358
1455
3415
1022
2140
1204
440
330
220
110
0
counts
11597
8698
5798
2899
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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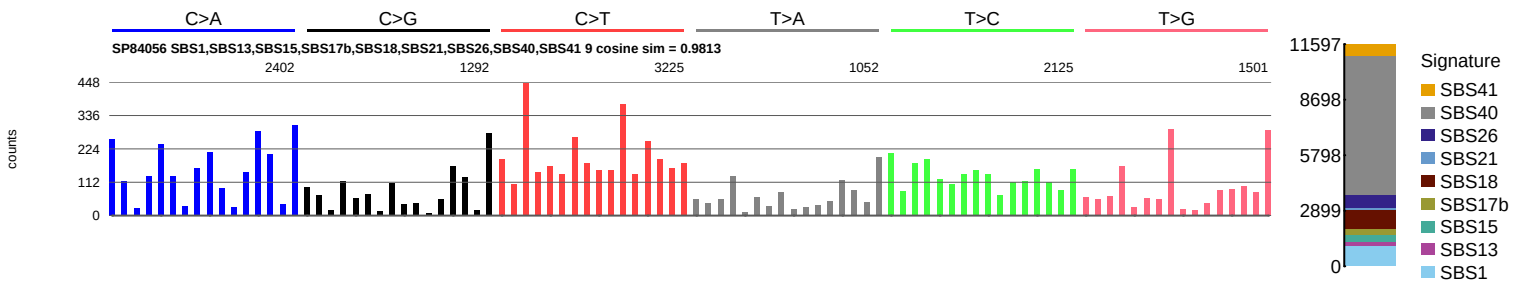
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98131
2336
1220
3267
1187
2127
1464
444
333
222
111
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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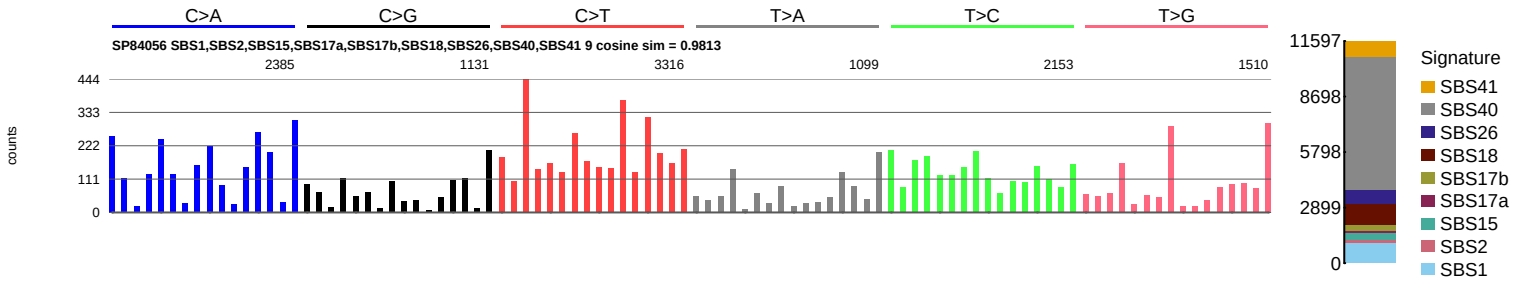
SP84056 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9813
2402
1292
3225
1052
2125
1501
448
336
224
112
0
counts
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS1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C>T
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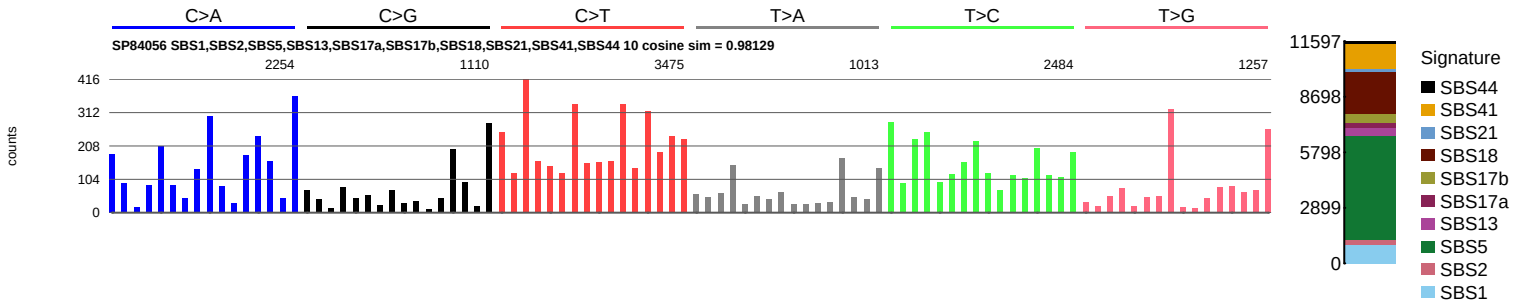
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9813
2385
1131
3316
1099
2153
1510
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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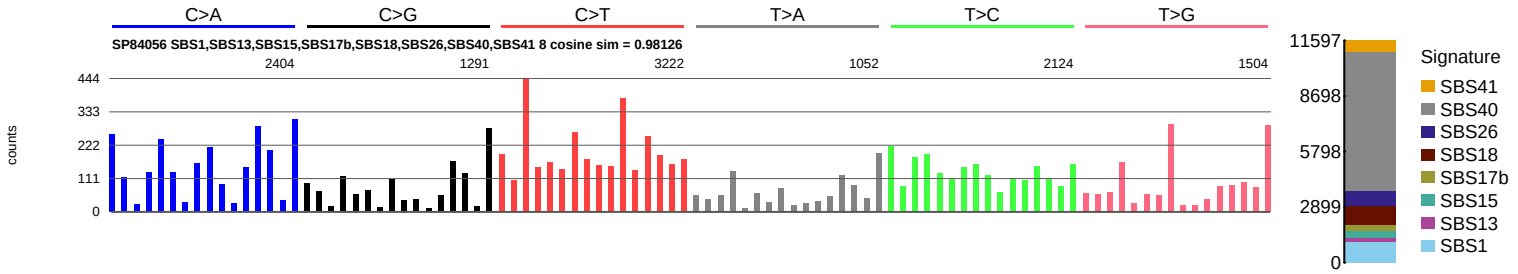
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98129
2254
1110
3475
1013
2484
1257
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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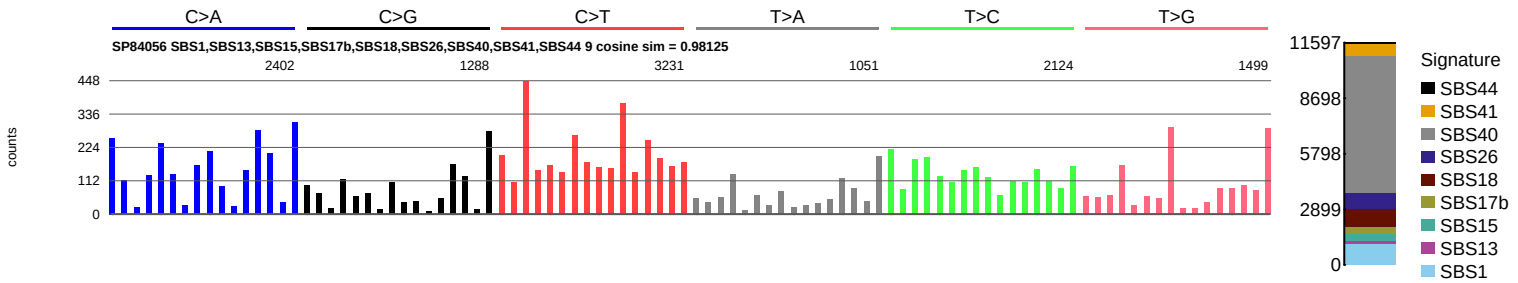
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98126
2404
1291
3222
1052
2124
1504
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS1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C>G
C>T
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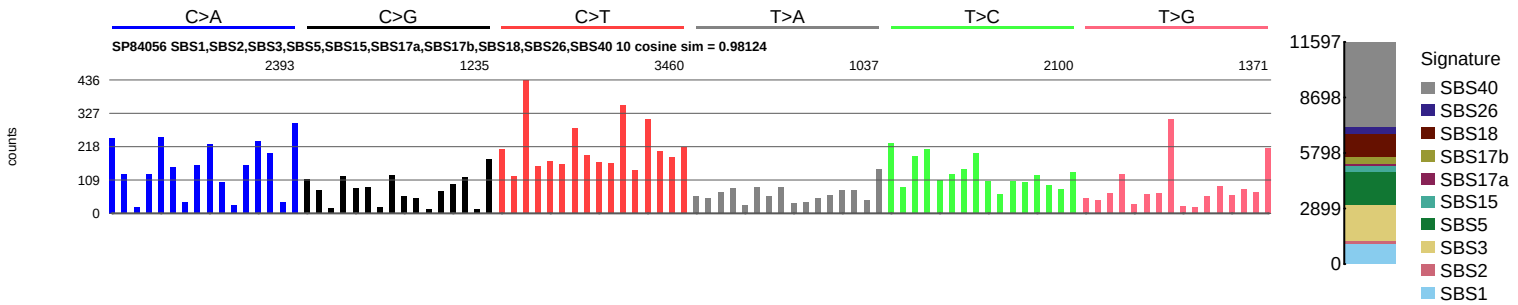
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98125
2402
1288
3231
1051
2124
1499
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS1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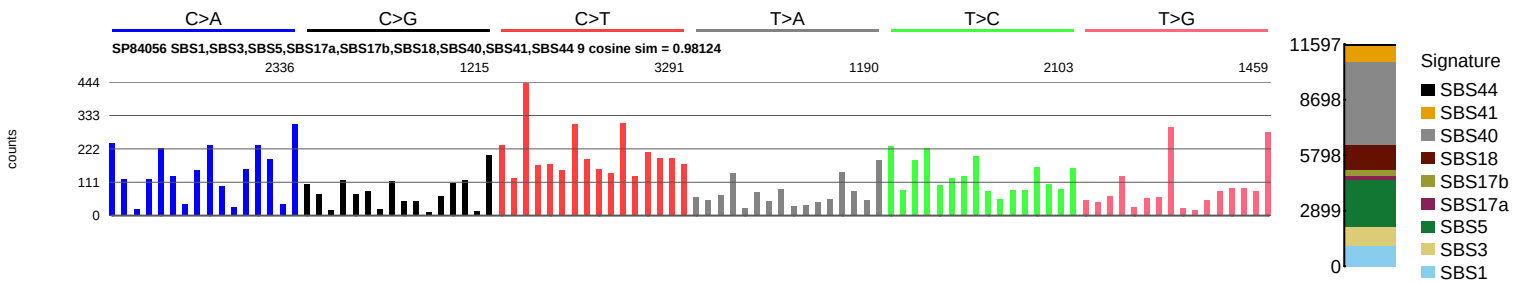
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98124
2393
1235
3460
1037
2100
1371
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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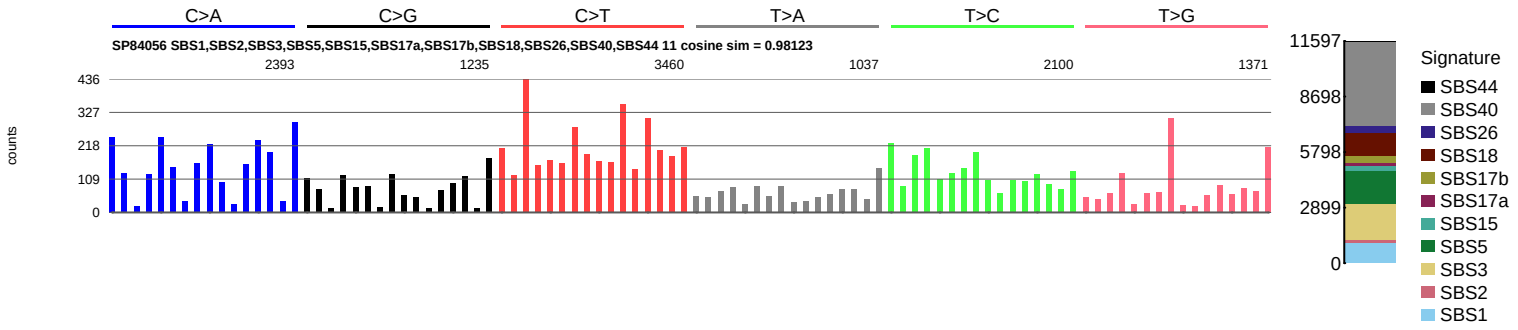
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98124
2336
1215
3291
1190
2103
1459
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
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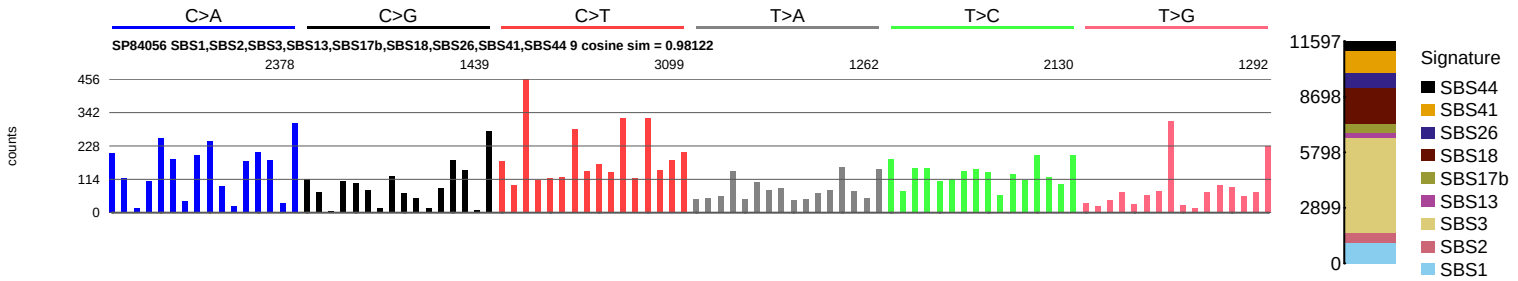
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98123
2393
1235
3460
1037
2100
1371
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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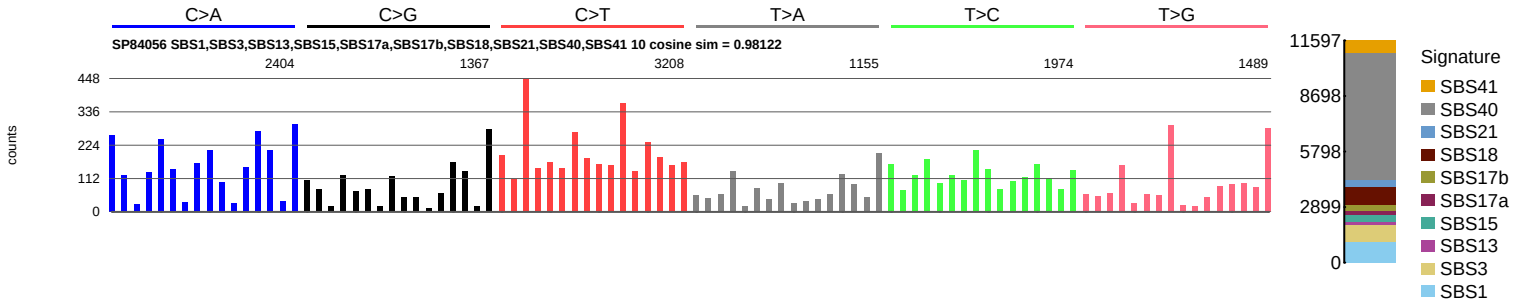
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98122
2378
1439
3099
1262
2130
1292
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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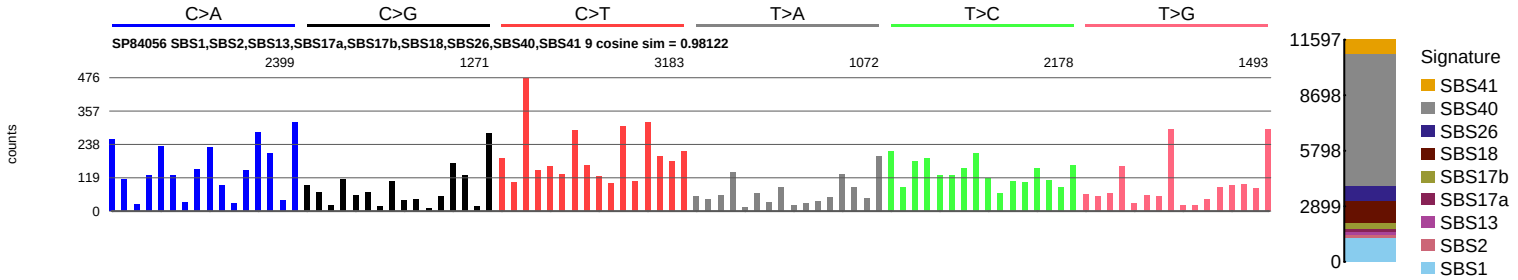
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98122
2404
1367
3208
1155
1974
1489
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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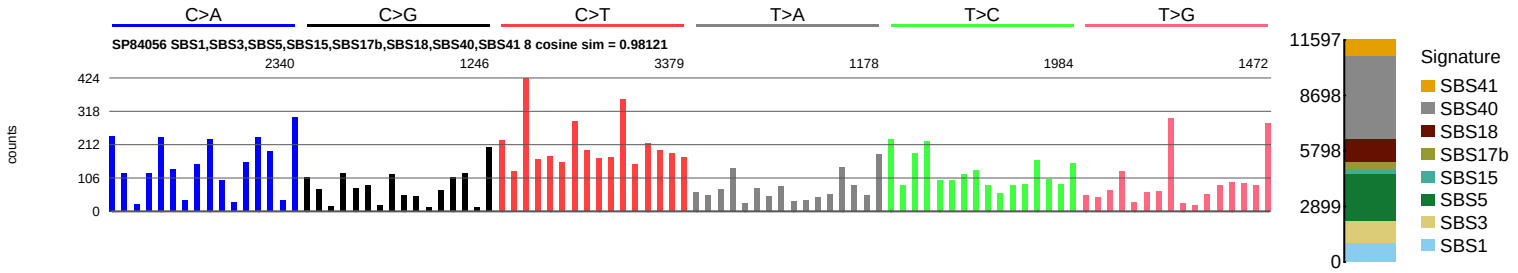
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98122
2399
1271
3183
1072
2178
1493
counts
476
357
238
119
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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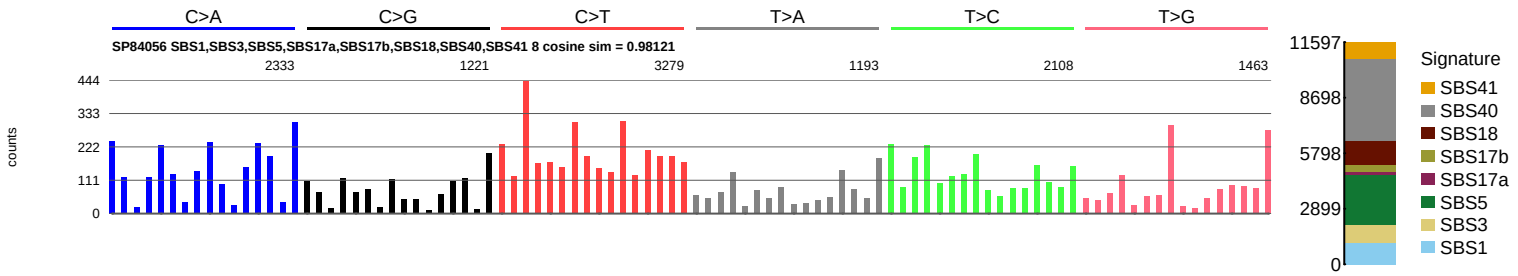
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98121
2340
1246
3379
1178
1984
1472
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1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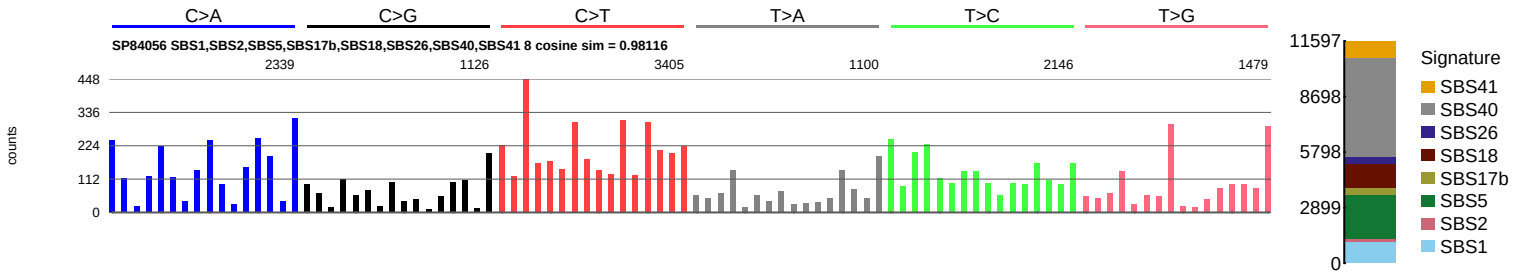
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98121
2333
1221
3279
1193
2108
1463
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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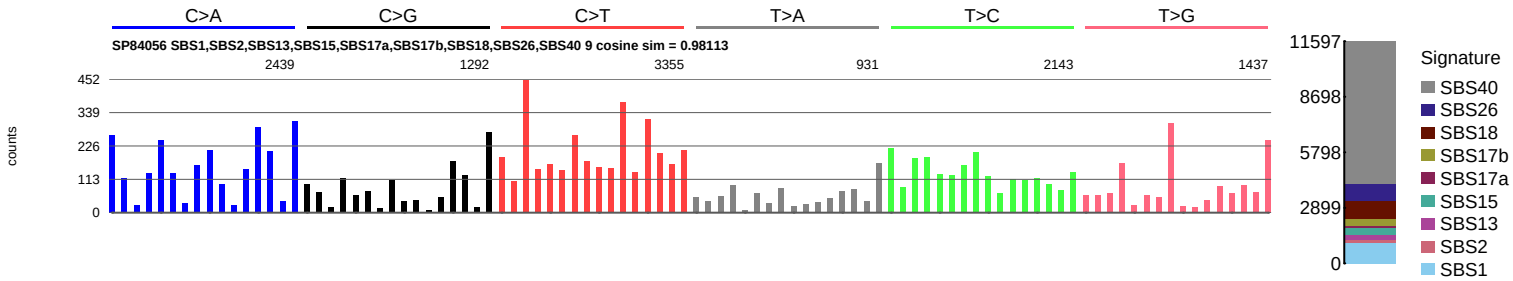
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98116
2339
1126
3405
1100
2146
1479
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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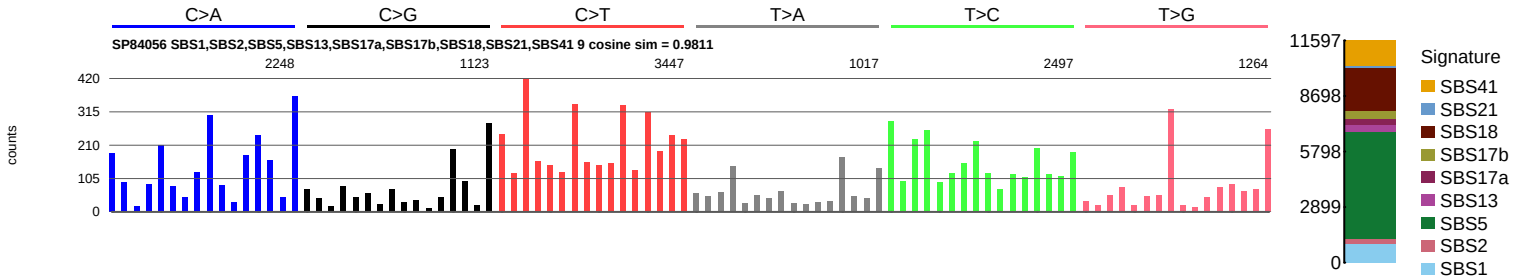
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98113
2439
1292
3355
931
2143
1437
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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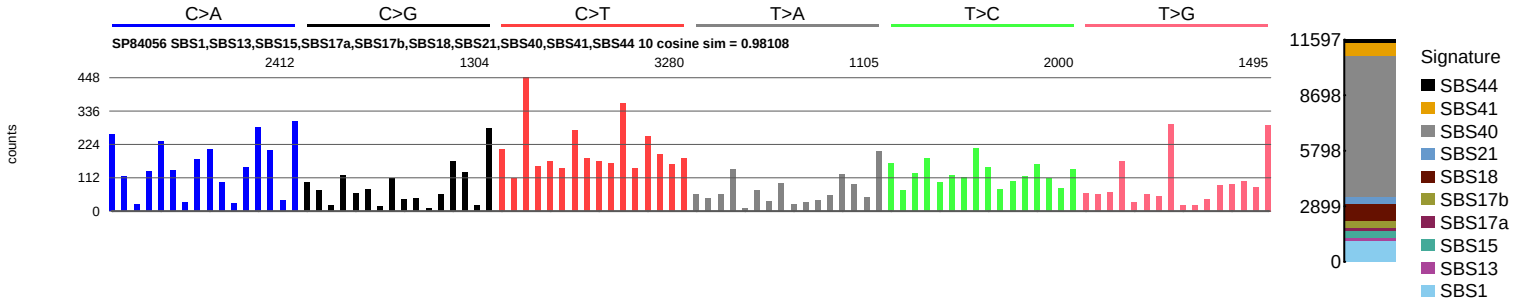
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9811
2248
1123
3447
1017
2497
1264
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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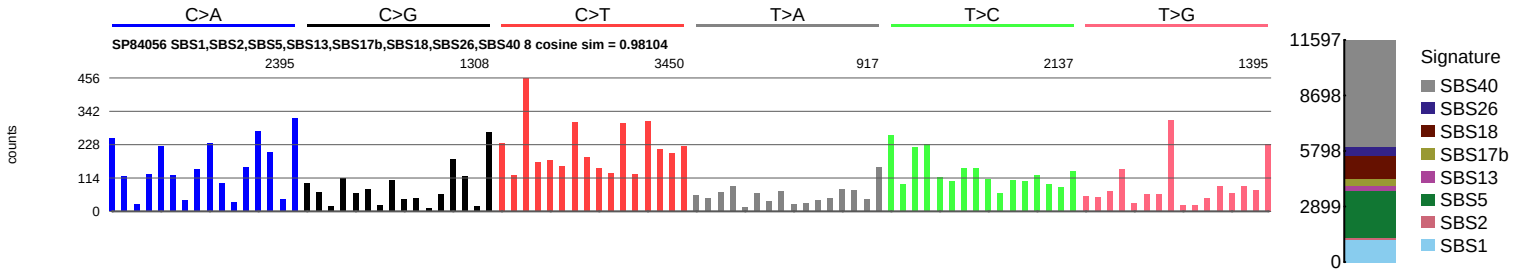
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98108
2412
1304
3280
1105
2000
1495
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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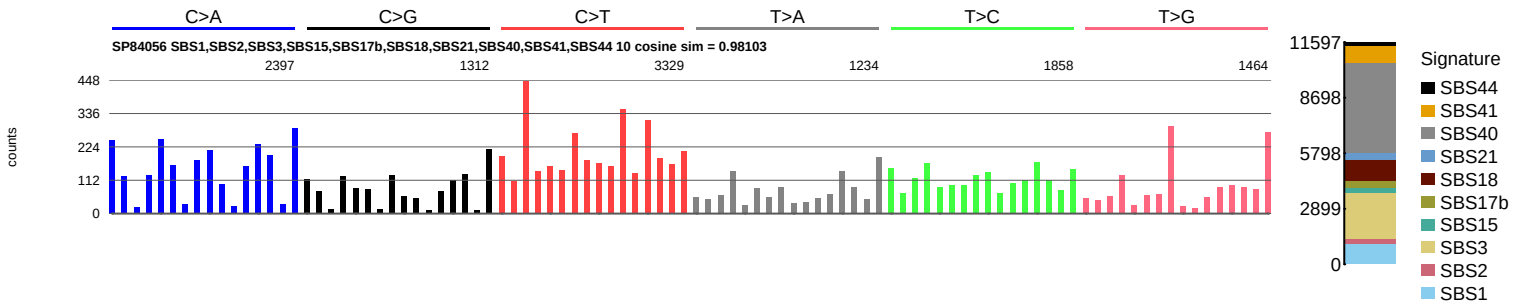
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98104
2395
1308
3450
917
2137
1395
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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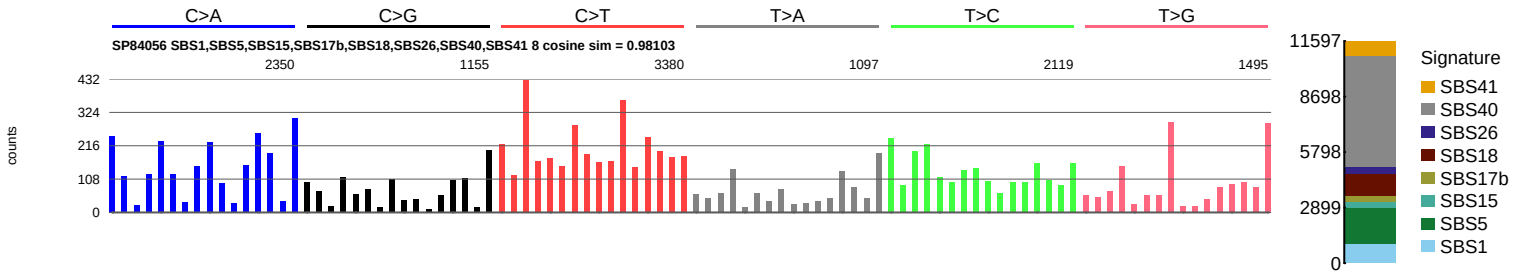
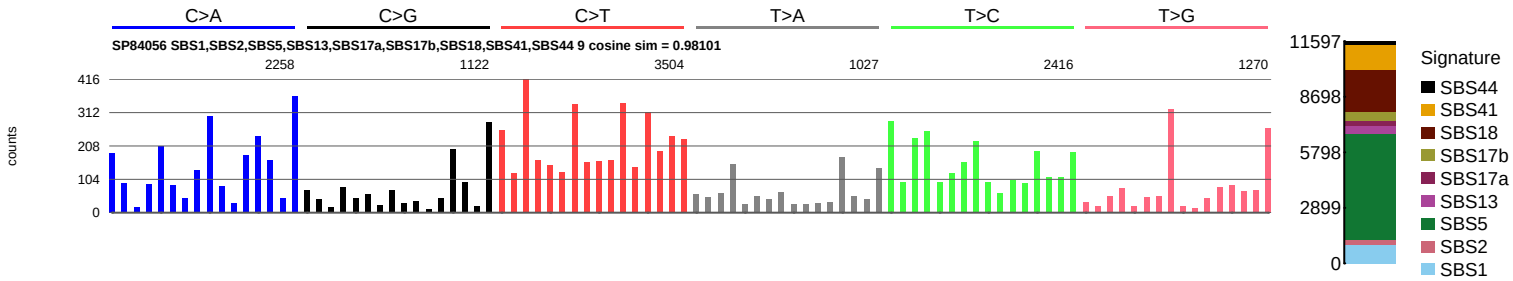
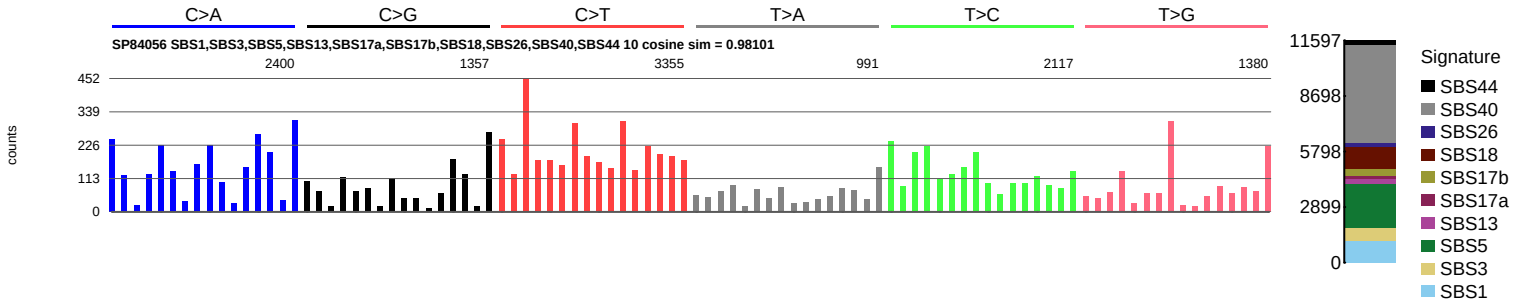
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98103
2397
1312
3329
1234
1858
1464
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98103
2350
1155
3380
1097
2119
1495
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98101
2258
1122
3504
1027
2416
1270
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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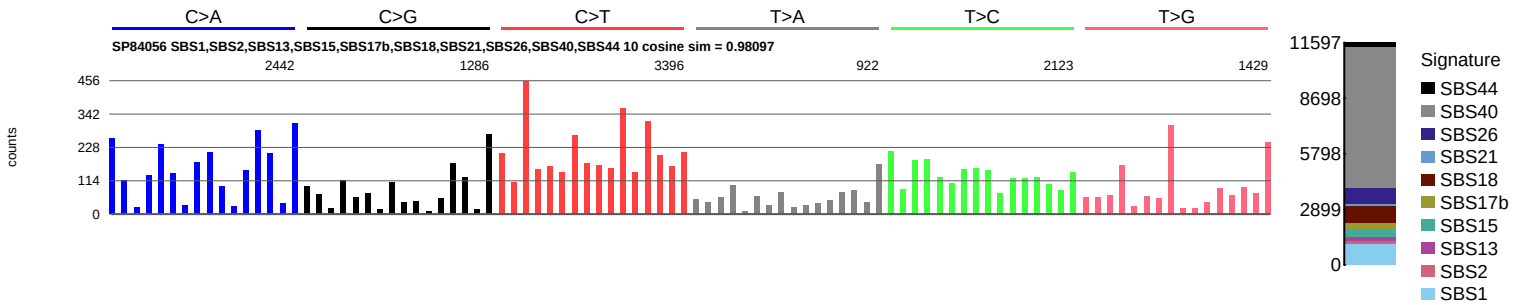
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98101
2400
1357
3355
991
2117
1380
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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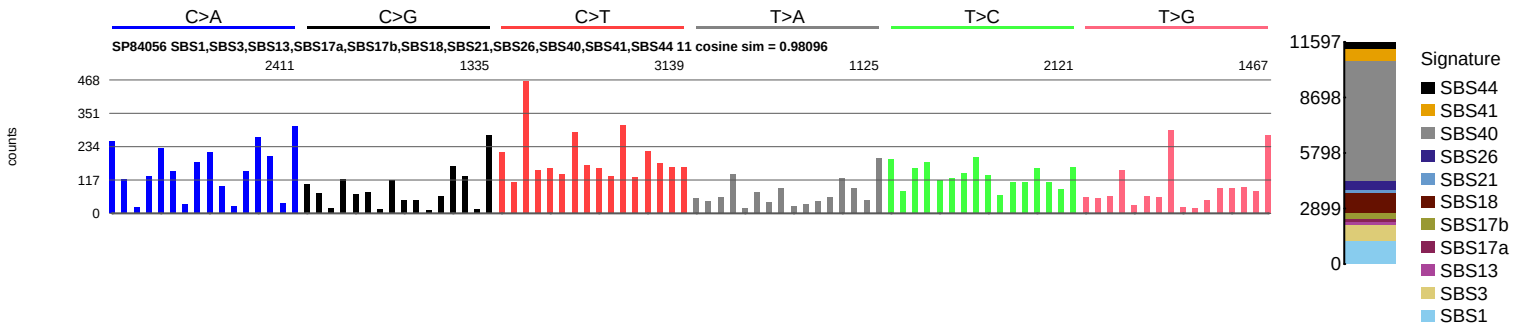
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98097
2442
1286
3396
922
2123
1429
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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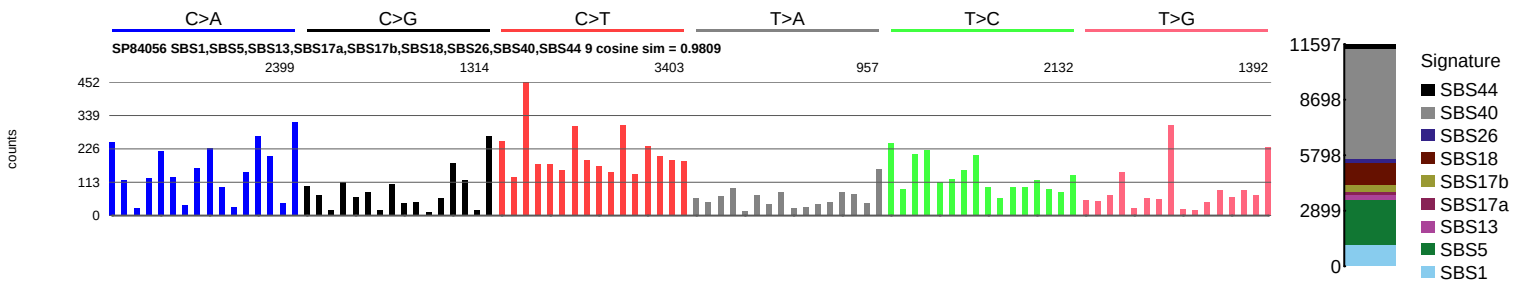
SP84056 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98096
2411
1335
3139
1125
2121
1467
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>T
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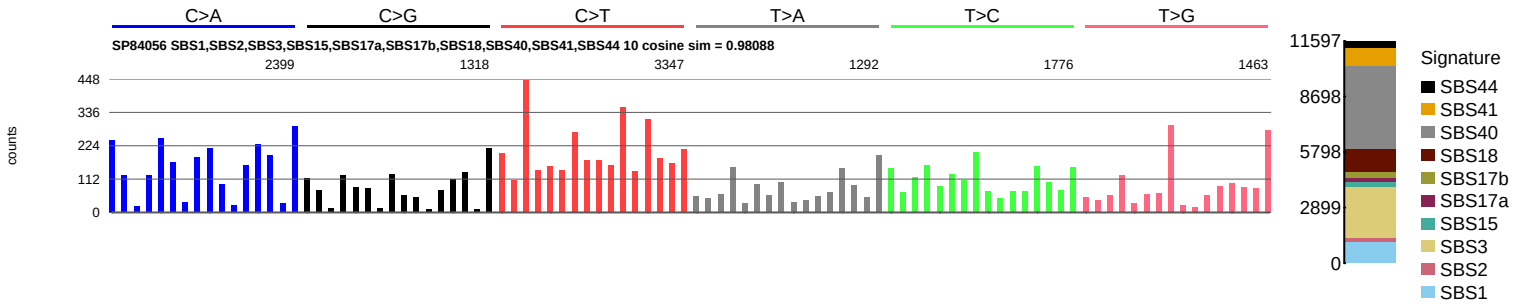
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9809
2399
1314
3403
957
2132
1392
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
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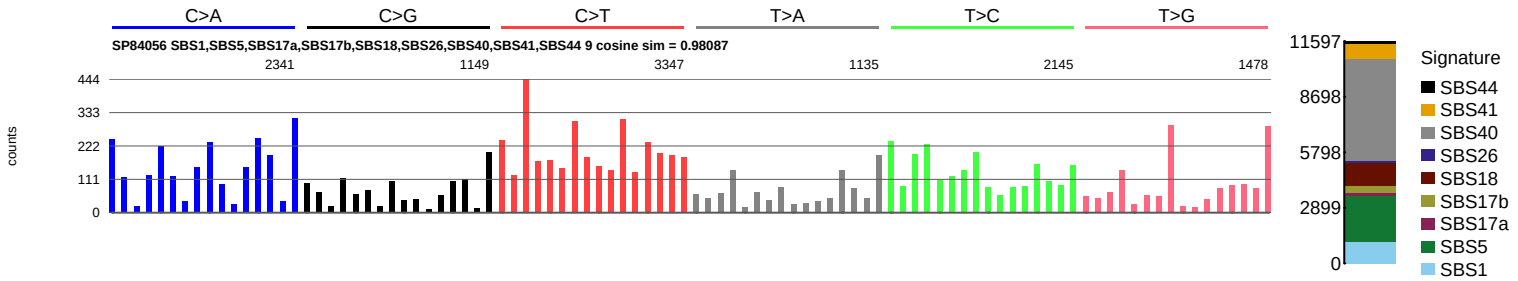
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98088
2399
1318
3347
1292
1776
1463
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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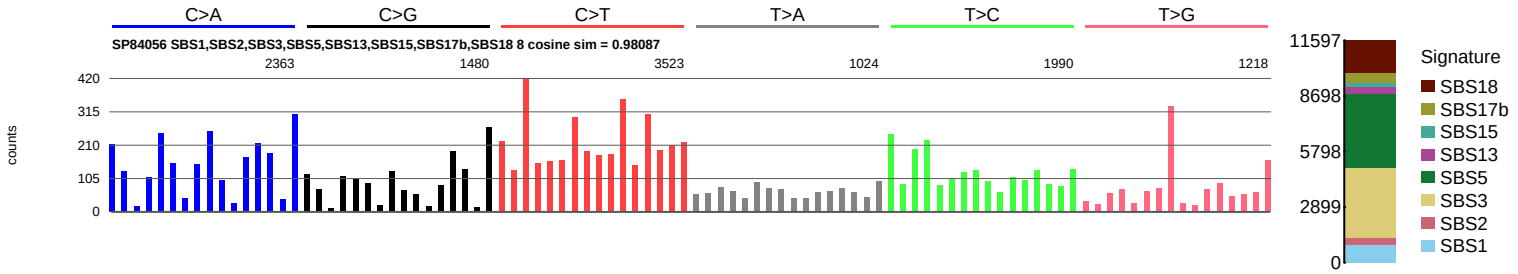
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98087
2341
1149
3347
1135
2145
1478
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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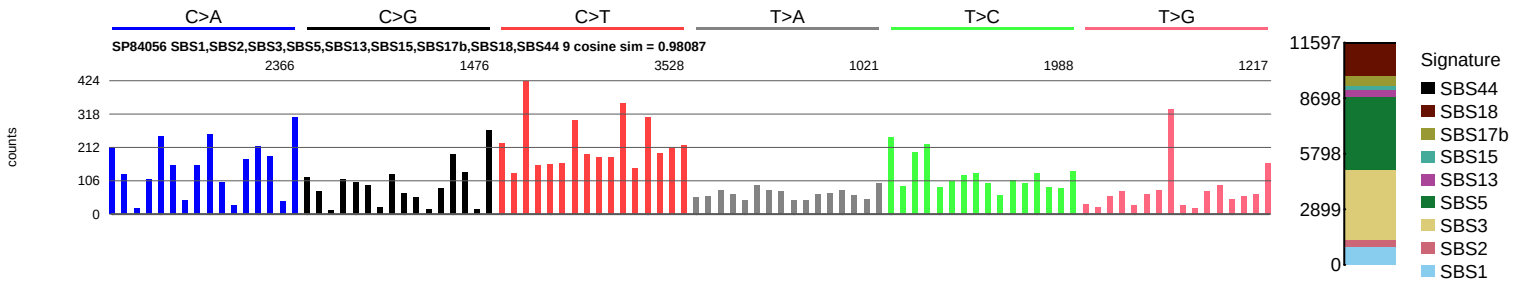
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98087
2363
1480
3523
1024
1990
1218
counts
420
315
210
105
0
11597
8698
5798
2899
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98087
2366
1476
3528
1021
1988
1217
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS44
SBS18
SBS17b
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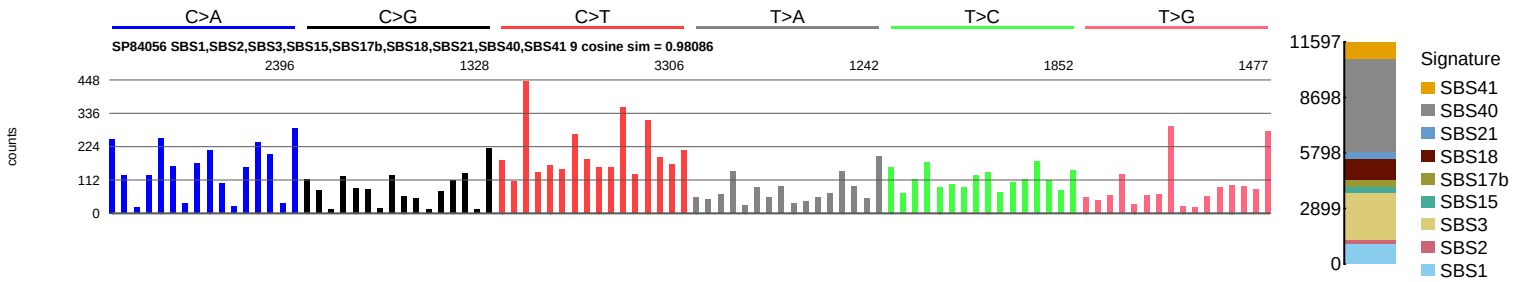
SBS3
SBS2
SBS1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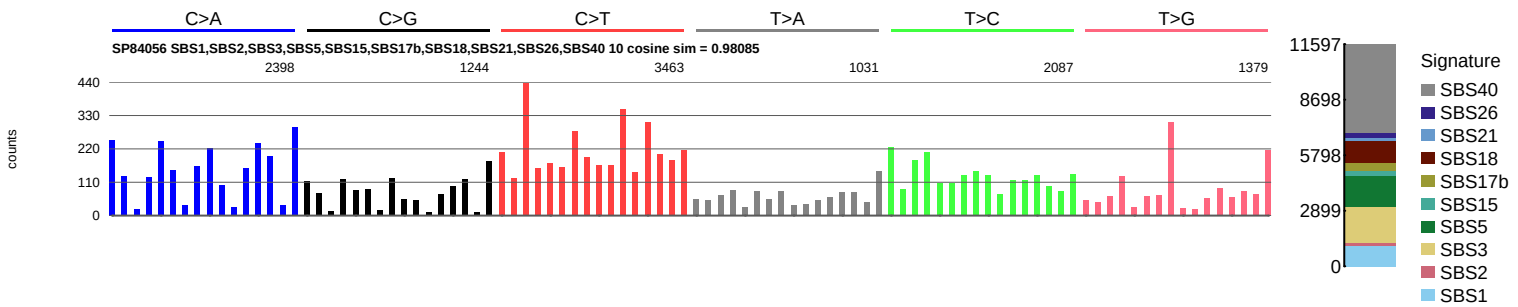
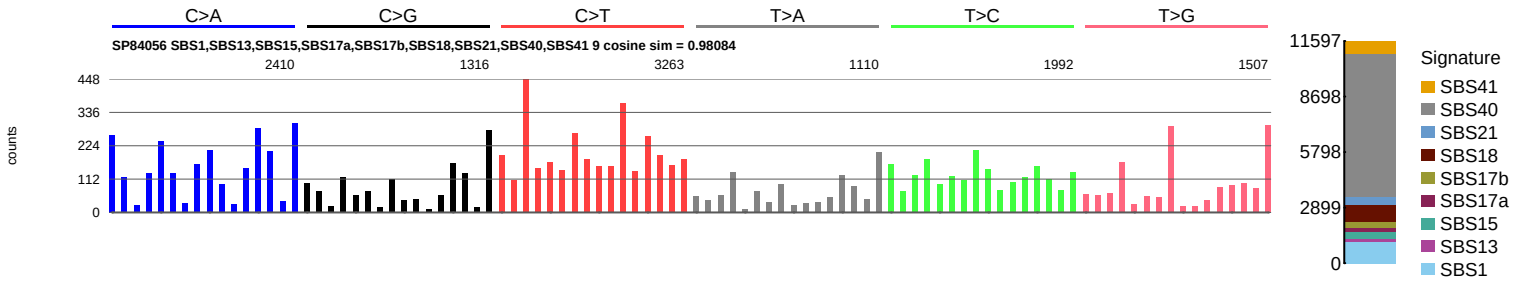
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98086
2396
1328
3306
1242
1852
1477
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98085
2398
1244
3463
1031
2087
1379
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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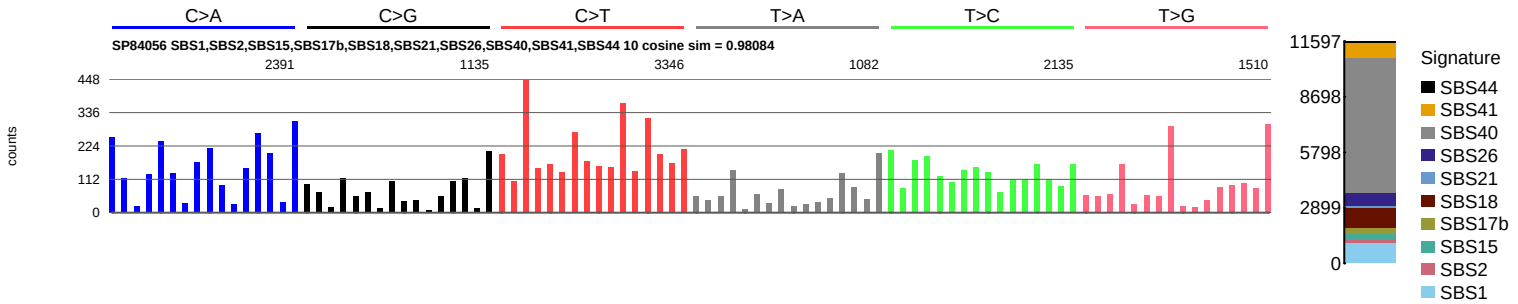
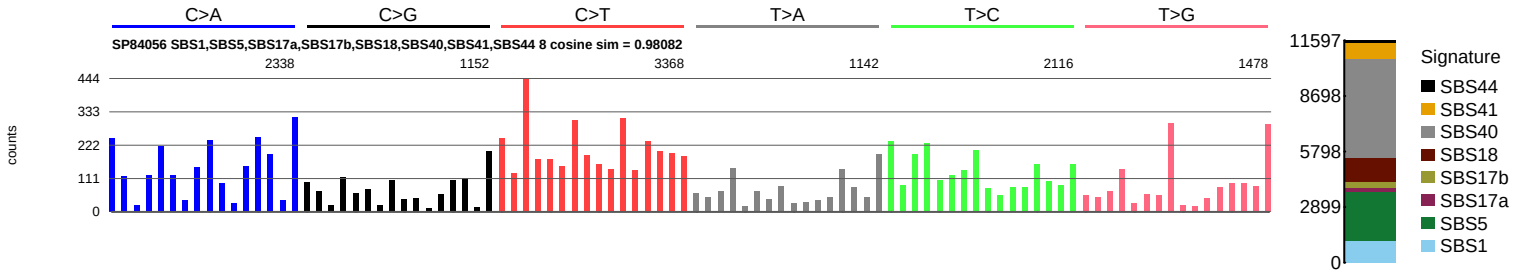
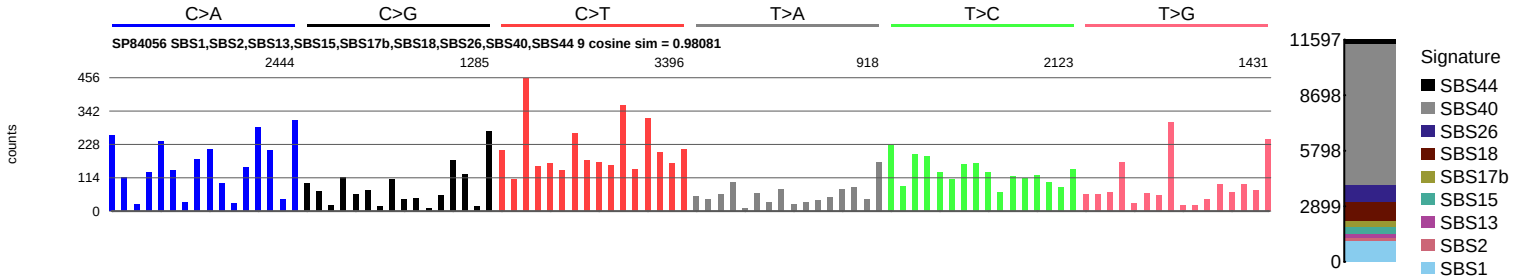
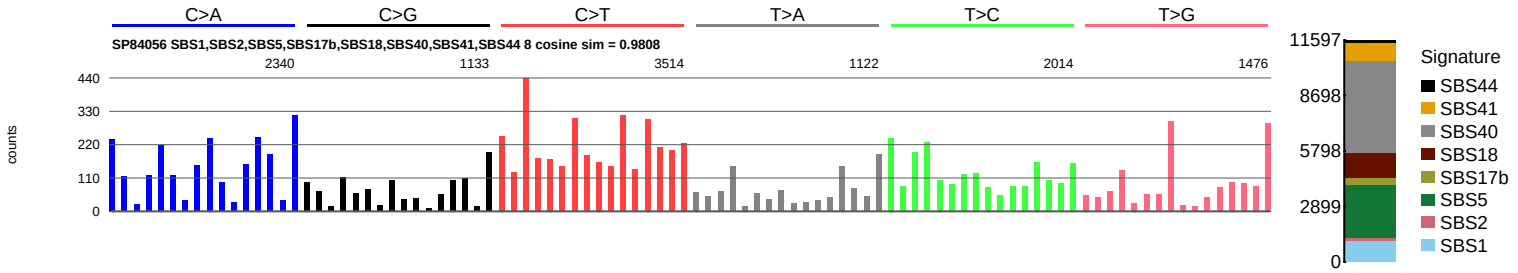
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98084
2410
1316
3263
1110
1992
1507
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98084
2391
1135
3346
1082
2135
1510
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98082
2338
1152
3368
1142
2116
1478
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98081
2444
1285
3396
918
2123
1431
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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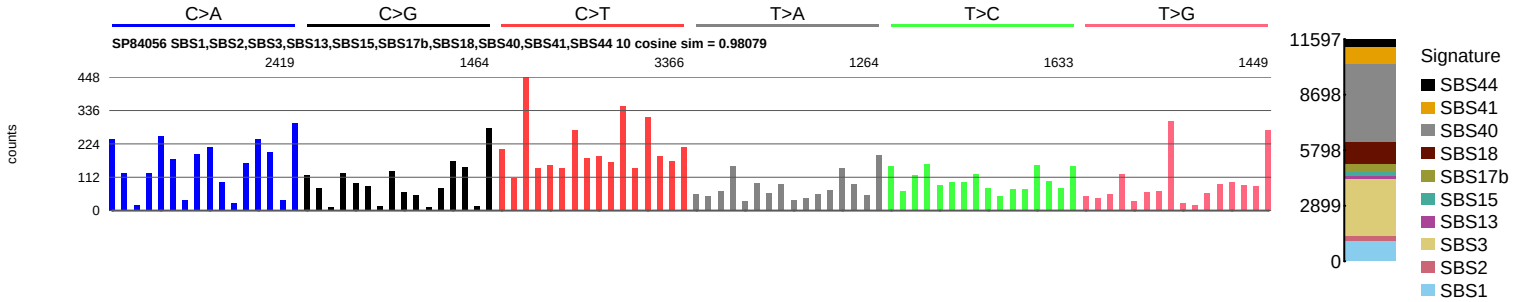
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9808
2340
1133
3514
1122
2014
1476
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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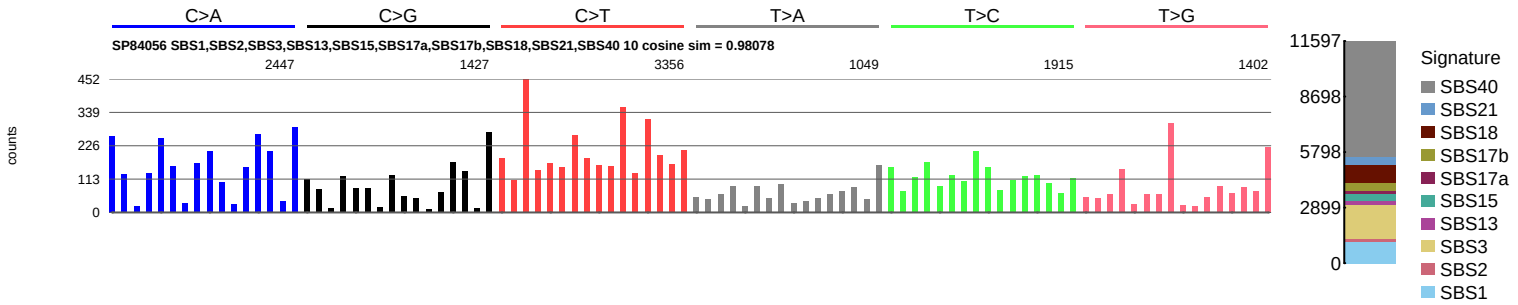
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98079
2419
1464
3366
1264
1633
1449
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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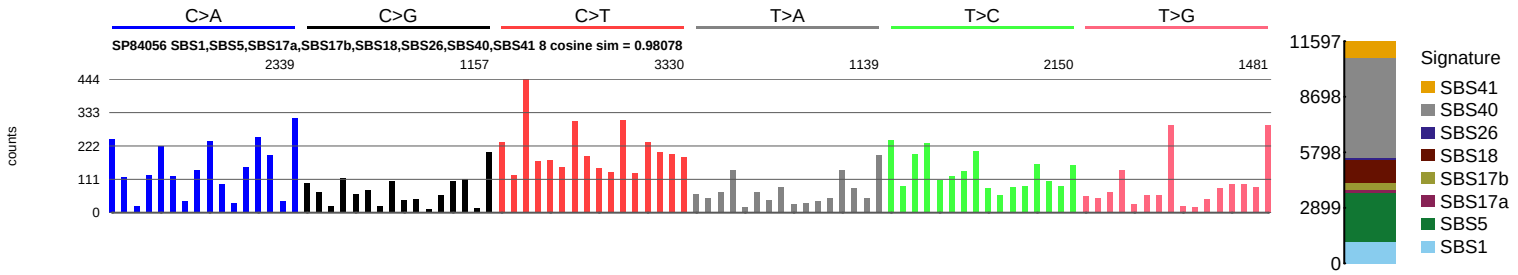
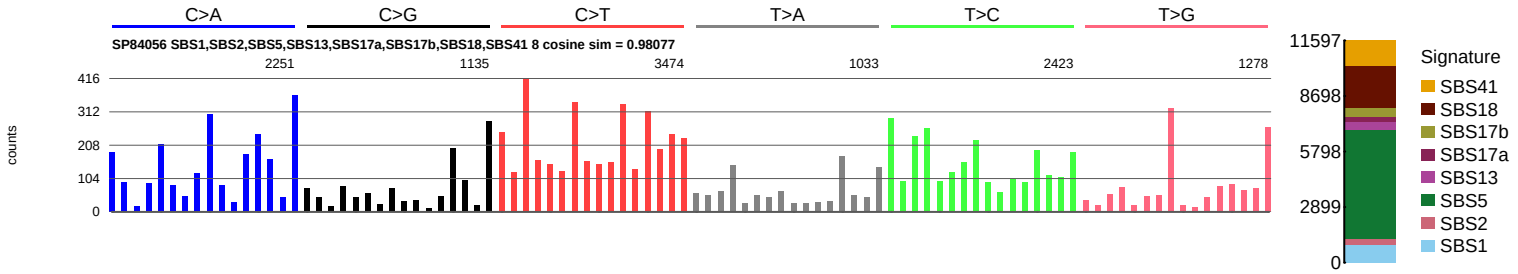
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98078
2447
1427
3356
1049
1915
1402
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98078
2339
1157
3330
1139
2150
1481
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98077
2251
1135
3474
1033
2423
1278
counts
416
312
208
104
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
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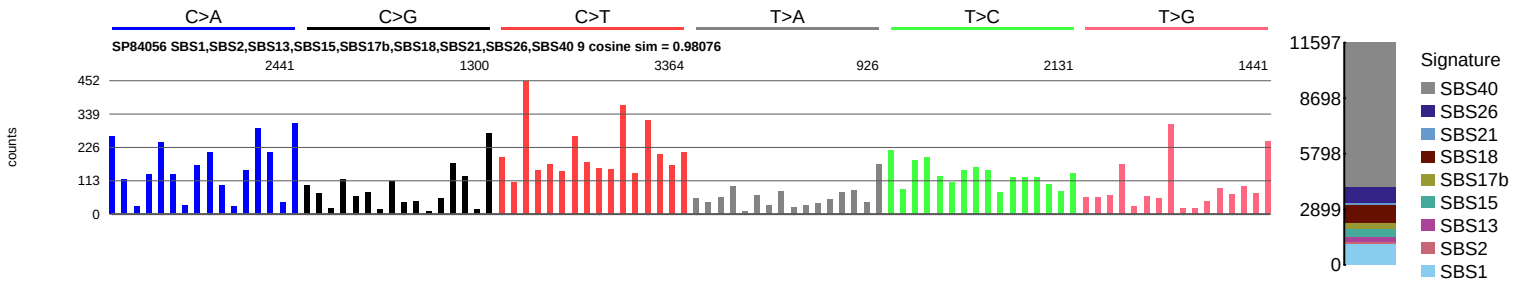
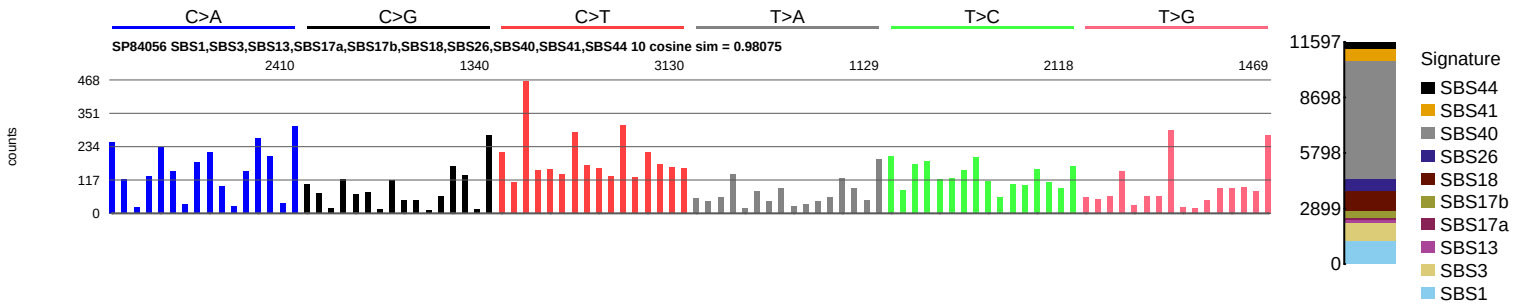
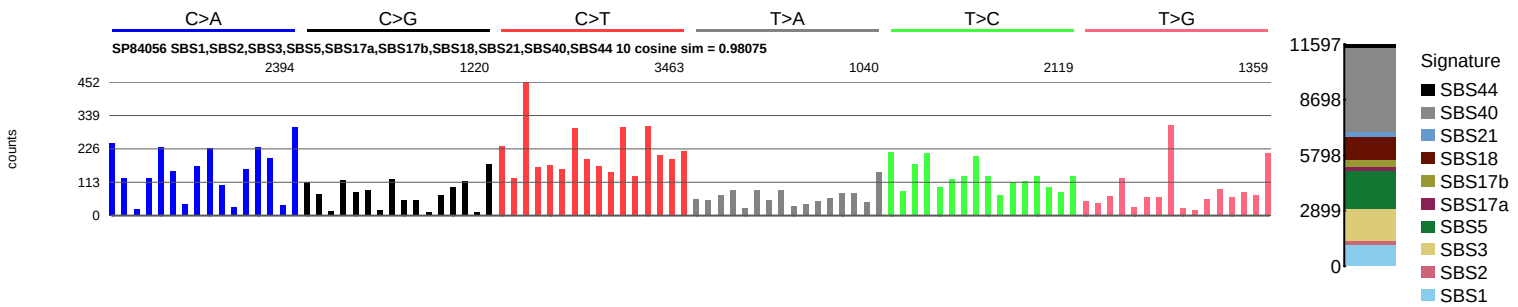
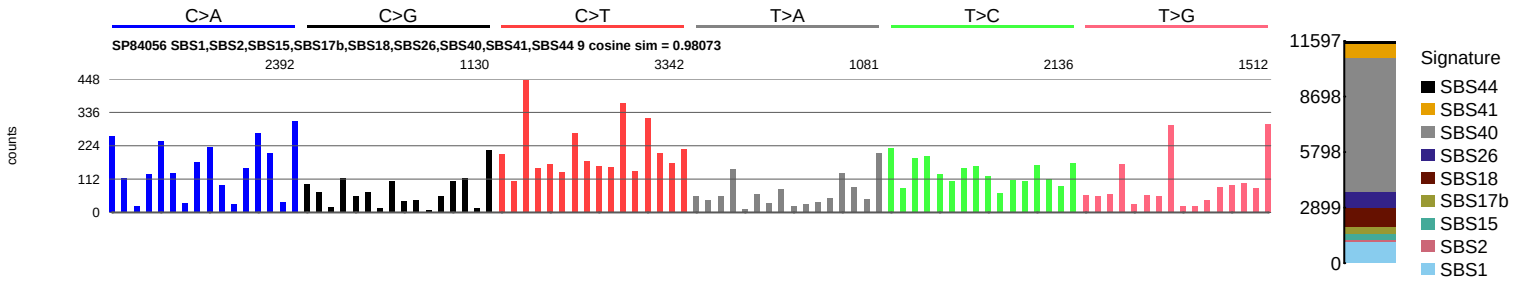
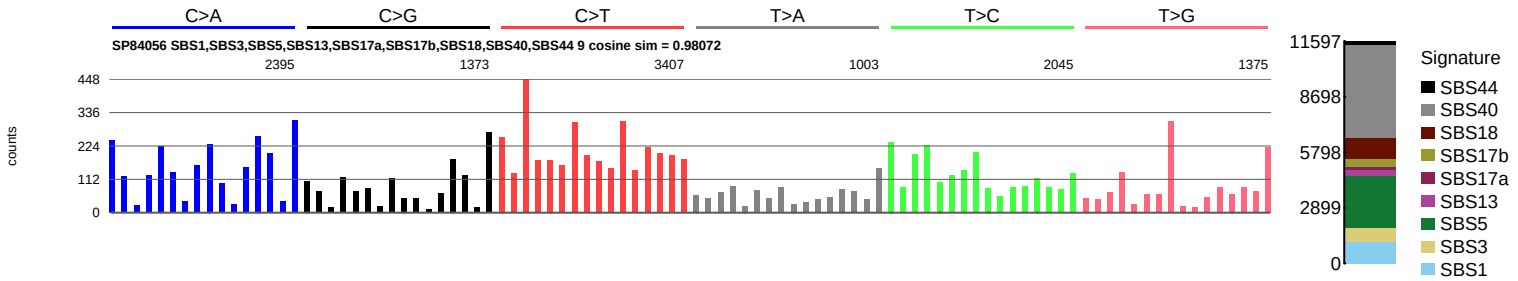
SBS13
SBS5
SBS2
SBS1

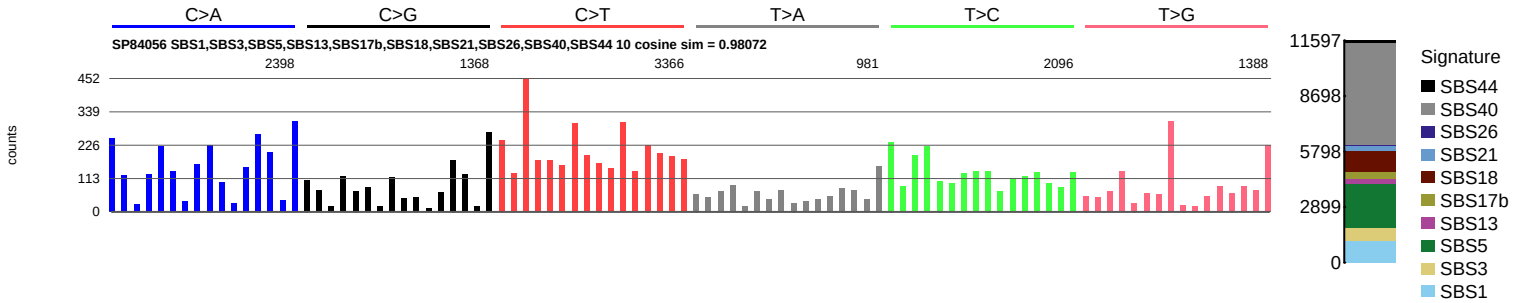

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98076
2441
1300
3364
926
2131
1441
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98075
2410
1340
3130
1129
2118
1469
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98075
2394
1220
3463
1040
2119
1359
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98073
2392
1130
3342
1081
2136
1512
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98072
2395
1373
3407
1003
2045
1375
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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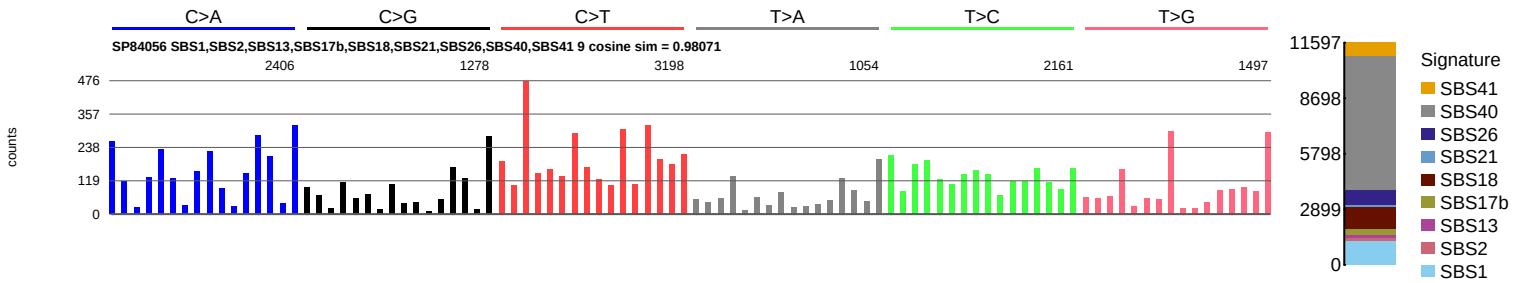
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98072
2398
1368
3366
981
2096
1388
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1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C>T
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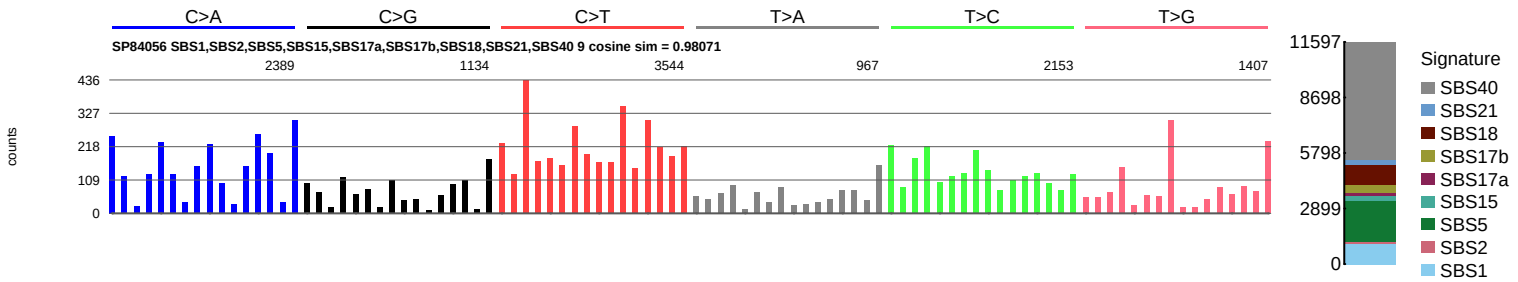
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98071
2406
1278
3198
1054
2161
1497
counts
476
357
238
119
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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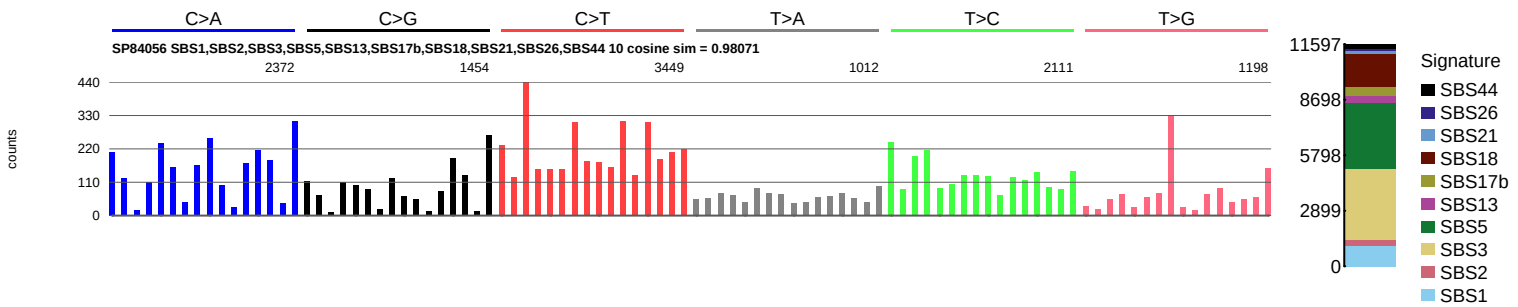
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98071
2389
1134
3544
967
2153
1407
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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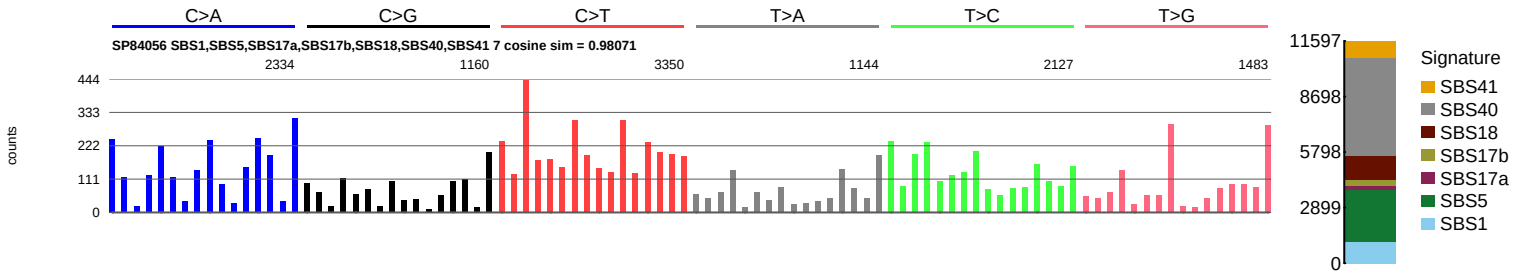
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98071
2372
1454
3449
1012
2111
1198
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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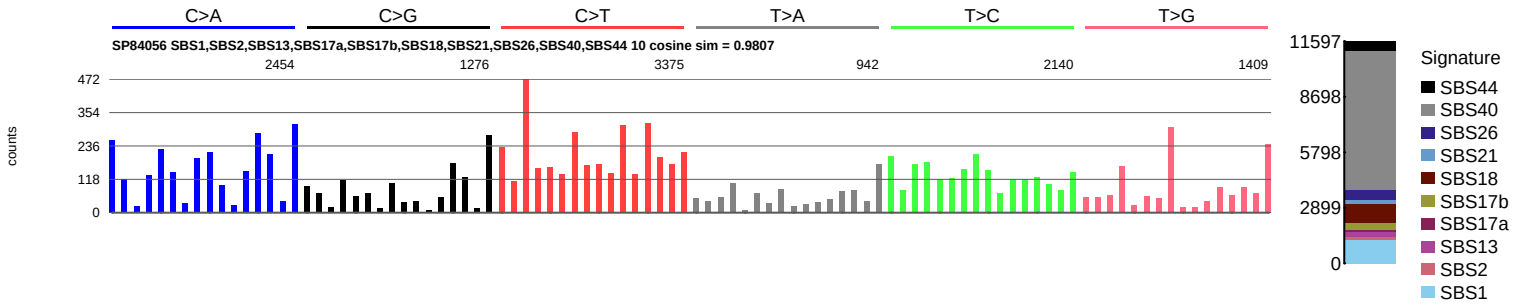
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98071
2334
1160
3350
1144
2127
1483
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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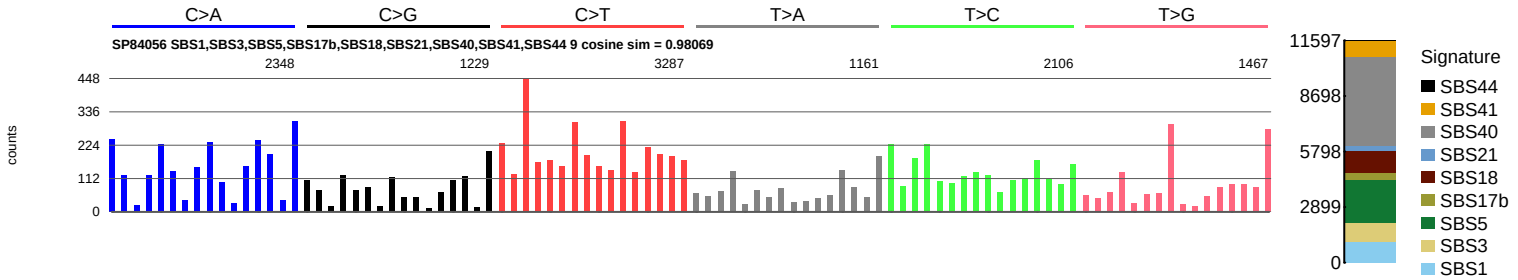
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9807
2454
1276
3375
942
2140
1409
counts
472
354
236
118
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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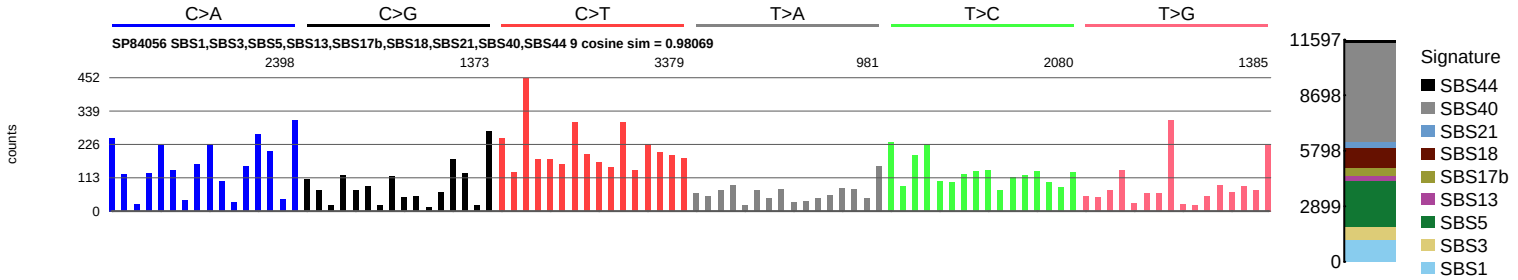
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98069
2348
1229
3287
1161
2106
1467
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98069
2398
1373
3379
981
2080
1385
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
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SBS21
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SBS17b
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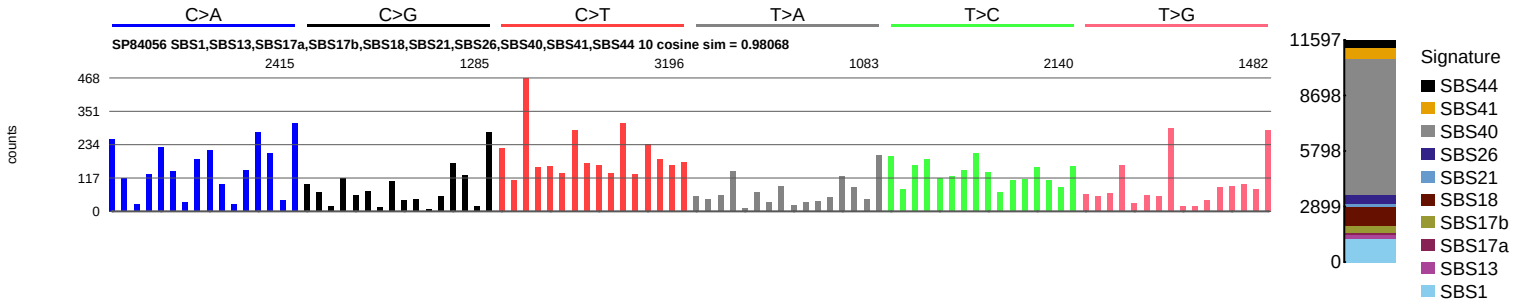
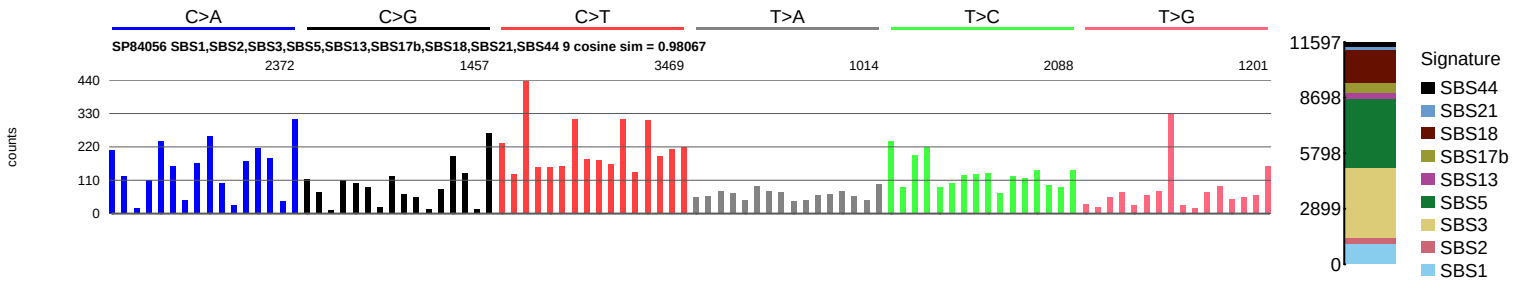
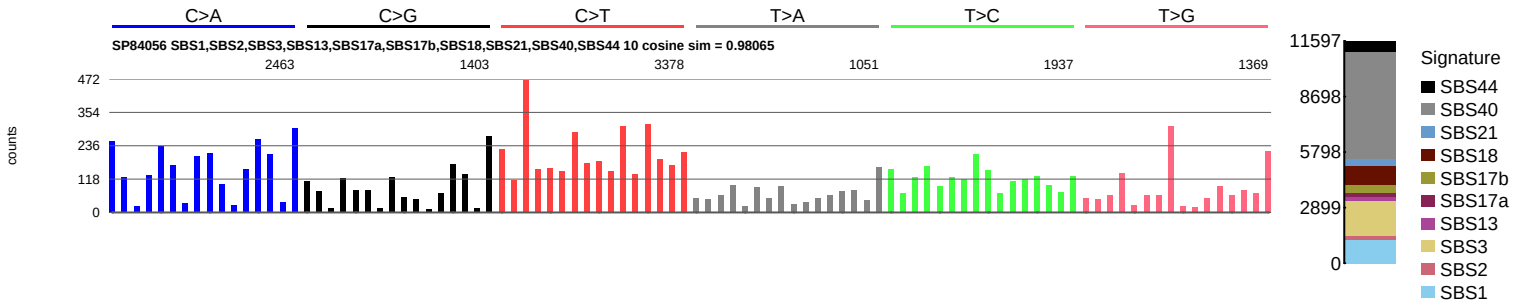
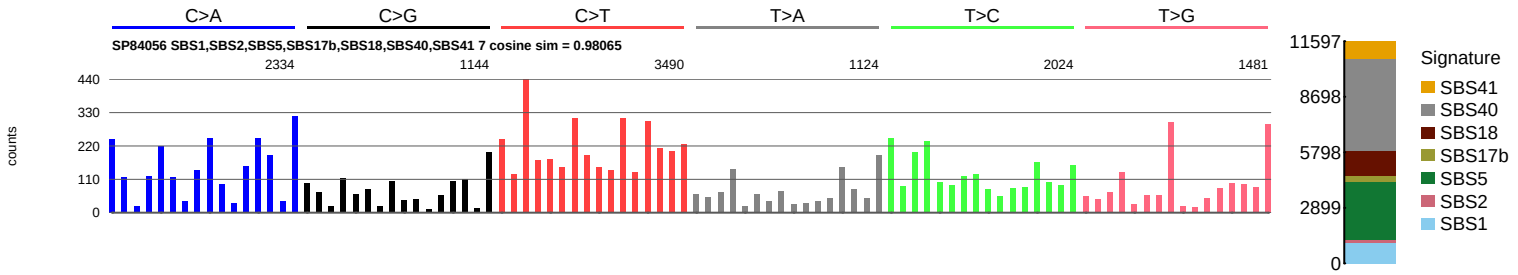
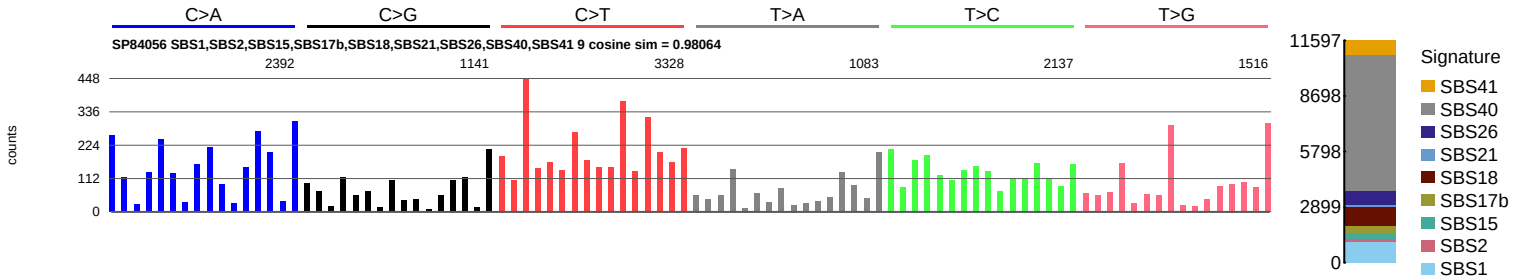
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98068
2415
1285
3196
1083
2140
1482
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98067
2372
1457
3469
1014
2088
1201
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98065
2463
1403
3378
1051
1937
1369
counts
472
354
236
118
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98065
2334
1144
3490
1124
2024
1481
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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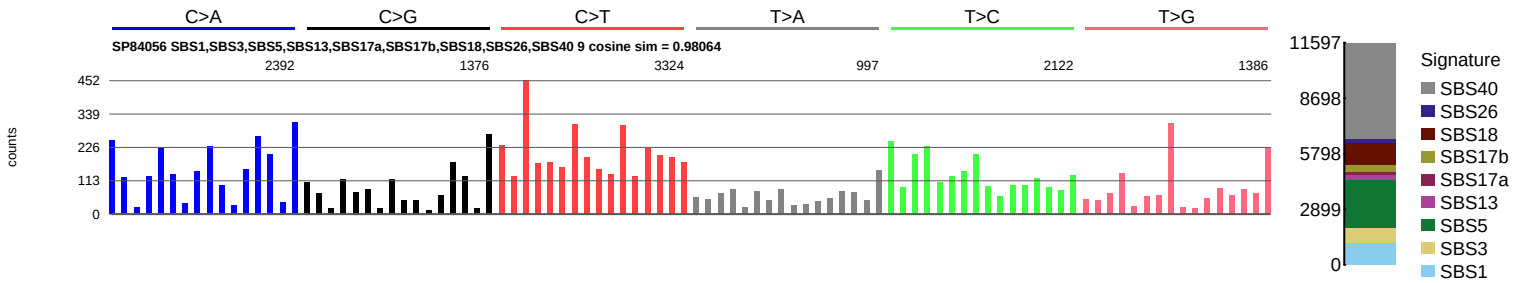
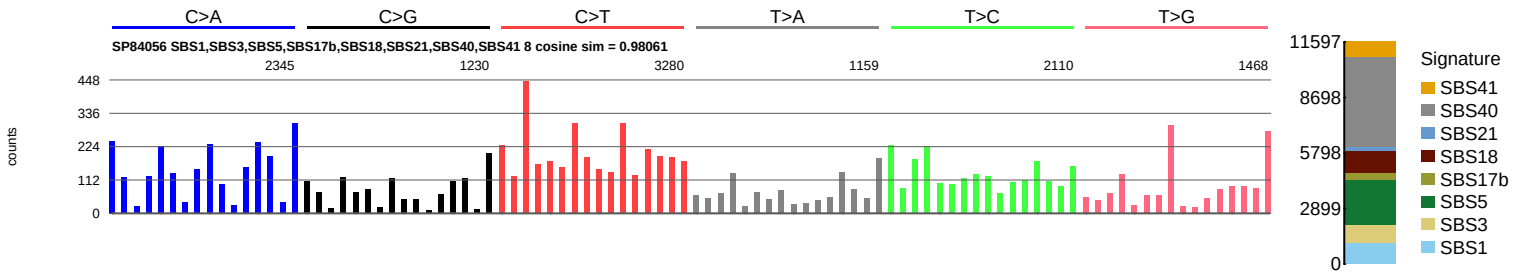
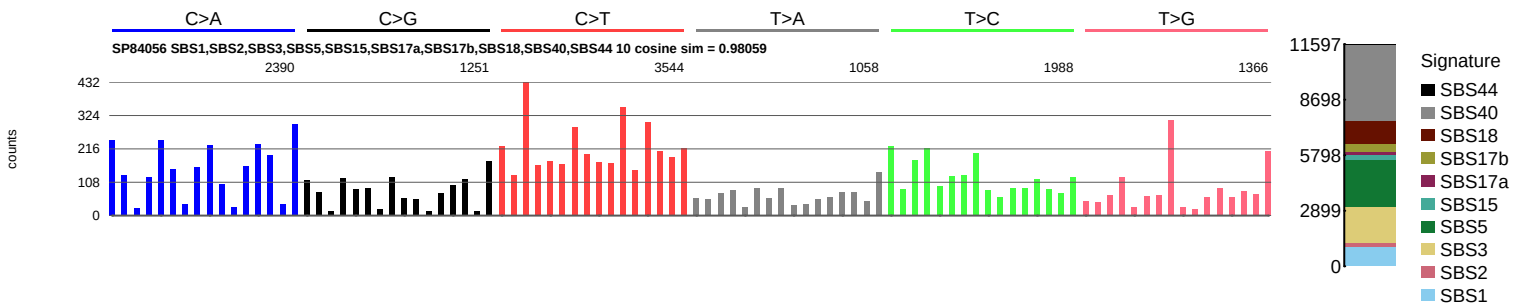
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98064
2392
1141
3328
1083
2137
1516
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98064
2392
1376
3324
997
2122
1386
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98061
2345
1230
3280
1159
2110
1468
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS1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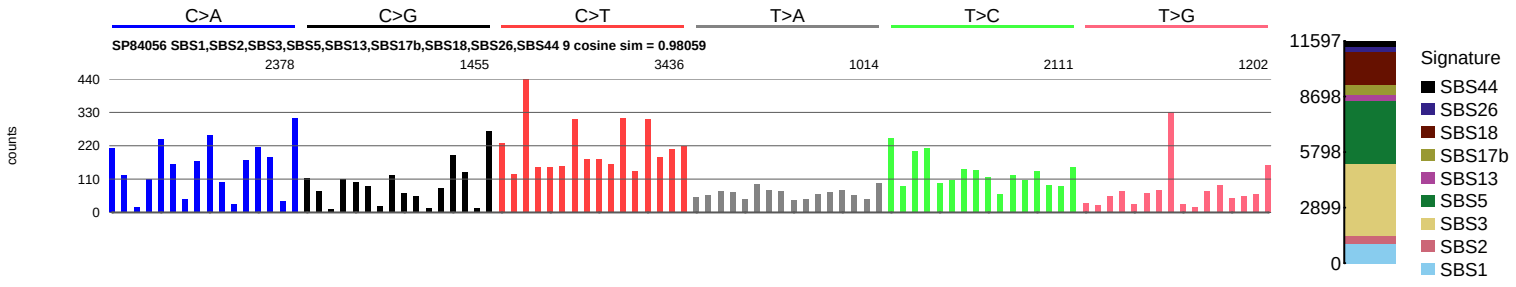
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98059
2390
1251
3544
1058
1988
1366
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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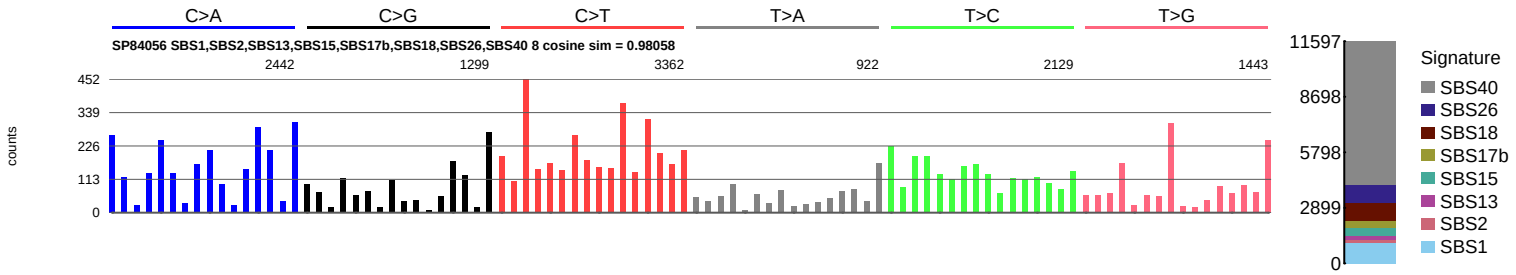
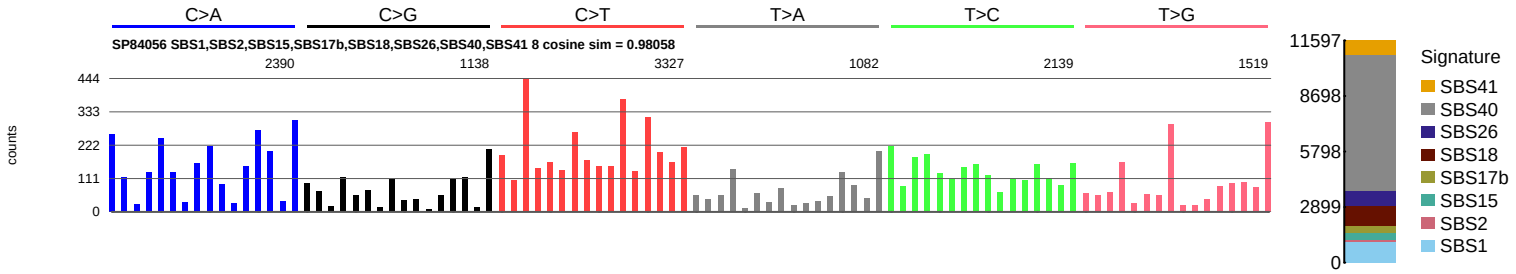
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98059
2378
1455
3436
1014
2111
1202
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98058
2442
1299
3362
922
2129
1443
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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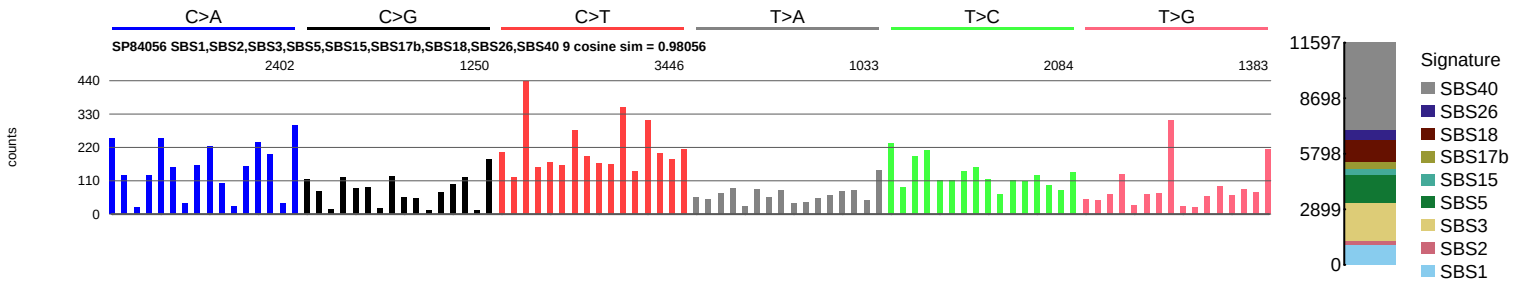
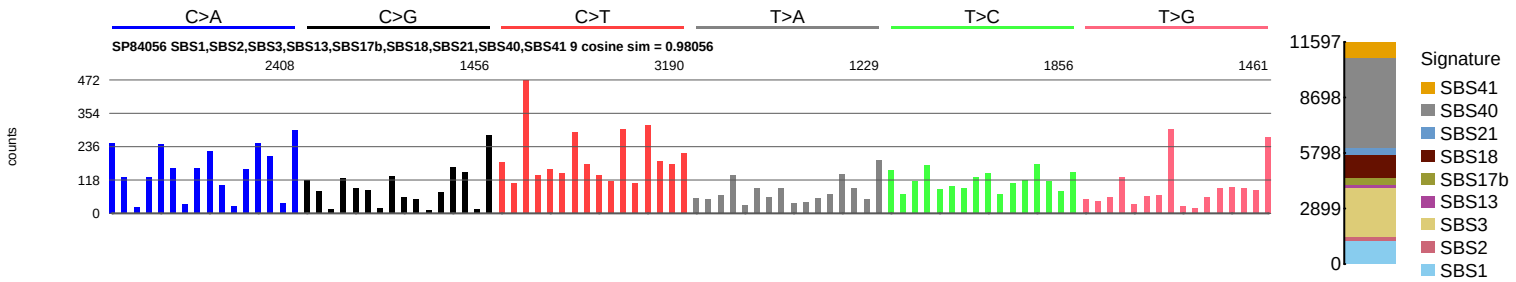
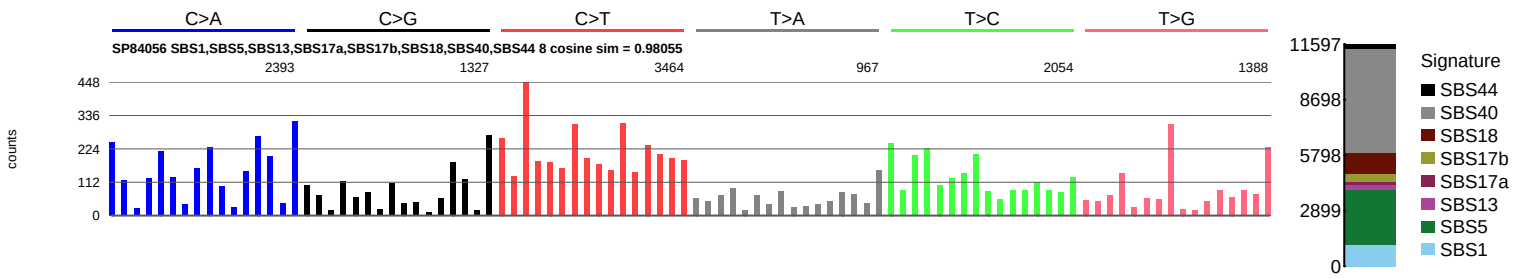
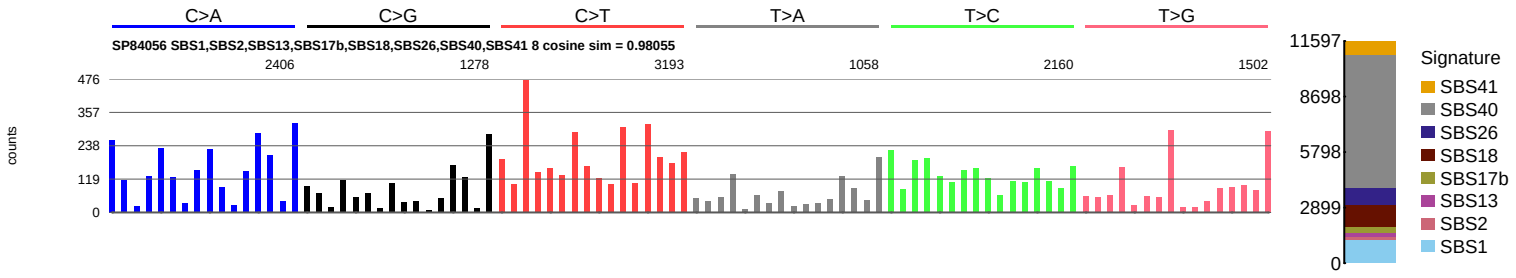
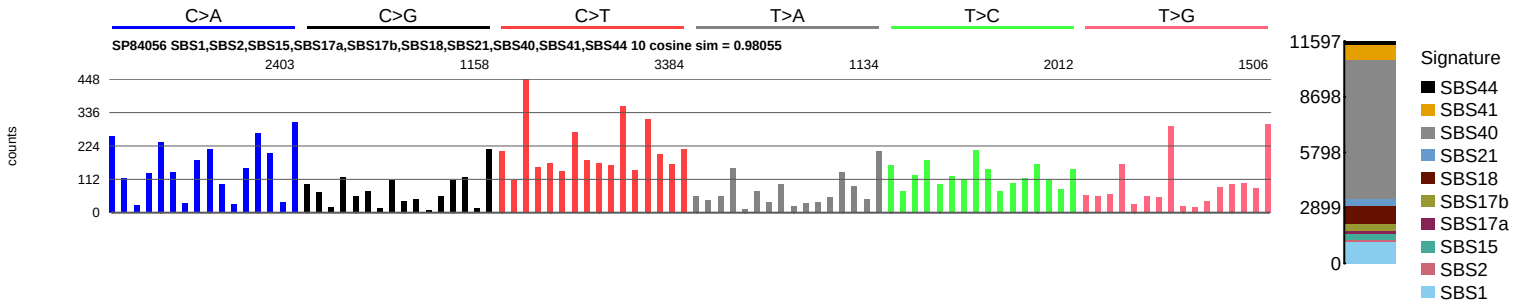
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98058
2390
1138
3327
1082
2139
1519
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

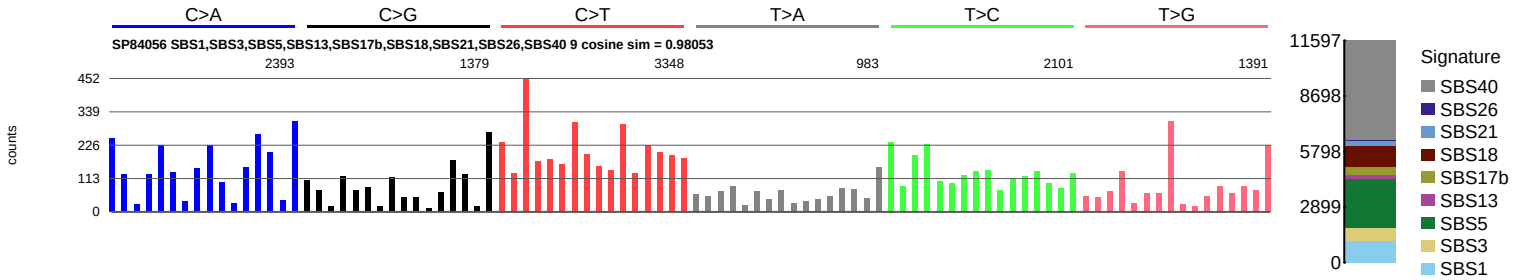

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98056
2402
1250
3446
1033
2084
1383
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

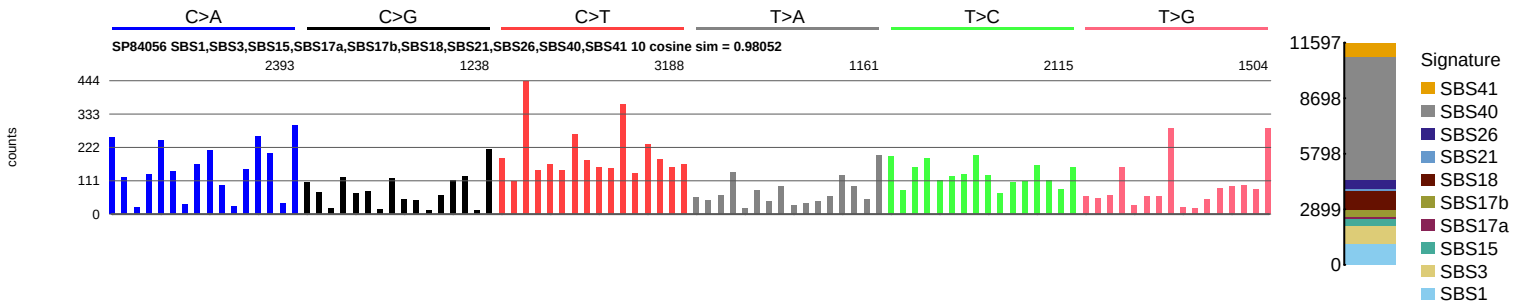

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98056
2408
1456
3190
1229
1856
1461
counts
472
354
236
118
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98055
2393
1327
3464
967
2054
1388
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98055
2406
1278
3193
1058
2160
1502
counts
476
357
238
119
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98055
2403
1158
3384
1134
2012
1506
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98053
2393
1379
3348
983
2101
1391
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1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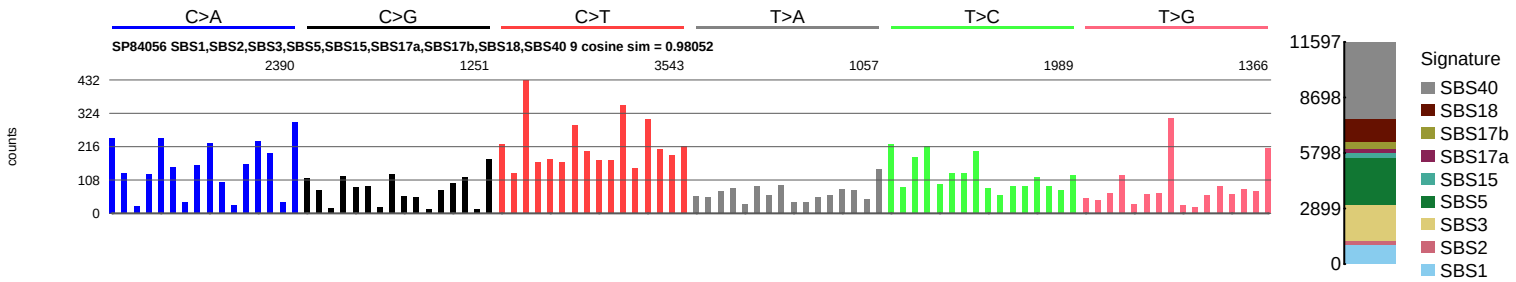
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98052
2393
1238
3188
1161
2115
1504
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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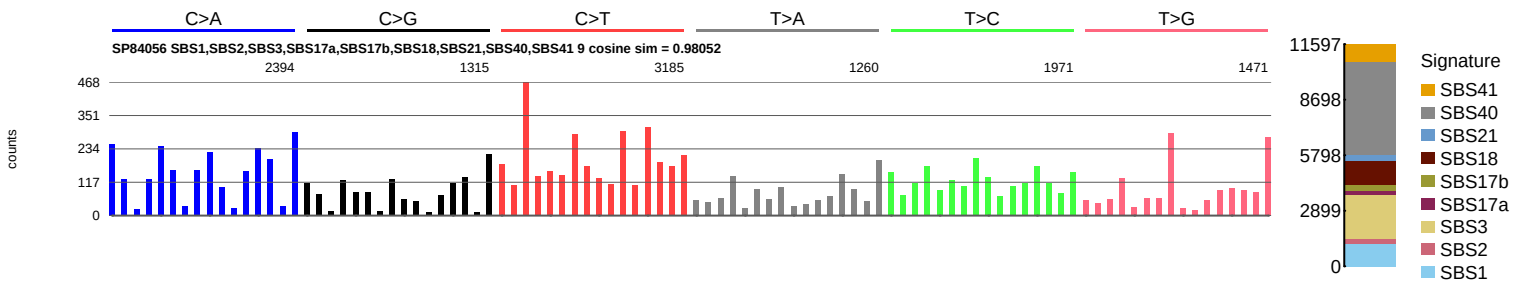
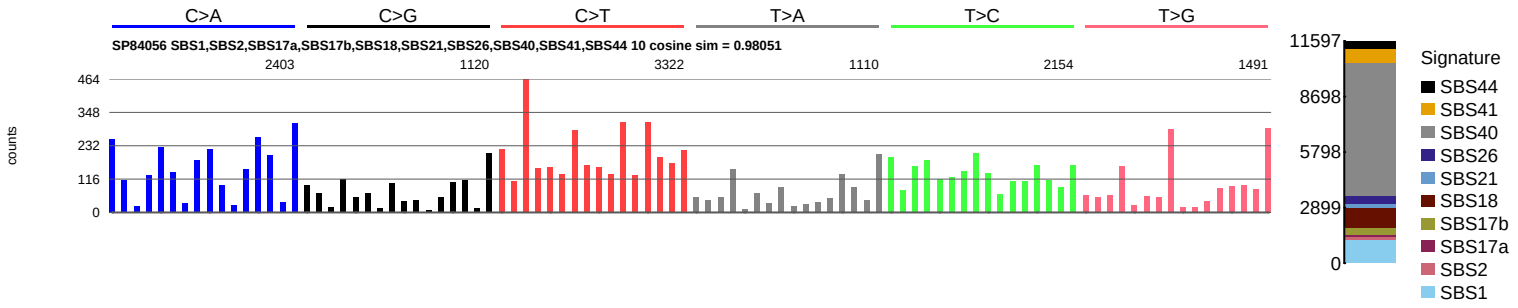
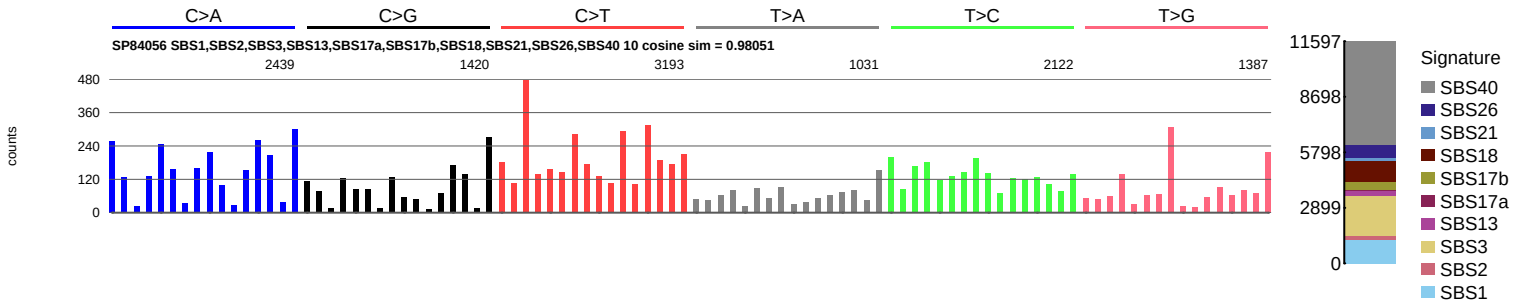
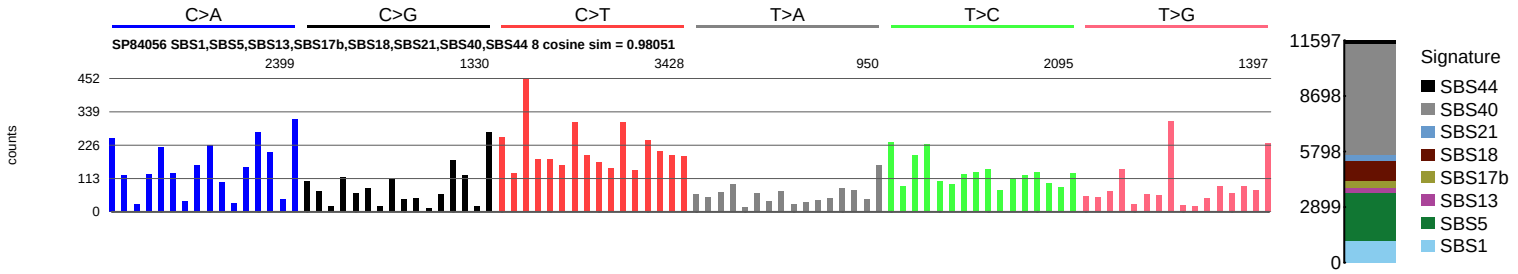
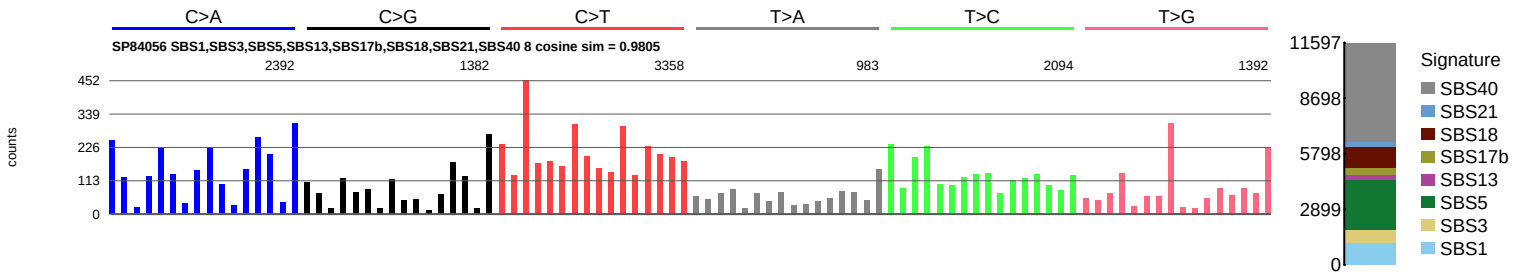
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98052
2390
1251
3543
1057
1989
1366
counts
432
324
216
108
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

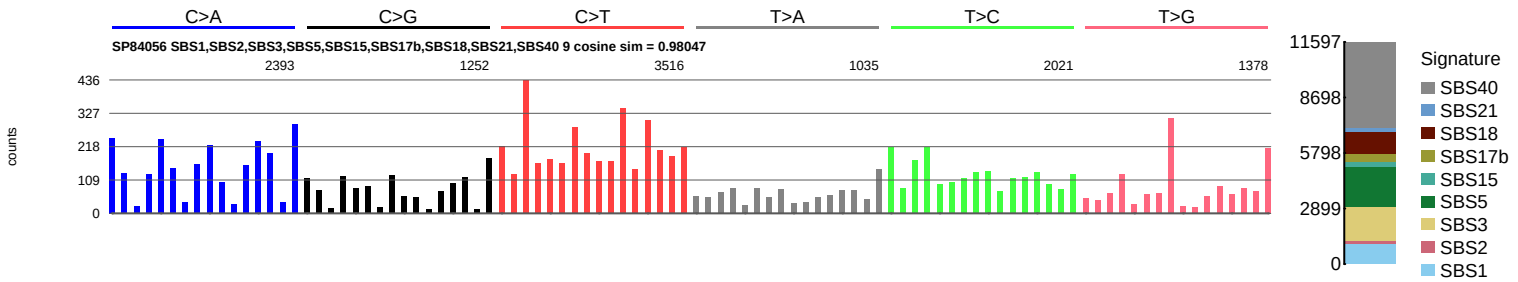

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98052
2394
1315
3185
1260
1971
1471
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

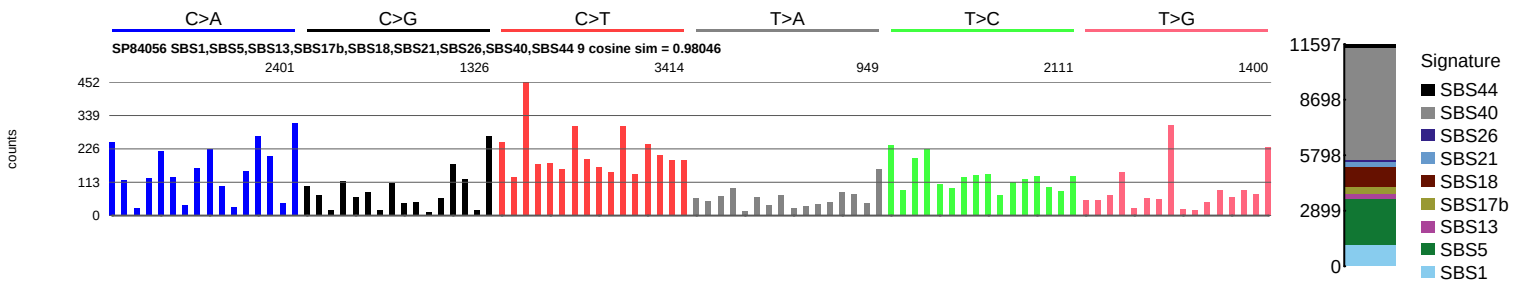

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98051
2403
1120
3322
1110
2154
1491
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

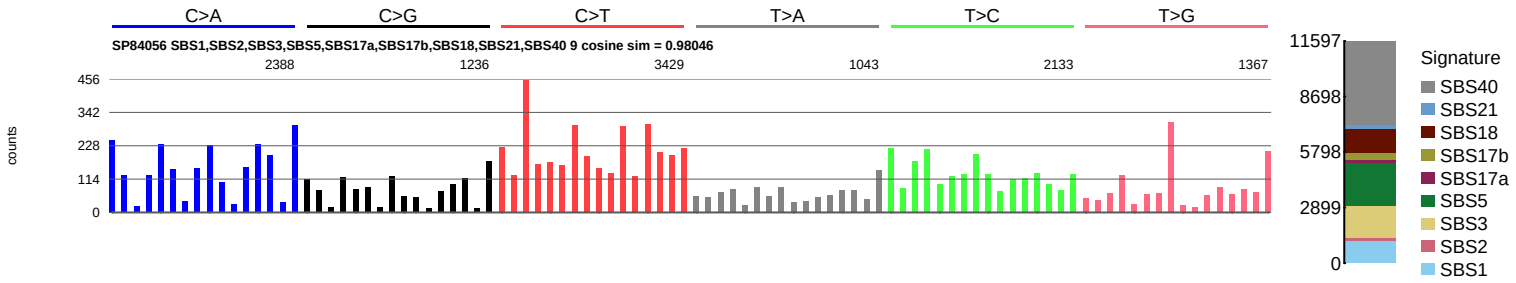

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98051
2439
1420
3193
1031
2122
1387
counts
480
360
240
120
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

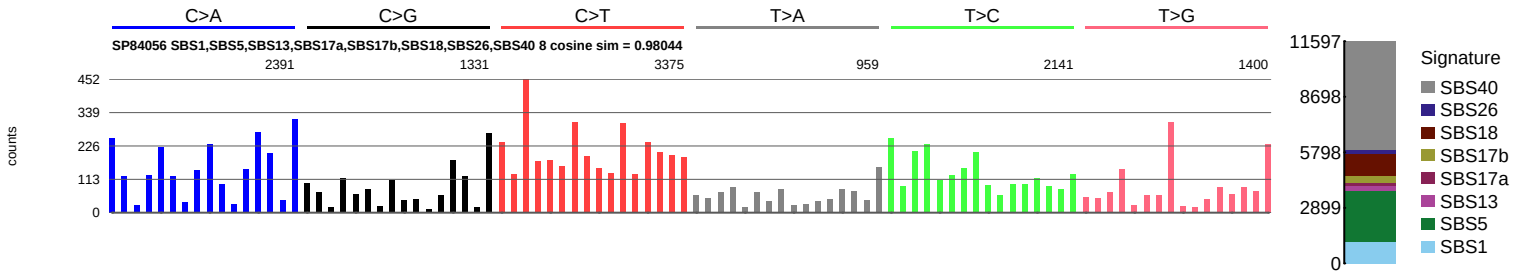

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98051
2399
1330
3428
950
2095
1397
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

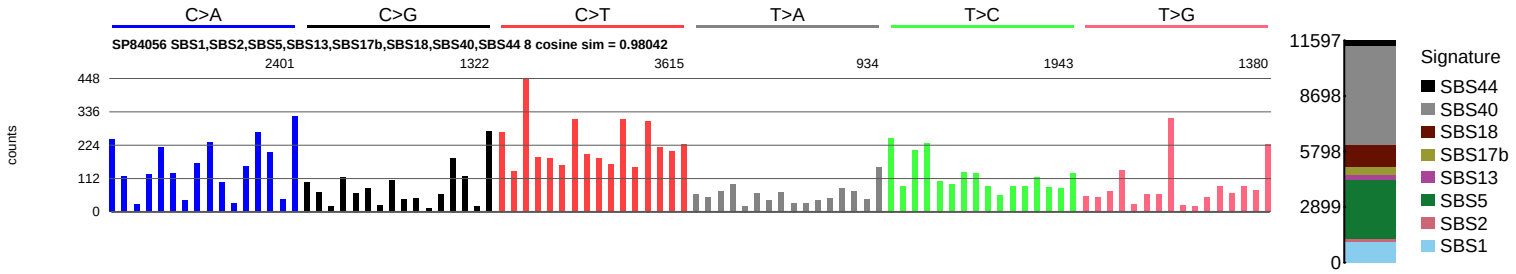

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9805
2392
1382
3358
983
2094
1392
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1

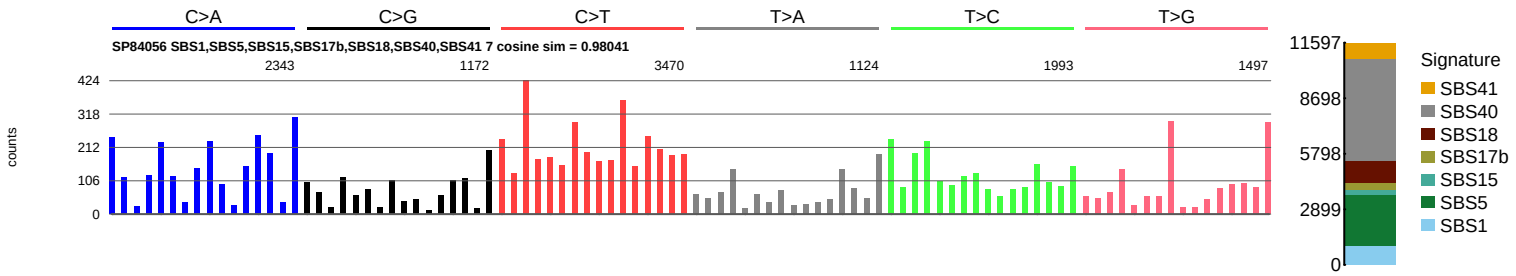

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98047
2393
1252
3516
1035
2021
1378
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

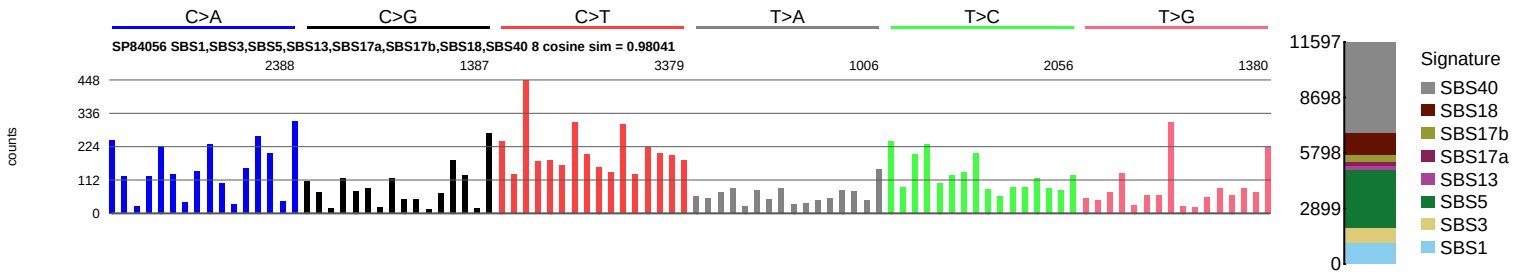

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98046
2401
1326
3414
949
2111
1400
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

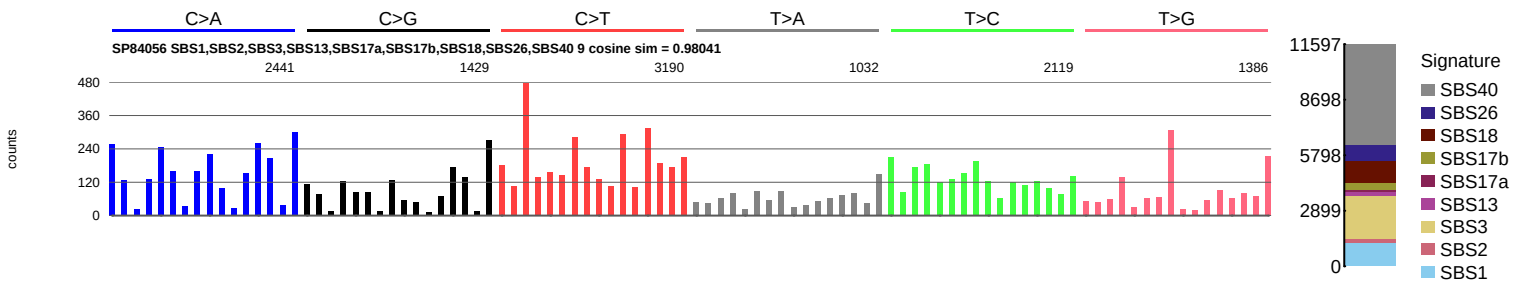

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98046
2388
1236
3429
1043
2133
1367
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

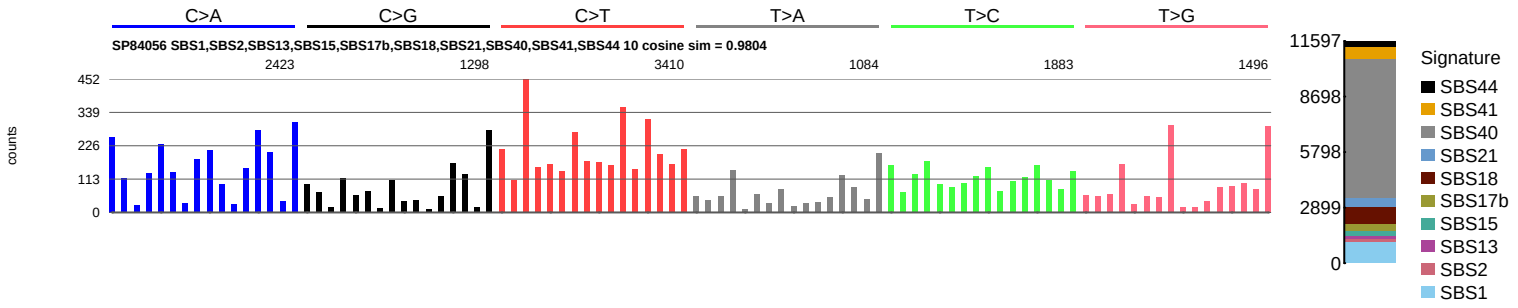

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98044
2391
1331
3375
959
2141
1400
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

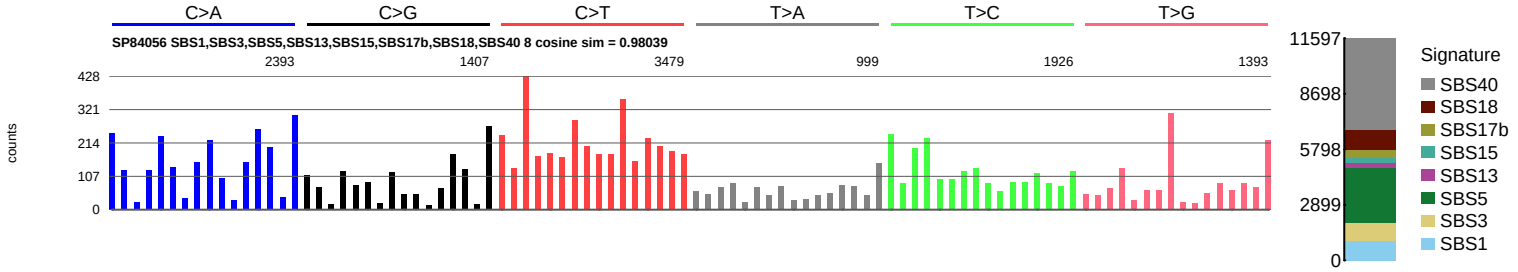

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98042
2401
1322
3615
934
1943
1380
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

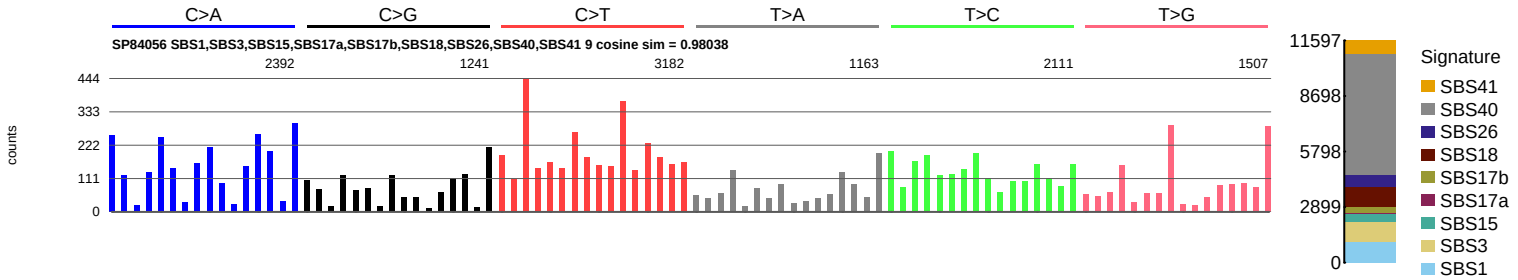

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98041
2343
1172
3470
1124
1993
1497
counts
424
318
212
106
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1

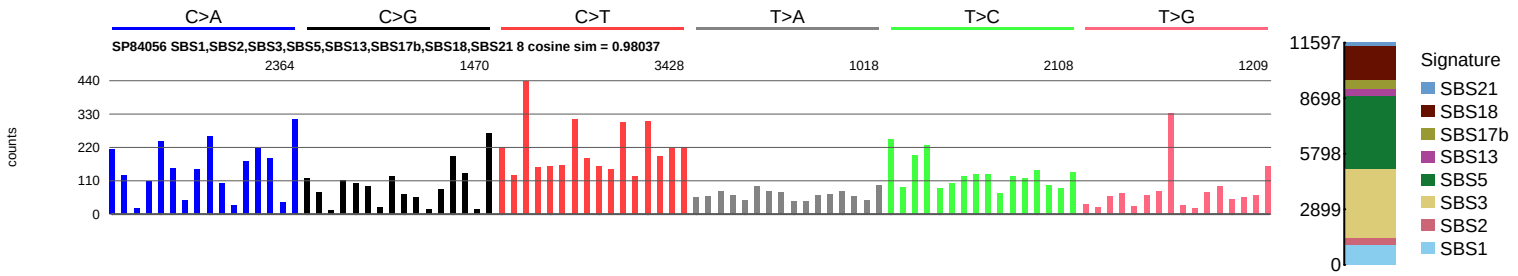

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98041
2388
1387
3379
1006
2056
1380
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

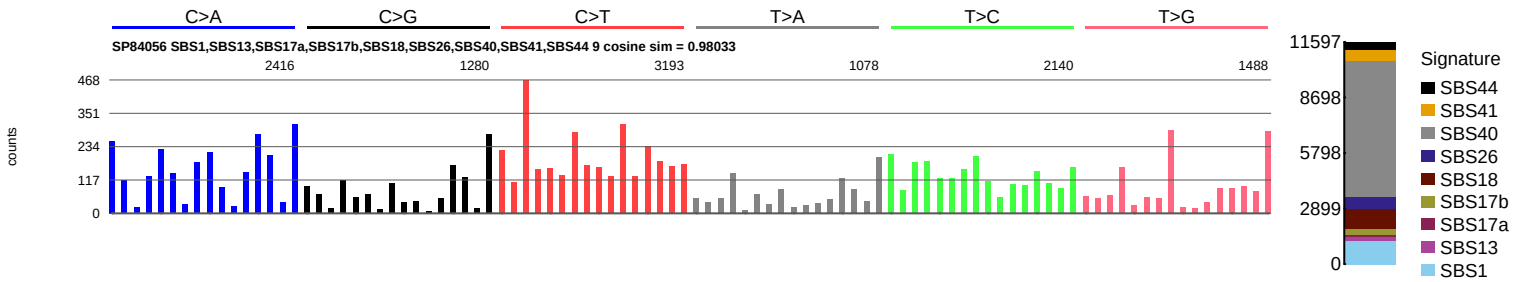

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98041
2441
1429
3190
1032
2119
1386
counts
480
360
240
120
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

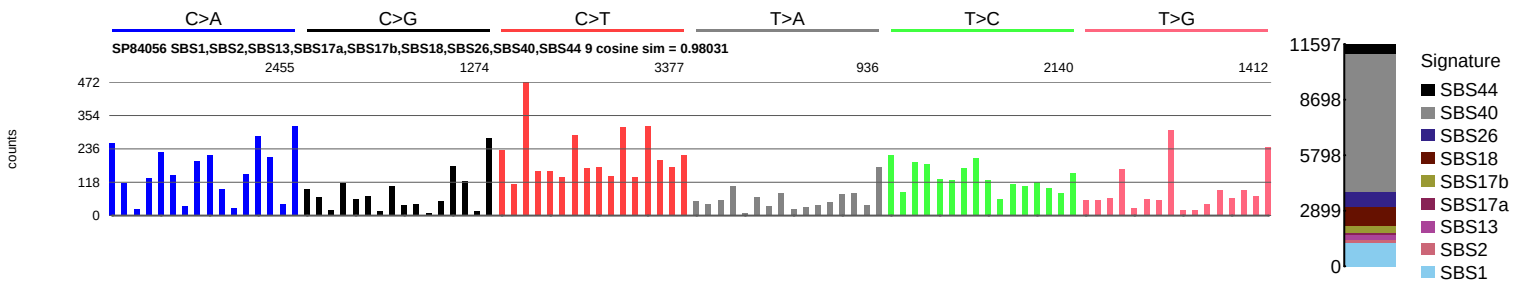

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9804
2423
1298
3410
1084
1883
1496
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

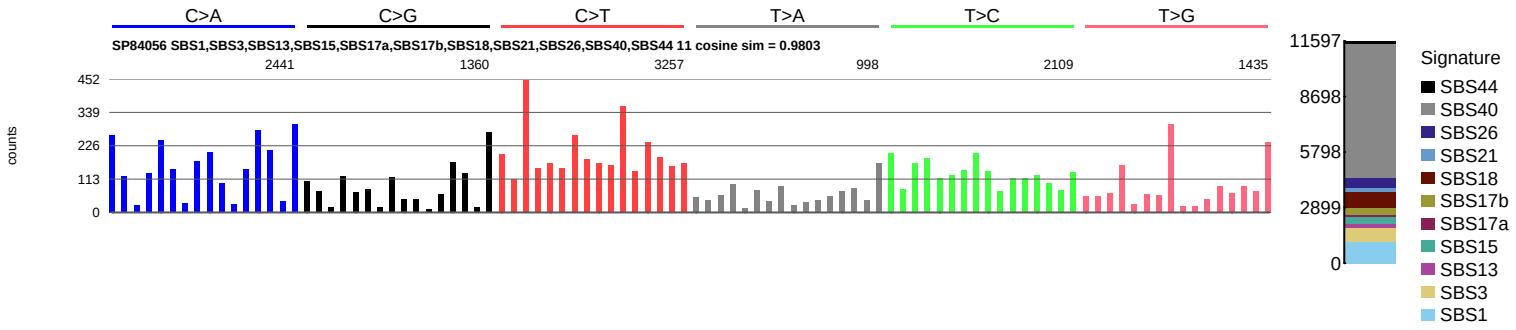

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98039
2393
1407
3479
999
1926
1393
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

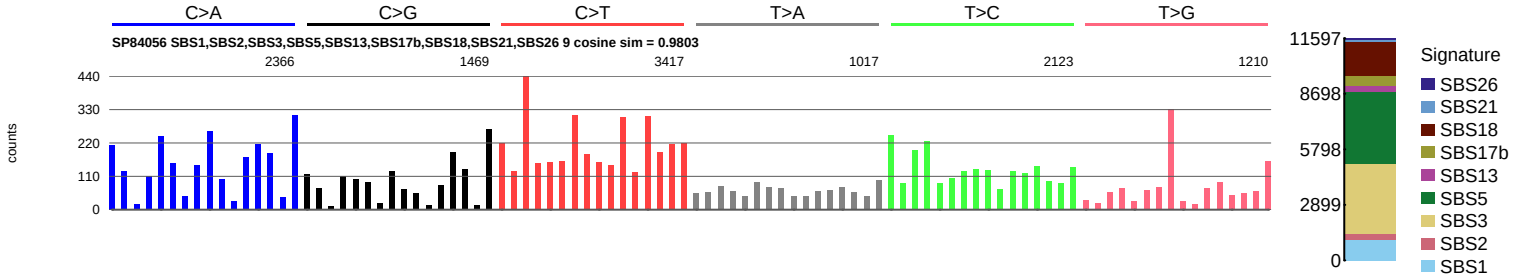

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98038
2392
1241
3182
1163
2111
1507
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

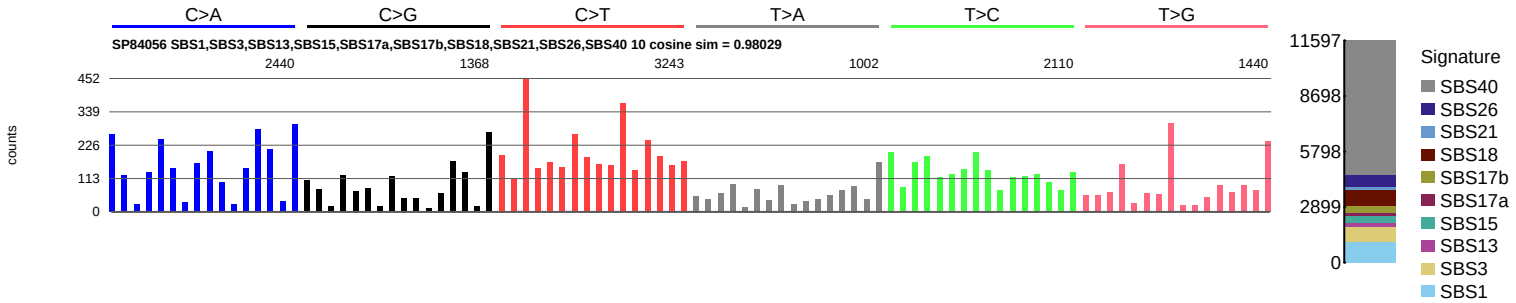

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98037
2364
1470
3428
1018
2108
1209
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98033
2416
1280
3193
1078
2140
1488
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98031
2455
1274
3377
936
2140
1412
counts
472
354
236
118
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9803
2441
1360
3257
998
2109
1435
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.9803
2366
1469
3417
1017
2123
1210
counts
440
330
220
110
0
11597
8698
5798
2899
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98029
2440
1368
3243
1002
2110
1440
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
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SBS26
SBS21
SBS18
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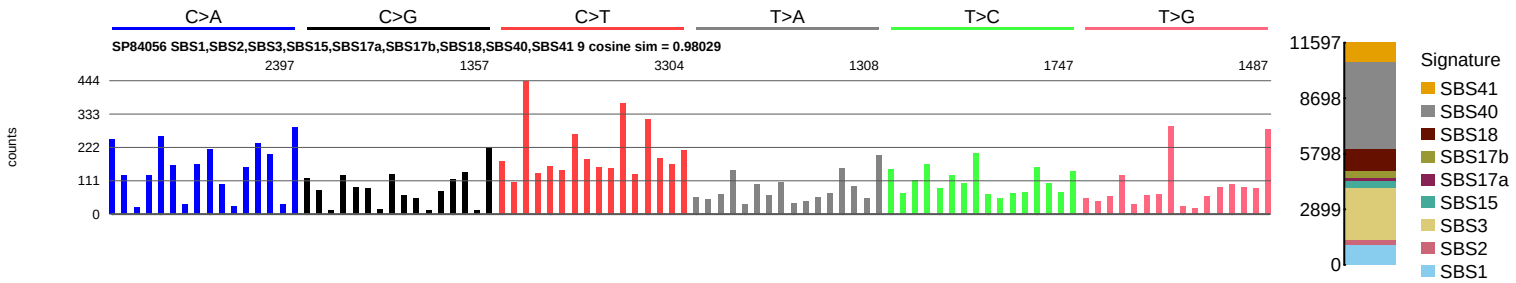
SBS17a
SBS15
SBS13
SBS3
SBS1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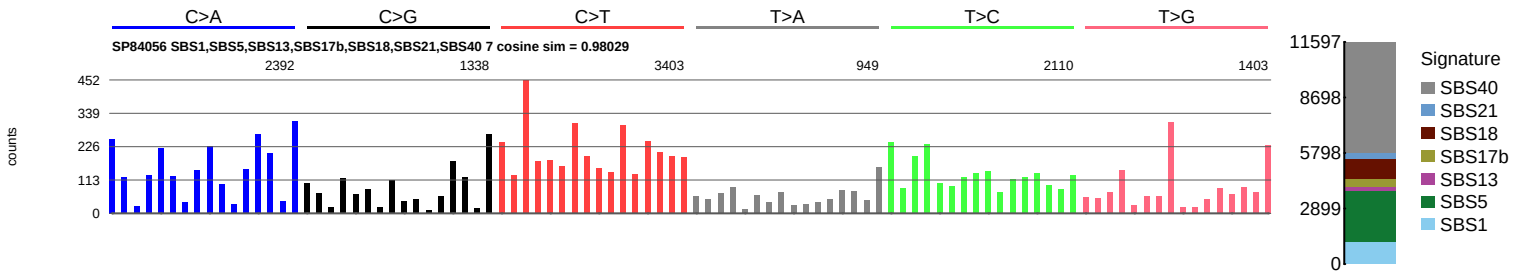
SP84056 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98029
2397
1357
3304
1308
1747
1487
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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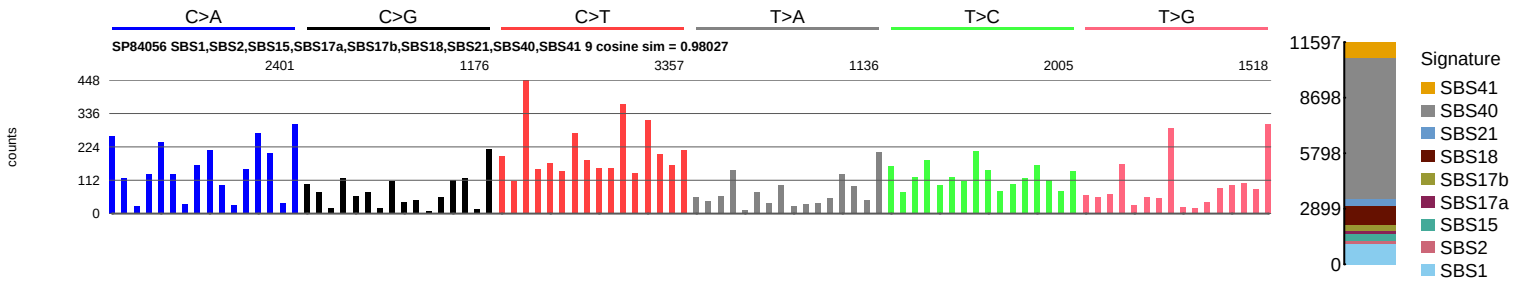
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98029
2392
1338
3403
949
2110
1403
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1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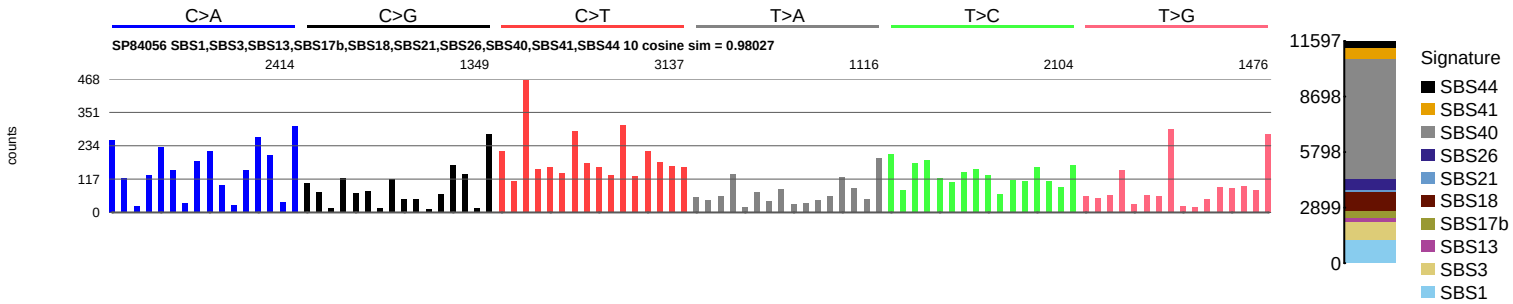
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98027
2401
1176
3357
1136
2005
1518
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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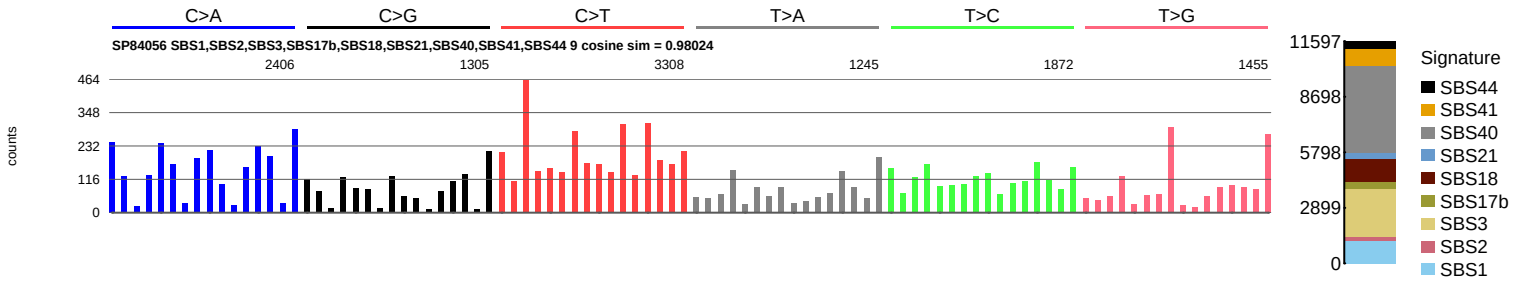
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98027
2414
1349
3137
1116
2104
1476
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS1


C>A
C>G
C>T
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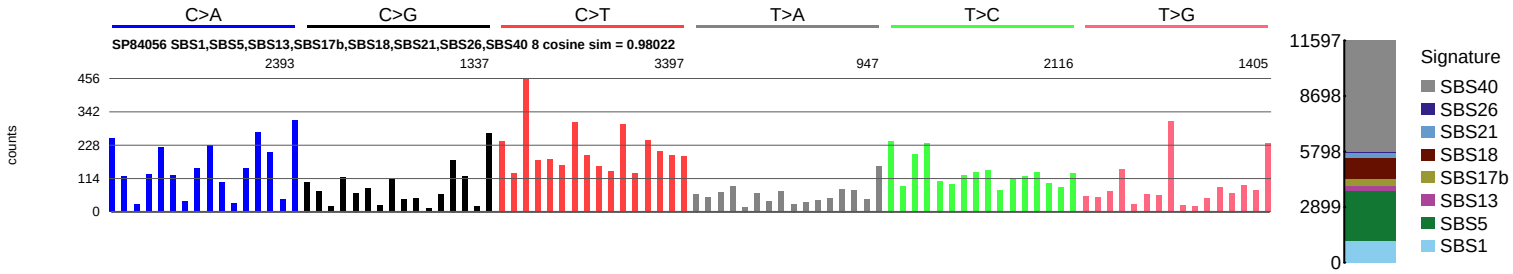
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98024
2406
1305
3308
1245
1872
1455
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1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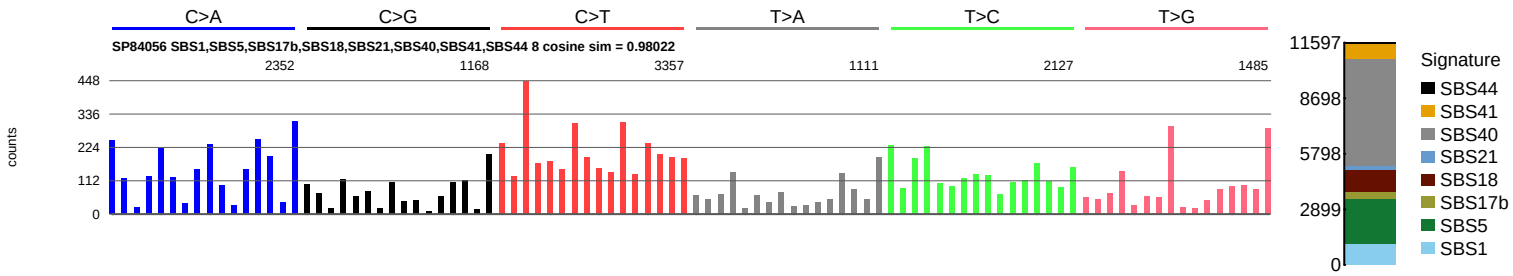
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98022
2393
1337
3397
947
2116
1405
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1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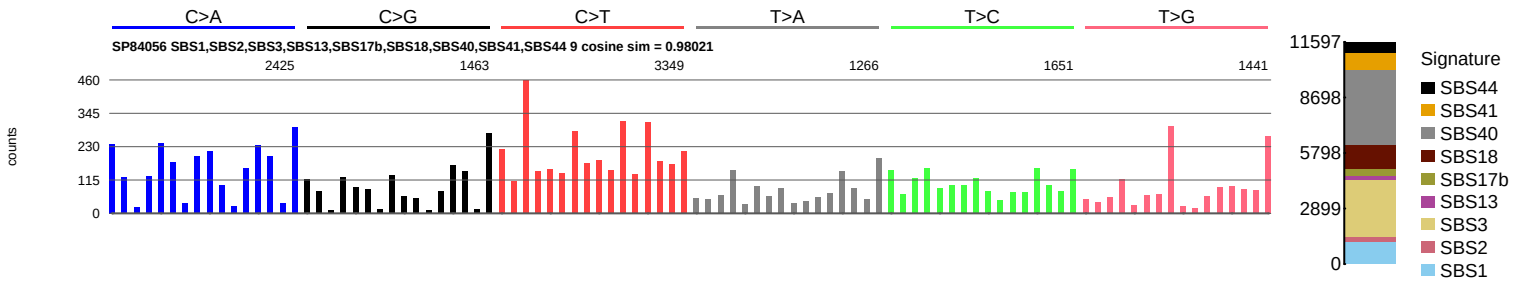
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98022
2352
1168
3357
1111
2127
1485
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1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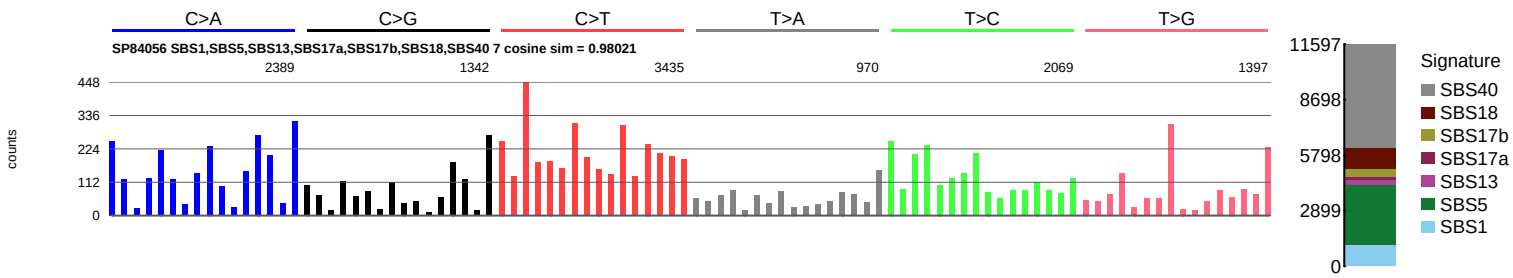
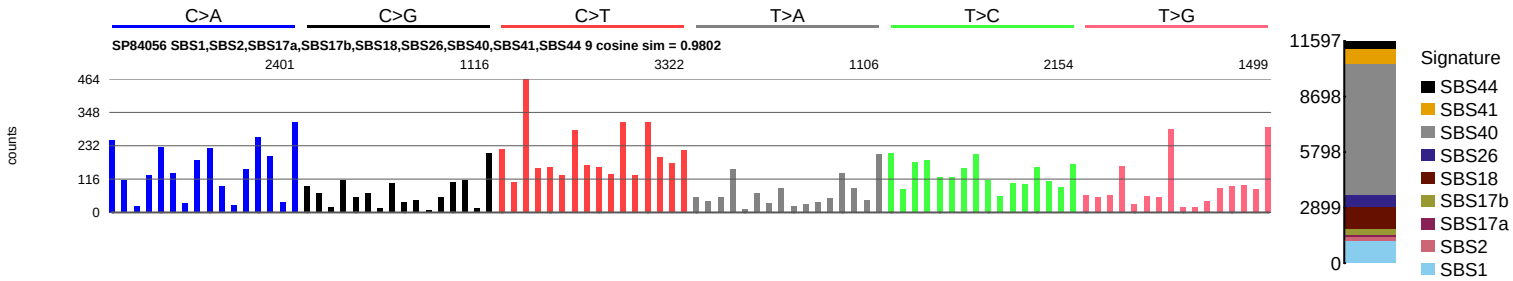
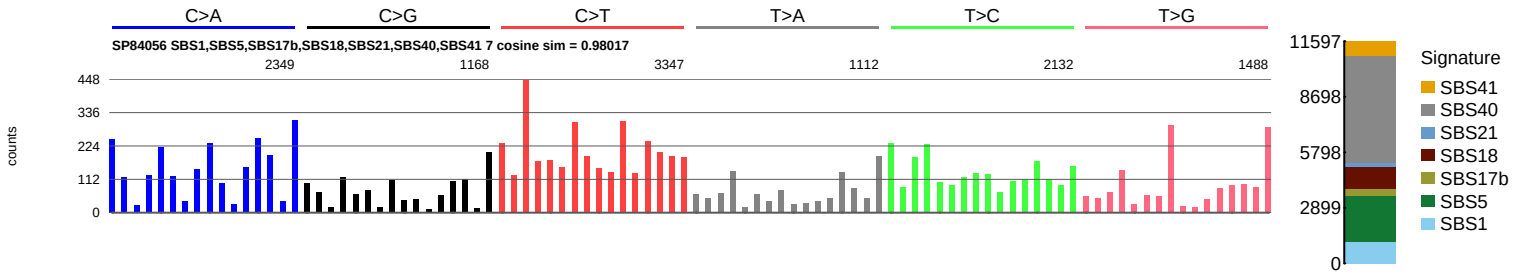
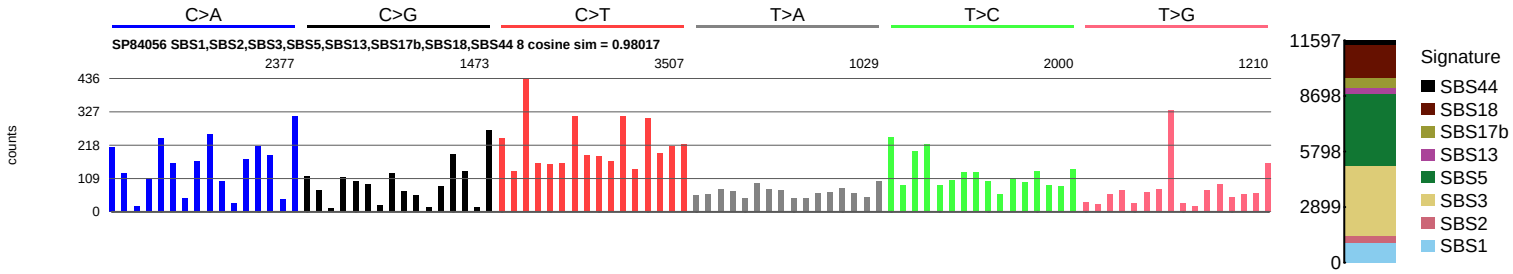
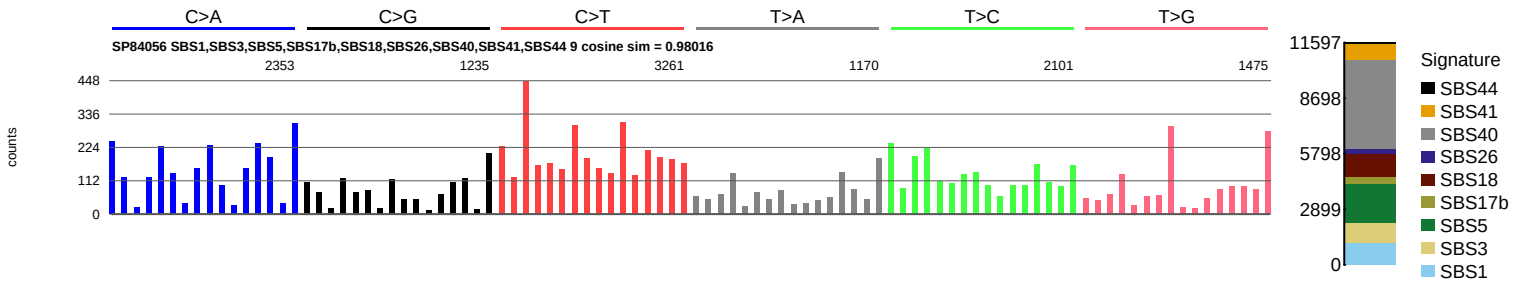
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98021
2425
1463
3349
1266
1651
1441
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

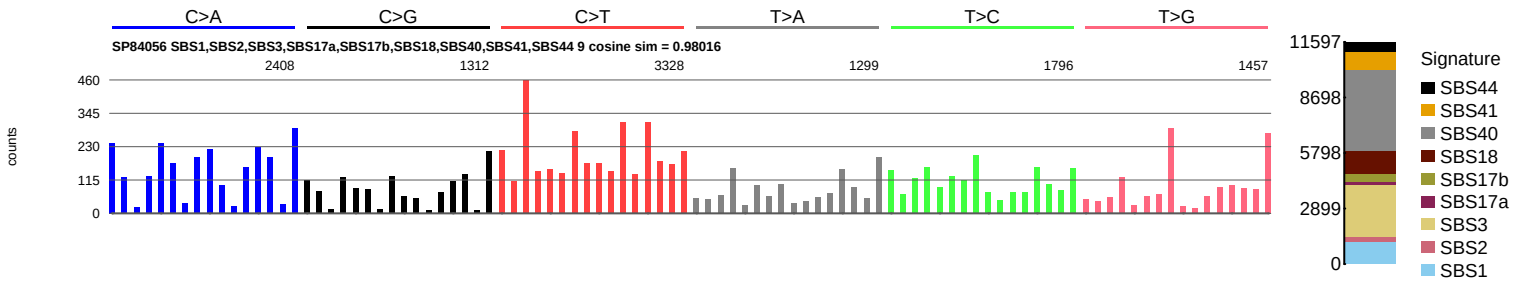

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98021
2389
1342
3435
970
2069
1397
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

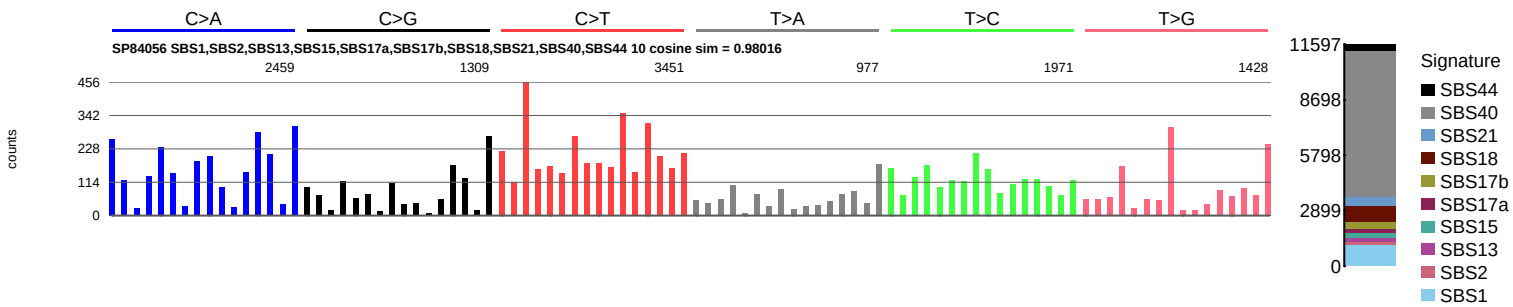

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9802
2401
1116
3322
1106
2154
1499
counts
464
348
232
116
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

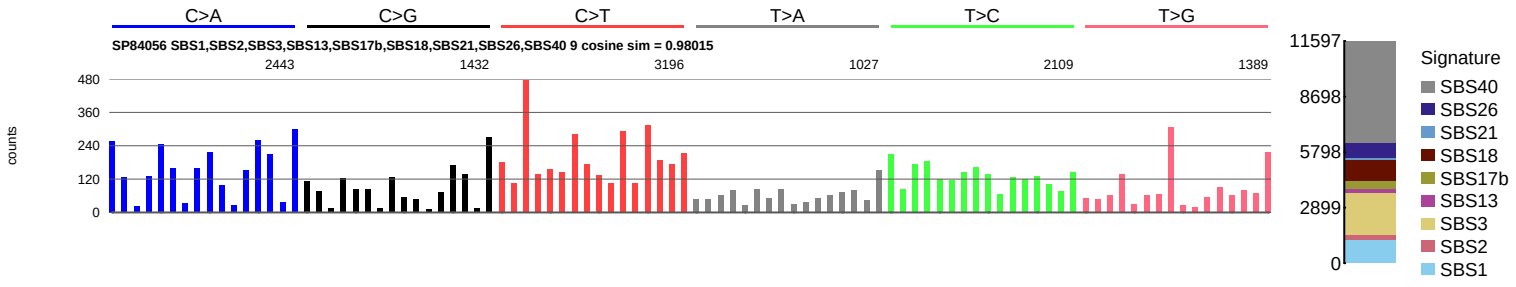

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98017
2349
1168
3347
1112
2132
1488
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1

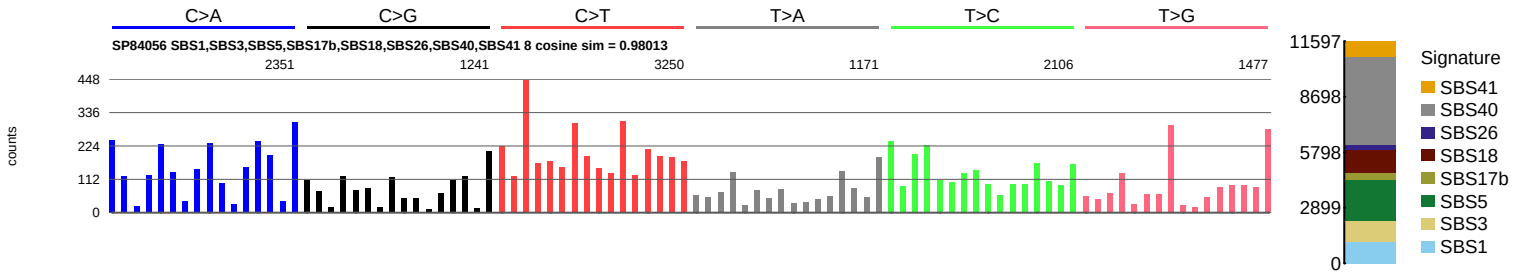

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98017
2377
1473
3507
1029
2000
1210
counts
436
327
218
109
0
11597
8698
5798
2899
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98016
2353
1235
3261
1170
2101
1475
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98016
2408
1312
3328
1299
1796
1457
counts
460
345
230
115
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98016
2459
1309
3451
977
1971
1428
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98015
2443
1432
3196
1027
2109
1389
counts
480
360
240
120
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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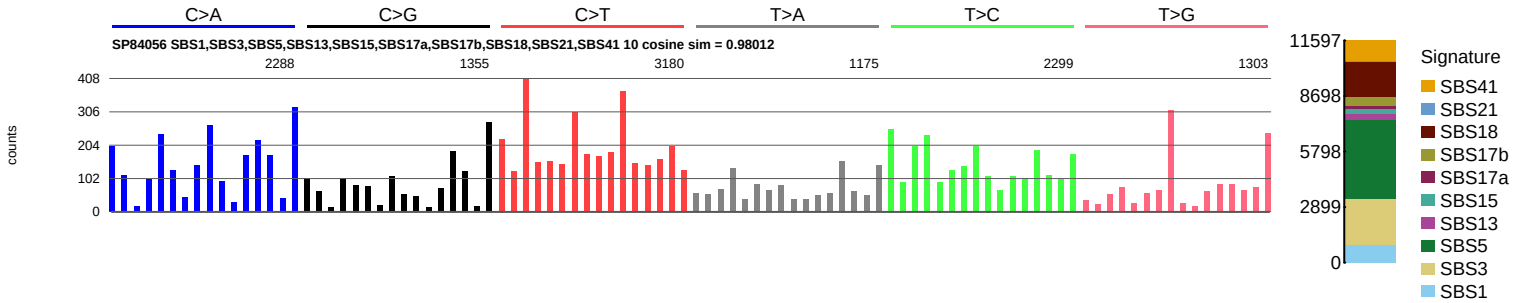
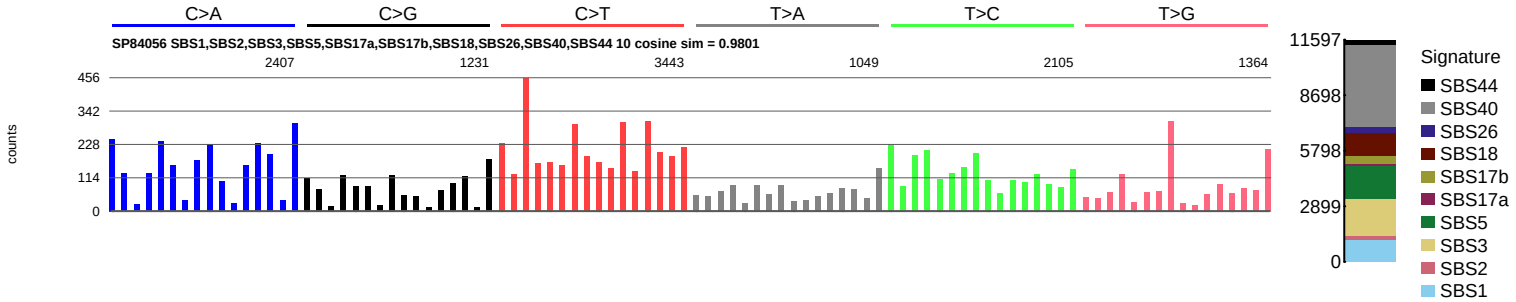
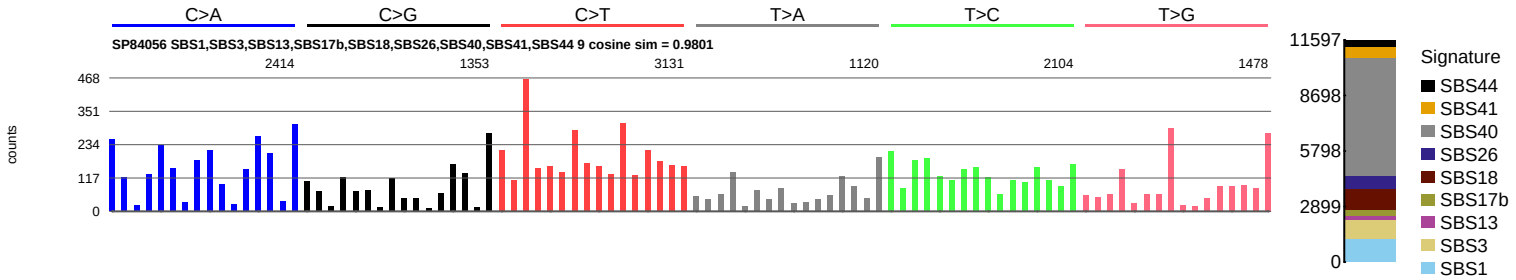
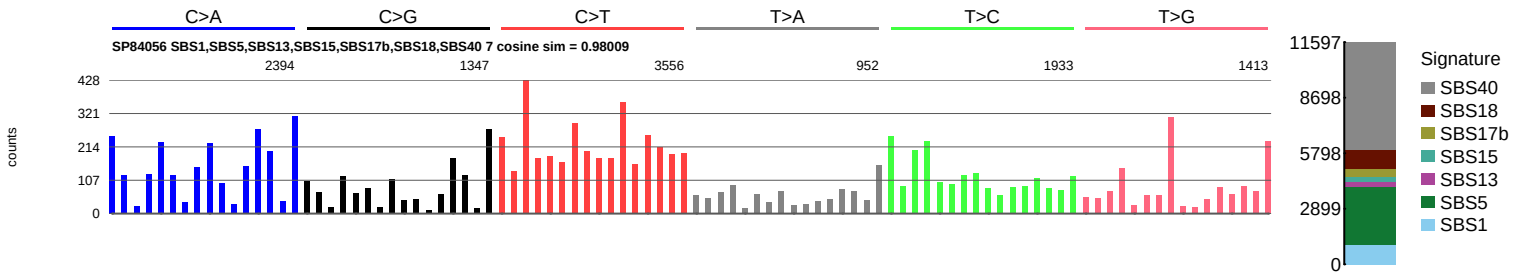
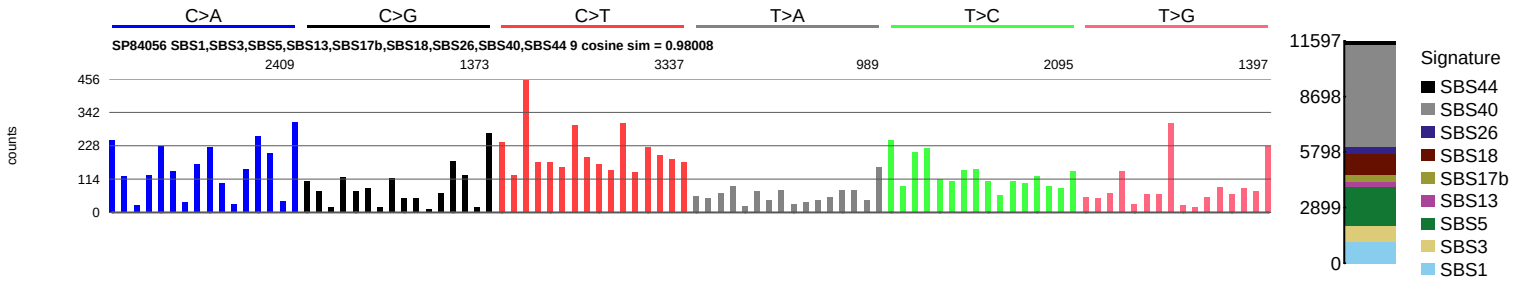
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98013
2351
1241
3250
1171
2106
1477
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS1

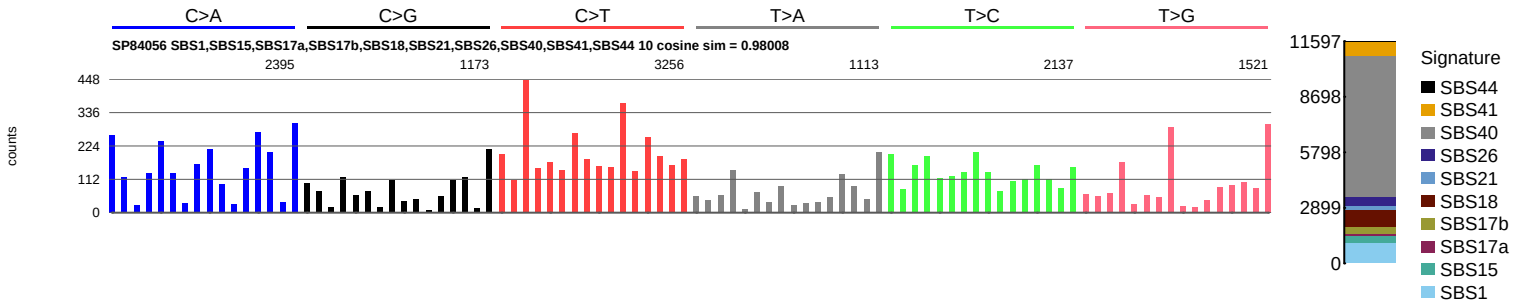

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98012
2288
1355
3180
1175
2299
1303
counts
408
306
204
102
0
11597
8698
5798
2899
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

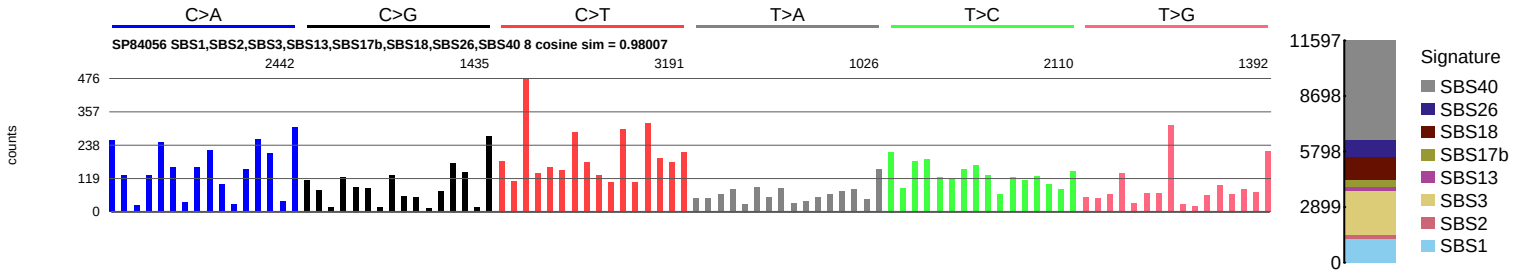

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9801
2407
1231
3443
1049
2105
1364
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9801
2414
1353
3131
1120
2104
1478
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98009
2394
1347
3556
952
1933
1413
counts
428
321
214
107
0
11597
8698
5798
2899
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98008
2409
1373
3337
989
2095
1397
counts
456
342
228
114
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98008
2395
1173
3256
1113
2137
1521
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98007
2442
1435
3191
1026
2110
1392
counts
476
357
238
119
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS18
SBS17b
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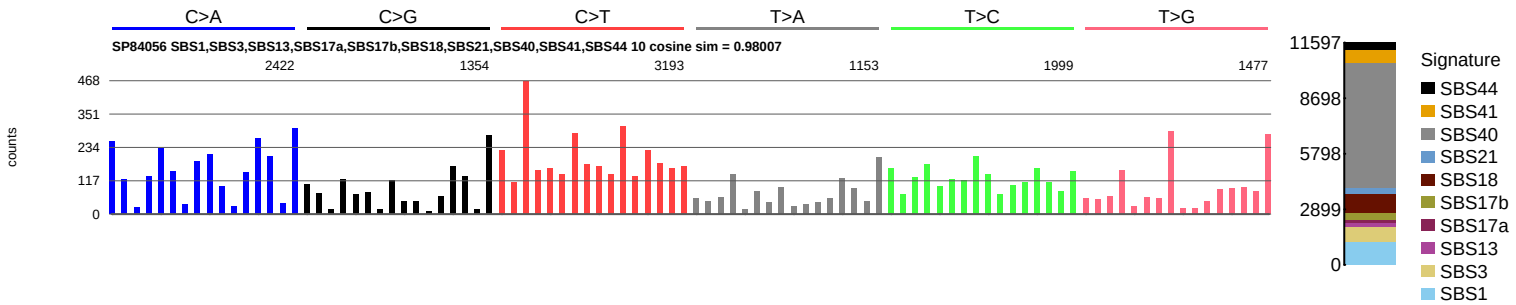
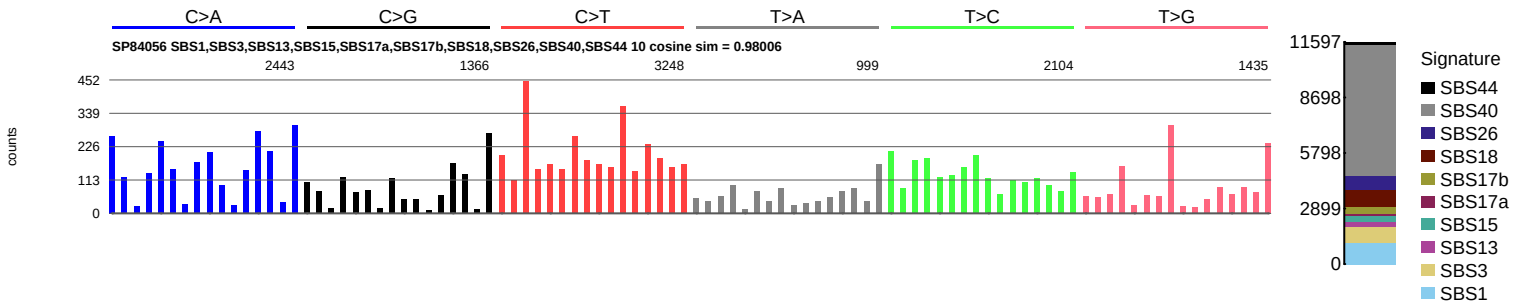
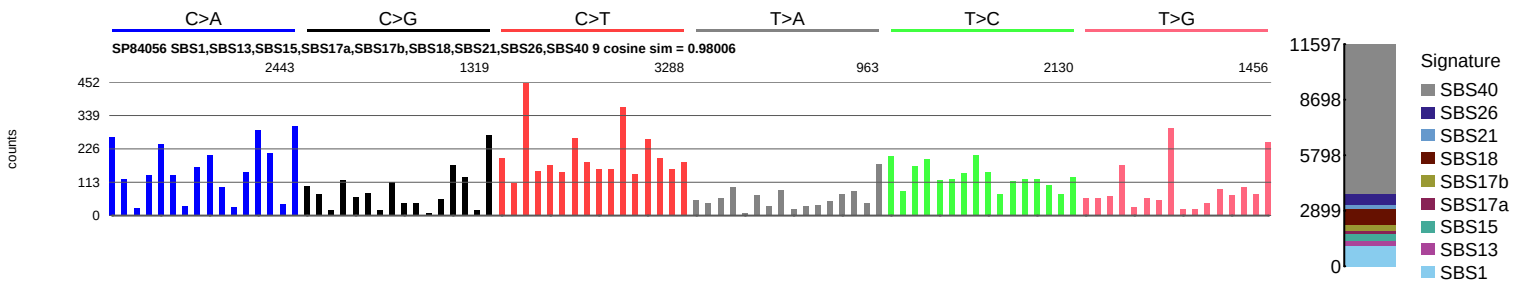
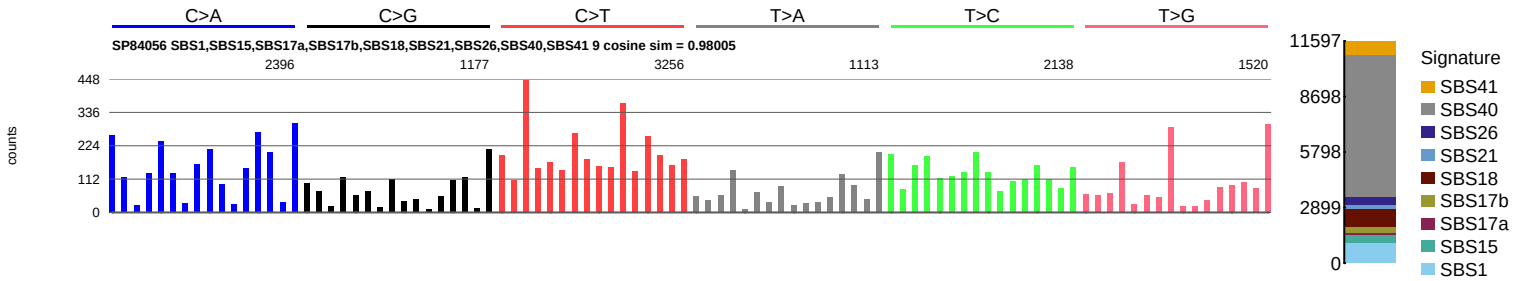
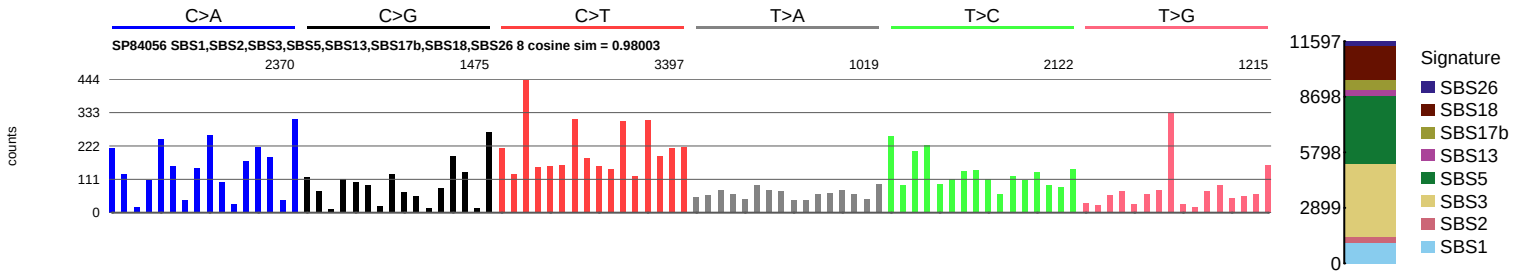
SBS3
SBS2
SBS1

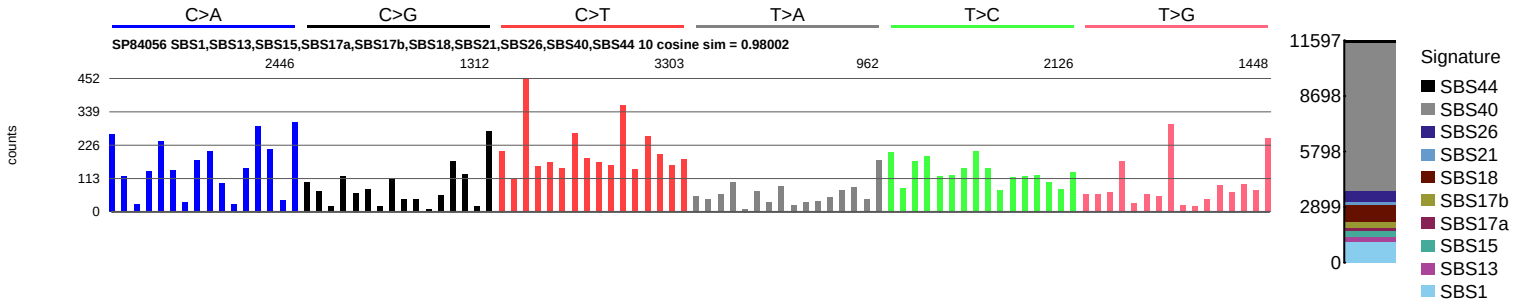

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98007
2422
1354
3193
1153
1999
1477
counts
468
351
234
117
0
11597
8698
5798
2899
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

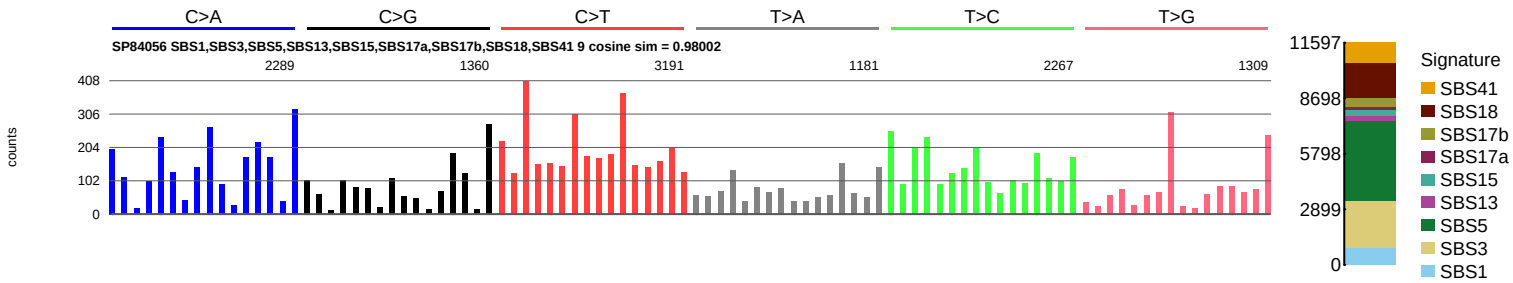

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98006
2443
1366
3248
999
2104
1435
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

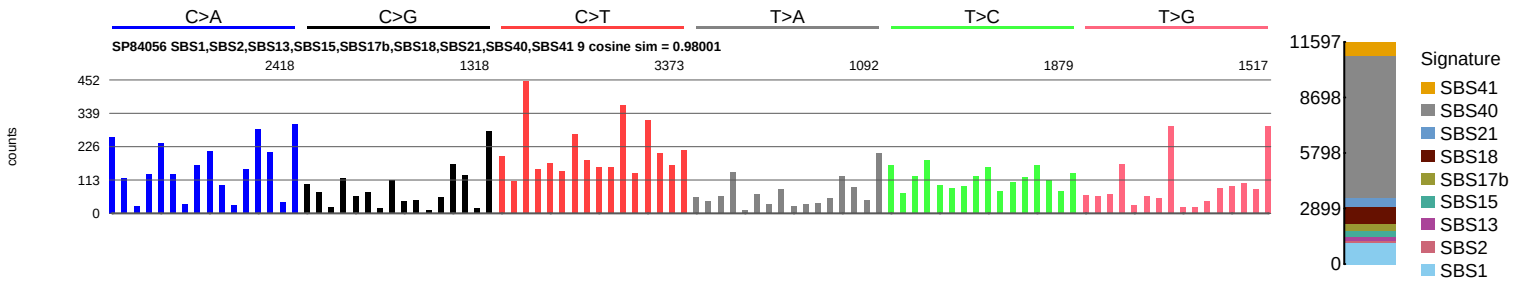

C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98006
2443
1319
3288
963
2130
1456
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98005
2396
1177
3256
1113
2138
1520
counts
448
336
224
112
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26 8 cosine sim = 0.98003
2370
1475
3397
1019
2122
1215
counts
444
333
222
111
0
11597
8698
5798
2899
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98002
2446
1312
3303
962
2126
1448
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98002
2289
1360
3191
1181
2267
1309
counts
408
306
204
102
0
11597
8698
5798
2899
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84056 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98001
2418
1318
3373
1092
1879
1517
counts
452
339
226
113
0
11597
8698
5798
2899
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1